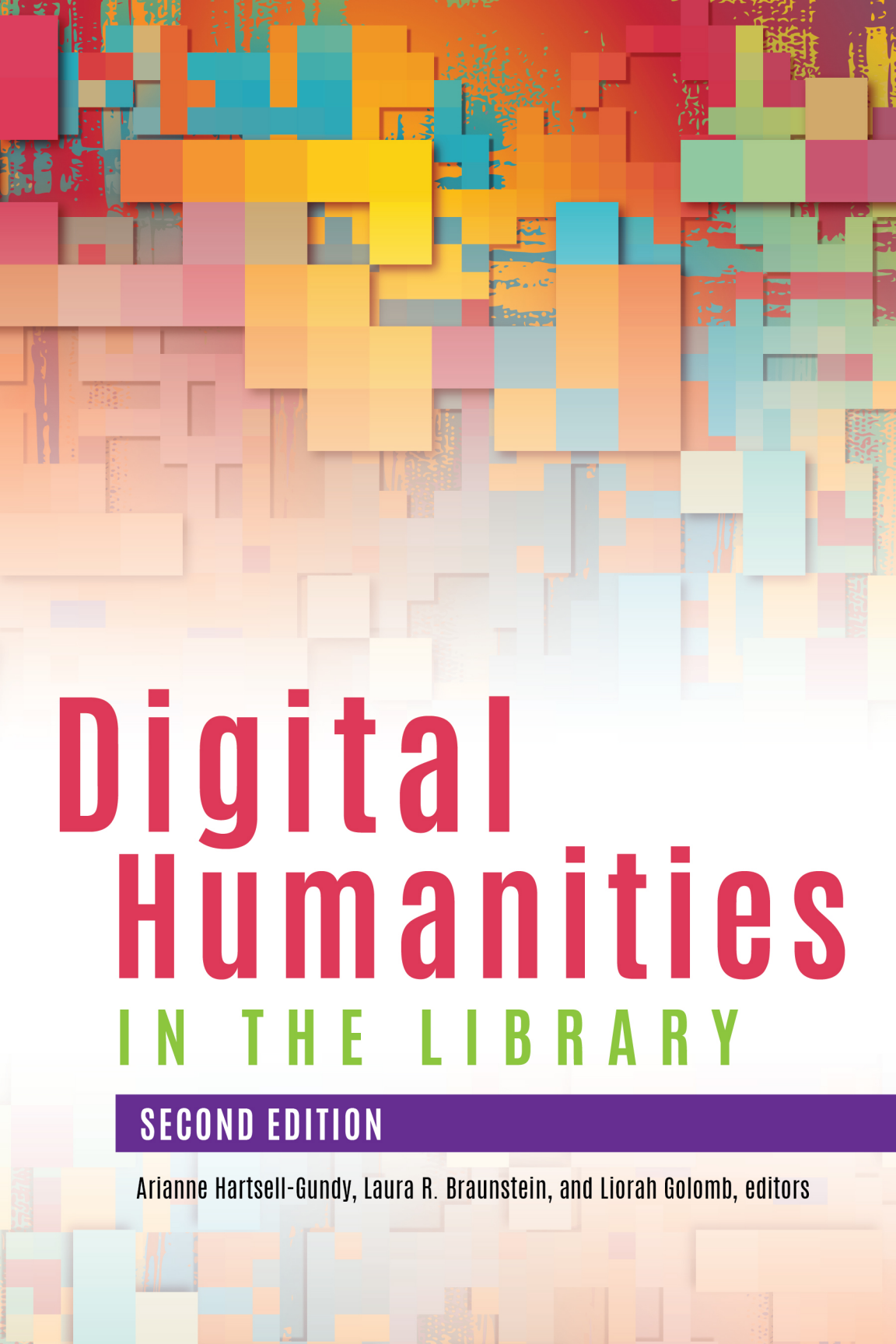

Digital
Humanities
IN THE LIBRARY
SECOND EDITION
Arianne Hartsell-Gundy, Laura R. Braunstein, and Liorah Golomb, editors

Digital Humanities

IN THE LIBRARY

SECOND EDITION

Editors

Arianne Hartsell-Gundy,
Laura Braunstein,
Liorah Golomb

Association of College and Research Libraries
A division of the American Library Association
Chicago, Illinois 2024

The paper used in this publication meets the minimum requirements of American National Standard for Information Sciences–Permanence of Paper for Printed Library Materials, ANSI Z39.48-1992. ∞

Library of Congress Control Number: 2024934618

Copyright ©2024 by the Association of College and Research Libraries.

All rights reserved except those which may be granted by Sections 107 and 108 of the Copyright Revision Act of 1976.

Printed in the United States of America.

28 27 26 25 24 5 4 3 2 1

This book is dedicated to the memory of David D. Oberhelman, an amazing librarian and wonderful colleague.

Contents

Foreword ix
Paige Morgan

Introduction xiii

Part 1. Theory

Chapter One 3
Distant Reading, Computational Stylistics, and Corpus Linguistics:
The Critical Theory of Digital Humanities for Literature Subject Librarians
David D. Oberhelman

Chapter Two 15
What Do Librarians Need to Know about Quantitative Methods in Digital Humanities?
Heather Froehlich

Chapter Three 29
Relearning Digital Humanities Librarianship
John Russell

Appendix 3A. "Introduction to Digital Humanities for Librarians" Syllabus 35

Appendix 3B. Digital Humanities and Information Literacy Workshop 40

Chapter Four 43
Centering our Values:
A Framework for Digital Humanities in the Library
Pamella R. Lach

Appendix 4A. DH@SDSU Shared/Community Values 58

Chapter Five 69

Inclusive Design:
A Method and Craft of Transforming Digital Humanities with User Experience

A. Miller

Part 2. Practice

Chapter Six 91

Into the Mix:
Subject Liaison Librarians and Digital Humanities Research

Mark Dahlquist, Katie Gibson, and Jenny Presnell

Appendix 6A. Practical Suggestions for Subject Librarians 102

Chapter Seven 107

Collaboration and CoTeaching:
Librarians Teaching Digital Humanities in the Classroom

Brian Rosenblum, Frances Devlin, Tami Albin, and Wade Garrison

Appendix 7A. Digital Humanities Assignment 121

Appendix 7B. Course Development Grants 123

Chapter Eight 127

Construction and Disruption:
Building Communities of Practice, Queering Subject Liaisons

Caro Pinto

Chapter Nine 139

A Checklist for Digital Humanities Scholarship:

Elizabeth Lorang and Kathleen A. Johnson

Chapter Ten 157

Moderating a Meaningful DH Conversation for Graduate Students in the Humanities

Kathleen A. Langan and Ilse Schweitzer VanDonkelaar

Chapter Eleven.....175

Spaces, Skills, and Synthesis

Anu Vedantham and Dot Porter

Chapter Twelve.....195

Sustaining the Digital Liberal Arts:
Institutional Challenges in Looking Beyond Grant Funding

Ginny Moran, Aisling Quigley, Brooke Schmolke, and Louann Terveer

Chapter Thirteen.....211

The Digital Humanities Summer Scholars Program:
Opportunities for Change During the COVID-19 Pandemic and Providing a Space for Diversity, Equity, and Inclusion in the Library

Angela Perkins

Chapter Fourteen.....233

The Restorative Potential of Library-Based DH:
Reconnecting Learning Communities in the Age of COVID-19

Emma Annette Wilson, Russell Hugh McConnell, Johanna Pang, and Maria Katsulos

Appendix 14A. Resources.....250

Appendix 14B. Sample DH Assignment.....252

Appendix. Tools and Resources Referenced in this Book.....263

About the Authors.....269

Foreword

IN EARLY 2013, William Pannapacker advised readers of the *Chronicle of Higher Education* (especially advocates and proponents of digital humanities) to "stop calling it 'digital humanities,'" explaining that "it seems more inclusive to call it digital liberal arts (DLA) with the assumption that we'll lose the "digital" within a few years, once practices that seem innovative today become the ordinary methods of scholarship."[1] Pannapacker wasn't the only researcher I can remember gesturing toward the point when the digital humanities (DH) would just be "the humanities." We are almost ten years on from that column, and while DH support is far more widespread than it was, it does not feel like we are on the verge of "losing the 'digital'" anytime soon, despite the efforts of many—including the practitioners collected in this volume. Why is that? And what might it indicate about the role that libraries and librarians play in supporting digital scholarship?

Answering these questions is partly about whether the needs around digital humanities support are settling enough that libraries can effectively plan for them. It looks to me as though at least some genres of DH seem to be on the way to stabilization: text mining is supported by free tools like Voyant and AntConc (neither of which require command-line knowledge or robust hardware), and has generated critiques in mainstream journals that are mounting complex arguments, rather than suggesting that the method is wholly invalid.[2] The genre of the digital exhibit/online collection/multimodal website has more variety in what it entails, rather than a set format—but there are tools like Omeka and WordPress that support this work and are fairly accessible; and there is enough activity to support more than two years of Jennifer Guiliano and Roopika Risam's *Reviews in DH* journal, which publishes on a monthly basis. But growing stabilization, in so many ways, means that more labor and support is needed, rather than less, because there is more interest in doing more complex work. An aspect of thinking about DH that doesn't often make it into print, but that certainly comes up in conversations at conferences, is the specter of success: what if our pilots succeed, and suddenly everyone (or even just a majority of researchers) wants to do this kind of work? Institutions have vastly different financial

CC BY NC

situations, depending on their locations, their endowments, their status as private or public, to name only a few influences. This means that while one university might have a whole team devoted to digital scholarship, another might have a single librarian, or a single librarian whose role it is to support DH as well as having a liaison role to support one or more academic subject programs. Even while DH support has become comparatively more available in the past ten years, I suspect the percent of colleges and universities with explicit support for digital scholarship is still vanishingly small. There is still relatively little effort on the part of academic institutions to share resources with each other, though I hope we will see the emergence of more proposals like A.J. Boston's "Read and Let Read," which offers a new way for higher education institutions to pay for public access to paywalled articles.[3] And sharing access to paywalled articles seems simple in comparison to sharing access to premium tools and sophisticated project management support.

However, this sort of complexity is not new to DH. We often reference the digital humanities as collaborative, highlighting collaborations as though they are evidence of our generosity, but it would be more accurate to call the digital humanities interdependent. A researcher, or even a team of faculty and graduate student researchers, relies on the library to provide access to tools (Omeka, Tableau, ArcGIS) and training, not just in how to use the tools, but how to develop and maintain a project and increase the likelihood that other scholars will find the project and use it. The expectations universities place on faculty to produce research, and on graduate students to produce research and pursue careers, do not make it possible for them to succeed without this support. Libraries, and library staff, are dependent on these faculty and student needs to show that they are contributing to research in ways that can not be replaced by a new subscription or license to an e-resource from Sage or Wiley. Both the student/faculty teams and the librarians depend on support from library and/or campus IT teams, and those teams' willingness to provide server space and to support installation of software and underlying dependencies (such as Java); they are all also dependent upon IT's willingness to allow installation of software that may not meet complex security requirements at a time when universities are especially concerned about ransomware and data security. These are not the highly visible hot button issues that DH depends on, but they take up space in meetings and generate lengthy email chains. Frequently, they can determine whether a project moves forward regardless of how interesting or valid its underlying research questions are.

It might seem counterintuitive to call attention to a vulnerability—in a society that prizes self-reliance, why call attention to the fact that we could not accomplish our work without help from others? But as we work to continue developing digital humanities in ways that make it more accessible and equitable, we should be looking for opportunities to be more transparent about the conditions that influence this work, and that make projects possible, or that create obstacles to be surmounted. This new edition of *Digital Humanities in the Library* is full of practitioners who are exploring vulnerabilities as they emerge—and finding ways for libraries and library staff to respond—and in doing so, shape not just digital humanities, but the work of libraries more broadly. They are illustrating why the digital humanities still require careful planning, and why they cannot easily be lumped into the humanities in general.

The biggest decisions about library resources are often made at the highest levels of library administrations—or even higher, at the level of campus administration. Centers and spaces are funded, labs are built, and salary lines are added or removed as new units are formed or reorganized. Our practitioners in front-facing roles can be some of the very best people to help library administrations know what sort of change is needed, which is just one of the reasons that we need to continue gathering their narratives about what they are seeing, and how they are responding to it. Assessment is never simple, and in digital humanities contexts, it's especially challenging, because the individual data is distinctive enough that it cannot easily be broken down into straightforward quantifiable indicators. There *are* commonalities, but they emerge and become clear only through careful reflection and dialogue. Any digital humanist who can enthuse about data can also tell you that computers alone cannot do the work—you need the thoughtfulness of a human expert to find the way forward. This collection can help us do that—and can help bring new library practitioners of DH into the community as we go on.

—Paige Morgan
Digital Publishing & Copyright Librarian
Head of Digital Scholarship and Publishing
University of Delaware Libraries, Museums and Press

Notes

1. Pannapacker, "Stop Calling It 'Digital Humanities,'" *The Chronicle of Higher Education*, February 18, 2013, https://www.chronicle.com/article/stop-calling-it-digital-humanities/.
2. Bode, "The Equivalence of 'Close' and 'Distant' Reading; or, Toward a New Object for Data-Rich Literary History," *Modern Language Quarterly*, (2017) 78 (1):

77–106, https://doi.org/10.1215/00267929-3699787; Da, "The Computational Case against Computational Literary Studies," *Critical Inquiry*, (2019) 45 (3), https://doi.org/10.1086/702594.

3. Boston, "Adding Equity to Transformative Agreements and Journal Subscriptions–The Read & Let Read Model," LSE Impact Blog. December 8, 2022, https://blogs.lse.ac.uk/impactofsocialsciences/2022/12/08/adding-equity-to-transformative-agreements-and-journal-subscriptions-the-read-let-read-model/.

Bibliography

Bode, Katherine. "The Equivalence of 'Close' and 'Distant' Reading; or, Toward a New Object for Data-Rich Literary History." *Modern Language Quarterly*. (2017) 78 (1): 77–106. https://doi.org/10.1215/00267929-3699787.

Boston, Arthur J. "Adding Equity to Transformative Agreements and Journal Subscriptions–The Read & Let Read Model." LSE Impact Blog. December 8, 2022. https://blogs.lse.ac.uk/impactofsocialsciences/2022/12/08/adding-equity-to-transformative-agreements-and-journal-subscriptions-the-read-let-read-model/.

Da, Nan Z. "The Computational Case against Computational Literary Studies." *Critical Inquiry*. (2019) 45 (3). https://doi.org/10.1086/702594.

Pannapacker, William. "Stop Calling It 'Digital Humanities.'" *The Chronicle of Higher Education*. February 18, 2013. https://www.chronicle.com/article/stop-calling-it-digital-humanities/.

Introduction

THE GENESIS of the first edition of *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists* was born from a need to explore more fully how libraries could support and engage with digital humanities and understand the role of the subject specialist in this work. There were many questions about how subject specialists could build effective relationships with functional specialists within the libraries and the larger university, learn relevant technology skills, effectively teach digital humanities related skills and concepts to students, and provide infrastructure and spaces as needed. Subject librarians are still navigating the roles they play in the creation, maintenance, and preservation of digital projects, but there are new questions to consider.

The field of digital humanities has expanded and evolved, and libraries and academic institutions have had to respond to global and national events that have had large scale impact, such as the Covid-19 pandemic and the racial reckoning of the summer of 2020. New scholarship in recent years, such as *Data Feminism* by Catherine D'Ignazio and Lauren F. Klein, *The Digital Black Atlantic* by Roopika Risam and Kelly Baker Josephs, and *Global Debates in the Digital Humanities* by Domenico Fiormonte, Sukanta Chaudhuri, and Paola Ricaurte, point to productive new ways to interrogate and expand the meaning of digital humanities. There has also been a movement in digital humanities in the past few years from what we might call "project-based DH" to "infrastructural" DH; thus, we aim to especially investigate what kinds of labor arrangements, in the library and in universities more broadly, make DH scholarship and production sustainable.

This expanded second edition includes reprints of key chapters from the first edition, along with new chapters that explore some of these new issues. New areas of focus in this edition include diversity, inclusion, and equity; issues of labor, precarity, and infrastructure; scholarly communication and taxonomies of credit; long-term sustainability, such as that advocated for by the Maintainers movement; and library digital humanities in the age of institutional austerity.

CC BY NC

We have chosen to divide this revised edition into two parts—Theory and Practice—to give the reader a grounding in the ideas and approaches influencing the work being done around digital humanities and then to follow that grounding with examples of how practitioners are supporting digital humanities in their everyday work. We also acknowledge that the boundaries between theory and practice can be permeable, and so some chapters disrupt that distinction. Just as in the first edition, our chapter authors come from a variety of institutions and work in many areas related to digital humanities, including subject specialists, digital scholarship center directors, UX experts, special collections librarians, and technical specialists. Though challenges and opportunities still exist, we have chosen to drop the subheading for this revised second edition to reflect how the conversations around digital humanities are evolving and expanding in libraries.

Part one begins with the reprint "Distant Reading, Computational Stylistics, and Corpus Linguistics: The Critical Theory of Digital Humanities for Literature Subject Librarians" by David D. Oberhelman, which addresses the need for subject librarians to understand the theoretical implications of DH. David passed away in January 2018; this edition is dedicated to his memory, in honor of his friendship and his many contributions to humanities librarianship.

Part one continues with "What Do Librarians Need to Know about Quantitative Methods in Digital Humanities?" by Heather Froehlich (University of Arizona). In her essay, she shows how librarians offering support to quantitative digital humanities projects are meeting humanities researchers where they are, which is often without a lot of underlying technical or mathematical knowledge. In chapter three, "Relearning Digital Humanities," John Russell (Pennsylvania State University) provides a chapter reflecting on his experience teaching an introduction to digital humanities course for librarians and how his understanding of the field has evolved over time. Pam Lach (San Diego State University) calls for academic libraries to be values-centered, as much as user-centered, when engaging in digital humanities work in chapter four, "Centering our Values: A Framework for Digital Humanities in the Library." This chapter demonstrates how being intentional about the values that guide our work helps build equitable partnerships, meet community needs, manage institutional challenges, engage in critical digital inquiry, and strive toward ethical praxis. The final chapter of this section, by A. Miller (Middle Tennessee State University), "Inclusive Design: A Method and Craft of Transforming DH with UX," examines how understanding user experience and design thinking can yield better insights into DH users' behaviors, wants,

and needs. By applying this human-centered thinking, DH researchers can better create and connect with their users.

Part two includes nine chapters. Chapter six is a revised chapter from our first edition by Mark Dahlquist, Katie Gibson, Jenny Presnell (Miami University). In "Into the Mix: Subject Liaisons Librarians and Digital Humanities Research," they examine the relationship of subject specialists to researchers and digital humanities centers. The authors identify the needs and skills of both researchers and digital centers and suggest a variety of roles that subject specialists can play in the development of a digital project. Chapter seven, "Collaboration and Co-Teaching: Librarians Teaching Digital Humanities in the Classroom" by Brian Rosenblum, Frances Devlin, Tami Albin, and Wade Garrison (University of Kansas Libraries) is a reprint that describes efforts by librarians with subject, instruction, and digital scholarship expertise to provide instruction and training in DH to graduate students and faculty. In chapter eight, "Construction and Disruption: Building Communities of Practice, Queering Subject Librarians," Caro Pinto examines in a reprint the possibilities of organizational change and the roles liaisons, archivists, and metadata specialists play. Chapter nine is a reprint from Elizabeth Lorang and Kathleen A. Johnson called "A Checklist for Digital Humanities Scholarship." It provides practical advice on beginning a project, describes librarian participation in University of Nebraska-Lincoln's Center for Digital Research in the Humanities, and includes valuable practical points to consider throughout a project's life cycle. Chapter ten's "Moderating a Meaningful DH Conversation for Graduate Students in the Humanities," by Kathleen A. Langan and Ilse Schweitzer VanDonkelaar (Western Michigan University), introduces us to the possibilities of using digital humanities to work with graduate students. This reprint is a case study of subject librarians helping to train graduate students in digital humanities, thus increasing both librarians' and students' professional skills. Chapter eleven is the final reprint of the volume. In "Spaces, Skills, and Synthesis," Anu Vedantham and Dot Porter describe how the creation of library spaces can facilitate collaboration in digital humanities. The authors describe the evolution of support for DH work at University of Pennsylvania through the library's adaptation of spaces, facilities, technical support, and faculty advising. Chapter twelve from Ginny Moran, Aisling Quigley, Brooke Schmolke, and Louann Terveer (Macalester College) is "Sustaining the Digital Liberal Arts: Institutional Challenges in Looking Beyond Grant Funding." It focuses on the institutional challenges of creating a sustainable support and innovation model for digital humanities, or, as they define it more broadly on their campus, digital liberal arts. In chapter

thirteen, "The Digital Humanities Summer Scholars (DHSS) Program: Opportunities for Change During the COVID-19 Pandemic and Providing a Space for DEI in the Library," Angela Perkins (Lafayette College) discusses her experience of converting her library's Digital Humanities Summer Scholars (DHSS) Program from an intimate, in-person process where she guided students through a journey in six weeks of original DH-based research to a completely virtual experience, in a little over two months, in response to the COVID-19 pandemic. Emma Annette Wilson, Russell Hugh McConnell, Johanna Pang, and Maria Katsulos (University of Alabama) finish the volume with "The Restorative Potential of Library-Based DH: Reconnecting Learning Communities in The Age of COVID-19," which examines the realities of teaching DH in the library in the age of COVID-19.

As many academic libraries transition from a subject-specialist model to a functional specialist model, some may ask if the discipline-specific librarian role will become obsolete. Our work on revising this book resoundingly suggests that subject librarians are not, and will not become, obsolete. There are opportunities for new cross-institutional collaborations and new approaches to the work being done. We hope that this new, revised edition continues valuable conversations about the future of our profession.

Digital Humanities

IN THE LIBRARY

PART 1. THEORY

CHAPTER ONE

Distant Reading, Computational Stylistics, and Corpus Linguistics:

The Critical Theory of Digital Humanities for Literature Subject Librarians

David D. Oberhelman

FOR LITERATURE librarians who frequently assist or even collaborate with faculty, researchers, IT professionals, and students on digital humanities (DH) projects, understanding some of the tacit or explicit literary theoretical assumptions involved in the practice of DH can help them better serve their constituencies and also equip them to serve as a bridge between the DH community and the more traditional practitioners of literary criticism who may not fully embrace, or may even oppose, the scientific leanings of their DH colleagues.* I will therefore provide a brief overview of the theory behind the technique of DH in the case of literature—the use of "distant reading" as opposed to "close reading" of literary texts as well as the use of computational linguistics, stylistics, and corpora studies—to help literature subject librarians grasp some of the implications of DH for the literary

* Literary scholars' opposition to the use of science and technology is not new, as Helle Porsdam has observed by looking at the current debates over DH in the academy in terms of the academic debate in Great Britain in the 1950s and 1960s between the chemist and novelist C. P. Snow and the literary critic F. R. Leavis over the "two cultures" of science and the humanities (Helle Porsdam, "'Too Much 'Digital,' Too Little 'Humanities'? An Attempt to Explain Why Humanities Scholars Are Reluctant Converts to Digital Humanities," Arcadia project report, 2011, DSpace@Cambridge, www.dspace.cam.ac.uk/handle/1810/244642).

critical tradition and learn how DH practitioners approach literary texts in ways that are fundamentally different from those employed by many other critics.* Armed with this knowledge, subject librarians may be able to play a role in integrating DH into the traditional study of literature.

Attempts to define DH in the modern academy tend to focus on DH as more of a series of practices and methods that utilize computer technology to examine objects studied by humanities such as literary texts.† As Kathleen Fitzpatrick, Director of Scholarly Communication for the Modern Language Association, has defined DH, it is "a nexus of fields within which scholars use computing technologies to investigate the kinds of questions that are traditional to the humanities, or, as is more true of my own work, ask traditional kinds of humanities-oriented questions about computing technologies."[1] The *Digital Humanities Manifesto 2.0* issued by several UCLA researchers maintains that DH is "an array of convergent practices" in which print is no longer the privileged medium of knowledge and "digital tools, techniques, and media have altered the production and dissemination of knowledge in the arts, humanities, and social sciences."[2] DH emerged out of what had been loosely called humanities computing starting in the 1940s and 1950s and extending through the early TEI encoding initiatives of the Internet age[3] and as such is usually described in terms of its practical applications.‡ Central to most definitions of DH is the basic tenet that DH is a dynamic process or methodology of using digital tools to study the humanities more than a

---

* There have not been many studies of how literature librarians can benefit from learning the basics of literary theory (apart from overall studies of the value of understanding the information needs of literary scholars), but Stephanie M. Mathson has argued that understanding and applying literary theory such as Wolfgang Iser's reader-response theory to librarian/patron service encounters can be beneficial (Stephanie M. Mathson, "Engaging Readers, Engaging Texts: An Exploration of How Librarians Can Use Reader Response Theory to Better Serve Our Patrons," *Library Philosophy and Practice* [August 2011]: 589, http://digitalcommons.unl.edu/libphilprac/589).

† For this study I will confine my exploration of DH projects to those dealing with literary topics that are textual in nature. There are many that examine the geography of regions associated with authors, historical content, economic data in literary texts, and many other aspects, but here I am concerned with projects that chiefly focus on the language and textual analysis of literary works.

‡ The loose and somewhat informal process by which DH received its name is indicative of its practice-based and experimental foundations; it was a term suggested by John Unsworth in coming up with a title for the 2004 Blackwell *Companion to Digital Humanities* because the editors and publisher wanted to avoid *humanities computing* or *digitized humanities* (Matthew G. Kirschenbaum, "What Is Digital Humanities and What's It Doing in English Departments?" *ADE Bulletin*, no. 150 [2010]: 55–61).

systematic theory of how to study the humanities or, in the case of literature, how to study literary texts. Yet to understand the role of DH in literature departments, it is necessary to explore the relationship between DH and the critical theories that inform how literary scholars approach texts. Although DH may appear to be more practice than theory, it is closely related to new theoretical trends with which literature librarians should become familiar so they can better work with their patrons.

Literary Theory in the Twentieth Century: A Brief Overview

Literary critical theory, which investigates the aesthetic, philosophical, political, and cultural assumptions underlying various techniques of reading and interpreting literary texts, has been an important aspect of study in English, foreign language, and comparative literature programs in American universities. Indeed, scholars and students of literature, including literature librarians who work with them, have since the mid–twentieth century had to contend with a multitude of theoretical schools of thought or the so-called "isms" (feminism, Marxism, structuralism/poststructuralism, etc.),[§] and debates over theories of reading have been featured prominently in courses, conference papers, and publications.

Many academics regard DH as an alternative to the theoretical, qualitative approaches to literature, one which brings the humanities more in line with the quantitative methods of other fields such as the social sciences and STEM disciplines. As Patricia Cohen has commented in a 2010 *New York Times* arts column, digital humanists often argue that data and big data sets should replace the competing political or philosophical systems as the key to approaching the humanities.[4] Literary theory, some DH proponents contend, is too narrow and inward-focused in its perspective on texts and therefore misses the big picture that DH methods can sketch by means of technology. Despite the insistence of DH practitioners that they are more concerned with practice than with theorizing about it, there are some fundamental theoretical presuppositions that inform the methods employed in the DH field, particularly in its literary application—presuppositions that in some way represent a theoretical paradigm shift of sorts for many critics that may

§ For concise, helpful overviews of history and concepts underlying the different schools of literary theory, see Terry Eagleton, *Literary Theory: An Introduction* (Minneapolis: University of Minnesota Press, 1983) and Paul H. Fry, *Theory of Literature*, Open Yale Course Series (New Haven, CT: Yale University Press, 2012).

explain why there is some hesitation about DH among humanities scholars (other than its obvious use of computers and scientific principles).

Central to the theory behind DH is the somewhat controversial theory the Italian-born critic Franco Moretti has termed "distant reading." In his theoretical texts and DH projects at the Stanford Literary Lab he co-founded with Matthew Jockers, Moretti articulates and models a framework for studying literature that breaks from over fifty years of theoretical tradition of close scrutiny of single texts or a delimited canon of literary works. I will offer a short summary of the theories based on close reading first, and then examine some of Moretti's theoretical pronouncements to illustrate how the "big data" approach DH takes to literature stems from a movement away from the privileging of close analysis of a single or limited group of texts.

From Close Reading to Distant Reading

Literary studies in the early decades of the twentieth century often focused on literary history, situating authors and texts in movements and establishing the canon of British, American, and other literatures. By the middle of the century, though, a group of Anglo-American literary critics such as I. A. Richards, William Empson, John Crowe Ransom, and Cleanth Brooks theorized that the literary text should be studied as an object of art with laws governing its aesthetic integrity. This formalist school, which looked at how literary meaning is conveyed by or is created by the form of the text (such as the poem), became known as the New Criticism, and it ushered in a theory of close reading that would, despite changes and the advent of French philosophical and linguistic theory in the latter decades of the century, become the basic theoretical model that several generations of critics and students in American literature classes learned. Perhaps the best example of their theoretical notions appears in Brooks's 1947 collection *The Well Wrought Urn*,[5] a volume containing essays that give detailed close readings of poems by John Donne, John Keats (whose "Ode on a Grecian Urn" gives the book its title), Shakespeare, and others. The New Critical technique of close reading, which outlines the carefully balanced tensions in meaning held together by symmetry in the poetic form, reflects the theoretical premise that the critic should give a detailed, almost microscopic analysis of the literary text to find its meaning. Subsequent generations of American literary critics, many of whom were trained in close reading, argued that the New Critics divorced literary works from their historical or cultural contexts and thereby ignored

important layers of meaning, but the basic belief that texts should be closely examined in terms of their own structure or logic persisted.

The appearance of French schools of literary criticism such as the structuralism of Roland Barthes, a school derived from the linguistic study of signs or semiotics of Ferdinand de Saussure and the anthropological study of myth as a system by Claude Lévi-Strauss, marked a point at which literary critics began to look at larger assemblages of texts, but close reading continued to be the dominant form of critical engagement with texts. In the 1970s and 1980s, deconstruction, the poststructuralist theory of reading texts based on the philosophical work of Jacques Derrida, had critics read for gaps, aporias (irresolvable contradictions), or traces of other concepts within their seemingly well-wrought tension of opposites that unhinge the integrity of the textual construct. For deconstructionist critics there is no "outside the text," meaning all ideas and concepts are bound to the faulty vehicle of (written) language that conveys them.* Although Derridan theory does not look just at one book—his notion of the text covers many writings beyond the covers of one book—its central premise is that the reader must turn a critical gaze to one instance of text at a time and read it closely. Although deconstruction represented a great departure from the New Critics' view that literary works are perfect, balanced objects of art, it, as Jonathan Culler and others have noted, takes some of the New Critics' approaches to reading to their ultimate conclusion,[6] and both theories are predicated upon an intense, microscopic attention to, or perhaps dissection of, a particular text or texts. In that sense, it is not surprising that American literary critics in particular picked up many of the deconstructive processes of close reading much like they had embraced the New Criticism decades earlier.

Other schools of literary theory emerged in that period that regarded close reading of texts to be too constraining, divorcing them from the culture and the political institutions that created them. Marxist theory, feminist theory, cultural materialism, and other more engaged schools also appeared, some using psychoanalysis, political philosophy, or the discourse analysis of philosopher Michel Foucault, but in the American university setting, the close reading of texts in the literary canon, even if that canon was changed, reinvented, or expanded, was still dominant. Indeed, many of these schools, or "isms" as they are often somewhat dismissively labeled, still rely upon a

* Derrida's famous aphorism in the original French is *il n'y a pas d'hors-texte*—literally, there is no outside-text, no meaning that transcends the imperfect text that expresses it (Jacques Derrida, *De la grammatologie* [Paris: Les Éditions de Minuit, 1967], 227).

close encounter with specific texts, often read in terms of different, interdisciplinary perspectives.

The dominance of close reading has been challenged in the twenty-first century, however, by the work of the DH circles that study vast assemblages of texts rather than home in on a single text. Franco Moretti, a Marxist scholar of the European novel born and trained in Italy prior to a distinguished academic career at Columbia and then Stanford University, has become the critical theorist most associated with the rise of DH and is the figure who best articulates the radical change in the approach to literary texts that the practice of DH entails. His concept of "distant reading" stands in marked opposition to the tradition of close reading, and his championing of a quantitative approach to texts that, in essence, does not involve having an individual critic read them at all gives a theoretical foundation to the disparate projects and data analyses that fall under the general moniker of DH.

Moretti's six books and numerous articles on different national traditions of the novel, the history of publishing, and related fields all reflect his conviction that literary scholars should not study individual great novels or even a narrow canon of novels, which is inevitably selective and omits the vast majority of the artistic output of any given literary time period, but rather should utilize computers to study large numbers of novels to gain a broader understanding of the genre and produce a better literary history. His seminal essay "Conjectures on a World Literature," reprinted in Moretti's award-winning 2013 collection *Distant Reading*, outlines the broad strokes of his critical theory of reading.[7] In it, Moretti questions whether scholars of literature should continue the vain task of trying to become well-read, for they will never be able to read but a miniscule, highly selective portion of the literary output of a given culture. Close reading may even be, by extension, an exercise in futility for Moretti, for picking one text to analyze or even deconstruct means hundreds of thousands have been left unread, so general conclusions about the concerns or thematic patterns in a literary age cannot legitimately be drawn. He changes the critical lens by rejecting the microscopic close reading of a single text and, instead, regards very large constellations of many literary works, which he describes a "planetary system," requiring a macroscopic approach.[8] The sheer number of texts—planets with their satellites and other bodies—defies the ability of any one critic to read them all, so Moretti calls for a new approach using alternate means to process and analyze the vast quantity of now-forgotten or unread books produced in a period or national literature. Readers must look at the big picture to discern how this system operates, and thus distant reading

gives a broader perspective on literary interpretation. As a result, Moretti hearkens back to the early theoretical aim to map out the historical contours of *Weltliteratur*, "world literature," recalling the style of literary history of the pre–New Critical era, albeit using the scientific principles of the twenty-first century to survey and categorize numbers of books that would have been impossible for the earlier literary historians to read and synthesize.

Moretti's 2005 book *Graphs, Maps, and Trees* explains how his theory of distant reading can construct a large-scale literary history from models derived from quantitative history (the graphs), geography (maps), and evolutionary theory (trees).[9] Here he shows his debt to the French Marxist *Annales* school of historiography, which focuses not on extraordinary events but on the *longue durée*, or long duration, of gradual historical change coming from the bottom up rather than from the great figures down. Here, as in other works on the novel in particular, Moretti uses computer data and digitized texts to construct graphical charts out of data from several thousand texts to trace massive patterns in book history, in the development of genres, titles, characters, and other features of novels in various national literature, all rendered in graphs, charts, and other graphical forms. He has extended this work into the DH projects he has worked on for the Stanford Literary Lab (http://litlab.stanford.edu) such as *Network Theory, Plot Analysis*, a comparison of dramatic patterns from over 300 Western plays, and "The Emotions of London," a project to create an emotional map of London derived from characters' emotional states in an array of novels from the 1700s to the 1800s.[10] Here the importation of social science methods of data manipulation into the literary critical domain becomes evident.

Moretti's vision of distant reading and his approach to literary study has been criticized for its faith in a scientific method, most notably by fellow Marxist scholar Christopher Prendergast.[11] The noted critical commentator Alan Liu has pointed out in a critique of Moretti and DH that even if we accept distant reading as a theoretical paradigm, digital humanists still need to formulate a "close reading 2.0" or "a method of micro-analysis in an era of big humanities"[12]—some provision for approaching individual texts while still regarding them as parts of big planetary systems. Other theorists, nevertheless, have embraced Moretti's big data concept of literary history and find his approach liberating, an invitation to move beyond the fear of not having read "everything" (or even enough) that so frequently haunts literary scholars, and consequently allows them to seek out new ways to approach the enormity of past and present literary output. The DH critic Stephen Ramsay, noting how critics today are faced with the old dilemma, so many books, so

little time, posits what he dubs the "hermeneutics of screwing around" to characterize how the great proliferation of online texts in the Internet age allows readers to engage in interpretive activity by just dipping in, following hyperlink trails, and exploring the various pathways through the mazes of webpages.[13] Moretti and his followers thus lay a foundation for DH practice that signals a departure from the literary critical theories based upon close reading and other theories that depend upon narrower canons of works rather than the enormous body of poems, plays, novels, and other literary output, most of which is now lost or forgotten.

Stylistics and Corpora

The other major theoretical foundation upon which DH rests is one that is largely derived not from literary studies per se, but from the importation of quantitative, especially computational linguistic models into the realm of literature. Language study and linguistics have long played a key role in literary criticism—and in late-twentieth-century theoretical circles, Saussure and other anthropological linguists were commonly read—but much of the data analysis that goes into the study of grammar, stylistics, and other aspects of linguistics has not been traditionally used for examining literary texts. Stylistics, the study of literary style (for author studies, period studies, and other applications), is one area in which some of the technical procedures of linguistics have been brought to bear upon creative texts and has gained greater importance with the rise of DH and its ability to let critics study the language of a text in a nonlinear fashion (as they must be in reading a printed book). In his 1986 book *Linguistic Criticism*, the noted British linguist Roger Fowler called for the application of linguistic stylistic analysis, especially the quantitative analysis of elements of language, to literary texts, arguing that there is no inherent distinction between "ordinary" uses of language and literary usage.*

Linguistic studies of literary stylistics are frequently associated with analyses of a linguistic "corpus" or large, usually digital, collection of texts as data to be scrutinized. Douglas Biber and Randi Reppen (2012) outline how "corpus stylistics" makes use of computational methods and computer technology to subject literary texts, either works of one author or a larger

* Roger Fowler, *Linguistic Criticism* (Oxford: Oxford University Press, 1986). For an early application of linguistic theory to prose literary texts, see also Geoffrey Leech and Mick Short, *Style in Fiction: A Linguistic Introduction to English Fictional Prose,* English Language Series (London: Longman, 1981).

constellation of texts in a corpus, to lexical, grammatical, and other linguistic investigation.[14] DH projects frequently create corpora for study and thus ally themselves with the computational linguists, who can provide a vocabulary and a theoretical underpinning for their work on stylistics. DH studies utilizing corpus stylistics can concentrate upon single authors, looking at their texts as whole body, or in distant reading fashion tackle the stylistic categories of a big array of digital texts much like the immense corpora of natural language gathered by linguists. One representative DH project currently under development that brings together the various strands of linguistic theory is the Tolkien Corpus Project, in which Robin Reid, an English faculty member specializing in literary stylistics, and Christian Hempelmann, a computational linguist, are in the process of creating a corpus database of the works of J. R. R. Tolkien to subject to corpus stylistics analysis.† Such work will be able to extend the research on Tolkien's style by putting his texts into a new framework for analysis and thus using computational linguistics to explore his syntax, grammar, and other elements, parsing the vast number of words and sentences of Tolkien into units that can be scrutinized. In their collection *Digital Literary Studies*, David L. Hoover, Jonathan Culpepper, and Kiernan O'Hallaran describe their corpus stylistics projects on Shakespeare, Henry James, Wilkie Collins, and other, larger groupings of novelists.[15] They make the case that corpus stylistics is a linguistic theory that is, compared to other linguistic theories, easy to master and light on jargon; they even provide a glossary of terms at the end of the volume. These approaches demonstrate how computational linguistics and corpus stylistics can provide a theoretical support digital humanists can turn to in their mining of the textual data they have amassed.

Conclusion: Literature Librarians and Theory

Moretti's theories of distant reading and computational corpus linguistics both represent realignments in the critical approach to literary texts and a movement away from the concentrated attention to the interplay of language, its structures, and their breakdowns in a limited number of texts toward a

† Robin Anne Reid and Christian Hempelmann, "The Tolkien Corpus Project," posted April 8, 2013, on the Tolkien Scholarship Project website, http://earendel.net/?q=node/4 [site discontinued]. Reid has also published some detailed analyses of Tolkien's style using the principles of linguistic stylistics and is extending her work by the development of a true corpus of many of Tolkien's writings to facilitate a larger, big-data study of his stylistics (Robin Anne Reid, "Mythology and History: A Stylistic Analysis of *The Lord of the Rings*," *Style* 43, no. 4 [Winter 2009]: 517–38).

much grander playing field in which big quantities of textual data are now the focus of study. These theories come into play in various forms for digital humanists collecting and analyzing their own databases, assemblages of texts, or other electronic objects. Literature subject librarians are generally trained in the social science and computer-based data analysis techniques of the LIS field with its own rules for gathering and interpreting information sets, but they also pay close attention to bibliographic detail in working with texts. In that sense, subject librarians, who frequently must shift between library jargon or theories of patrons' information-seeking behavior and the language of scholars or students as they conceptualize their own research methods, are adept at code switching—to use the terms Thomas Bartscherer and Roderick Coover employ to characterize interdisciplinary dialogue in the DH world[16]—so they may be able to promote code switching among their patrons who have their own divergent theoretical approaches to literature.

Thus understanding some of the literary critical theory behind DH and how it embodies a different relationship between critic and text can be very beneficial to subject specialists, for they can help serve as intermediaries or even interpreters who stand between the DH scholars and practitioners and their world of planetary systems and big data and the other literary scholars trained in the "isms" and focused on the close reading of texts, seeking to find ways to bridge the divide between these scholars as they do with collection- and item-level bibliographic control. By knowing the respective theoretical universes from which their various constituencies come, subject librarians can help find ways to encourage the ongoing integration of DH in its many forms into the literature and language departments with which they work.

Notes

1. Kathleen Fitzpatrick, "The Humanities, Done Digitally," in *Debates in the Digital Humanities*, edited by Matthew K. Gold (Minneapolis: University of Minnesota Press, 2012), 12.
2. *Digital Humanities Manifesto 2.0*, posted June 22, 2009, on *Humanities Blast* (blog) by Todd Presner, 2, www.humanitiesblast.com/manifesto/Manifesto_V2.pdf.
3. Susan Hockey, "The History of Humanities Computing," in *A Companion to Digital Humanities*, edited by Susan Schreibman, Ray Siemens, and John Unsworth (Oxford: Blackwell, 2004), www.digitalhumanities.org/companion.
4. Patricia Cohen, "Digital Keys for Unlocking the Humanities' Riches," *New York Times*, November 10, 2010, C1.
5. Cleanth Brooks, *The Well Wrought Urn: Studies in the Structure of Poetry* (New York: Harcourt, Brace and World, 1947).
6. Jonathan Culler, *On Deconstruction: Theory and Criticism after Structuralism* (Ithaca, NY: Cornell University Press, 1982).

7. Franco Moretti, "Conjectures on a World Literature," in *Distant Reading* (London: Verso, 2013), 43-62.
8. Ibid., 45.
9. Franco Moretti, *Graphs, Maps, and Trees* (London: Verso, 2005).
10. Franco Moretti, *Network Theory, Plot Analysis,* pamphlet 2 (Stanford, CA: Stanford Literary Lab, May 1, 2011), http://litlab.stanford.edu/LiteraryLabPamphlet2.pdf; "The Emotions of London," Stanford Literary Lab website, Projects page, accessed December 22, 2014, http://litlab.stanford.edu/?page_id=13.
11. Christopher Prendergast, "Evolution and Literary History: A Response to Franco Moretti," *New Left Review*, no. 34 (July–August 2005), http://newleftreview.org/II/34/christopher-prendergast-evolution-and-literary-history.
12. Alan Liu, "The State of Digital Humanities: A Report and Critique," *Arts and Humanities in Higher Education* 11, no. 1–2 (February/April 2012): 26.
13. Stephen Ramsay, "The Hermeneutics of Screwing Around; or What You Do with a Million Books," in *Pastplay: Teaching and Learning History with Technology*, edited by Kevin Kee (Ann Arbor: University of Michigan Press, 2014), 111–120.
14. Douglas Biber and Randi Reppen, "Corpus Linguistics," in *Corpus Linguistics*, edited by Douglas Biber and Randi Reppen (London: Sage, 2011), xix-vliii.
15. David L. Hoover, Jonathan Culpepper, and Kieran O'Hallaran, *Digital Literary Studies: Corpus Approaches to Poetry, Prose, and Drama,* Routledge Advances in Corpus Linguistics (New York: Routledge, 2014).
16. Thomas Bartscherer and Roderick Coover, "Introduction," in *Switching Codes: Thinking through Digital Technology in the Humanities and the Arts*, ed. Thomas Bartscherer and Roderick Coover (Chicago: University of Chicago Press, 2011), 1–11.

Bibliography

Bartscherer, Thomas, and Roderick Coover. "Introduction." In *Switching Codes: Thinking through Digital Technology in the Humanities and the Arts*, edited by Thomas Bartscherer and Roderick Coover, 1–11. Chicago: University of Chicago Press, 2011.

Biber, Douglas and Randi Reppen, eds. *Corpus Linguistics*. London: Sage, 2011.

Brooks, Cleanth. *The Well Wrought Urn: Studies in the Structure of Poetry*. New York: Harcourt, Brace & World, 1947.

Cohen, Patricia. "Digital Keys for Unlocking the Humanities' Riches." *New York Times*, November 10, 2010, C1.

Culler, J. D. *On Deconstruction: Theory and Criticism after Structuralism*. Ithaca, NY: Cornell University Press, (1982).

Derrida, Jacques. *De la grammatologie*. Paris: Les Éditions de Minuit, 1967.

Digital Humanities Manifesto 2.0. Humanities Blast (blog), posted June 22, 2009 by Todd Presner. www.humanitiesblast.com/manifesto/Manifesto_V2.pdf.

Eagleton, Terry. *Literary Theory: An Introduction*. Minneapolis: University of Minnesota Press, 1983.

Fitzpatrick, Kathleen. "The Humanities, Done Digitally." In *Debates in the Digital Humanities*, edited by Matthew K. Gold, 12–15. Minneapolis: University of Minnesota Press, 2012.

Fowler, Roger. *Linguistic Criticism*. Oxford: Oxford University Press, 1986.

Fry, Paul H. *Theory of Literature*, Open Yale Course Series. New Haven, CT: Yale University Press, 2012.

Hockey, Susan. "The History of Humanities Computing." In *A Companion to Digital Humanities*, edited by Susan Schreibman, Ray Siemens, and John Unsworth. Oxford: Blackwell, 2004. www.digitalhumanities.org/companion.

Hoover, David L., Jonathan Culpepper, and Kieran O'Hallaran. *Digital Literary Studies: Corpus Approaches to Poetry, Prose, and Drama,* Routledge Advances in Corpus Linguistics. New York: Routledge, 2014.

Kirschenbaum, Matthew G. "What Is Digital Humanities and What's It Doing in English Departments?" *ADE Bulletin*, no. 150 (2010): 55–61.

Leech, Geoffrey, and Mick Short. *Style in Fiction: A Linguistic Introduction to English Fictional Prose,* English Language Series. London: Longman, 1981.

Liu, Alan. "The State of Digital Humanities: A Report and Critique." *Arts and Humanities in Higher Education* 11, no. 1–2 (February/April 2012): 8–41.

Mathson, Stephanie M. "Engaging Readers, Engaging Texts: An Exploration of How Librarians Can Use Reader Response Theory to Better Serve Our Patrons." *Library Philosophy and Practice* (August 2011): 589. http://digitalcommons.unl.edu/libphilprac/589.

Moretti, Franco. *Distant Reading*. London: Verso, 2013.

———. *Graphs, Maps, and Trees: Abstract Models for a Literary History*. London: Verso, 2005.

———. *Network Theory, Plot Analysis,* pamphlet 2. Stanford, CA: Stanford Literary Lab, May 1, 2011. http://litlab.stanford.edu/LiteraryLabPamphlet2.pdf.

Porsdam, Helle. "Too Much 'Digital,' Too Little 'Humanities'? An Attempt to Explain Why Humanities Scholars Are Reluctant Converts to Digital Humanities." Arcadia project report, 2011. DSpace@Cambridge. www.dspace.cam.ac.uk/handle/1810/244642.

Prendergast, Christopher. "Evolution and Literary History: A Response to Franco Moretti." *New Left Review*, no. 34 (July–August 2005). http://newleftreview.org/II/34/christopher-prendergast-evolution-and-literary-history.

Ramsay, Stephen. "The Hermeneutics of Screwing Around; or What You Do with a Million Books." In *Pastplay: Teaching and Learning History with Technology,* edited by Kevin Kee, 111–20. Ann Arbor: University of Michigan Press, 2014.

Reid, Robin Anne. "Mythology and History: A Stylistic Analysis of *The Lord of the Rings*." *Style* 43, no. 4 (Winter 2009): 517–38.

Reid, Robin Anne, and Christian Hempelmann. "The Tolkien Corpus Project." Posted April 8, 2013, on the Tolkien Scholarship Project website. http://earendel.net/?q=node/4 [site discontinued].

Stanford Literary Lab. "The Emotions of London." Stanford Literary Lab website, Projects page. Accessed December 22, 2014. http://litlab.stanford.edu/?page_id=13.

CHAPTER TWO

What Do Librarians Need to Know about Quantitative Methods in Digital Humanities?

Heather Froehlich

Introduction

Digital Humanities (DH) librarianship is not necessarily a one-size-fits-all experience. As Zhang, Su, and Hubschman argue, DH positions cover a range of skills, many of which have not been fully codified into standard library positions.[1] These different skills can be drawn upon in a variety of ways: some may be more quantitative, some may be driven by narrative and focus on specific publishing platforms (such as Omeka), some may focus more on visualization, and some positions require a little bit of all these skills. While many DH librarians might have training in more than one particular method (such as network analysis, geospatial mapping, text and data mining, and/or digital editing), DH librarianship is built on librarians' assumed ability to learn several methods well enough to teach themselves the basics of emergent data practices to sufficiently guide humanities researchers through the process of using these resources.

The practice of quantitative DH librarianship is not limited to collaborations with humanities scholars; they will often interface with researchers in disciplines like communication or business, who are very interested in observing language in practice.[2] DH librarians are involved in instruction and consultations; they collaborate on projects as partners; and they often manage sustainability and the project life cycle for researchers. This is

CC BY SA

important for humanities researchers, because they have discipline-specific needs, unique across the rest of the university. Most humanities researchers produce solo-authored, sustained arguments that rely upon direct quotes from primary and secondary source materials as evidence for a theoretically-supported claim. This is very different than the more experimental modes of the sciences or social sciences, which tend to begin with a theory and test that theory with a population of data using specifically qualitative, quantitative, or mixed-methods approaches. As Zhang, Su, and Hubschman argue, several other positions in academic libraries can cover aspects of these skillsets.[3] Crucially, though, we must avoid being too techno-positivistic in defining the DH librarian skillset, as the methods undergirding such projects often borrow advanced concepts from the social sciences and information sciences and retrieval.[4]

I argue that humanities data and humanities-driven projects imply specific, and often somewhat bespoke, approaches to librarians' roles. These may traditionally be described as "data librarian," "metadata specialist," "scholarly communications librarian," "statistical consultant," or "curator." Often, quantitative DH projects begin with scholars knowing what they want to investigate, but they may need help acquiring, curating, and applying methods to their imagined data set. In this essay, I will discuss how librarians who offer support to quantitative DH projects are meeting humanities researchers where those researchers are, which is often without a lot of underlying technical or mathematical knowledge. I begin with a literature review discussing various definitions of DH librarians' jobs. In the second section, I outline two distinct learning outcomes that typify quantitative DH projects and explore how DH librarians provide services to collaborators depending on which of the two learning outcomes they fall under. Finally, I summarize what librarians with other specialties need to know about how quantitative methods in DH look and operate differently from other kinds of research data services.

Literature Review

Sula's "Digital Humanities and Libraries" and Vandegrift and Varner's "Evolving in Common" each argue that DH services are a natural extension for libraries to take on.[5] Vandegrift and Varner specifically note that "the roles and responsibilities of research librarians are shifting to encompass the broadening scope of scholarship, especially involving digital archival and special collections, digital tools and progressive service models."[6] Sula

considers libraries the natural laboratory for humanities scholars, and despite concerns about institutional barriers to success,[7] it is clear that universities have largely bought in to the need for DH (or digital scholarship) services.

In their content analysis of DH job advertisements for academic libraries, Zhang, Su, and Hubschan identify seventy job postings for DH librarians from 2006 through 2018 in the United States. Their study implies that these jobs have permeated librarianship. The author's interest in DH librarianship suggests that the field has largely moved past Nowviskie's early-2010s model of "scholar-practitioners"[8] and to a place where at least some digital practices are subsumed into librarian training in graduate school; of their sample, only 7% of the listings required a PhD in a relevant discipline while 83% required a master's degree.[9]

Indeed, the first edition of this collection discusses many of the ways these roles are enacted in practice. For example, in their chapter "Traversing the Gap," Gibson, Ladd, and Presner describe bridging the gap between technologists and subject specialists, as well as the role DH librarians take on connecting researchers to other campus experts with relevant skillsets. They argue that "in libraries, this takes the form of making existing structures and services, such as metadata and repositories, available to scholars. The library acts in a supportive capacity, but it is not necessarily an active participant in the research."[10] A common service model is "to help researchers on campus learn about DH methods [and] foster campus-wide discussions on the topic."[11] Under this model, subject librarians are available to help interpret results but require a certain level of methodological knowledge to be effective, which requires regular knowledge upkeep. Other librarians, such as Rosenblum, Devlin, Albin, and Garrison, discuss how different kinds of instructional practices can connect with digital scholarship more generally in a collaborative model.[12] Vedantham and Porter focus on how different collaborative spaces also inform teaching and learning service models.[13] There has been increased discussion of consultations and trainings in practice since this volume was first published, including Russell and Hensley on pedagogy and instruction,[14] Morgan on consultations,[15] and Johnson and Dehmlow with a survey of DH librarianship's general portfolio (as a representative selection).[16]

While Johnson and Dehmlow offer a survey of DH outcomes, Zhang, Su, and Hubschan ultimately argue that these positions cover a range of scholarly practices while lacking a set of core competencies to measure against.[17] This particular arm of librarianship lacks a certain level of coherency: "Though 'DH Librarian' is now an actual job title and position in academic libraries, the American Library Association (ALA) has not developed a framework of

core competencies for this emerging profession, and few LIS degree programs and professional organizations are providing sufficient support for students and information professionals to gain the skills and knowledge required to be successful in the profession."[18] The ALA has established core competencies for all librarians, but different subspecialties (science librarianship, collections and technical services, archivists, law librarians, etc.)[19] have their own established communities of practice and shared expectations that allow for benchmarking success across institutions. The concern is that without a coherent sense of DH competencies, it is difficult to justify and assess why these positions are needed. Walker problematizes this somewhat, claiming "what we [DH practitioners] are doing is merely an extension of what we [libraries] have been doing for years."[20] Varner and Vandegrift would agree, arguing that this is a natural progression to best serve the evolving needs of humanities researchers.

In what follows, I build on these discussions of portfolio and praxis by suggesting that quantitative methods in DH librarianship offer two major discrete learning outcomes. I suggest that while we may lack a set of ALA-authorized core competencies, we provide a specific kind of service which is largely dependent on two sorts of patron needs. In each instance, our audience can develop quantitatively-driven skills and confidence without getting stuck in a cycle of endless referrals, where they are likely to receive mixed messages regarding next steps.

Two Kinds of Consultations

In general, DH librarians supporting quantitative research methods have two discrete kinds of consultations, which I frame in terms of their individual learning outcomes. I use 'learning outcomes' specifically because the DH librarian's role is to facilitate knowledge exchange between librarian and researcher. We *could* perform the task for the researcher, but that removes the possibility for situated learning to take place. Our two main learning outcomes are as follows, and echo Gibson, Ladd, and Presnell's discussion of "first-order" and "second-order" projects:[21]

1. Researchers will test an idea about performing a quantitative experiment by discussing best approaches with the DH librarian and decide on a suitable outcome.
2. Researchers will state a desire to learn a particular method or approach for a specific research question and develop a sustained course of action with the DH librarian to execute the project.

As DH research practices become more embedded in teaching and scholarship, there is the potential for additional learning outcomes to be introduced.

In this next section, I will discuss the potential routes the DH librarian will take to execute each of these learning outcomes, using prototypical examples derived from actual consultations with researchers.

In the first learning outcome, researchers come to the meeting with a sense of curiosity, wanting to explore the possibility of doing a difficult to execute or otherwise complex analysis. Depending on disciplinary background, this user rarely needs heavy support in command-line based programmatic practices. In many cases, they believe they are approaching some kind of impossible ask or imagine their needs will be met with complexity beyond the level of interest they have in the question. Often, they are not well-versed in what their available options are for doing this kind of work. These kinds of researchers could be looking to do something as simple as wanting to find out if a particular concept undergoes semantic change during a specific period, or if there is a way to track specific vocabulary in a small set of texts without having to set up codes with software such as NVivo. These are lower-stakes questions for both researcher and librarian, ones that usually require one or two consultations to figure out a reasonable plan to best help the researcher achieve their goals. Often, they are simply curious about what their options are and want to know what the best approach would be to achieve their task, without sinking a ton of training time into this activity.

This researcher needs a simple, straightforward solution. One example of this first kind of learning outcome is when a researcher is interested in semantic change of a word or concept over time. They may have heard of Google NGrams,[22] but are unclear on the underlying data and how to access this resource. In this case, the librarian can explain the pros and cons of using the Google Books collection in contrast to other platforms (including HathiTrust's historical collections[23] and the curated historical corpora from Brigham Young University's English Corpora[24] as two alternatives). From there, we establish the best resource for the task and the librarian teaches the researcher the basics of using the interface. After meeting, the researcher should be able to develop their own queries and interpret the outcomes for themselves. Should they decide they want to analyze their own collection of texts or need help developing a more specific query, they may return to seek advice with another targeted question.

This category of learning outcomes includes consultations with researchers who want to extract major themes from their own collection of texts, such as a survey, and can comfortably use entry-level platforms (like Voyant

Tools[25]) after one consultation to observe major themes and linguistic patterns in their textual data. If the researcher has previous training in quantitative methods, their needs can often be met by one-off consultations. The patron understands the stakes and has the skills they need to complete the task. However, these consultations may move into the second learning outcome if the researcher develops a sustained interest in developing quantitative analytic skills more deeply.

The second learning outcome described above requires more focused and ongoing support. When researchers state a desire to learn a particular method or approach, it is typically for sustained scholarly inquiry. Rarely do we see interest at this level "just to find out" what is happening. Therefore, the DH librarian will develop a plan of action which begins by prescribing a particular software or method the researcher should use to answer their stated question, and we will train them in the basics of this prescribed approach. From there, the researcher and librarian will collaboratively assess results from this process, with iteration being a key part of this second learning outcome. As the researcher proceeds, more complex concepts may be introduced, especially those related to statistical or other forms of numeric thinking. As the researcher gets comfortable with these quantitative methods, we continue to iterate and expand on the project.

In this second outcome, the researcher may have attended a conference where they encountered topic models they want to perform on their own texts, or they have become interested in tracing specific linguistic features in their texts at scale. These activities—even if they still involve using off-the-shelf resources such as Voyant Tools, AntConc, or MALLET—will require more hands-on learning to understand the outputs of these platforms. While tutorials are available, researchers often desire specific guidance on how to interpret and understand the outcomes of these processes. This specifically is what often leads to a more sustained series of consultations: the researcher wants targeted, specific help understanding the output of these processes.

In each learning outcome, support for quantitative DH requires deep and significant knowledge of which platforms, techniques, and methods are available for users at different stages of their digital methods journey. In many ways, the DH librarian's role is to provide direct assistance, rather than referring the researcher to various specialists around the library. In no particular order, we might consult on these areas of specialist expertise in our consultations: copyright, metadata, data management, literature review for methods, methods training and assessment, and data visualization. Thus, our real task is to assess what is available for the researcher according to

their current confidence with technology and develop a plan to help them execute their desired project. These varied informational and digital literacies are often the most complex parts of our roles, and are not a one-size-fits-all approach. For instance, a researcher coming in with advanced computational and/or digital literacies (such as experience programming in Python or R) can be given more advanced advice in the very first instance.

Some researchers are just curious about what is available to them as off-the-shelf platforms for performing simple word frequency analysis, and some want to launch a major methodological overhaul. Our consultations therefore contain several features of reference interviews, such as the following inquiries:

- What are your file types (or what files do you need), how were they selected, what do they represent, which dates do they cover, who wrote the articles? (Do you need to do this yourself, or can we use someone else's work?)
- Do you need a graphical user interface to familiarize yourselves with the concept at hand, or can you use a Python package to complete this task and simply need advice on best practices?
- What kinds of statistical methods are appropriate for this particular task?
- Do you need to visualize your data in a complex, bespoke way, or will something simple from an off-the-shelf tool suffice? Do you need to visualize *any* of your data?
- Where does the data go when we are done with the project? Do you have the rights to share it more widely? How will you preserve the data for future projects?

Rolled up in these questions is an assessment of which approach(es) would be best to solve the problem at hand. As with all reference interviews, the biggest part of our role as librarian is to provide interpretive support. We coach researchers through understanding what their data does, what it looks like, and how it operates within the scope of the platform we have selected for the project. Generally, DH librarians all have a range of trusted solutions for most tasks presented to us in these kinds of interviews. Answers provided in these discussions shape the directions of our conversations: for example, if developing a corpus for analysis becomes a key feature of the project, earlier phases of both kinds of consultations are often preoccupied with introducing the idea of metadata and data curation.

The researcher ultimately guides this process and we respond to what they bring to the table. We must, of course, be mindful of providing training and enabling our researchers to succeed. For humanities scholars in particular, we might be their first contact moving into more complex computational processes than they might be used to performing. Based on these answers, we might become an iterative partner in helping the researcher understand what their output looks like and how it functions. We might be able to send someone off to do their own analyses with a graphic user interface, but they may require additional support after working independently for a while to get them closer to the information they are seeking.

As Morgan writes, "while I can make strong recommendations, the scholars I work with are the ultimate decision makers; and this is why my work *is* emotional labor, as opposed to pure technical advice. The way that I provide guidance and information will have an emotional impact, and people will make better decisions if they are feeling steady, resilient, and energized, rather than depleted, by their work."[26] Morgan describes this as emotional labor in that it has a lot to do with setting expectations and being reasonable about the time and energy required to do this kind of scholarship. And sometimes DH librarians underestimate how much support a researcher might need, requiring further energy, labor, and effort. Quantitative DH methods often borrow methods and techniques from the social sciences and computer science. We may be already used to doing these tasks, so accessing the mindset of a novice is always helpful.

It is also in our best interest to avoid being too technocentric in these discussions. Our patrons may be approaching something mathematically complex like topic modelling for the very first time. We become involved in teaching the implications of the statistics for their particular purposes—but (generally) by keeping them away from the actual calculus underlying the processes. This is not about hiding the details of a chosen technique, but rather breaking tasks down into discrete tasks to understand quantitative digital approaches in a meaningfully scaffolded format. This builds researcher confidence in their ability to develop further skills. And, ultimately, we let the researchers set their comfort levels and respond accordingly. Quantitative methods in DH librarianship cover a wide range of formats and needs. Many projects can suffice with keyword-in-context or collocational analysis, but researchers might have encountered more complex methods such as topic modeling or sentiment analysis and be interested in applying these tasks to their research projects.

This section has outlined two major learning outcomes for scholars looking to get involved in implementing quantitative DH methods. In both learning outcomes, the librarian and the researcher collaborate to establish what they need to accomplish this project (a corpus) and introduce them to a software package to conduct their analysis. As necessary, we also train them to develop good metadata practices to trace vocabulary across the topics, as well as encourage them to think about language as data being presented in a wholly non-linear fashion, adjust their approaches to reflect the needs of the project, and iterate the process as needed. Next, I discuss how DH librarians use their skills and knowledge to execute these two specific learning outcomes as a larger discussion.

Discussion / Assessment

In each of the two learning outcomes described above, relationships between librarian and researcher are built on sustained trust. This trust is bilateral: first, that the librarian will not wildly overestimate the amount of time and energy the researcher can devote to this project and that the material is pitched appropriately for the researcher. In exchange, the researcher will perform the follow-through to learn the skills being presented to them. From the librarian side, it is crucial that someone coming from philosophy or history should not be expected to suddenly learn a programming language to perform a simple word frequency task, or have to familiarize themselves with advanced calculus to implement a topic model, *until they are ready to.* One danger of referring researchers to different specialists in the library is that not all colleagues may be familiar with a researcher's learning context and level of experience. Thus, for instance, scaffolding learning about regular expressions after establishing a practice of keyword-in-context searching may make the experience less daunting for someone trying out quantitative methods. Working with one DH librarian can help establish situated knowledge.

Having to refer the researcher around the library would mean having to constantly explain their project and goals to different colleagues. This also risks the researcher being given potentially conflicting advice from a range of different people. An even worse outcome would be if every library staff member they meet assumes the patron already has all the other necessary background knowledge for quantitative research. DH librarians can refer our patrons to appropriate channels should they need more support than we can offer them—having one centralized person to speak to is a strong benefit for the patron seeking advice. DH librarians are, therefore, more like

subject specialists, with specific knowledge of the methods available and the questions that come to bear on these methods. And, as the above learning outcomes show, working with a DH librarian is not a one-size-fits all experience. Researchers at different stages of developing quantitative projects in the humanities have unique needs.

As Zhang, Su, and Hubschan show, the range of skills DH librarians possess encompasses a range of methods as well as focused disciplinary knowledge. The most common analogous position to DH librarians might be a social science data librarian; researchers working with qualitative and quantitative social science have similar needs.[27] However, social science researchers have distinct needs from humanities researchers, including a requirement for methods in their degree programs. We understand the specific needs of humanities researchers: patrons from the humanities have such specific data-driven needs (including convincing them to think about their processes as data-driven in the first place). We use more generalizable approaches, such as text mining (a blanket term that covers a huge range of tasks, from machine learning techniques to keyword-in-context analysis) and data visualization in the sense of identifying what is data and what needs to be illustrated.

Conclusion

At the beginning of this essay, I referred to the range of skills and roles ascribed to DH librarians.[28] This article argues that the lack of consistency in DH roles is potentially troubling, as there is not a complete or otherwise specifically coherent set of skills anticipated in these roles, following on a lack of core competencies.[29] I have argued that the necessary skills and knowledge for DH librarians intersect with many different library positions. Unless they are specifically trained as DH practitioners, humanities researchers rarely (if ever) have this level of training and rely on the expertise of others. Quantitative and other DH methods involve meeting researchers where they are and providing them scalable, accessible resources. I argue that our consultations fall under one of two major tasks: trying out a hypothesis by performing a quantitative experiment, or wanting to develop specific research skills.

The major goal of these consultations is not to make these concepts seem impossible to implement. Instead, we provide quantitative support to humanities researchers to empower them to prototype or otherwise dip a toe into new-to-them methods while meaningfully growing their confidence. DH librarianship may lack an ALA-approved set of core competencies and a shared vocabulary, but our hybridity is ultimately a benefit to our scholarly communities.

Notes

1. Yin Zhang, Fangli Su, and Barbara Hubschman, "A Content Analysis of Job Advertisements for Digital Humanities-Related Positions in Academic Libraries," *The Journal of Academic Librarianship* 47, no. 1 (2021): 1–11, https://doi.org/10.1016/j.acalib.2020.102275.
2. This, of course, has lots to do with how individual colleges are structured and which departments are part of which collection of disciplines.
3. Zhang, Su, and Hubschman, "A Content Analysis," 1.
4. Paige C. Morgan, "The Consequences of Framing Digital Humanities Tools as Easy to Use," *College & Undergraduate Libraries* 25, no. 3 (3 July 2018): 211–231, https://doi.org/10.1080/10691316.2018.1480440.
5. Chris Alen Sula, "Digital Humanities and Libraries: A Conceptual Model," *Journal of Library Administration* 53, no. 1 (January 1, 2013): 10–26, https://doi.org/10.1080/01930826.2013.756680; Micah Vandegrift and Stewart Varner, "Evolving in Common: Creating Mutually Supportive Relationships Between Libraries and the Digital Humanities," *Journal of Library Administration* 53, no. 1 (January 1, 2013): 67–78, https://doi.org/10.1080/01930826.2013.756699.
6. Micah Vandegrift and Stewart Varner, "Evolving in Common," 68.
7. Miriam Posner, "What Are Some Challenges to Doing DH in the Library?," Miriam Posner's Blog (blog), August 10, 2012, http://miriamposner.com/blog/what-are-some-challenges-to-doing-dh-in-the-library/; Michael Furlough, "Some Institutional Challenges to Supporting DH in the Library," Mike Furlough (blog), August 16, 2012, http://www.mikefurlough.net/?p=51; Isabelle Galina Russell, "The Role of Libraries in Digital Humanities (El Papel de Las Bibliotecas En Las Humanidades Digitales)," World Library and Information Congress: 77th IFLA General Conference and Assembly, San Juan, Puerto Rico, 2011, 1–6, https://www.ifla.org/past-wlic/2011/104-russell-en.pdf; Trevor Muñoz, "Digital Humanities in the Library Isn't a Service," Trevor Muñoz, accessed July 15, 2022, https://trevormunoz.com/archive/posts/2012-08-19-doing-dh-in-the-library/.
8. Bethany Nowviskie, "Skunks in the Library: A Path to Production for Scholarly R&D," *Journal of Library Administration* 53, no. 1 (1 January 2013): 53, https://doi.org/10.1080/01930826.2013.756698.
9. That said, 22% of all postings described a PhD as a "preferred" or otherwise desirable qualification, and "the majority of job positions did not require a doctoral degree" (Zhang, Su, and Hubschman , 2021, 5). So while those with PhDs in digital humanities are encouraged to apply, a terminal master's in library science represents minimum requirement for 93% of these positions.
10. Katie Gibson, Marcus Ladd, and Jenny Presnell. "Traversing the Gap: Subject Specialists Connecting Humanities Researchers and Digital Scholarship Centers," in *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, eds. Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb (Association of College and Research Libraries, 2015), 7.
11. Gibson, Ladd, and Presnell, "Traversing the Gap," 8.
12. Brian Rosenblum, Frances Devlin, Tami Albin, and Wade Garrison, "Librarians Teaching Digital Humanities in the Classroom," In *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, eds. Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb (Association of College and Research Libraries, 2015), 151–176.

13. Anu Vedantham and Dot Porter, "Spaces, Skills, Synthesis," in *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, eds. Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb (Association of College and Research Libraries, 2015), 183.
14. John E. Russell and Merinda Kaye Hensley, "Beyond Buttonology: Digital Humanities, Digital Pedagogy, and the ACRL Framework," *College & Research Libraries News* 78, no. 11 (December 4, 2017): 588-600, https://doi.org/10.5860/crln.78.11.588.
15. Morgan, "The Consequences of Framing," 211–31.
16. Daniel Johnson and Mark Dehmlow, "Digital Exhibits to Digital Humanities: Expanding the Digital Libraries Portfolio," in *New Top Technologies Every Librarian Needs to Know: A LITA Guide*, ed. Kenneth J. Varnum (American Library Association, 2019).
17. Zhang, Su, and Hubschman, "A Content Analysis," 1.
18. Zhang, Su, and Hubschman, "A Content Analysis," 1.
19. See ALA's "Core Competences," Text, Education & Careers, June 10, 2008, https://www.ala.org/educationcareers/careers/corecomp/corecompetences. For a complete list of specialist core competencies, see "Knowledge and Competencies Statements," Text, Education & Careers, August 3, 2006, https://www.ala.org/educationcareers/careers/corecomp/corecompspecial/knowledgecompetencies.
20. Judy Walker, "Digital Humanities for the Rest of Us," in *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, eds. Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb (Association of College and Research Libraries, 2015), 130.
21. Katie, Ladd, and Presnell, "Traversing the Gap," 3.
22. Yuri Lin, Jean-Baptiste Michel, Erez Aiden Lieberman, Jon Orwant, Will Brockman, and Slav Petrov, "Syntactic annotations for the Google Books Ngram Corpus," Proceedings of the ACL 2012 system demonstrations, 2012: 169–174.
23. HathiTrust Research Center, accessed January 21, 2024, https://www.hathitrust.org/htrc.
24. Mark Davies, "The Corpus of Contemporary American English: 450 Million Words, 1990–Present," https://www.english-corpora.org//coca/, 2008, accessed April 2021; Mark Davies and Robert Fuchs, "Expanding Horizons in the Study of World Englishes with the 1.9 Billion Word Global Web-Based English Corpus (GloWbE)," *English World-Wide* 36, no. 1 (2015): 1–28; Mark Davies, "The Corpus of Historical American English (COHA): 400 Million Words, 1810–2009," https://www.english-corpora.org//coha/, 2010, accessed April 2021.
25. Stéfan Sinclair and Geoffrey Rockwell, *Voyant Tools*, 2016, http://voyant-tools.org/.
26. Paige C. Morgan, "Not Your DH Teddy Bear, or Emotional Labor is not going away," DH + Lib blog, June 29, 2016, https://acrl.ala.org/dh/2016/07/29/not-your-dh-teddy-bear/.
27. James Mahoney and Gary Goertz, "A Tale of Two Cultures: Contrasting Quantitative and Qualitative Research," *Political Analysis* 14, no. 3 (2006): 227–49, https://doi.org/10.1093/pan/mpj017; Michael S. Lewis-Beck, Alan Bryman, and Tim F. Liao, *The SAGE Encyclopedia of Social Science Research Methods*, 1st edition (Thousand Oaks, Calif: SAGE Publications, Inc, 2003).
28. Zhang, Su, and Hubschman, "A Content Analysis," 1.
29. Zhang, Su, and Hubschman, "A Content Analysis," 11.

Bibliography

Davies, Mark. "The Corpus of Contemporary American English: 450 Million Words, 1990–Present." https://www.english-corpora.org//coca/, 2008. Accessed April 2021.

Davies, Mark. "The Corpus of Historical American English (COHA): 400 Million Words, 1810–2009." https://www.english-corpora.org//coha/, 2010. Accessed April 2021.

Davies, Mark and Robert Fuchs. "Expanding Horizons in the Study of World Englishes with the 1.9 Billion Word Global Web-Based English Corpus (GloWbE)," *English World-Wide* 36, no. 1 (2015): 1–28.

Furlough, Michael. "Some Institutional Challenges to Supporting DH in the Library." Mike Furlough (blog). August 16, 2012. http://www.mikefurlough.net/?p=51.

Gibson, Katie, Marcus Ladd, and Jenny Presnell. "Traversing the Gap: Subject Specialists Connecting Humanities Researchers and Digital Scholarship Centers." In *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*. Edited by Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb, 7. Chicago: Association of College and Research Libraries, 2015.

Johnson, Daniel and Mark Dehmlow. "Digital Exhibits to Digital Humanities: Expanding the Digital Libraries Portfolio." In *New Top Technologies Every Librarian Needs to Know: A LITA Guide.* Edited by Kenneth J. Varnum. Chicago: American Library Association, 2019.

Lewis-Beck, Michael S., Alan Bryman, and Tim F. Liao. *The SAGE Encyclopedia of Social Science Research Methods*, 1st edition. Thousand Oaks, Calif: SAGE Publications, Inc, 2003.

Lin, Yuri, Jean-Baptiste Michel, Erez Aiden Lieberman, Jon Orwant, Will Brockman, and Slav Petrov. "Syntactic annotations for the Google Books Ngram Corpus." Proceedings of the ACL 2012 system demonstrations. 2012. 169–174.

Mahoney, James and Gary Goertz. "A Tale of Two Cultures: Contrasting Quantitative and Qualitative Research." *Political Analysis* 14, no. 3 (2006): 227–49. https://doi.org/10.1093/pan/mpj017.

Morgan, Paige C. "Not Your DH Teddy Bear, or Emotional Labor is not going away". DH + Lib (blog), June 29, 2016. https://acrl.ala.org/dh/2016/07/29/not-your-dh-teddy-bear/.

Morgan, Paige C. "The Consequences of Framing Digital Humanities Tools as Easy to Use." *College & Undergraduate Libraries* 25, no. 3 (3 July 2018): 211–231. https://doi.org/10.1080/10691316.2018.1480440.

Munoz, Trevor. "Digital Humanities in the Library Isn't a Service." Trevor Munoz (blog), August 19, 2012. https://trevormunoz.com/archive/posts/2012-08-19-doing-dh-in-the-library/.

Nowviskie, Bethany. "Skunks in the Library: A Path to Production for Scholarly R&D." *Journal of Library Administration* 53, no. 1 (1 January 2013): 53–66.

Posner, Miriam. "What Are Some Challenges to Doing DH in the Library?" Miriam Posner's Blog (blog), August 10, 2012. http://miriamposner.com/blog/what-are-some-challenges-to-doing-dh-in-the-library/.

Rosenblum, Brian, Frances Devlin, Tami Albin, and Wade Garrison. "Librarians Teaching Digital Humanities in the Classroom." In *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*. Edited by Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb, 151–176. Chicago: Association of College and Research Libraries, 2015.

Russell, Isabelle Galina. "The Role of Libraries in Digital Humanities (El Papel de Las Bibliotecas En Las Humanidades Digitales)." World Library and Information Congress:

77th IFLA General Conference and Assembly. San Juan, Puerto Rico, 2011: 1–6. https://www.ifla.org/past-wlic/2011/104-russell-en.pdf.

Russell, John E. and Merinda Kaye Hensley. "Beyond Buttonology: Digital Humanities, Digital Pedagogy, and the ACRL Framework." *College & Research Libraries News* 78, no. 11 (4 December 2017): 588–600. https://doi.org/10.5860/crln.78.11.588.

Sinclair, Stéfan and Geoffrey Rockwell. *Voyant Tools*, 2016. http://voyant-tools.org/.

Sula, Chris Alen. "Digital Humanities and Libraries: A Conceptual Model," *Journal of Library Administration* 53, no. 1 (January 1, 2013): 10–26. https://doi.org/10.1080/01930826.2013.756680.

Vandegrift, Micah and Stewart Varner. "Evolving in Common: Creating Mutually Supportive Relationships Between Libraries and the Digital Humanities." *Journal of Library Administration* 53, no. 1 (January 1, 2013): 67–78. https://doi.org/10.1080/01930826.2013.756699.

Vedantham, Anu and Dot Porter. "Spaces, Skills, Synthesis." In *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, edited by Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb, 177–99. Chicago: Association of College and Research Libraries, 2015.

Walker, Judy. "Digital Humanities for the Rest of Us." In *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, edited by Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb, 129-150. Chicago: Association of College and Research Libraries, 2015.

Zhang, Yin, Fangli Su, and Barbara Hubschman, "A Content Analysis of Job Advertisements for Digital Humanities-Related Positions in Academic Libraries." *The Journal of Academic Librarianship* 47, no. 1 (2021): 1–11. https://doi.org/10.1016/j.acalib.2020.102275.

CHAPTER THREE

Relearning Digital Humanities Librarianship

John Russell

THE DIGITAL humanities can be a problematic and unstable field of practice to define, so when someone says "let's train librarians in digital humanities," what do they mean? Do they mean "let's pick some tools that I've heard are important and upskill some librarians?" What are the goals and learning outcomes for such training, and how do those goals map onto the lived experience of digital humanities librarians? Digital humanities librarianship is an array of practices, yet in some contexts it is reduced to software training. My experience over the past seven years of teaching an introduction to digital humanities course for librarians has changed my thinking about training librarians in DH. In this chapter, I will reflect on my experience with the course and how my own understanding of the field has evolved.

In my four-week online introduction to the digital humanities for librarians (see Appendix 3A), we typically spent half the time discussing the definition of digital humanities, based on selected readings and digital projects. When I first began teaching this course, I was very much a "DH is the computational analysis of humanities objects" person. The digital projects we studied pointed towards a different reality, mainly as a foil ("where is the analysis here?"). But through iterations of the course, my thinking changed and I backed away from a clear-cut definition.

Reflecting more on my own experience with needs assessment for digital scholarship and the experience of other librarians doing the same (in large part because environmental scans and needs assessments came up frequently in course discussions), I realized that I preferred a more nominalist approach to the "what is DH?" question. In the course forum, I once wrote "DH is just the digital things that humanists in your community are doing" in jest, a non-definition to elicit a chuckle from the students. But I started to take

this non-definition seriously, as not just a defensible assertion but as the best approach to coming to terms with the multiplicity and plurality of what gets called "digital humanities."

Unsurprisingly, many librarians taking the course over the years have balked at this non-definition of DH. The whole point of taking an introduction to DH, they argued, is learning what DH is! But the reality is that any definition is problematic, unsatisfying, vague, or silly. The "What is DH?" X (formerly known as Twitter) account has been tweeting a different definition of DH every day since October 2018 and Jason Heppler's "What is Digital Humanities?" website[1] currently has 817 definitions that it presents in randomized display. As Clifford Anderson notes, it is difficult to define DH because of "the lack of shared methodologies," and while he is thinking mostly about computational methods, scholarship that has served to highlight the social justice potential of DH (for instance, via feminism, postcolonial theory, or Black studies) further expands and complicates the methodological horizons of DH.[2]

The middle part of my intro to DH course focused on digital humanities tools. Students learned Voyant, text encoding with TEI, and had an additional option of tackling a *Programming Historian*[3] tutorial should they wish to try something different. Perhaps surprisingly, I do not get very conceptual with this section as I just wanted students to try out a couple of tools and talk about their experiences. Voyant has been very popular, as it's fairly easy to get started with it; text encoding was an often maddening experience for students who struggled with the rules or found encoding tedious.

A lot of DH trainings for librarians take a primarily tool-based approach. The ARL Digital Scholarship Institute covers Omeka, mapping, and text encoding.[4] In 2012, an ALA Annual pre-conference DH workshop covered text analysis and text encoding. The HathiTrust Research Center facilitated a text analysis "road show."[5] When I first started in digital humanities librarianship, I signed up for workshops like these because I believed that the way to gain expertise was to master software. I doubt I am the only librarian who has felt that way and it is not hard to understand why some librarians who want to learn DH might feel a slightly panicked obsession with trying to learn all the "computer stuff" all the time.

Reading Matthew Lincoln's essay "Tool Trouble" helped crystallize for me the problem with this way of thinking. Lincoln describes a disconnect during a workshop where the attendees expected "that I would be talking about 'plug-and-play' tools, when in fact I obstreperously kept asking about their research methodologies, and how exactly they were hoping to make

those methodologies interact with the methodologies offered by geographic and network paradigms."[6] Lincoln's audience seemed to believe that learning DH was a matter of learning how to use software. What we can observe from this attitude is how the concept of "tool" is also a way to manage the indeterminacy of DH. If we cannot define DH or define it broadly ("computational analysis"), but we must become conversant in it, one could resolve that indeterminacy by reducing DH to a suite of tools to master.

Working with digital tools is often how librarians understand the field of DH librarianship. We offer workshops and consultations and build collections to help DH researchers identify and learn digital tools. These practices are very visible—and quantifiable—activities (along with "digital projects") that can be listed in annual reports and demonstrate to administrators (and faculty and students) that the library is doing its part to promote digital humanities. However, workshops are a species of one-shot instruction and unsatisfying in similar ways, detached from a curriculum. Decontextualized DH workshops, like one-shot instructional sessions, are often transactional services where "content is requested and then deposited into students' minds with a focus on measurable skills" at the expense of "deeper learning, critical thinking, and inclusive pedagogy."[7] The workshop model lends itself to learning the mechanics of a technical process but isn't always set up to help people understand why and in what circumstances this technical process is useful, or how it might help students and scholars address their research questions.

My intro to DH course ends with a focus on librarianship. In the first years that I taught it, my perspective was very much about librarians doing digital work as partners and not as a service. As a wider range of librarians from different types of libraries enrolled in the course, there was more resistance to this model. For librarians balancing multiple sets of duties and who were not expected to do research, the focus on doing digital projects was alienating and foreign to their day-to-day experience. I was sympathetic and their reactions got me thinking about our profession's equation of DH with digital projects. When we consult with researchers to help them organize their data or work with instructors to incorporate web-based projects into their courses, are we not also "doing DH?" I came to realize—and I admit it took me a while to understand this—that the week of reading about librarians doing needs assessments, consultations, and "collections as data"[8] work was just as much a part of addressing the "what is DH" question as the first two weeks' focus on definitions.

There's been a strain of auto-exoticism in DH discourse in librarianship: that it's new, it's the future, it's disruptive, it's not "traditional" librarianship,

and that it requires up- or re-skilling of librarians. Trevor Muñoz argued for de-centering the service model of DH librarianship because digital humanities "involves research and teaching and building things and participating in communities both online and off. In libraries, digital humanities should involve those same things."[9] For Micah Vandegrift, even where DH "shares the most basic goal with the library," that of information access, it's still something other, something external to be embraced, a "calling of digital work in contrast to the vocation of servitude [librarianship]."[10] I don't know that we have moved beyond this mode of thinking—I certainly hadn't when I first began to teach the introduction to DH course, and, as I note above, it took me a long time to recognize librarianship as a way of doing DH.

This drive to assert novelty, I argue, has warped our professional approach to training. Even where such trainings explicitly center the heterogeneity of digital humanities practice, the overall structure of our trainings still rests on assumptions about DH—tools, projects—that reifies a DH that exists outside of librarianship toward which we should move. How weird would it be to have a digital humanities bootcamp for librarians that was primarily about doing environmental scans and needs assessments? If the digital humanities are a multitude of practices, shouldn't the focus for librarians be on figuring out their local scene and getting a handle on the digital things that humanists on their campus are doing? Or on making connections between digital humanities and information literacy? But I am not sure that many people would want to attend a DH institute where no one learns any software.

I want to be clear that I am not against learning software or even learning to code. It is a magical and extraordinarily sublime thing to make computers do things you could not make them do before. Teaching people how to scrape web pages or divide data into subsets or generate a map excites people. Often they can't believe they've achieved something that had previously seemed so impossible and foreign. It can lead to wonderful feelings of accomplishment for both teacher and students. However, I believe that there needs to be greater intention and transparency around what it is librarians need to learn. Maybe we should not reflexively assume, for example, that learning text analysis is part of the core curriculum for DH librarianship and instead have broader discussions about the specificity of DH at various kinds of institutions and build the DH curriculum from there.

When we think of digital humanities librarianship in the greater context of librarianship practice, it opens new possibilities. Much of the hype about the digital humanities lending a culture of experimentation or innovation moves in one direction, a sublation of librarianship into digital humanities.

As a result of some of the conversations that happened in my introduction to digital humanities for librarians course, I asked what would happen if I looked at digital humanities work through the lens of our practices of instruction, consultation, and collection management.

One outcome of this way of thinking was the creation of a workshop in 2017, one I created with my colleague Rebecca Miller Waltz. We brought together librarians to discuss digital humanities, the ACRL *Framework for Information Literacy in Higher Education*, and to see what could be learned from bringing together the digital humanities and information literacy. How might digital humanities instruction, with its software-oriented instructional practices, benefit from a focus on information literacy principles? How might digital humanities methods and projects help build information literacy skills in non-digital humanities instructional settings?

The five-hour workshop began with overviews of both the digital humanities and the *Framework*, a discussion of three readings relating to DH and instruction, a classroom activity, and then breakouts where participants created a lesson plan (with learning outcomes) for an instructional activity incorporating some aspect of digital humanities (see Appendix 3B). Instruction librarians saw excellent opportunities for helping students to understand how to generate and think about search terms within digital humanities projects. The attendees also had engaging conversations about approaches to teaching technology in general. This intensive focus on the *Framework* certainly led me to question more deeply how I approached DH training and whether teaching software—or "buttonology"—was always the best approach.[11]

The practice of digital humanities librarianship is not new—"Digital Humanities in Libraries: New Models for Scholarly Engagement," a highly-cited special issue of the *Journal of Library Administration*, came out in 2013,[12] the first edition of this volume was published in 2015,[13] and the *dh+lib*[14] celebrates its tenth anniversary in 2023—but it feels like we're still revolving around (and arguing about) a vision of digital projects and tools, grounded in novelty, that ignores the connections between digital humanities and librarianship. For example, a recent article asserts the importance of understanding data to the digital humanities, positing threshold concepts for data literacy to "provide a lens through which we can explore a piece of digital humanities scholarship. Defining threshold concepts will be useful in communication among digital humanists and our daily pedagogical work."[15] Hopefully, we will see further examples of these types of connections. For me, focusing on broader issues in librarianship has helped me relearn DH

librarianship. By trying to teach my professional colleagues about the digital humanities, I came to recognize and revise my assumptions about what DH is, and as well, to rethink what it means to be a librarian in the constellation of DH theory and practice.

Appendix 3A. "Introduction to Digital Humanities for Librarians" Syllabus

The "Introduction to Digital Humanities for Librarians" course plan changed a little bit every time I taught it for Library Juice Academy, often in response to student comments but sometimes out of restlessness, and the following is the syllabus for the final session of the course in early 2022.

Week 1

This week we'll get a sense of some of the issues that motivate the digital humanities. None of these readings are particularly specialized and are intended to be suggestive of different aspects of digital humanist activities. Be looking for bigger themes:

- Why use computational methods in the humanities?
- What is the relationship between scale and the need for computational methods?
- What kinds of questions can humanists bring to computational work?
- How do the digital humanities help engage academics or institutions with their publics?

Please read the following articles:

- Taylor Arnold and Lauren Tilton, "New Data? The Role of Statistics in DH," in *Debates in Digital Humanities 2019*. https://dhdebates.gc.cuny.edu/read/untitled-f2acf72c-a469-49d8-be35-67f9ac1e3a60/section/a2a6a192-f04a-4082-afaa-97c76a75b21c#ch24
- Kim Gallon, "Making a Case for the Black Digital Humanities," in *Debates in Digital Humanities 2016*. https://dhdebates.gc.cuny.edu/read/untitled/section/fa10e2e1-0c3d-4519-a958-d823aac989eb#ch04
- Tim Sharratt, "Unremembering the Forgotten," in *Debates in Digital Humanities 2019*. https://dhdebates.gc.cuny.edu/read/untitled-f2acf72c-a469-49d8-be35-67f9ac1e3a60/section/be608100-95b6-4e48-bfd5-a82a588da8f1#ch12

Once you've read these articles, write a short post (3-5 paragraphs) defining digital humanities as you see it represented in these readings. How do digital tools/approaches support/augment/enable humanities scholarship? Was

there anything unfamiliar or unexpected? What questions do you have about the digital humanities?

Week 2

This week will be devoted to exploring (at a basic level) how digital humanities are done. You'll take a look at a digital project to try to get a broader sense of DH practice while also thinking about what goes into making a DH project.

If you run into any problems, use the General Discussion forum to ask questions. Any reflections on these activities should also go in that forum.

Summary Assignment Requirement for this week:

- The digital project review
- Either option 1 (Voyant) OR option 2 (Text Encoding) OR option 3 (Programming Historian tutorial)

Choose one of the DH projects below, take a look at it, and then write up a short review for this forum. What I want you to think about is what one might look for in a digital project (you'll also be getting exposure to different types of digital projects). Some questions to consider (these are only here to stimulate your thoughts, feel free to pick a couple or ignore in your review):

- Where does this fit into humanities scholarly communication? Does it make a scholarly argument?
- Is it easy to use or navigate? Are you running into any problems with usability?
- What software/hardware was involved in creating this project?
- Can you access and re-use the data from the project? Why might this be an important consideration?
- What was most engaging/interesting about this project?
- What was least engaging/interesting?
- What questions might you ask of students looking at these projects? (Or, how might you use this kind of exercise in the context of an instruction session?)

Once you've written your review (and again, these don't need to be very long), choose a review done by one of your fellow students, read it & comment on it.

- Slave Voyages, https://slavevoyages.org/
- Chinese Text Project, https://ctext.org/

- Mining the [Richmond Daily] Dispatch, https://dsl.richmond.edu/dispatch/
- Vincent Van Gogh: The Letters, https://vangoghletters.org/vg/
- Colored Conventions Project, http://coloredconventions.org/
- On the Books: Jim Crow and the Algorithms of Resistance (Collections as Data cohort 1 project), https://onthebooks.lib.unc.edu/

Option 1 (Voyant)
Here are some interviews on the digital humanities by Melissa Dinsman published in the *L.A. Review of Books* (https://lareviewofbooks.org/feature/the-digital-in-the-humanities/; there are more than these 5, so feel free to explore or collect all of them):

- Laura Mandell
- Ted Underwood
- Pamela Fletcher
- Jessica Marie Johnson
- Marisa Parham

Use Voyant (http://voyant-tools.org/) to create a word cloud of these texts and then, when you have a word cloud that you feel expresses an answer to the "what is DH?" question, export that image and upload it here.

Also, write a short (2 or 3 paragraphs) reflection in the General Discussion forum about how this visualization compares to the work you did in week 1 (reading articles on digital humanities and then teasing out themes). Why, or in what contexts, would a word cloud be a better way to explain a topic than an essay? What is this kind of distant reading really good at? What is it not very good at?

You are welcome to use a different set of texts if you wish!

Tutorial options:

- Voyant's "Getting Started" page, http://voyant-tools.org/docs/#!/guide/start
- Miriam Posner's tutorial for Voyant, https://github.com/miriamposner/voyant-workshop/blob/master/investigating-texts-with-voyant.md

See also: Heather Froehlich, "Exploring Recipe Transcriptions with Voyant in the Classroom," https://emroc.hypotheses.org/2596

Option 2 (Text Encoding)

TEI by Example (http://teibyexample.org/) is an online tutorial for learning how to encode texts using TEI (the Text Encoding Initiative). There is a lot there and you are welcome to go through as much of it as you wish.

For this assignment, I'd like for you to read Tutorials 0 (Introduction to Text Encoding and the TEI) and 1 (Common Structure and Elements). Take the test for Common Structure and Elements and make a screenshot of your completed test, then do the Common Structure and Elements exercise (or as much of it as you can).

Upload your screenshot and then write a short reflection piece (1–3 paragraphs) in the General Discussion forum on your experience with text encoding: what did you like? what was unpleasant or difficult? what is "digital humanities" about this activity?

Option 3 (Programming Historian)

If you'd like an extra challenge, you can do so by working through one (or more) of the lessons at The Programming Historian (https://programminghistorian.org/en/lessons/; note that there are lessons available in English, Spanish, and French).

You'll get credit for the assignment by uploading (or typing in here) a few paragraphs that discuss what you learned in your tutorial, what challenges (if any) you had with this kind of static tutorial, and/or how you might improve the tutorial.

If you're not sure where to start, try one of these:

- R Basics with Tabular Data
- Reshaping JSON with jq
- Corpus Analysis with Antconc
- Getting Started with Topic Modeling and MALLET

Week 3

This week we'll consider the role of libraries and librarians in the digital humanities.

- Jennifer Vinopal and Monica McCormick, "Supporting Digital Scholarship in Research Libraries: Scalability and Sustainability." *Journal of Library Administration* 53, no. 1 (2013): 27–42.
- Lotte Wilms. "Digital Humanities in European Research Libraries: Beyond Offering Digital Collections." *LIBER Quarterly: The Journal of the Association of European Research Libraries*, 31, no. 1.

- Sarah Ames. "Transparency, provenance and collections as data: The National Library of Scotland's Data Foundry." *LIBER Quarterly: The Journal of the Association of European Research Libraries*, 31, no. 1.
- E. Leigh Bonds, "Preparing, Facilitating, Assessing: A Reflection on Digital Humanities Consultations," *International Information & Library Review* (2019).

Based on these readings, what does digital humanities in libraries look like? How does this compare with your readings in week 1? Write a short forum post addressing these questions or other thoughts you had about the readings.

Week 4

Now that you've gotten this far, it's time to think about what you might do in your own work. Your final project can be a research guide for DH or for some particular aspect of it (e.g., a guide to text mining), an outline of a training program for some population at your institution, a professional development plan, a needs assessment plan, a digital humanities project plan, or whatever you can dream up.

My goal is to get all of you thinking concretely about what we've covered up to now and how it is applicable to your own situation. Whatever you decide to do for your final project, keep it manageable (do something that you can get done by the end of the course!). You don't need to have a finished project, so having something that you've started and done enough to get some meaningful feedback from me is going to be sufficient. I want you to get started on a journey, not complete one.

If you're not sure what you want to do, contact me and we can come up with something together.

Appendix 3B. Digital Humanities and Information Literacy Workshop

Readings:

Andrea Baer, "Critical Information Literacy in the College Classroom: Exploring Scholarly Knowledge Production Through the Digital Humanities," in *Information Literacy and Social Justice: Radical Professional Praxis*, eds. Lua Gregory and Shana Higgins, 99–120. Los Angeles: Library Juice Press, 2013. https://scholarworks.iu.edu/dspace/handle/2022/17236

Stewart Varner, "Library Instruction for Digital Humanities Pedagogy in Undergraduate Classes," in John W. White and Heather Gilbert, eds., *Laying the Foundations: Digital Humanities in Academic Libraries*, West Lafayette, IN: Purdue University Press, 2016.

Zoe Borovsky and Elizabeth McAulay, "Digital Humanities Curriculum Support Inside the Library," in Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb, eds. *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*. Chicago: Association of College and Research Libraries, 2015.

Workshop Outcomes:

- After this workshop, participants will be able to:
- Situate digital humanities within information literacy and library frameworks;
- Integrate digital humanities tools into the library classroom;
- Develop an artifact (e.g., a lesson plan) exemplifying the use of digital humanities tools within an information literacy context;
- Participate in and contribute to an authentic community of practice surrounding digital humanities and information literacy.

8:45–9:00: Coffee, tea, and continental breakfast

9:00–9:15: Welcome, introductions, and housekeeping

9:15–10:15: Formal talks:

  - 9:15–9:45: Digital Humanities: An Overview
  - 9:45–10:15: Information Literacy: An Overview

10:15–11:00: Journal club and conversation (discussion of readings in groups)

11:00–11:15: BREAK

11:15–12:00: Digital Humanities classroom activity

12:00–1:00: Lunch and lesson plan brainstorming

1:00–1:30: Report out

1:30–2:00: Wrap up, next steps, and workshop evaluation

Notes

1. Jason A. Heppler, "What Is Digital Humanities?," accessed July 10, 2022, https://www.whatisdigitalhumanities.com/.
2. Clifford B. Anderson, "Introduction," in *Digital Humanities and Libraries and Archives in Religious Studies: An Introduction*, ed. Clifford B. Anderson (Boston: De Gruyter, 2022), 3.
3. "The Programming Historian," last modified July 8, 2022, https://programminghistorian.org/.
4. Association for Research Libraries, "ARL Digital Scholarship Institute," last modified August 3, 2021, https://github.com/tech-at-arl/Digital-Scholarship-Institute.
5. HathiTrust Research Center, "HTRC Digging Deeper, Reaching Further—Libraries Empowering Users to Mine the HathiTrust Digital Library Resources," accessed January 21, 2024, https://teach.htrc.illinois.edu/.
6. Matthew D. Lincoln, "Tool Trouble: Thoughts on *differences*," Matthew Lincoln, PhD (blog), April 25, 2014, https://matthewlincoln.net/2014/04/25/tool-trouble.html.
7. Nicole Pagowsky, "The Contested One-Shot: Deconstructing Power Structures to Imagine New Futures," *College & Research Libraries* 82, no. 3 (2021): 301.
8. Thomas Padilla, et al., "Always Already Computational—Collections as Data," accessed July 10, 2022, https://collectionsasdata.github.io/.
9. Trevor Muñoz, "Digital Humanities in the Library Isn't a Service," Trevor Muñoz (blog), August 19, 2012, https://trevormunoz.com/archive/posts/2012-08-19-doing-dh-in-the-library/.
10. Micah Vandegrift, "What Is Digital Humanities and What's It Doing in the Library?" *In the Library with the Lead Pipe*, June 27, 2012, https://www.inthelibrarywiththeleadpipe.org/2012/dhandthelib/.
11. John E. Russell and Merinda Kaye Hensley, "Beyond Buttonology: Digital Humanities, Digital Pedagogy, and the ACRL Framework," *College & Research Libraries News* 78, no. 11 (2017): 588–91, 600.
12. Barbara Rockenbach, ed., "Digital Humanities in Libraries: New Models for Scholarly Engagement," special issue, *Journal of Library Administration* 53, no. 1 (2013).
13. Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb, eds., *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, (Chicago: Association of College and Research Libraries, 2015).
14. "dh+lib: where digital humanities and libraries meet" (blog), accessed January 21, 2024, https://dhandlib.org/.

15. Kayla Abner, "Data Literacy as Digital Humanities Literacy," *dh+lib* (blog), June 22, 2020, https://dhandlib.org/2020/06/22/data-literacy-as-digital-humanities-literacy-exploration-of-threshold-concepts/.

Bibliography

Abner, Kayla. "Data Literacy as Digital Humanities Literacy: Exploration of Threshold Concepts." *dh+lib*, (blog), June 22, 2020. https://dhandlib.org/2020/06/22/data-literacy-as-digital-humanities-literacy-exploration-of-threshold-concepts/.

Anderson, Clifford B. "Introduction." In *Digital Humanities and Libraries and Archives in Religious Studies: An Introduction*. Edited by Clifford B. Anderson, 1–10. Boston: De Gruyter, 2022. https://doi.org/10.1515/9783110536539-002.

Association for Research Libraries. "ARL Digital Scholarship Institute." Last modified August 3, 2021. https://github.com/tech-at-arl/Digital-Scholarship-Institute.

Hartsell-Gundy, Arianne, Laura Braunstein, and Liorah Golomb. *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*. Chicago: Association of College and Research Libraries, 2015.

HathiTrust Research Center. "HTRC Digging Deeper, Reaching Further—Libraries Empowering Users to Mine the HathiTrust Digital Library Resources." Accessed January 21, 2024. https://teach.htrc.illinois.edu/.

Lincoln, Matthew D. "Tool Trouble: Thoughts on *differences*." Matthew Lincoln, PhD (blog), April 25, 2014. https://matthewlincoln.net/2014/04/25/tool-trouble.html.

Muñoz, Trevor. "Digital Humanities in the Library Isn't a Service." Trevor Muñoz (blog), August 19, 2012. https://trevormunoz.com/archive/posts/2012-08-19-doing-dh-in-the-library/.

Padilla, Thomas, Laurie Allen, Stewart Varner, Sarah Potvin, Hannah Frost, and Elizabeth Russey Roke. "Always Already Computational—Collections as Data." Accessed July 10, 2022. https://collectionsasdata.github.io/.

Pagowsky, Nicole. "The Contested One-Shot: Deconstructing Power Structures to Imagine New Futures." *College & Research Libraries* 82, no. 3 (2021): 300–309. https://doi.org/10.5860/crl.82.3.300.

"The Programming Historian." Accessed January 21, 2024. https://programminghistorian.org/.

Rockenbach, Barbara, ed. "Digital Humanities in Libraries: New Models for Scholarly Engagement." Special issue, *Journal of Library Administration* 53, no. 1 (2013). https://www.tandfonline.com/toc/wjla20/53/1.

Russell, John E., and Merinda Kaye Hensley. "Beyond Buttonology: Digital Humanities, Digital Pedagogy, and the ACRL Framework." *College & Research Libraries News* 78, no. 11 (2017): 588–91, 600. https://doi.org/10.5860/crln.78.11.588.

Vandegrift, Micah. "What Is Digital Humanities and What's It Doing in the Library?" *In the Library with the Lead Pipe*, June 27, 2012. https://www.inthelibrarywiththeleadpipe.org/2012/dhandthelib/.

What is DH (@What_is_DH). X (formerly Twitter). Accessed July 10, 2022. https://twitter.com/What_is_DH.

CHAPTER FOUR

Centering our Values:
A Framework for Digital Humanities in the Library

Pamella R. Lach

ON NOVEMBER 29, 2021, nearly two years into the COVID-19 global pandemic, members of the Digital Humanities Initiative at San Diego State University gathered over Zoom to imagine new futures. *Reimagining DH@SDSU: A Conversation about Radical Design for the DH Initiative & DH Center* was inspired by dual impulses: resistance to a post-pandemic "return to normal" coupled with a call for radical transformation wrought by the social, political, and environmental upheavals of 2020/2021. Taking as its starting point the design sessions from the original planning of SDSU Library's DH Center (which involved an interdisciplinary group of faculty, students, and library folks), this event reframed those earlier sessions for the pandemic and beyond. Asked to respond to several prompts, faculty, staff, and student participants from across disciplines and geographic boundaries brainstormed possibilities related to the values, work, and people of SDSU's DH Initiative.

Through this conversation we recommitted to our shared values of anti-racism, an ethos of care, and centering humans over technology. Attendees also emphasized the need for creative hybridity in the structuring of our spaces, programs, and resources, driven in part by the desire to extend the DH Center's physical walls to create new communities and networks of exchange. While this was only the beginning of a much larger conversation, one that would be held with administrators and strategic partners in the coming weeks and months, it reminded us of the importance of coming together to imagine new possibilities. And it underscored the importance of the DH Center as both a physical space in the library that facilitates collaborative and ethical DH engagement and a conceptual/virtual space that spans disciplinary boundaries and geographic borders.

CC BY NC SA

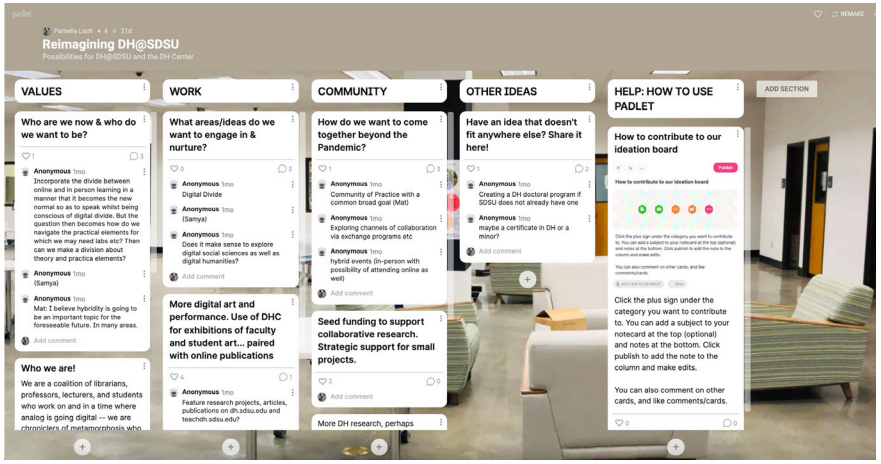


Figure 4.1. *Screenshot of collaborative Padlet developed for Reimagining DH@SDSU, https://padlet.com/plach1/mucsuocb7azvcsr.*

This chapter calls for academic libraries to be values-centered as much as user-centered when engaging in digital humanities work. Taking as its case study the growth of DH at San Diego State University (SDSU) before and during the pandemic, this chapter demonstrates how being intentional about the values that guide our work helps build equitable partnerships, meet community needs, manage institutional challenges, engage in critical digital inquiry, strive toward anti-oppressive praxis, and imagine new possibilities for building, growing, and renewing ethical digital humanities programs.

Engaging in Ethical and Values-Based Digital Humanities Librarianship: A Review of the Literature

Values-oriented library work might seem like a natural extension of a user-centered approach that recognizes the communities we serve are comprised of individuals with unique perspectives and values. Yet there is resistance to articulating a set of academic library values beyond providing access to information, teaching information literacy, and supporting the research enterprise. Such resistance is a product of the racist, colonialist, and imperialist history of US libraries, much of which is obscured by and for library workers.[1] The lack of knowledge about how libraries functioned as spaces for encouraging and forcing assimilation to white cultural norms, or as white exclusionary spaces, translates into an "inability to articulate race

and racial privilege in LIS"[2] as well as a "vocational awe" resulting in "notions that libraries as institutions are inherently good, sacred notions, and therefore beyond critique."[3] If we cannot critique libraries, we have little hope of dismantling their racist structures and systems for workers and patrons alike.

Clinging to the myth of neutrality—the notion that libraries provide objective and unbiased access to information—makes change more difficult.[4] As Ferretti notes, "Neutrality upholds white supremacy, the dominant narrative in our society.… Neutrality or deliberate ignorance speaks as loudly as directly addressing social issues. Neutrality is an opinion…. Neutral librarianship intentionally ignores marginalized communities and experiences."[5] In January 2021, at its Midwinter virtual meeting, the American Library Association (ALA) acknowledged the link between structural racism and library neutrality. In the "Resolution to Condemn White Supremacy and Fascism as Antithetical to Library Work," the organization offered a critique of the codification of vocational awe and neutrality[6] when resolving to acknowledge "the role of neutrality rhetoric in emboldening and encouraging white supremacy and fascism."[7] Though an important step in the "paradigm shift to begin to dismantle racism" that Chiu, Ettarh and Ferretti call for,[8] the ALA's Library Bill of Rights remains unchanged, and debates over neutrality continue to roil the profession.[9]

That such a step came so recently underscores the structural problems BIPOC, queer, and disabled library employees face in hostile work environments. There has been increased attention directed toward structural and systemic racism in libraries, particularly in the wake of George Floyd's 2020 murder.[10] And yet, there is an overwhelming reliance on superficial diversity efforts, as Leung and López-McKnight note in their introduction to *Knowledge Justice: Disrupting Library and Information Studies through Critical Race Theory*.[11] Thus inequitable and oppressive practices, as well as a lack of representation, persist in libraries and professional organizations alike. Many BIPOC library workers regularly call out the microaggressions and racial gaslighting they experience at ALA events.[12] In March 2021, the Black Caucus of the ALA spoke out against racism experienced in the ALA Council and in September 2021 a group of BIPOC librarians released a "Statement Against White Appropriation of Black, Indigenous, and People of Color's Labor." The authors condemned the "continued adherence to white supremacist culture and ideals" that continually harms BIPOC library folx, even as white library workers claim to value and fight for racial social justice.[13] Such statements are matched by public resignations of BIPOC library workers from academic and public libraries, citing hostile work environments.[14]

In a profession that is overwhelmingly white, retaining BIPOC library employees by creating anti-oppressive workplaces demands values-oriented approaches that embrace changing the culture of library work through critical and radical librarianship. As more academic libraries develop diversity statements and equity plans,[15] and as equity, diversity and inclusion (EDI) work is increasingly integrated into job descriptions (as is the case at my current and former institutions), we must remember that meaningful EDI work cannot be siloed from other professional duties. Working within an EDI frame demands that we put our anti-oppressive values at the forefront of our labor.

Just as mainstream academic librarianship sharpened its focus on anti-racist social justice work, many digital humanists recommitted to these values in recent years. The Association for Computers and the Humanities (ACH), a leading professional DH organization, released a statement "on Black Lives Matter, Structural Racism, and Our Organization," that included tangible "actions that will deepen the organization's commitment to anti-racist principles and that will help us understand and undo ACH's role in anti-Blackness."[16] Even as a newer field steeped in disruption and innovation, DH nonetheless replicates and advances racist, hegemonic academic norms in its subjects, processes, and scholarly products.[17] Intersectional feminist and postcolonial DH scholars are developing liberatory DH praxis whose values are visible in a range of scholarship and public-facing projects that seek to re-center DH away from the Global North and recover the lived experiences of marginalized people.[18]

As an emergent and interdisciplinary field, many digital humanists take a values-informed approach. Lisa Spiro's influential essay, "'This Is Why We Fight': Defining the Values of the Digital Humanities" articulated a statement of values for DH—a way "to develop a keener sense of what it stands for and what is at stake in its work." Her proposed values of openness, collaboration, collegiality and connectedness, diversity, and experimentation seek to position DH work as ethical, expansive, and broadly representative.[19] Likewise, the Humane Metrics Initiative (HuMetricsHHS) models values-inflected approaches for the humanities in its values framework which emphasizes equity, openness, collegiality, soundness, and community.[20] Reflecting on their efforts to develop their framework, Agate et al. argue for a radical reshaping of academic culture that transforms how academics are evaluated, dismantles hypercompetition, and redefines what success can look like. "Shifting from abstract standards of excellence to enacting values-inflected purposeful work has the capacity to transform the culture of higher education—making scholarship more meaningful for those who undertake it, more valuable for

the communities engaged by it, and more transparent to those who need to evaluate it."[21] These values shape my own praxis in building a community that can come together in the library to engage in ethical DH work and which is built on equitable collaboration between faculty and library folks. Key to these relationships is a shift from the traditional library service model toward a structure rooted in partnerships created through shared purpose and values.[22]

A similar reorientation is underway in many tech spaces, as researchers and makers call for ethical approaches to design that account for the human biases that get baked into AI algorithms, predictive technologies, and the unintended consequences of systems otherwise designed to do no harm. The Design Justice Network seeks to create a "growing community of designers, developers, technologists, scholars, educators, community organizers, and many others who are working to examine and transform design values, practices, narratives, sites, and pedagogies so that they don't continue to reinforce interlocking systems of structural inequality." Their design principles span technology, pedagogy, and the built environment.[23] Likewise, researchers Batya Friedman and David G. Hendry's Value Sensitive Design (VSD) framework encourages designers to consider the unintended consequences and possible harm of the technology they create. They offer tools, such as the VSD cards, to help designers "scaffold moral and technical imagination around the design of new technologies" in service to international justice.[24] These cards can be adapted for broader use; I used them both to teach undergraduate students about the digital project lifecycle and to encourage digital scholarship practitioners in the ethical design and growth of library spaces.[25]

My approach to designing, launching, and growing the DH Center in SDSU's library is rooted in the understanding that our values, as digital humanists and librarians, matter. Rejecting the long-held notion that libraries are neutral, this approach insists that we lean into our shared values to shape our work. For DH@SDSU, that work has always been about anti-racist social justice. Our values push us to continually question the value of the digital, to center humans—and human connections—in the digital age, and to make visible the disparate impacts of the digital as new digital divides are created.

DH@SDSU: Building a Values-Focused Approach in the Library

A large, public R2 institution with R1 aspirations, SDSU has a long history of social justice curriculum and scholarly innovation. This commitment is shaped by our membership in the California State University (CSU) system,

our status as both a Hispanic-Serving (HSI) and Asian American and Native American Pacific Islander (ANNAPISI) Serving Institution, and our positionality occupying the unceded lands of the Kumeyaay people situated near the constructed U.S.-Mexico border. SDSU's Digital Humanities Initiative, DH@SDSU, began within this institutional context as a grassroots, faculty-led community. SDSU Library was central to the growth of DH@SDSU, dedicating space, resources, and staffing to create a home for the initiative in what would become the Digital Humanities Center (DHC). A key partner from the earliest days, library members participated in cross-disciplinary conversations that engendered meaningful partnerships with teaching faculty across departments and colleges. Over time, DH@SDSU grew into a formalized initiative that is attentive to its members, intentional in its collaborations, and explicit about its anti-racist values, which are embodied physically and operationally in the DHC.

The DHC began in fall 2016 as a conversation among faculty and students about their hopes for DH@SDSU. Rather than focus ideation sessions on specific DH projects, activities, or tools, we explored who we wanted to be as a community, and how the DHC could support an emergent set of values. These values-based design sessions helped us articulate a shared vision of a physical space that centered humans and their relationships in the digital age. Our vision embraced critical digital literacy as its guiding design principle, seeking ways to manifest our critical stance towards the uneven and often harmful consequences of digital technologies, particularly for historically marginalized and vulnerable communities. These dual stances shaped the formation of the DHC as the hub for the DH community, influencing the space's design and programmatic services.

Rather than configure the space and its services according to existing models for digital humanities labs and library-based digital scholarship centers,[26] the DHC's flexible and iterative model was designed around the specific needs and desires of our DH community, envisioned to respond and adjust to shifts in the community's focus. When we launched the space in early 2018, it was still something of a blank canvas with a minimalist design and incomplete spaces to accommodate future growth. To underscore DH@SDSU's foundational values of human connection, interdisciplinarity, and community, we minimized the technology in plain sight, relying on our various digital studios (podcasting, electronic literature, virtual reality) and flexible learning spaces to be the sites of technological encounters. From the beginning, the design process was iterative, with subsequent community-based design activities continuing long after the launch.

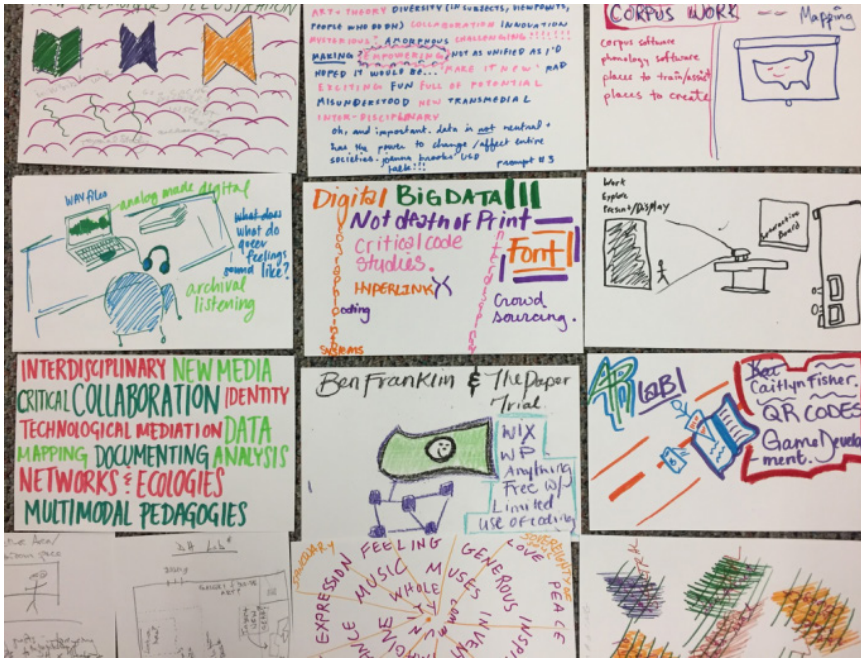


Figure 4.2. *Sketches produced in Fall 2016 DHC community design sessions.*

We adopted the same values-based, community-driven, iterative approach for designing programmatic services in the DHC that could seed deep, intentional, and meaningful collaborations. Rather than explore transactional services, such as digitization and data management (handled by other library units), the DHC sought opportunities to partner across campus to advance critical digital engagement. The space intentionally lacks a permanent service desk to reinforce this approach. Programmatic services—from public talks and workshops to research groups and showcases—and instruction sessions modeled collaboration, co-learning, and a process-oriented embrace of experimentation and risk-taking. Moreover, from the beginning the content and subject matter of our public-facing activities reflected and advanced the critical anti-racist values of our community. We continue to invite a broad range of speakers whose work addresses the intersection of technology, anti-racism, and social justice. Likewise, we employ a critical stance in our workshop series, which emphasize critical digital literacy alongside skill building. This same approach frames my DH LibGuide, which balances information about tools with carefully curated intersectional

readings and projects.[27] In short, both the design of the physical space and its programmatic services were shaped by, and continue to help shape, the values of DH@SDSU.



Figure 4.3. *Chicana and Chicano Studies Professor Jay Garza workshops digital projects with his students, March 07, 2019.*



Figure 4.4. *Professor Elizabeth A. Pollard introduces her HIST 503: Ancient Rome class to their final project showcase, April 30, 2019. Dr. Pollard and I collaborated over several semesters to scaffold digital humanities into this class. See https://sites.google.com/sdsu.edu/hist503/.*

Being values-driven helped the DHC weather several challenges over the years, including the perception of being little more than a free event venue and our perennial staffing shortages. Our values informed our decision-making about usage requests, empowering us to turn down requests too far afield.[28] These values have similarly helped us articulate, and re-articulate, the importance of DH in our library, both to library folks and campus, by giving us a framework and vocabulary for translating our work into more familiar forms of library services. And these values served as a beacon for our ever-shifting staffing model. Because of high staff turnover, usually driven by low institutional wages, we regularly revisit our essential work in order to maintain the most important programs and partnerships. Our values are considered in the crafting of revised position descriptions and guide us in finding ethical ways of relying on temporary and student staffing to fill gaps. Prior to the pandemic, we explored position reclassifications to provide more opportunities for advancement. To fill unexpected vacancies, we hired former students in gap semesters into full-time, short-termed emergency positions. I approached these short-term contracts as a paid internship, providing temporary staff with training and opportunities to support their educational and career goals. Throughout these challenges, we continued to redefine the space and programs in ways that the DH@SDSU community most needed.

Over the past few years, I began to formalize these values as a shared articulation of what it means to do DH at SDSU and in the wider profession. For example, I modeled these efforts in my work with the Association of College & Research Libraries (ACRL) Digital Scholarship Section (DSS). The DSS embedded values of equity, diversity, inclusion, anti-racism, and anti-oppression in its structures and operations when it was founded in 2017.[29] While I did not widely circulate early iterations of DH@SDSU's values statement, each version helped me better articulate those values and, equally important, informed strategic decision making for growing DH in the library, particularly during the COVID-19 pandemic (see Appendix 4A for the original statement and Fall 2020 update). At its core, DH@SDSU's shared values describe who we are, what we do, and who we want to be as a community. Each re-articulation centers our feminist ethos for engaging with each other and our work, emphasizing collaboration, cooperation, and connectedness; distributed and non-hierarchical expertise; an orientation towards process and experimentation; equitable and sustainable labor practices; care for ourselves and one another; and the drive to organize ourselves and our pursuits around anti-racist social justice.

Work Less, Care More: Re-Centering our Values in a Global Pandemic

When the global pandemic forced us to shut the DHC's doors and shift to remote services, we returned to our founding values as a guidepost. DH@SDSU had already suffered an unexpected loss in fall 2019 with an extended closure to replace the DHC's floor. During those months of closure, we opted to delay all public programming to signal the importance of the space to the community. Trying to maintain a virtual community via our blog and newsletter, however, proved insufficient. As people began returning to the DHC in November 2019, and making plans to collaborate with us in spring 2020, we realized we would have to rebuild a community that had been disjointed, distanced, and disengaged.

When we closed our doors again a few months later due to the COVID-19 lockdown, we knew we would have to be more attentive to the community as we virtualized DHC operations. We would need to privilege care—care for ourselves and each other—at a time of rapid pivots, rising anxiety, and fear of the longer-term ramifications of living during a pandemic. Unlike the maintenance closure a few months earlier, we knew we could not suspend operations, but we also knew that the virtual work would have to be intentional, meaningful, and mindful of the myriad of challenges our community members faced. We tested a few approaches in low-stakes virtual programming in April and May 2020, and spent the summer revising and rearticulating our shared values to shape the work for the fully virtual 2020-2021 academic year. Virtualizing our operations in alignment with our values proved generative, even if virtual interactions could not replace the loss of community joined by a shared space. Our values, in short, provided a buoy in the shifting waters of constant pivoting, re-training to prepare for virtual instruction, and the uncertainties of future budget and staffing cuts during a time of social upheaval.

Revisiting our values enabled us to stay focused on the essential work of supporting and maintaining our community and meeting their most pressing needs while remaining attentive to ethical labor practices by resisting the pressure to do more with less. Rather than try to be all things to all people, we scaled back our efforts so that we could tend to our community of faculty, staff, and students. Our guiding mantra became "Work Less, Care More," and this guided our pandemic choices. Rather than develop new projects, we concentrated our efforts on supporting virtual instruction and developing virtual events to showcase existing expertise. The importance

of protecting and nurturing the relationships at the heart of DH@SDSU girded our work, as we modeled care for ourselves and each other, resisted out-of-scope requests, and refused to engage in work unless it focused on anti-racist social justice.

With this renewal of our shared values, adapted for the pandemic, we set to work. We began by creating *Teaching During Quarantine*, an aggregation of existing campus DH resources and experience to support virtual instruction.[30] Created through an equitable partnership between DH@SDSU Co-Director Dr. Jessica Pressman and myself, the site is a model of the DHC's approach to its pandemic work. Designed as a showcase of DH pedagogy at SDSU, the site consists of three major parts: resources for incorporating tools into instruction, approaches to teaching students to think critically about digital technology and culture, and a calendar of virtual book talks and workshops for the 2020–2021 academic year (and subsequently updated through spring 2023). We launched the website July 1, 2020 and quickly began growing the site in response to specific community requests. Conversations with Dr. Pressman, for instance, inspired the addition of the "Setting the Tone" page.[31] Discussing her own concerns about creating a productive and generous learning community over Zoom, we realized that our values-informed approach to virtual interactions could be a model for other faculty. We continued to grow the site even as campus increasingly returned to in-person operations. In fall 2021 we rebranded the site as *Teaching Beyond Quarantine* and began making plans to improve its structure and usability. In spring 2022 we began planning to restructure and make more visible the peer-to-peer tool tutorials DHC student assistants created during the pandemic.

While the DHC informally created tutorials and other learning objects before the pandemic, the intentional production of virtual tutorials during the pandemic was explicitly designed with ethical labor practices in mind. With our permanent staff searches on hold due to a CSU-wide "hiring chill," we relied on temporary staff and student assistants to keep DHC's operations virtual. While I focused on instruction, consultations, collaborative programming, and workshops, the DHC staff engaged in tutorial creation. We envisioned the tutorial project as a professional growth opportunity (rather than simply busy work) for our graduate and undergraduate student assistants to gain tech skills while creating tangible objects that they could show future employers. We established a workflow for creating tool-based tutorials to support virtual instruction, DH instruction, and the campus' adoption of Adobe Creative Cloud for student use in 2020–2021. Under the direction of

the temporary DHC staff, first Cassie Tanks and then Cathy Qiu, we encouraged DHC students and staff to identify a tool of interest, teach themself the tool, develop a project, and then create slides and a video tutorial to teach others, including their peers. Rather than insist on a standard look and feel, the tutorials reflect the creator's individual interests, skills, and creative abilities. While the tutorials range in quality and content, all embody our core values of enacting ethical labor practices, giving credit where credit is due, emphasizing process over product, decentering expertise, and emphasizing co-learning. We released the tutorials as a set of shared Google Drive folders and embedded them in *Teaching During Quarantine* and my DH LibGuide.[32] Though the first phase of this project reflects its just-in-time nature, tutorial production continues well beyond campus repopulation, responding to community needs while providing tangible professional development for the team. Under DHC Programs Specialist Romain Delaville's direction, we migrated the tutorials to the DH@SDSU YouTube channel for greater accessibility and findability in summer 2022, while a new group of student workers developed additional tutorials.[33]

In addition to creating tool tutorials, the DHC virtualized its public programs. Prior to the pandemic, the DHC hosted a range of public lectures, workshops, and showcases that emphasized experimentation, co-learning, and co-creation. We transformed those activities into a robust set of online events held over Zoom. While attending events via Zoom could never replicate the experience of coming together in the DHC, event setup was significantly easier and online attendance was higher than most of our face-to-face events (geographic proximity was no longer a barrier to attendance). Zoom's live transcription feature supported accessibility, and the chat function facilitated participation. We encouraged audience members to keep cameras off and ask questions via chat if they were uncomfortable being seen and heard. Moreover, for folks unable to attend in real time (whether due to scheduling conflicts or wifi issues), we developed a repository of recordings, posted to our YouTube channel and at TeachDH.[34] These videos could be used as future resources—such as extra credit for classes—to keep our community engaged. Indeed, my fall 2020 text analysis workshop was assigned in spring 2021 at CSU in San Marcos. As of June 2022, it had been viewed nearly 950 times and my introduction to digital pedagogy workshop for the humanities over 300 times.

Our efforts to virtualize DHC operations centered on the need to maintain our DH community (in ways we failed to during the 2019 facilities closure) by providing critical areas of support, developing opportunities to come together in virtual spaces, and being mindful of the cost of extra labor

during the pandemic. To keep us focused, and ready to defend this position at a moment's notice, we developed a pandemic version of our shared values to protect us against unreasonable requests (see Appendix 4A). With *work less, care more* to guide us, we intentionally slowed down our work, focusing on the most essential elements, and refused to ask our community members to take on more labor. Rather than do more and more, we turned inward, using our values to contemplate who we are and where we wanted to go during and beyond the pandemic. This allowed us to be open to new futures and new possibilities, such as coordinating our work with other CSU campuses in order to grow DH across the system in ways that are unique and much needed for Californians. And we used our pandemic values to safeguard against the anticipated uncertainties of the future—budget cuts, reduced staffing, and long-term limits to our capacity. This helped protect ourselves against institutional demands that ran counter to our own priorities of ethical, collaborative, anti-racist digital humanities praxis.

We encapsulated this approach in a March 19, 2021 statement about why we chose *not* to hold our annual DH project showcase as a virtual event, which received an overwhelmingly positive response among DH@SDSU folks and the broader DH community. Before Covid our showcases were the culminating event of our academic year—a way to celebrate both the projects our faculty and students created and the physical space that facilitated those accomplishments. But we recognized the visible and hidden costs of asking faculty to recruit students to participate during the too-long pandemic year. And so we declared, "After careful thought, we have decided that the way to honor our community is to *not* ask more of our weary faculty; *not* to ask our students to take the lesser option once again. Instead, we acknowledge the continued and cumulative effects of pandemic anxiety, Zoom fatigue, and the myriad of life stressors this past year has wrought."[35]

Our decision to cancel our showcase, driven by our values, also recognized the importance of the DHC to the community. We had come to view our showcase as a celebration of the library space as much as the work that happens in and through it. And so, as I argued in a DH symposium organized by the ACRL DSS Digital Humanities Discussion Group in March 2021, "Despite successful virtualization, we continue to maintain that *space matters*. There is something *irreplaceable* in bringing people together to explore, create, and critique the digital."[36] After more than a year of operating virtually, we know that the DHC is more than just a set of programs, instruction sessions, consultations, and resources. But the work of redefining what is possible will continue long after the COVID-19 global pandemic.

Beyond Quarantine: A Framework for Reimagining DH in the Library

In the early days of the pandemic, it was already clear that we would need to radically re-envision the DHC for a post-pandemic world. Anticipating long-term staffing reductions (which are beginning to manifest) and tightened budgets that would hamper operations, capacity, and growth, we worked to reframe the losses of the pandemic into opportunities for creative reinvention—a chance to fix things that did not work well before and imagine new possibilities for a leaner and more intentional operational model. Centering our shared values throughout this period of forced disruption allowed us to focus on the work that mattered most to our community—to scale down and, rather than try to do more with less, do less with greater impact.[37]

This approach shaped our fall 2021 hybrid reopening, which included by-appointment access to our digital studios, small-scale in-person instruction, and virtual programs. We fully reopened the space in spring 2022 with a hybrid programming approach that allowed for in-person activities when necessary and appropriate, and continuing virtual (and/or hybrid) events to ensure greater access to our programs. It is my intention that we maintain a hybrid programming approach well beyond the pandemic to remain agile in the face of future crises, whether environmental, political, social, or health-related. While a hybrid model runs the risk of a "space grab" due to perceived underutilization, such an approach nonetheless insists on a more intentional use of space, and is the approach our DH community explicitly wants. We must constantly interrogate the competing needs of bringing people together physically with ensuring access to distant DH communities whose interests and values align and intersect with ours. In short, this approach asks us to redefine what *capacity* and *usage* mean in a post-pandemic climate. We can continue making the case for investing in DH library spaces, even and especially in a more hybridized world, by grounding our work in shared values that balance community needs with institutional priorities.[38]

To borrow from Miriam Posner,[39] rather than invest in technology, furniture, and other objects that live in a library DH space, we need to invest in the people—the relationships, the passions, the values—that make up the core of DH in a particular place. Focusing on the community and their shared values allows DH library programs to weather institutional shifts and commitments by focusing on the work that matters most to the people who matter most. Aligning the work to the community's values, just as a library's strategic plan might align with the overarching campus strategic plan, allows

us to make the case for DH in the library by being not simply user-centered, but values-centered. It allows us to say "yes" to some work and "no" to other projects that go against our values. And it empowers us to advocate for ethical labor practices, to pause some work rather than burden our partners with extra work that does not advance our values. Radical transformation is possible, but only within a framework of shared values of ethical, anti-racist, social justice DH work that is truly collaborative, generous, and generative, and that centers humans and humaneness in the work.

Appendix 4A. DH@SDSU Shared/Community Values

The following is an intentional articulation of who we are, what we do, and who we want to be as a community. These shared values are a set of guiding principles that shape our current and future work. They are inspired by early DH@SDSU efforts, including the successful application to become an Area of Excellence, as well as efforts of DH@SDSU members to work for equity, diversity, inclusion (EDI), and social justice, both on campus and in our professional lives. These values are not set in stone; we commit as a community to amend these as we continue to grow and explore new scholarly and pedagogic directions. (From Summer/Fall 2019.)

We, the DH community of SDSU, value:

- The importance of feminist collaboration and cooperation, which emphasizes non-hierarchical consensus and acknowledges and values individuals' labor, including emotional labor and so-called "soft" skills;
- Giving credit where credit is due;
- Care work as equal to, if not greater measure than, code work;
- The process of creating as much as the products we create;
- Our position as co-learners with each other and our students;
- The importance of people and human connection over technology;
- A shared commitment to focus our labor on EDI and social justice, both locally and globally;
- Non-hierarchical forms of academic labor;
- Equitable relationships and visible labor practices that seek non-exploitative conditions, particularly for our students;
- Our Initiative as a network of individuals, rather than a rigid or predefined set of core members; with a leadership model that is flexible/responsive, extends outwards, invites new members, and welcomes new and shared leadership.

DH Center: Values Driven

The DH Center is a visual and physical representation of these core values, in particular: equity, visible labor, care work, giving credit, being process-oriented, and co-learning opportunities.

The DHC design process was values-driven: early design sessions were developed and implemented in order to identify shared values and shared vision for the DHC and DH@SDSU. That initial language was used to design the DHC, both as a space, as a set of programmatic services, and as a community (Fall 2016):

> *The Digital Humanities Center at SDSU Library is envisioned as a site that values the importance of the humanities, and the interactions of humans in real time and space, in the digital age. It will serve as the home for our Digital Humanities Initiative and Area of Excellence, and function as a hub for interdisciplinary collaborative inquiry that interrogates the intersections and divergences of the human and the digital. The Center will support work that seeks to leverage the digital—in research, teaching, and learning—to promote diversity, inclusivity, and social justice across uneven boundaries of race, class, gender, sexuality, nationality, ability, age, religious identity, and spoken language, among many other human factors. We welcome faculty, staff, students, and community members from across the disciplines and spanning the technological spectrum, from the skeptics, to the curious, to the consumers and producers/makers/civic hackers of digital technologies. The space will provide opportunities for faculty, staff, students, and the community to create connections and collaborations across disciplinary and technological boundaries; to share and showcase their work to inspire ever-more creative digital research; to experiment, use, adapt, and create new technologies for the social good; and to question and critique the impact of the digital in the Information Age.*

Practicing our Values

Our core values represent the type of community we want to be and the work we want to do. To that end, let's think intentionally about how our actions, research questions, and interactions may support or detract from these values. How will we, as individuals and members of a community, commit to upholding our values?

Acknowledgments

These values were inspired by similar efforts being undertaken in professional societies and scholarly organizations, including:

- the HuMetrics HSS Code of Conduct, https://humetricshss.org/code-of-conduct/
- the ACRL Digital Scholarship Section Community Agreement, http://tiny.cc/dss-agreements
- the Digital Library Federation Code of Conduct, https://www.diglib.org/about/code-of-conduct/

Update: DH Values for a Global Pandemic (October, 2020)

With the ongoing global pandemic, we recommit to these values as a way to focus our energy and renew our purpose. With anticipated long-term staffing shortages and budget austerity, we insist that we respond strategically to requests to "do more with less" by being thoughtful in the work we take up, making sure our commitments align with our values, advance our partnerships across campus and the community, and meet our individual and collective capacities.

We remain ever mindful of the added stressors of the pandemic, including homeschooling young children, the breakdown of work/life balance, and (invisible) pressures to maintain pre-pandemic levels of productivity—all of which are amplified by larger social challenges, including the fight for racial justice, environmental crises, and political uncertainty. We cannot and should not ignore these factors, which can overshadow and fully displace our professional focus. We must reserve our energy for the work that matters; we resist taking on unnecessary work.

As such, we embrace the following principles as an extension of our core values. We will:

- Help our community with virtual instruction and other **immediate** needs while reminding faculty that they need not reinvent their pedagogic praxis for the ongoing pandemic
- Engage in equitable and collaborative work that meaningfully advances our values without placing undue burdens on ourselves or our DH

colleagues, recognizing that our ability to perform "extra" labor might be less capacious than in non-pandemic times
- Showcase and amplify current and ongoing work of the DH community before taking on new projects
- Resist calls to "do more with less" and instead strategically choose to do things with greater (potential) impact
- Center the importance of relationships and community by modeling care for ourselves and each other
- Remember that we do not strive to be all things to all people. Our purpose, particularly during the pandemic, is to fill needs and niches that are most relevant to our broader DH community
- Insist that all DH efforts focus on anti-racism and social justice, that those efforts strive to avoid creating new digital divides, and that they resist supporting digital surveillance efforts, whether in learning spaces (through edtech) or in public spaces.

Notes

1. Honma, "Trippin' Over the Color Line: The Invisibility of Race in Library and Information Studies," *InterActions: UCLA Journal of Education and Information Studies* 1, no. 2 (2005), https://escholarship.org/uc/item/4nj0w1mp; Sierpe, "Confronting Librarianship and its Function in the Structure of White Supremacy and the Ethno State," *Journal of Radical Librarianship* 5 (2019): 84–102, https://journal.radicallibrarianship.org/index.php/journal/article/view/39.
2. Honma, n.p.
3. Ettarh, "Vocational Awe and Librarianship: The Lies We Tell Ourselves," *In the Library with the Lead Pipe* January 10, 2018, https://www.inthelibrarywiththeleadpipe.org/2018/vocational-awe/.
4. Ferretti, "Neutrality is Hostility: The Impact of (False) Neutrality in Academic Librarianship," *Medium*, February 13, 2018; https://medium.com/@CityThatReads/neutrality-is-hostility-the-impact-of-false-neutrality-in-academic-librarianship-c0755879fb09; Sierpe, "Confronting Librarianship and its Function in the Structure of White Supremacy and the Ethno State."
5. Ferretti, "Neutrality is Hostility."
6. Chiu, Ettarh, and Ferretti, "Not the Shark, but the Water: How Neutrality and Vocational Awe Intertwine to Uphold White Supremacy," in *Knowledge Justice: Disrupting Library and Information Studies through Critical Race Theory*, edited by Sofia Y. Leung and Jorge R. López-McKnight, 49–71, Cambridge, MA: MIT Press, 2021.
7. American Library Association, "Resolution to Condemn White Supremacy and Fascism as Antithetical to Library Work," accessed January 20, 2021, https://www.ala.org/aboutala/sites/ala.org.aboutala/files/content/ALA%20CD%2043%20Resolution%20to%20Condemn%20White%20Supremacy%20and%20Fascism%20as%20Antithetical%20to%20Library%20Work%20Final_0.pdf.
8. Chiu, Ettarh, and Ferretti, "Not the Shark," 67.

9. Cooke et al., "Once More for Those in the Back: Libraries Are Not Neutral," *Publishers Weekly*, June 10, 2022, https://www.publishersweekly.com/pw/by-topic/industry-news/libraries/article/89576-once-more-for-those-in-the-back-libraries-are-not-neutral.html.
10. Ossom-Williamson et al., "Starting with I: Combating Anti-Blackness in Libraries," *Journal Articles: Leon S. McGoogan Health Sciences Library* 9 (December 2020), https://digitalcommons.unmc.edu/mcgoogan_articles/9; Burke, "Moving Beyond the Statements: The Need for Action to Address Structural Racism at Predominantly White Institutions," *International Journal of Multidisciplinary Perspectives in Higher Education* 5, no. 1 (August 15, 2020): 174–179, https://www.ojed.org/index.php/jimphe/article/view/2632; Hurley, Kostelecky and Townsend, "Cultural Humility in Libraries," *Reference Services Review* 47, no. 4 (2019): 544-555, https://doi.org/10.1108/RSR-06-2019-0042; Fox, ed., "Disorientation Guide to Librarianship," October 21, 2021, http://violetbfox.info/disorientation/. Since Floyd's 2020 murder, there was a growth in LibGuides, syllabi, and shared reading lists on anti-racism in libraries. See, for example, this public Zotero group to which members continue to add content: https://www.zotero.org/groups/2513245/anti-racism_readings_for_academic_libraries.
11. Leung and López-McKnight, "Introduction: This is Only the Beginning," in *Knowledge Justice: Disrupting Library and Information Studies through Critical Race Theory*, 3, Cambridge, MA: MIT Press, 2021.
12. Hathcock, "Goodbye, ALA," At The Intersection: Blog about the Intersection of Libraries, Law, Feminism, and Diversity, June 29, 2021, https://aprilhathcock.wordpress.com/2021/06/29/goodbye-ala/; Hathcock, "White Librarianship in Blackface: Diversity Initiatives in LIS," *In the Library with the Lead Pipe*, October 7, 2015, http://www.inthelibrarywiththeleadpipe.org/2015/lis-diversity/.
13. Black Caucus of the American Library Association, "Press Release: BCALA Statement to ALA Council Against Racism," March 17, 2021, https://www.bcala.org/press-release-bcala-statement-to-ala-council-against-racism.
14. See, for instance, Ogihara Resignation Letter, January 29, 2021; and Booker, ILF Resignation Letter, March 31, 2021.
15. The ACRL Digital Scholarship Section's EDI Toolkit for Digital Scholarship provides a sampling of library diversity plans. Since the toolkit's launch in April, 2021, there are likely far more library diversity plans than what is reflected here: https://dsseditoolkit.hcommons.org/resources/#library_diversity_plans.
16. Association for Computers and the Humanities, "ACH Statement on Black Lives Matter, Structural Racism, and Our Organization," accessed June 18, 2020, https://ach.org/ach-statement-on-black-lives-matter-structural-racism-and-our-organization/.
17. McPherson, "Why Are the Digital Humanities So White? or Thinking the Histories of Race and Computation," in *Debates in the Digital Humanities*, edited by Matthew K. Gold, Minneapolis: University of Minnesota Press, 2012, https://dhdebates.gc.cuny.edu/read/untitled-88c11800-9446-469b-a3be-3fdb36bfbd1e/section/20df8acd-9ab9-4f35-8a5d-e91aa5f4a0ea#ch09; Liu, "Toward a Diversity Stack: Digital Humanities and Diversity as Technical Problem," *PMLA/Publications of the Modern Language Association of America* 135, no 1 (2020): 130–151, https://doi.org/10.1632/pmla.2020.135.1.130; Risam, "Beyond the Margins: Intersectionality and the Digital Humanities," *Digital Humanities Quarterly* 9, no. 2 (2015), http://www.digitalhumanities.org/dhq/vol/9/2/000208/000208.html; and Losh and Wernimont, eds., *Bodies of Information: Intersectional Feminism and Digital Humanities*, Minneapolis:

University of Minnesota Press, 2018, https://dhdebates.gc.cuny.edu/projects/bodies-of-information.

18. See, for instance, Risam, *New Digital Worlds: Postcolonial Digital Humanities in Theory, Praxis, and Pedagogy*; D'Ignazio and Klein, *Data Feminism*; and Risam and Josephs eds., *The Digital Black Atlantic: Exploring the Intersections of Digital Humanities and African Diaspora Studies*. Projects include Torn Apart/Separados, http://xpmethod.columbia.edu/torn-apart/volume/1/; the Colored Conventions Project, https://coloredconventions.org/; and the Mukurtu cultural heritage content management system, https://mukurtu.org/.
19. Spiro, "'This Is Why We Fight': Defining the Values of the Digital Humanities," in *Debates in the Digital Humanities*, edited by Matthew K. Gold, Minneapolis: University of Minnesota Press, 2012, https://dhdebates.gc.cuny.edu/read/untitled-88c11800-9446-469b-a3be-3fdb36bfbd1e/section/9e014167-c688-43ab-8b12-0f6746095335#ch03.
20. Humane Metrics Initiative (HuMetricsHHS), "Values Framework," accessed April 26, 2021, https://humetricshss.org/our-work/values/.
21. Agate et al., "The Transformative Power of Values-Enacted Scholarship," *Humanities and Social Sciences Communications* 7, article no. 165 (2020), https://doi.org/10.1057/s41599-020-00647-z.
22. Muñoz, "Digital Humanities in the Library Isn't a Service," Trevor Muñoz (blog), April 19, 2012, http://trevormunoz.com/notebook/2012/08/19/doing-dh-in-the-library.html; Muñoz, "In Service? A Further Provocation on Digital Humanities Research in Libraries," *dh+lib* Mini-Series, June 19, 2013, http://acrl.ala.org/dh/2013/06/19/in-service-a-further-provocation-on-digital-humanities-research-in-libraries/; Vandegrift and Varner, "Evolving in Common: Creating Mutually Supportive Relationships Between Libraries and the Digital Humanities," *Journal of Library Administration* 53, no. 1 (2013): 67–78, https://doi.org/10.1080/01930826.2013.756699; Posner, "No Half Measures: Overcoming Common Challenges to Doing Digital Humanities in the Library," *Journal of Library Administration* 53, no. 1 (2013): 43–52, https://doi.org/10.1080/01930826.2013.756694; Risam, Snow, and Edwards, "Building an Ethical Digital Humanities Community: Librarian, Faculty and Student Collaboration," *College and Undergraduate Libraries* 24, no. 2–4 (2017): 337–349, https://doi.org/10.1080/10691316.2017.1337530; and Risam and Edwards, "Transforming the Landscape of Labor at Universities through Digital Humanities," in *Digital Humanities, Libraries, and Partnerships: A Critical Examination of Labor, Networks, and Community*, Robin Kear and Kate Joranson, eds., 3–17, Cambridge, MA: Chandos Publishing, 2018.
23. Costanza-Chock, *Design Justice: Community-Led Practices to Build the Worlds We Need*, preface, Cambridge, MA: MIT Press, 2020, https://designjustice.mitpress.mit.edu/; Design Justice Network design principles, https://designjustice.org/read-the-principles.
24. Friedman and Hendry, *Value Sensitive Design: Shaping Technology with Moral Imagination*, Cambridge, MA: MIT Press, 2019. VSD cards are available at https://vsdesign.org/toolkits/.
25. Chen et al., "Incorporating Values Sensitive Design into Crowdsourcing Methodologies for Knowledge Collaboration," *Proceedings of the Association of Information Science and Technology* 54 no. 1 (2017): 545–548, https://asistdl.onlinelibrary.wiley.com/doi/10.1002/pra2.2017.14505401068. As co-convener of the ACRL Digital Scholarship Section's Digital Scholarship Center Discussion Group, Rebecca Metzger and I used VSD for "Designing Accessible Digital Scholarship Spaces and Services," at ALA

Annual 2019 in Washington, D.C., https://www.eventscribe.com/2019/ALA-Annual/fsPopup.asp?Mode=presInfo&PresentationID=519136 [site discontinued].

26. Vinopal and McCormick, "Supporting Digital Scholarship in Research Libraries: Scalability and Sustainability," *Journal of Library Administration* 53, no. 1 (2013): 27–42, https://doi.org/10.1080/01930826.2013.756689; Goldenberg-Hart, "Report of a CNI-ARL Workshop: Planning a Digital Scholarship Center 2016," Coalition for Networked Information and Association of Research Libraries, August 12, 2016, https://library.educause.edu/resources/2016/8/report-of-a-cni-arl-workshop-planning-a-digital-scholarship-center-2016; Miller, "DS/DH Start-Ups: A Library Model for Advancing Scholarship through Collaboration," *Journal of Web Librarianship* 10, no. 2 (2016): 83–100, https://doi.org/10.1080/19322909.2016.1149544; and Anne et al., "Building Capacity for Digital Humanities: A Framework for Institutional Planning," ECAR Working Group Paper. Louisville, CO: ECAR, May 31, 2017, https://library.educause.edu/resources/2017/5/building-capacity-for-digital-humanities-a-framework-for-institutional-planning.
27. Lach, Digital Humanities LibGuide, https://libguides.sdsu.edu/digitalhumanities.
28. The DHC usage policy, updated regularly, is available at https://library.sdsu.edu/digital-humanities-center/usage-request-policy.
29. The DSS EDI Committee's Toolkit for Digital Scholarship provides an overview of DSS efforts to engage in anti-oppressive and anti-racist work: https://dsseditoolkit.hcommons.org/edi-work-in-the-acrl-digital-scholarship-section/.
30. *Teaching During Quarantine*, https://teachdh.sdsu.edu/.
31. "Setting the Tone," *Teaching During Quarantine*, https://teachdh.sdsu.edu/creating-community/.
32. "Tools," Teaching During Quarantine, https://teachdh.sdsu.edu/tools/; DH LibGuide, https://libguides.sdsu.edu/digitalhumanities/create.
33. SDSU DH Initiative YouTube Channel, Tool Tutorials Playlist, https://www.youtube.com/playlist?list=PLbsjQnZdzhODquq_ZUW7gdLbbokYMlv3A.
34. Virtual events were posted at https://teachdh.sdsu.edu/programming/ and archived for re-use on the DH Initiative YouTube channel, https://www.youtube.com/channel/UC5_1muxwS1HDXkReMtQiEZQ/videos.
35. Pressman and Lach, "Our DH Showcase Statement: Honoring our Experience as Rationale for Cancellation," DH Initiative Blog, March 19, 2021, https://dhblog.sdsu.edu/dh-news/our-dh-showcase-statement.
36. Lach, "Space Still Matters: Advocating for Dedicated DH Library Spaces in a Virtual World," ACRL Digital Humanities Discussion Group Online Symposium, March 29, 2021, https://youtu.be/kZUubAcTGvU.
37. Lach, "Beyond Quarantine: Reimagining Remote Possibilities in a Library Digital Humanities Center," ACRL Digital Humanities Discussion Group Online Symposium, March 29, 2021, https://youtu.be/kZUubAcTGvU.
38. Lach, "Space Still Matters."
39. Posner, "Commit to DH People not DH Projects," Miriam Posner's Blog (blog), March 18, 2014, http://miriamposner.com/blog/commit-to-dh-people-not-dh-projects/.

Bibliography

ACRL Digital Scholarship Section. "Equity, Diversity, and Inclusion Toolkit." Updated March 17, 2021. https://dsseditoolkit.hcommons.org/.

Agate, Nicky, Rebecca Kennison, Stacy Konkiel, Christopher P. Long, Jason Rhody, Simone Sacchi, and Penelope Weber. "The Transformative Power of Values-Enacted Scholarship." *Humanities and Social Sciences Communications* 7, article no. 165 (2020). https://doi.org/10.1057/s41599-020-00647-z.

American Library Association. "Resolution to Condemn White Supremacy and Fascism as Antithetical to Library Work." 2020–2021 ALA CD#43. 2021 ALA Virtual Midwinter Meeting. Accessed January 20, 2021. https://www.ala.org/aboutala/sites/ala.org.aboutala/files/content/ALA%20CD%2043%20Resolution%20to%20Condemn%20White%20Supremacy%20and%20Fascism%20as%20Antithetical%20to%20Library%20Work%20Final_0.pdf.

Anne, Kirk M., Tara Carlisle, Quinn Dombrowski, Erin Glass, Tassie Gniady, Jason Jones, Joan Lippincott, et al. "Building Capacity for Digital Humanities: A Framework for Institutional Planning." ECAR Working Group Paper. Louisville, CO: ECAR, May 31, 2017. https://library.educause.edu/resources/2017/5/building-capacity-for-digital-humanities-a-framework-for-institutional-planning.

Association for Computers and the Humanities. "ACH Statement on Black Lives Matter, Structural Racism, and Our Organization." Accessed June 18, 2020. https://ach.org/ach-statement-on-black-lives-matter-structural-racism-and-our-organization/.

Black Caucus of the American Library Association. "Press Release: BCALA Statement to ALA Council Against Racism." March 17, 2021. https://www.bcala.org/press-release-bcala-statement-to-ala-council-against-racism.

Booker, Latrice. ILF Resignation Letter, March 31, 2021. Accessed April 1, 2021. https://indiana-my.sharepoint.com/:w:/g/personal/lbooker_iu_edu/EZRXYtiV3rNFg_0RnfR02MMBmE4FNzGo2lE-3p8kMPSADg?rtime=PR5wzVr12Eg.

Brown, Alexandria, James Cheng, Isabel Espinal, Brittany Paloma Fiedler, Joyce Gabiola, Sofia Leung, Nisha Mody, Alanna Aiko Moore, Teresa Y. Neely, and Peace Ossom-Williamson. "Statement Against White Appropriation of Black, Indigenous, and People of Color's Labor." *WOC+LIB*. September 3, 2021. https://www.wocandlib.org/features/2021/9/3/statement-against-white-appropriation-of-black-indigenous-and-people-of-colors-labor.

Burke, Monica. "Moving Beyond the Statements: The Need for Action to Address Structural Racism at Predominantly White Institutions." *International Journal of Multidisciplinary Perspectives in Higher Education* 5, no. 1 (August 15, 2020): 174–179. https://www.ojed.org/index.php/jimphe/article/view/2632.

Chen, Annie T., Samantha J. Kaplan, Pamella Lach, and Lu Xiao. "Incorporating Values Sensitive Design into Crowdsourcing Methodologies for Knowledge Collaboration." *Proceedings of the Association of Information Science and Technology* 54 no. 1 (2017): 545–548. https://asistdl.onlinelibrary.wiley.com/doi/10.1002/pra2.2017.14505401068.

Chiu, Anastasia, Fobazi M. Ettarh, and Jennifer A. Ferretti. "Not the Shark, but the Water: How Neutrality and Vocational Awe Intertwine to Uphold White Supremacy." In *Knowledge Justice: Disrupting Library and Information Studies through Critical Race Theory*, edited by Sofia Y. Leung and Jorge R. López-McKnight, 49–71. Cambridge, MA: MIT Press, 2021.

Cooke, Nicole A., Renate Chancellor, Yasmeen Shorish, Sarah Park Dahlen, and Amelia Gibson. "Once More for Those in the Back: Libraries Are Not Neutral." *Publishers Weekly*, June 10, 2022. https://www.publishersweekly.com/pw/by-topic/industry-news/libraries/article/89576-once-more-for-those-in-the-back-libraries-are-not-neutral.html.

Costanza-Chock, Sasha. *Design Justice: Community-Led Practices to Build the Worlds We Need*. Cambridge, MA: MIT Press, 2020. https://designjustice.mitpress.mit.edu/.

D'Ignazio, Catherine and Lauren F. Klein. *Data Feminism*. Cambridge, MA: MIT Press, 2020. https://data-feminism.mitpress.mit.edu/.

Ettarh, Fobazi. "Vocational Awe and Librarianship: The Lies We Tell Ourselves." *In the Library with the Lead Pipe*, January 10, 2018. https://www.inthelibrarywiththeleadpipe.org/2018/vocational-awe/.

Ferretti, Jennifer A. "Neutrality is Hostility: The Impact of (False) Neutrality in Academic Librarianship." *Medium*, February 13, 2018. https://medium.com/@CityThatReads/neutrality-is-hostility-the-impact-of-false-neutrality-in-academic-librarianship-c0755879fb09.

Fox, Violet B., ed. "Disorientation Guide to Librarianship." October 21, 2021. http://violetbfox.info/disorientation/.

Friedman, Batya and David G. Hendry. *Value Sensitive Design: Shaping Technology with Moral Imagination*. Cambridge, MA: MIT Press, 2019.

Goldenberg-Hart, Diane. "Report of a CNI-ARL Workshop: Planning a Digital Scholarship Center 2016." Coalition for Networked Information and Association of Research Libraries, August 12, 2016. https://library.educause.edu/resources/2016/8/report-of-a-cni-arl-workshop-planning-a-digital-scholarship-center-2016.

Hathcock, April. "Goodbye, ALA." *At The Intersection: Blog about the Intersection of Libraries, Law, Feminism, and Diversity*, June 29, 2021. https://aprilhathcock.wordpress.com/2021/06/29/goodbye-ala/.

Hathcock, April. "White Librarianship in Blackface: Diversity Initiatives in LIS." *In the Library with the Lead Pipe*, October 7, 2015. http://www.inthelibrarywiththeleadpipe.org/2015/lis-diversity/.

Honma, Todd. "Trippin' Over the Color Line: The Invisibility of Race in Library and Information Studies." *InterActions: UCLA Journal of Education and Information Studies* 1, no. 2 (2005). https://escholarship.org/uc/item/4nj0w1mp.

Humane Metrics Initiative (HuMetricsHHS), "Values Framework." Accessed April 26, 2021. https://humetricshss.org/our-work/values/.

Hurley, David A., Sarah R. Kostelecky and Lori Townsend. "Cultural Humility in Libraries." *Reference Services Review* 47, no. 4 (2019): 544-555. https://doi.org/10.1108/RSR-06-2019-0042.

Lach, Pamella R. "Beyond Quarantine: Reimagining Remote Possibilities in a Library Digital Humanities Center." CNI Digital Scholarship Planning 2020 Webinar Series, September 15, 2020. https://www.cni.org/events/cni-workshops/digital-scholarship-planning-2020-webinar-series/supporting-digital-scholarship-during-the-pandemic-whats-possible.

Lach, Pamella R. "Space Still Matters: Advocating for Dedicated DH Library Spaces in a Virtual World." ACRL Digital Humanities Discussion Group Online Symposium, March 29, 2021. https://youtu.be/kZUubAcTGvU.

Leung, Sofia Y. and Jorge López-McKnight. "Introduction: This is Only the Beginning." In *Knowledge Justice: Disrupting Library and Information Studies through Critical Race Theory*, edited by Leung and López-McKnight, 1–41. Cambridge, MA: MIT Press, 2021.

Liu, Alan. "Toward a Diversity Stack: Digital Humanities and Diversity as Technical Problem." *PMLA/Publications of the Modern Language Association of America* 135, no 1 (2020): 130–151. https://doi.org/10.1632/pmla.2020.135.1.130.

Losh, Elizabeth and Jacqueline Wernimont, eds. *Bodies of Information: Intersectional Feminism and Digital Humanities*. Minneapolis: University of Minnesota Press, 2018. https://dhdebates.gc.cuny.edu/projects/bodies-of-information.

McPherson, Tara. "Why Are the Digital Humanities So White? or Thinking the Histories of Race and Computation." In *Debates in the Digital Humanities*, edited by Matthew K. Gold. Minneapolis: University of Minnesota Press, 2012. https://dhdebates.gc.cuny.edu/read/untitled-88c11800-9446-469b-a3be-3fdb36bfbd1e/section/20df8acd-9ab9-4f35-8a5d-e91aa5f4a0ea#ch09.

Miller, A. "DS/DH Start-Ups: A Library Model for Advancing Scholarship through Collaboration." *Journal of Web Librarianship* 10, no. 2 (2016): 83–100. https://doi.org/10.1080/19322909.2016.1149544.

Muñoz, Trevor. "Digital Humanities in the library Isn't a Service." Trevor Muñoz (blog), April 19, 2012. http://trevormunoz.com/notebook/2012/08/19/doing-dh-in-the-library.html.

Muñoz, Trevor. "In Service? A Further Provocation on Digital Humanities Research in Libraries." *dh+lib* Mini-Series, June 19, 2013. http://acrl.ala.org/dh/2013/06/19/in-service-a-further-provocation-on-digital-humanities-research-in-libraries/.

Ogihara, Ellen. Resignation Letter, January 29, 2021. Accessed April 15, 2021. https://www.theolafmessenger.com/2021/research-and-instruction-librarian-ellen-ogihara-resigns-from-college-citing-bias-and-discrimination-within-library-and-information-technology-services/.

Ossom-Williamson, Peace, Jamia Williams, Xan Goodman, Christian I.J. Minter and Ayaba Logan. "Starting with I: Combating Anti-Blackness in Libraries." *Journal Articles: Leon S. McGoogan Health Sciences Library* 9 (December 2020). https://digitalcommons.unmc.edu/mcgoogan_articles/9.

Posner, Miriam. "Commit to DH People not DH Projects." Miriam Posner's Blog. March 18, 2014. http://miriamposner.com/blog/commit-to-dh-people-not-dh-projects/.

Posner, Miriam. "No Half Measures: Overcoming Common Challenges to Doing Digital Humanities in the Library." *Journal of Library Administration* 53, no. 1 (2013): 43–52. https://doi.org/10.1080/01930826.2013.756694.

Pressman, Jessica and Pam Lach. "Our DH Showcase Statement: Honoring our Experience as Rationale for Cancellation." *DH Initiative Blog*, March 19, 2021. https://dhblog.sdsu.edu/dh-news/our-dh-showcase-statement.

Risam, Roopika. "Beyond the Margins: Intersectionality and the Digital Humanities." *Digital Humanities Quarterly* 9, no. 2 (2015). http://www.digitalhumanities.org/dhq/vol/9/2/000208/000208.html.

Risam, Roopika. *New Digital Worlds: Postcolonial Digital Humanities in Theory, Praxis, and Pedagogy*. Evanston, IL: Northwestern University Press, 2019.

Risam, Roopika and Susan Edwards. "Transforming the Landscape of Labor at Universities through Digital Humanities." In *Digital Humanities, Libraries, and Partnerships: A Critical Examination of Labor, Networks, and Community*, edited by Robin Kear and Kate Joranson, 3–17. Cambridge, MA: Chandos Publishing, 2018.

Risam, Roopika and Kelly Baker Josephs, eds. *The Digital Black Atlantic: Exploring the Intersections of Digital Humanities and African Diaspora Studies*. Minneapolis: Minnesota University Press, 2021. https://dhdebates.gc.cuny.edu/projects/the-digital-black-atlantic.

Risam, Roopika, Justin Snow and Susan Edwards. "Building an Ethical Digital Humanities Community: Librarian, Faculty and Student Collaboration." *College and Undergraduate Libraries* 24, no. 2–4 (2017): 337–349. https://doi.org/10.1080/10691316.2017.1337530.

Shirazi, Roxanne. "Reproducing the Academy: Librarians and the Question of Service in the Digital Humanities." Roxanne Shirazi (blog), July 15, 2014. http://roxanneshirazi.com/2014/07/15/reproducing-the-academy-librarians-and-the-question-of-service-in-the-digital-humanities/.

Sierpe, Eino. "Confronting Librarianship and its Function in the Structure of White Supremacy and the Ethno State." *Journal of Radical Librarianship* 5 (2019): 84–102. https://journal.radicallibrarianship.org/index.php/journal/article/view/39.

Spiro, Lisa. "'This Is Why We Fight': Defining the Values of the Digital Humanities." In *Debates in the Digital Humanities*, edited by Matthew K. Gold. Minneapolis: University of Minnesota Press, 2012. https://dhdebates.gc.cuny.edu/read/untitled-88c11800-9446-469b-a3be-3fdb36bfbd1e/section/9e014167-c688-43ab-8b12-0f6746095335#ch03.

Vandegrift, Micah and Stewart Varner. "Evolving in Common: Creating Mutually Supportive Relationships Between Libraries and the Digital Humanities." *Journal of Library Administration* 53, no. 1 (2013): 67–78. https://doi.org/10.1080/01930826.2013.756699.

Vinopal, Jennifer and Monica McCormick. "Supporting Digital Scholarship in Research Libraries: Scalability and Sustainability." *Journal of Library Administration* 53, no. 1 (2013): 27–42. https://doi.org/10.1080/01930826.2013.756689.

CHAPTER FIVE

Inclusive Design:

A Method and Craft of Transforming Digital Humanities with User Experience

A. Miller

PROJECT DESIGN in user experience and project design in digital humanities (DH) share many of the same motivating questions. When setting up surveys as part of your research, do you consider the language or purpose of the question? Are there leading questions? Have you considered the background or demographics of your audience? When setting up tasks for user-testing a research project, do you strategize how to recruit users with a diverse approach or just try to get as many as possible? These are just a few of the questions user experience (UX) professionals would consider when creating a digital project. These questions should also be asked by the creators of DH projects.

Libraries, and the academy in general, have specialists in various disciplines, including accessibility, information technology, DH, UX, Open Educational Resources (OER), and subject specialists (such as for the social sciences or humanities). User experience research and design methods can help these specialists bridge gaps or push the boundaries of their work in DH.

This chapter will provide an introduction to the field of UX, an overview of the user-centered design (UCD) phases and applications in UX, an overview of UX research and design phases, the importance of applying UX to DH, and will culminate in an inclusive design checklist that can help root the UX process in digital projects, regardless of the role of the subject specialist or reader of this book. This chapter is neither comprehensive nor definitive but is intended to be helpful to those who are new to UX and inclusive design in digital projects.

CC BY NC

Introduction

UX research protocols can be applied to DH workflows to enhance the experience of DH practitioners and DH project end-users. Using the stages of human-centered design—empathy, define, ideate, prototype, test, implement[1]—DH projects have the chance to become more inclusive in their design and use. For example, UX research design practices can help get to root problems with carefully planned surveys, card sort activities, moderated interviews, or observational sessions where unbiased, open-minded, and respectful practices are the norm (some of these practices are discussed later in the chapter).

UX can also enhance the user interface (UI) with sound design strategies around how user-friendly a product is, such as typography, layout, navigation, accessibility, and usability. A research-based prototype can be user-tested before launch, with screeners and tasks in place to see how intuitive the product is for users. For example, a spatial analysis research project may yield a GIS map or a curated database of archival findings. The resulting website or project UI may have any number of challenges to various users, including typeface, color palette, wording, wayfinding, or responsive features of content. Users may be accessing the project on various devices, have varying cognitive or physical abilities and even different purposes for accessing or using the product (DH project).

UX is becoming more normative in DH programs and services offered at many institutions and libraries. Groups and individuals are seeking ways to enhance or become involved in DH, and UX is a great way to help people get started with strategies, such as design thinking, that can yield better insights into user behavior, wants, and needs. By applying this human-centered thinking, DH researchers can better create and connect with their users. More importantly, DH practitioners (and libraries in general) that use UX practices can reach wider audiences and help their current audiences even more with inclusive, human-centered practices that use empathy, definition, ideation, prototyping, and testing.

The Problem

Digital humanities projects are commonplace across academia and DH scholars increasingly seek ways to advance their activities. However, many DH projects lack a critical vantage point: user-focused design and usability.[2] While arguing that DH is a system with users, Charlie Edwards compels the

DH community and tool builders to acknowledge that the system has serious usability problems, affecting both new and expert users.[3]

Early DH projects focused on the system's raw functionality rather than on the user's experience, where a DH project's weak front-end (UI) would create difficulty for a user navigating the system.[4] Edwards proposes that "DH and its possible futures are likely to be shaped, delimited, or advanced by how DHers choose to design and build their conceptions of the user, and the extent to which *all* users can participate in that process."[5]

Burdick, as cited by Caviglia et al. (2009) argued that DH disciplines hardly recognize design as a discipline, rather just an addition used by people without design backgrounds.[6] Burdick additionally noted that most scholars design their own digital projects or outsource the development to research assistants or IT staff. As Caviglia et al. continue, "Moreover, the lack of designers and, especially design researchers, leads often to an appropriation of the theoretical aspects of the discipline by humanities scholars."[7] When non-UX/non-UI designers create DH projects, problems may occur that affect usability and project sustainability. Typical problems across products and services include: ambiguous and inconsistent vocabularies; disregard of graphical conventions; opacity of the system status; missing documentation; missing strategies to avoid mistakes; and disregard of convention for workflows.[8]

These problems could be solved if users were a part of the design process. Although Klaus Thoden et al. (2017) would argue that many projects in DH do address usability and integrate user-centered design practices, "the resulting tools are often not easy to use or are not self-explanatory. Although usability guidelines and heuristics exist, many DH tools fail to even comply with the simplest rules."[9] DH scholars are not necessarily UX/UI designers and most certainly are not the totality of possible end-users. All projects need to be designed with users and for users.

Definitions

User experience designers should never assume what a user knows or thinks. For consistency in this chapter, the definition of DH will be broadly defined, within the larger umbrella term of digital scholarship.

- "Digital scholarship is scholarship enhanced by the design of digital projects, incorporation of digital tools, collaboration among digital partners, and dissemination through digital platforms."[10]

- As a subset of digital scholarship, DH is a dynamic and interdisciplinary field that intersects humanistic inquiry with design and computing. The definition of DH largely varies, as seen by continually refreshing the page at http://whatisdigitalhumanities.com. Regardless of definition, or choice of technology, DH projects begin with a research question. Solving that research question can involve any number of people, technologies, and project goals. Approaching the project with a focus on the user experience can help increase successful outcomes.
- User experience (UX) "encompasses all aspects of the end-user's interaction with the company, its services and its products."[11] It is a broad concept that covers a user's feelings, attitude, and behavior while using a system, service, product, or space. User experience research is the process of collecting information about the behaviors, needs, and motivations of users for a particular product, service, or space.[12] UX can be divided up into research and design, but it is often an iterative process to understand the impact on an audience. UX design is "the process of supporting user behavior through usability, usefulness, and desirability provided in the interaction with a product" or service.[13] This step happens after the UX research, where the research is synthesized and applied to form a design.
- Usability is a quality attribute of a "user's experience when interacting with products or systems, including websites, software, devices, or applications. Usability is about effectiveness, efficiency, and overall satisfaction.[14] Usability is a subset of UX.
- Inclusive design is a design process that enables and draws on the full range of human diversity, such as ability, language, culture, gender, and age. The goal is to have a product, service, or environment that is "designed to be usable by as many people as reasonably possible, without the need for specialized adaptations."[15] Inclusive design is a subset of UX.
- Accessibility is related to inclusive design as they both focus on "making products or services available to a wider range of people. However, accessibility is more limited in scope as it is focused on "specific accommodations to ensure that people with disabilities have access to products, services, or environments."[16]
- User interface (UI) design (also called interaction design) studies how users interact with a webpage, application, or product with a focus on "anticipating what users might need to do and ensuring that the

interface has elements that are easy to access, understand, and use to facilitate those actions."[17] A UI is a view of an object, page, or program.
- User-centered design (UCD), human-centered design, and/or design thinking are all approaches and mindsets that enable a user-focused framework for designers to understand, explore, and materialize a meaningful design solution that involves users in the design process.[18]

Overview of User-Centered Design

UCD, also known as design thinking, is an ideology or mindset supported by various practitioners in different industries. Although the process and terminology used can vary, at its core, design thinking is a framework that is hands-on and user-centric. The *Design Thinking for Libraries Toolkit* offers a brief introduction of the non-linear phases:

- Empathy: A key principle in the design thinking and human-centered design process that represents the user's perspective (research, listen, and observe users).
- Define: Combine research and observations to identify the problem, target users, and identify the constraints to work within. Ultimately looking at the users' needs to define a design challenge.
- Ideate: To generate (creative) ideas to solve the user challenge as part of an exploratory activity that encourages imagination (quantity supersedes quality in the ideate phase). Brainstorm prompts and identify themes or patterns.
- Prototype: A rough, mocked-up, tangible manifestation of your idea. Prototypes can be paper, physical, digital, or a combination of low- or high-fidelity.
- Test: Iterate your ideas by testing the prototypes with users, asking for feedback. Iterate the feedback into the concept and reevaluate or adapt the prototype.
- Implement: Getting to scale by telling your story, developing a roadmap, and stewarding the project with a plan for long-term stewardship and continuity (for example, who will continue the project or maintain it?).[19]

For more on how this user-centered design process can be specifically applied to libraries, consider reviewing the *Design Thinking for Libraries Toolkit* (https://designthinkingforlibraries.com). For additional resources on design thinking, see the Additional Resources section at the end of the chapter.

Applications of UCD and UX

Several examples of how user experience has been incorporated into DH projects can serve as illustrations of these principles in practice. Mia Ridge and her team curated a DH project focused on deep maps and spatial narratives, and found UX to be paramount to its design:

> The process of creating wireframes and prototype screens—and more importantly, the discussions and debates they initiated—helped us understand the complexity of the requirements for deep maps, the design tensions they would have to accommodate, and the importance of descriptive language in interdisciplinary platforms. Our prototyping process lead us to devise a conceptual model that helped clarify relationships between spatial narratives and deep maps and encompass the different definitions of deep maps encountered at the Institute.[20]

Similarly, Ashley Rojas describes how her project team centered the project's focus on the user. They wanted the DH project's database to be easy to navigate and to contain content the audience would find valuable and useful.[21] An example of how Rojas' team put the user first was in their design and testing:

> To ensure easier readability for individuals who use screen readers, we constructed the HTML to begin with the first item of the menu and when the 'tab' button is pressed the item selected to activate goes through each of the items, left to right, before going through the main body of text.[22]

This process enables the use of the tab key to run through items selected (title, menu, footer, etc.) in a left to right format. This is a design practice that some DH scholars might miss if designing without UX in mind.

According to Melinda Baumann, the DH@UVA site owners conducted user testing on the DH certificate program to assess the understanding of certificate requirements, the application process, clarity on what counts as electives, and exploring the initial perceptions of the site organization, among others.[23] The methodology used and sheer reporting of the results on their website are great examples of UCD and UX being applied to DH.

Overview of UX Research and Design Phases

There are numerous online resources to learn about the UX research and design phases. Some of these sources include the Humans in User Experience at UCLA, User Experience Professionals Association, Usability.gov, IDEO, the Nielsen Norman Group, and UX at University of Arizona Libraries.[24]

The purpose of UX research is to understand how a user thinks, feels, and acts when interacting with a product or service. There are many ways to conduct user research depending on the project. According to the 18F UX Guide, the goals or activities within UX research may involve exploring possibilities, interacting with the community, testing assumptions, and investigating tools and systems.[25] Examples of some of the more popular UX research methods and tools include:

- Focus groups (a moderated discussion, typically with 5–10 participants)
- Individual interviews
- Personas (researched, reliable, and realistic representations of the user audience)
- Surveys (structured questionnaire)
- Task/scenario analysis (observe users conduct tasks related to the project)
- Content inventory (list of all the content on a site to gain insight: text, images, documents)
- Organization schemes/structures (how the site categorizes content and relationships within site)
- Market analysis (compare competitive or related projects)

Additional examples of UX research methods are available at Usability.gov and 18F Methods.[26]

The purpose of UX design is to synthesize research and apply it in a design form. UX design can be broken down into the design of the information architecture (content layout, navigation, terminology), interaction design (calls to action, prompts, intuitive user flow), UI design, visual design, and usability, among others. According to Susan Farrell, the goals or activities within UX design may involve designing for universal access, giving users control, preventing errors, providing helpful defaults, mapping features to user needs, considering diverse contexts, considering social implications, tracking usability bugs, keeping data safe, and making user documentation.[27] Examples of some of the more popular UX design methods and tools include:

- Wireframes (two-dimensional illustrations of page UI, focusing on prioritization of content, space allocation, and intended behaviors of UI elements)
- UI elements (checkboxes, radio buttons, dropdown lists, sliders, breadcrumbs, search fields, icons, notifications, progress bars, etc.)
- Card sorting (which helps evaluate the information architecture of the site)
- Prototypes (a draft version of product, low- or high-fidelity)
- Color basics (a color wheel illustrates relationships between color levels)
- Usability testing (to evaluate a product or service by testing with representative users)
- Accessibility evaluation
- Scenarios (stories and context for users of the site or product)
- User documentation (including how-tos, instructions, FAQs, etc)

For additional examples of UX design methods, see Rohrer (2014) for help determining which UX research method to use in a project; Usability.gov (2021) "Methods" for help choosing design methods for content strategy, UI design, usability evaluation, information architecture, interaction design, and visual design; and the UX Cookbook[28] for various UX design recipes to replicate. If time or resources are limited and only one activity can be performed, Susan Farrell recommends a qualitative (think-aloud) usability test.[29] For this, it is best to recruit 3–5 representative users and give them tasks to perform while you silently listen and watch them perform the tasks while they think out loud. This method will allow you to learn about misinterpreted design elements and get insights into user behaviors that can be advantageous to the redesign.

Designing for and testing against the user experience of DH projects will not only make better projects for more users, but it will help DH practitioners reconsider their own purpose and audience for their projects as they iteratively design with UX in mind.

Importance of UX in DH

While technology professionals in education and commercial applications have made significant progress in meeting the needs of users, scholars creating digital projects often fail to take users' needs into account.[30] It is wrong to assume everyone approaches information or websites with the same goal or ability.

In an ideal world, a DH project or website would not only be compatible with desktop computers, laptops, smart phones, and tablet devices, but "the materials we create should also work well with such tools as refreshable braille displays, digital talking book devices, screen reader applications, and screen magnification software."[31] This is important because by working to meet the needs of people with disabilities and conducting usability testing with the disability community, the DH community "will also benefit significantly as it rethinks its assumptions about how digital devices could and should work with and for people."[32] This raises the importance of addressing the audience of DH projects. The intended audience may be humanities scholars or technically savvy users, but the potential audience is far larger. "Prioritizing a wider audience can help further adoption of tools in general, and thus further acceptance of their use and development as comprising legitimate scholarly work."[33]

Even when DH scholars focus on an intended audience, DH tools still tend to neglect the typical humanities user in their design and documentation. "Builders of digital humanities tools, especially those that deal with technically more sophisticated techniques, like text mining and visualization, could considerably increase their tools' visibility and speed of adoption with more attention to their user interface and clear instructions with example use cases."[34] Regardless of discipline, users want the benefits or purpose of a DH project described up front, ideally with the UI and documentation erring on the side of obvious rather than clever.[35]

According to Klaus Thoden et al., the goal is to "find consensus on what good user experience and usability means in the Digital Humanities, what practices this might entail and what factors influence these practices."[36] Clair Warwick urges scholars to think about "use" before the DH tool is built; this is done by studying potential users.[37] There is no substitution for this process, as creators, designers, and researchers are not users. Personas and use cases are tools that can be applied in the design process but must "be used with care in a multimethod user study, and should never be a substitute for other, more time-intensive methods."[38] This is how we limit bias in the research design process, with mixed methods and testing.

Simplicity is also important to UX and DH. For example, consider the terminology used in a DH project or even the name of the DH tool. A resource with a confusing, unusual, or hard to pronounce title may not be found nor understood by potential users. User testing could help limit that risk. Similarly, to increase the UX of a DH project, even a little usability

testing is better than none, and a simple way to achieve this includes providing sample data and good documentation.[39]

Finally, one of the most important UX applications you can do is usability testing. Even when done with just five users, you will discover about 85 percent of the design problems in the system.[40] This is important to DH not only because it will help find and solve design problems, but it can help design for the larger potential audience.

Inclusive Design

This section features three examples of academic institutions that highlight the need to create and build with inclusive design in mind. Inclusive design is an iterative process that keeps the diversity and uniqueness of a user forefront, especially users at the margins who are often left out of hypothetical average-based solutions.[41] There are many resources for inclusive design models, but these three were specifically chosen for the academic perspective, as that is relatable to many academic-focused DH projects.

The Inclusive Design Research Centre at the Ontario College of Art and Design University (OCAD U) created an online guide to inclusive strategies. According to its *Inclusive Design Guide*,[42] the insights, practices, tools, and activities of the guide can be applied to digital designs as well as to the design of services for physical products and built environments. Many of the tools described in the guide are applicable to the creation of DH projects, like the tools referenced in the overview of UX research and design phases earlier in this chapter.

The Inclusive Design Toolkit,[43] created by the Inclusive Design Group in the Department of Engineering at the University of Cambridge, emphasizes that every design decision can affect the user experience. According to the Inclusive Design Group, "understanding the range of user capabilities and the impact they have on product use is vital to producing inclusively designed products." The Inclusive Design Toolkit includes examples of the range of user capabilities, such as vision, hearing, thinking, reach and dexterity, and mobility.[44]

Kim Christen, professor and director of Digital Technology Culture at Washington State University, describes the use of UX research and design methods when system designers created Mukurtu.[45] Mukurtu is an open source, digital content management system originally developed in collaboration with Warumungu Aboriginal community members and

built upon Warumungu knowledge systems. Christen describes the challenge of curating Indigenous communities' cultural heritage materials, as well as solutions and outcomes. Among its many development steps, Mukurtu collaborators and developers used cognitive walkthroughs (individuals were asked a series of tasks) and user stories (short descriptions of a software function from the end-user perspective). Part of the success of Mukurtu was building in relationships with the Indigenous community (the end-users) in the development process, which is a good framework to model when creating an inclusive DH project.[46]

This chapter does not attempt to reinvent an inclusive design toolkit or guide. Rather, the purpose of this chapter is to highlight relevant work being done and resources for user-centered design, user experience research and design, and inclusive design. The following Inclusive Design Checklist is meant to serve as a resource when developing DH projects. This can be seen as a compilation of curated qualities that are critical to the various methodologies presented in this chapter (UX research, UX design, UCD, design thinking, inclusive design, usability, etc.).

The framework for this Inclusive Design Checklist is to help DH project collaborators create and design for 1) the potential audience of a DH project, not just the intended audience, and 2) both the expert user and casual user. Designing with users and addressing this framework will help increase many aspects of inclusion in DH projects. To help stay on track with this framework, consider adapting the checklist below.

Inclusive Design Checklist

This is not an ordered list, rather it is diverse, complex, and iterative in nature. There are two versions, a brief checklist for reference, and an expanded checklist with detailed explanations.

BRIEF CHECKLIST

- You ≠ user
- Avoid bias
- Be mindful
- Prioritize content
- Be consistent
- Offer choice
- Document

EXPANDED CHECKLIST

You Are Not the User

No matter what your role is (DH specialist, librarian, researcher, designer, etc.), it is best to identify users (of varying kinds) and design for intended and potential users of a product or service rather than design based on your own interests or preferences. When conducting user research or usability testing, be sure to avoid asking leading questions to users, as this can also create bias (a leading question encourages the participant to answer in a particular way). Examples of leading questions to avoid include: Do you find this feature frustrating to use? Would you rather use this version or that version? A lot of people don't like this website, what do you think? (There are other ideas in the Additional Resources section.) The effectiveness of a design improves when this mindset is applied (and undergoes user testing!), which helps save time and money in the long run.

Avoid Bias and Generalizations

It is important to recognize unconscious and conscious bias and generalizations and purposefully avoid them in the design. Making presumptions and assumptions that are not based on user research is not an effective design method. Therefore, it is critical to recruit users for research, design, and testing phases, to base the design on real information from a diverse group of users.

Be Mindful of Images, Text, and Color

Avoid unnecessary elements and keep the language clear. A simple, uncluttered interface or product is always the best route. Any extraneous information can detract from the relevant or important content. Use color strategically to draw attention toward or away from items, while staying away from color palettes that may be challenging for people with vision difficulties. Select images that are appropriately labeled and have meaning and use alt text and captions. Select a typeface for the text that will help increase scan-ability and readability. (Content style and visual design guides are available in the Additional Resources section.)

Prioritize Content

Content should be navigable and easy to find. Consider the spatial relationships among items on a page (or product) and structure the layout based on importance (also known as information architecture). Keep in mind the variety of users, many of whom may scan the content rather than read

thoroughly. Carefully place and prioritize content to give attention to important pieces near the top of a webpage or product. (Information architecture guides are available in the Additional Resources section.)

Be Consistent

Try to follow industry standards (i.e. common UI elements) and develop and document your own standards to follow for consistency. The content style and visual design section of the Additional resources list provides a link to common UI elements. Be consistent with words and actions so users do not have to guess if something means the same thing. Develop a style guide for fonts, typeface size, colors, and structure such as headings and captions, and use this to keep all content consistently formatted. This helps with branding and marketing, as well as user satisfaction.

Offer Choice, Give Control

People appreciate options; that is one reason why it is important to design for both the expert and casual user. Options may include the ability to select all and filter results, the ability to toggle the closed captions on or off for an embedded video, the ability to zoom in or out of a page, or the ability to go back, undo, or exit an unwanted action. Options like these give users a choice and a sense of control, which benefits the user experience.

Document

The best products don't need explanation. However, if a feature can help one user, then it will likely help others. Provide open documentation on the project website. Do not archive it where it cannot be found or accessed. Keep user documentation concise and communicate updates with users (if applicable). Other forms of documentation may include instructions on how to maintain or update the product, README files, enhancements, use cases, outcomes, and others. Use strategies to keep track of documentation such as collection archives, project management software, or code repositories.

Summary

There is so much that can be learned from the methodologies described in this chapter: design thinking or UCD, UX research, UX design, usability, accessibility, and inclusive design. They are interconnected, important, and useful to solving problems that affect DH, the academy, social change, industry standards, and everyday interactions.

Don Norman, author of "The Design of Everyday Things," highlights how common everyday things such as tiny fonts on labels, low-contrast instructional manuals, or door handles should not be just designed for one population (i.e., the elderly or people with disabilities), but to use inclusive design for a simple reason: it helps everyone. Norman also describes how "curb cuts were meant to help people who had trouble walking, but it helps anyone wheeling things: carts, baby carriages, suitcases."[47] In recent years, the world has taken a closer look at the lack of inclusive measures, and we have seen how non-inclusive designs were used to create CPR training mannequins, spacesuits, military equipment, car crash dummies, smartphones, sports attire, science gear, and office spaces.[48] Certain products are starting to be re-designed with a focus on human diversity—the variety of users needing to use those products. These same diverse users are also the potential audience of DH projects and websites. As designers of those DH projects, we have the responsibility to transform digital humanities into a positive user experience.

In its most simple form, a user-centered design, regardless of the problem, is most likely to be the best solution. Involving users from the start in the design and testing of a product or service will help create trust in the product or service, increase usability, and prolong sustainability. It can also help save time and money because inclusive designs are "less brittle, easier to update, requires fewer accessibility patches, fewer service calls, shorter training, and generally lasts longer."[49] Inclusive design and UX are worthwhile investments for the DH and beyond.

Groups and individuals who seek ways to enhance or become involved in DH and UX can get started learning strategies, such as design thinking, that can yield better insights into user behavior, wants, and needs. By applying this user-centered thinking, DH researchers can better connect with users. More importantly, DH practitioners (and libraries in general) that use UX practices can reach wider audiences and help their current audiences even more with inclusive, human-centered practices that use the elements in this chapter and in the inclusive design checklist. Adapting a UX mindset can help connect libraries and DH practitioners with the transformations taking place in their fields (and the world in general). UX and inclusive design can help bridge the gap or push the boundaries of their work, creating better user experiences for everyone.

Resources

- d.School Starter Kit, Stanford University. https://dschool.stanford.edu/resources/dschool-starter-kit
- Hemingway App. https://hemingwayapp.com/
- Humans in User Experience. https://hue.humspace.ucla.edu/resources/
- IDEO, Design Thinking Defined. https://designthinking.ideo.com
- Inclusive Design Research Centre, The Inclusive Design Guide. https://guide.inclusivedesign.ca/
- Accessibility guidelines for UX Designers. https://uxdesign.cc/accessibility-guidelines-for-a-ux-designer-c3ba775539be
- Nielsen Norman Group. https://www.nngroup.com
- UX@UA, podcast listing. https://uxua.arizona.edu/learn-ux/our-guide-ux/getting-started
- Vox Media, Accessibility Guidelines: The Checklist. https://accessibility.voxmedia.com/
- W3C Web Accessibility Initiative, Diverse abilities and barriers. https://www.w3.org/WAI/people-use-web.
- WebAIM, Contrast checker. https://webaim.org/resources/contrastchecker
- WebAIM, WAVE Web Accessibility Evaluation Tool. https://wave.webaim.org
- Web Content Accessibility Guidelines. https://www.w3.org/WAI/standards-guidelines/wcag

Resources by Topic

Content Style and Visual Design (Typeface, Color, Shapes, Alt Tags, Captions)

- Content style guides. https://theuxcookbook.com/content-style-guides
- Visual Design Basics. https://www.usability.gov/what-and-why/visual-design.html
- Designing for Web Accessibility. https://www.w3.org/WAI/tips/designing/
- Design System. https://ux-guide.18f.gov/design/use-a-design-system/
- Images Tutorial. https://www.w3.org/WAI/tutorials/images/
- UI Elements. https://www.usability.gov/how-to-and-tools/methods/user-interface-elements.html

- Principles of Inclusive Design. https://uxua.arizona.edu/learn-ux/inclusive-design-accessibility

Human-Centered Design

- Design Kit. https://www.designkit.org/
- Human-centered design methods. https://methods.18f.gov/
- UX Design Guide. https://ux-guide.18f.gov/
- Inclusive Design Guide. https://guide.inclusivedesign.ca/

Information Architecture

- Information Architecture Basics. https://www.usability.gov/what-and-why/information-architecture.html
- Tree testing. https://theuxcookbook.com/tree-testing
- Card sorting. https://theuxcookbook.com/card-sorting

Interaction Design

- Complete Beginner's Guide to Interaction Design. https://www.uxbooth.com/articles/complete-beginners-guide-to-interaction-design/
- Interaction Design Basics. https://www.usability.gov/what-and-why/interaction-design.html

Planning

- Gerry McGovern, "What Really Matters: Focusing on Top Tasks." http://alistapart.com/article/what-really-matters-focusing-on-top-tasks/
- Methods. https://methods.18f.gov/
- Project Management Basics. https://www.usability.gov/what-and-why/project-management.html
- Research planning. https://theuxcookbook.com/research-planning

UX Research

- Usability.gov, User Research. https://www.usability.gov/how-to-and-tools/methods/user-research/index.html.
- 18F Methods. https://methods.18f.gov/

UX Writing

- Alejandra Rodriguez, "UX Writing: Creating Microcopy That Speaks to Users." https://www.uxmatters.com/mt/archives/2019/08/ux-writing-creating-microcopy-that-speaks-to-users.php
- Writing for Web Accessibility. https://www.w3.org/WAI/tips/writing/
- Writing for the Web. https://theuxcookbook.com/writing-for-the-web

Usability and Accessibility

- Accessibility. https://accessibility.18f.gov/
- Making Audio and Video Media Accessible. https://www.w3.org/WAI/media/av/
- Usability.gov. https://www.usability.gov/what-and-why/index.html

For Questions to Apply to User Interviews, Usability Testing, or Leading Questions to Avoid

- User Interviews. https://theuxcookbook.com/user-interviews
- Usability Testing. https://theuxcookbook.com/usability-testing
- Running a Usability Test. https://www.usability.gov/how-to-and-tools/methods/running-usability-tests.html

Notes

1. See *Human Centered Design* sources listed under "Resources by Topic" at the end of the bibliography.
2. Giorgio Caviglia et al., "Communication Design and the Digital Humanities. Visualizations and Interfaces for Humanities Research," Proceedings of the Fourth International Forum for Design as a Process, Belo Horizonte-MG, Brazil, September 19–22, 2012; Charlie Edwards, "The Digital Humanities and Its Users," in *Debates in the Digital Humanities*, Matthew K. Gold, ed. (Minneapolis: University of Minnesota Press, 2012); Klaus Thoden et al., "User-Centered Design Practices in Digital Humanities—Experiences from DARIAH and CENDARI," *ABI Technik* 37, no. 1, (2017): 2–11.
3. Edwards, "The Digital Humanities and Its Users."
4. Ibid.
5. Ibid.
6. Giorgio Caviglia et al., "Communication Design and the Digital Humanities. Visualizations and Interfaces for Humanities Research."
7. Ibid.
8. Thoden et al. "User-Centered Design Practices," 4.
9. Ibid., 9.
10. This brief definition of digital scholarship is provided by the Digital Scholarship Initiatives (DSI) at the Walker Library, Middle Tennessee University. It is available on the DSI website at http://dsi.mtsu.edu/about.
11. Nielsen Norman Group, "User Experience," accessed January 21, 2024, https://www.nngroup.com/articles/definition-user-experience/.
12. Usability.gov, "User Research Basics," accessed January 21, 2024, https://www.usability.gov/what-and-why/user-research.html.
13. Wikipedia.org, "User Experience Design," accessed January 21, 2024, https://en.wikipedia.org/wiki/User_experience_design.
14. Usability.gov, "Usability Evaluation Basics," accessed January 21, 2024, https://www.usability.gov/what-and-why/usability-evaluation.html.
15. Wikipedia.org, "Inclusive Design," accessed January 21, 2024, https://en.wikipedia.org/wiki/Inclusive_design.

16. Wikipedia.org, "Accessibility," accessed January 21, 2024, https://en.wikipedia.org/wiki/Accessibility.
17. Usability.gov, "User Interface Design Basics," accessed January 21, 2024, https://www.usability.gov/what-and-why/user-interface-design.html.
18. Nielsen Norman Group, "Design Thinking 101," accessed January 21, 2024, https://www.nngroup.com/articles/design-thinking; UXPA International, "Definitions of User Experience and Usability," accessed January 21, 2024, https://uxpa.org/definitions-of-user-experience-and-usability/.
19. Chicago Public Library and Aarhus Public Libraries, "Design Thinking for Libraries Toolkit," accessed January 21, 2024, https://designthinkingforlibraries.com/.
20. Ridge et al, "Creating Deep Maps and Spatial Narratives through Design," *International Journal of Humanities & Arts Computing* 7, no. 1–2 (2013), 187.
21. Ashley Rojas, "Heritage Reconstructed: Virtualizations of Archaeological Sites in Peril," DHUM 70002 Digital Humanities: Methods and Practices (Spring 2020) (blog), May 19, 2020, https://dhpraxis20.commons.gc.cuny.edu/heritage-reconstructed-virtualizations-of-archaeological-sites-in-peril/.
22. Ibid.
23. Melinda Baumann, "Research Results: Usability Testing on New Design for DH@UVA site," December 19, 2018, https://ux.library.virginia.edu/usability-testing-on-new-design-for-dhatuva-site/ [site discontinued].
24. UCLA, "Humans in User Experience," accessed January 21, 2024, https://hue.humspace.ucla.edu/; UXPA International, "UX Education," accessed January 21, 2024, https://uxpa.org/about-ux/; Usability.gov, "How To & Tools," accessed January 21, 2024, https://www.usability.gov/how-to-and-tools/index.html; IDEO, "Designing for Access," accessed January 21, 2024, https://www.ideo.org/; Nielsen Norman Group, "Home," accessed January 21, 2024, https://www.nngroup.com; The University of Arizona, "UX@UA," accessed January 21, 2024, https://uxua.arizona.edu.
25. 18F, "User Experience Design Guide," accessed January 21, 2024, https://ux-guide.18f.gov/research/.
26. Usability.gov, "How To & Tools"; 18F, "Methods," accessed January 21, 2024, https://methods.18f.gov/.
27. Susan Farrell, "UX Research Cheat Sheet," Nielsen Norman Group, February 12, 2017, https://www.nngroup.com/articles/ux-research-cheat-sheet/.
28. Christian Rohrer, "When to Use Which User-Experience Research Methods," Nielsen Norman Group, July 17, 2022, https://www.nngroup.com/articles/which-ux-research-methods/; https://www.usability.gov/how-to-and-tools/methods/user-research/index.html; University of Arizona Libraries, "UX Cookbook," accessed January 21, 2024, https://theuxcookbook.com/.
29. Farrell, "UX Research Cheat Sheet."
30. George H. Williams, "Disability, Universal Design, and the Digital Humanities," in *Debates in the Digital Humanities,* Matthew K. Gold, ed. (Minneapolis: University of Minnesota Press, 2012).
31. Ibid.
32. Ibid.
33. Fred Gibbs and Trevor Owens, "Building Better Digital Humanities Tools: Toward Broader Audiences and User-Centered Designs," *Digital Humanities Quarterly* 6, no. 2 (2012): para. 36, http://digitalhumanities.org/dhq/vol/6/2/000136/000136.html.
34. Ibid., para 26.
35. Ibid.

36. Thoden et. al, "User Centered Design Practices," 2.
37. Warwick, "Studying Users in Digital Humanities," in *Digital Humanities in Practice*, Claire Warwick, Melissa Terras, and Julianne Nyhan, eds., chapter 1. (London: Facet Publishing, 2011).
38. Ibid., 6.
39. Thoden et. al, "User-Centered Design Practices."
40. Edwards, "The Digital Humanities and Its Users"; Janet M. Six and Ritch Macefield, "How to Determine the Right Number of Participants for Usability Studies," UXmatters (blog), January 4, 2016, https://www.uxmatters.com/mt/archives/2016/01/how-to-determine-the-right-number-of-participants-for-usability-studies.php; Jakob Nielsen, "Why You Only Need to Test with 5 Users," Nielsen Norman Group, March 18, 2000, https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/.
41. Inclusive Design Research Centre (IDRC), "Philosophy," accessed December 20, 2021, https://idrc.ocadu.ca/about/philosophy/
42. OCAD University, "The Inclusive Design Guide," accessed January 21, 2024, https://guide.inclusivedesign.ca.
43. University of Cambridge, "Inclusive Design Toolkit," accessed January 21, 2024, http://www.inclusivedesigntoolkit.com.
44. Ibid., "About Users: Overview," http://www.inclusivedesigntoolkit.com/usercapabilities/usercap.html.
45. Washington State University Center for Digital Scholarship and Curation, "Welcome to Mukurtu CMS," accessed January 21, 2024, mukurtu.org.
46. Kim Christen, "Indigenous Knowledge Systems and Mukurtu CMS," The Design for Diversity Learning Toolkit, October 12, 2018, https://des4div.library.northeastern.edu/indigenous-knowledge-systems-and-mukurtu-cms/.
47. Don Norman, "I Wrote the Book on User-Friendly Design. What I See Today Horrifies Me." Fast Company (blog), May 8, 2019, https://www.fastcompany.com/90338379/i-wrote-the-book-on-user-friendly-design-what-i-see-today-horrifies-me.
48. Ritu Prasad, "Eight Ways the World is Not Designed for Women," BBC News, June 5, 2019, https://www.bbc.com/news/world-us-canada-47725946.
49. Jutta Treviranus, "The Three Dimensions of Inclusive Design, Part Two," Medium, March 29, 2018, https://medium.com/@jutta.trevira/the-three-dimensions-of-inclusive-design-part-two-7cacd12b79f1.

Bibliography

Baumann, Melinda. "Research Results: Usability testing on new design for DH@UVA site." User Experience at the UVA Library. Last modified December 19, 2018. https://ux.library.virginia.edu/usability-testing-on-new-design-for-dhatuva-site/. Site discontinued.

Caviglia, Giorgio, Paolo Ciuccarelli, and Nicole Coleman. "Communication Design and the Digital Humanities. Visualizations and Interfaces for Humanities Research." Proceedings of the Fourth International Forum for Design as a Process, Belo Horizonte-MG, Brazil. September 19–22, 2012.

Christen, Kim. "Indigenous Knowledge Systems and Mukurtu CMS." The Design for Diversity Learning Toolkit, Northeastern University Library. Published October 12, 2018. https://des4div.library.northeastern.edu/indigenous-knowledge-systems-and-mukurtu-cms/#more-1237.

Edwards, Charlie. "The Digital Humanities and Its Users." In *Debates in the Digital Humanities*, edited by Matthew K. Gold, chapter 13. Minneapolis: University of Minnesota Press, 2012. https://dhdebates.gc.cuny.edu/read/untitled-88c11800-9446-469b-a3be-3fdb36bfbd1e/section/27116231-77fa-4f88-90cd-2bf1b4261a18#ch13.

Gibbs, Fred, and Trevor Owens. "Building Better Digital Humanities Tools: Toward Broader Audiences and User-Centered Designs." *Digital Humanities Quarterly* 6, no. 2 (2012). http://digitalhumanities.org/dhq/vol/6/2/000136/000136.html.

Gold, Matthew K. *Debates in the Digital Humanities*. Minneapolis: University of Minnesota Press, 2012. https://dhdebates.gc.cuny.edu/projects/debates-in-the-digital-humanities.

Nielsen, Jakob. "Why You Only Need to Test with 5 Users." Nielsen Norman Group, March 18, 2000. https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/.

Norman, Don. "I Wrote the Book on User-Friendly Design. What I See Today Horrifies Me." Fast Company (blog), May 8, 2019. https://www.fastcompany.com/90338379/i-wrote-the-book-on-user-friendly-design-what-i-see-today-horrifies-me.

Prasad, Ritu. "Eight Ways the World is Not Designed for Women." BBC News. June 5, 2019. https://www.bbc.com/news/world-us-canada-47725946.

Ridge, Mia, Don Lafreniere, and Scott Nesbit. "Creating Deep Maps and Spatial Narratives through Design." *International Journal of Humanities & Arts Computing* 7, no. 1–2 (2013): 176–189. DOI: 10.33/66/ijhac.2013.0088.

Rohrer, Christian. "When to Use Which User-Experience Research Methods." Nielsen Norman Group, July 17, 2022. https://www.nngroup.com/articles/which-ux-research-methods/.

Rojas, Ashley. "Heritage Reconstructed: Virtualizations of Archaeological Sites in Peril." DHUM 70002 Digital Humanities: Methods and Practices (Spring 2020) (blog), May 19, 2020. https://dhpraxis20.commons.gc.cuny.edu/heritage-reconstructed-virtualizations-of-archaeological-sites-in-peril/.

Six, Janet M., and Ritch Macefield. "How to Determine the Right Number of Participants for Usability Studies." UXmatters (blog), January 4, 2016. https://www.uxmatters.com/mt/archives/2016/01/how-to-determine-the-right-number-of-participants-for-usability-studies.php.

Terras, Melissa, Julianne Nyhan, and Edward Vanhoutte. *Defining Digital Humanities: A Reader.* London: Routledge, 2013. https://doi.org/10.4324/9781315576251.

Thoden, Klaus, Juliane Stiller, Natasa Bulatovic, Hanna-Lena Meiners, and Nadia Boukhelifa. "User-Centered Design Practices in Digital Humanities—Experiences from DARIAH and CENDARI." *ABI Technik* 37, no. 1, (2017): 2–11. doi.org/10.1515/abitech-2017-0002.

Treviranus, Jutta. "The Three Dimensions of Inclusive Design, Part Two." Medium, March 29, 2018. https://medium.com/@jutta.trevira/the-three-dimensions-of-inclusive-design-part-two-7cacd12b79f1.

Warwick, Clair. "Studying Users in Digital Humanities." In *Digital Humanities in Practice*, edited by Claire Warwick, Melissa Terras, and Julianne Nyhan, chapter 1. Facet Publishing, 2011.

Williams, George H. "Disability, Universal Design, and the Digital Humanities." In *Debates in the Digital Humanities*, edited by Matthew K. Gold, chapter 12. Minneapolis: University of Minnesota Press, 2012. https://dhdebates.gc.cuny.edu/projects/debates-in-the-digital-humanities.

Digital Humanities

IN THE LIBRARY

PART 2. PRACTICE

CHAPTER SIX

Into the Mix:

Subject Liaison Librarians and Digital Humanities Research

Mark Dahlquist, Katie Gibson, and Jenny Presnell

Introduction

While much contemporary research in the humanities relies incidentally upon digital tools and collections, digital humanities (DH) research "employs computer-based technologies with the aim of exploring new areas of inquiry in the humanities," seeking to develop novel insights and to communicate in new ways by exploring the affordances of digital methods and technologies.[1] This reliance on digital tools and emphasis on innovation contributes to the character of DH as a challenging and frequently "intensely collaborative business," which relies upon "a wide range of skills, from specialized humanities research to metadata and management skills, rare books and archival librarianship and IT expertise."[2] Increasingly, DH projects also involve collaboration with members of communities from outside the university, especially as libraries strive to increase the availability of digitized materials and scholarly conversation that reflect the experiences of historically underrepresented populations. This chapter considers the roles that liaison or subject area librarians can play in supporting DH research, and in fostering conversations that connect researchers to potential collaborators and available information resources.

The original seventeenth-century meaning of the English term "liaison" provides a useful metaphor for understanding the role that subject area librarians can play in DH research and instruction. The earliest use of the term recorded in the Oxford English Dictionary occurs in a 1669 cookbook, in which the term "liaison" describes the use in cooking of ingredients such as

CC BY NC ND

eggs and flour as binding agents to bring about a "liaison and thickening" of a mixture to produce a more savory dish.[3] A related and perhaps even more fitting definition was proposed by John Dryden in 1665, who imported the word from the French. For him, a "liaison" referred to the unity of a play that results from good conversational mix, in which all a play's characters are involved with one another. A "*Liaison des Scenes*," he writes, "tis a good mark of a well-contrived Play when all the Persons are known to each other, and every one of them has some affairs with all the rest."[4]

Describing for Dryden a kind of conversational cooking, the term "liaison" seems to have entered English as a fitting metaphor for the challenging work liaison librarians can undertake to bring together the people and parts that comprise DH projects, thickening conversations among digital scholarship centers and teams.[5] This chapter will highlight some of the separations that often present challenges for the successful implementation of DH research; gaps that can emerge from different DH support models (service, lab and network); and the variety of skill sets that are often needed to plan and implement DH projects. We will suggest that the fundamental skills of the liaison librarian—engagement, collection development, and instruction—provide opportunities for subject area librarians to foster conversations that create an open kitchen-space or hearth where cooking and communication brings the dish together.

Digital Humanities Models and the Subject Librarian

The differences between the DH service models that exist at different colleges and universities and academic libraries present a range of challenges and opportunities to liaison librarians. Some universities maintain physical centers (often but not always located within libraries), while others do not. The Ithaka S+R report, "Sustaining the Digital Humanities: Host Institution Support Beyond the Startup Phase" outlines three common models ("service," "lab," and "network") found at institutions with established digital humanities programs.[6] In the service model, "the service unit," whether the unit is free-standing or under a library or IT umbrella, "seeks to meet the demand expressed by faculty, often with a strong focus on meeting an individual's research needs."[7] In libraries this can take the form of making existing structures and services, such as repositories and metadata, available to scholars. The library acts in a supportive capacity but is not an active participant in the research. Rather, the service model emphasizes "helping faculty to execute the projects they have conceived and for which they have

received funding."[8] One criticism of the service model is that many librarians "see their work not as supporting research, but as research" and that in DH centers operating under a service model, "partnerships of equals" among librarians and research faculty are not encouraged.[9]

In a lab DH model, the DH-oriented organization functions more like a university science laboratory, relying heavily on grant funding in order to allow the leaders and project managers of such DH labs to maintain a sharper focus when developing distinctive and ambitious digital projects.[10] Well-known DH labs include Scholars' Lab at the University of Virginia, or the Roy Rosenzweig Center for History and New Media at George Mason University; such labs may employ full-time developers in creating and supporting innovative software applications. Teams in the lab model form to address needs as they arise and can grow to bring in additional people as grant funding and need allow. Because this model is flexible and brings together people in a project-centered collaboration, there is great variety in lab model collaborations. However, the lab model can often exclude liaison librarians from being involved in projects. This issue can arise in connection with many types of DH centers[11] but is especially likely to occur where lab model DH research is strongly or exclusively promoted.

Finally, the network model is a more organic network of services and resources on a campus, in which DH support grows over time to meet a variety of needs, with each network element providing resources to contribute to the success of digital humanities projects. At Miami University, the authors' institution, digital humanities resources generally fall into this model, with support coming from the Libraries' Center for Digital Scholarship, the Miami Humanities Center, Research Computing Support, and from university IT Services. Within this network Miami University subject liaison librarians operate in variety of ways; for example, in the evaluation of digital humanities funding proposals submitted to the Miami University Humanities Center, by planning (again, in partnership with the Humanities Center) an annual Digital Humanities Forum, and by participating on a voluntary basis in regular library-based Digital Humanities interest group meetings, which provide opportunities for discussion and practical skill development among subject specialists and others. This is less the result of a top-down determination that "it is appropriate to cross-train subject specialists,"[12] and more a commitment to sustaining a conversation that permits subject librarians and technologists alike to develop and expand their digital literacies, under a "see one, do one, teach one" approach to bringing DH skills to faculty and students.

No matter the local arrangement, subject librarians have the potential to play important roles in developing successful DH projects. The skills cultivated by subject liaison and other librarians in the course of their regular duties, including skills "often found in the backrooms of our libraries," can provide key contributions to the success of digital humanities projects.[13] Librarians with more specialized skills can contribute these to digital humanities projects, assisting with tasks such as text encoding,[14] metadata creation,[15] and preservation and long-term sustainability of completed DH projects.[16] But, while there is a clear role for librarians, there is a need to recognize some distinctions between types of librarians and the contributions each might make.

Creating the Team

Many medium and larger-sized DH projects require the creation of teams possessing the right mix of members in order to bring digital humanities projects to life. While some DH projects (including many that rely on collections and tools acquired by the library and supported through library-created guides and documentation) are completed without assistance from the libraries or a DH center, others require a range of assistance and planning. In such cases, a librarian with project management skills can connect individuals within and outside the library and keep project participants on schedule. Teams that integrate research and technology skills benefit from including a variety of members, including a principal investigator (or researcher), specialized DH librarians, subject liaison librarians, and technologists.[17]

Researchers

Commonly the initiator of a scholarly DH project is a faculty researcher from outside the libraries, although some projects are initiated by librarians or by library units seeking to highlight their unique collections. The scholar may bring their own data and request support in identifying and learning the appropriate tools,[18] or may have a question and require support in finding or creating data in a usable format. In many collaborative projects, the researcher is primarily responsible for determining the scope of the project and any associated corpus, and establishing the research question or goal. Research scholars know how to structure an inquiry appropriate to their discipline and have a depth of knowledge in the content area. But their knowledge of the potential uses of data analytics and information architecture is often an area in which assistance from others can be helpful.

DH Librarians

Many types of librarians and library staff have a large toolkit of skills to bring to DH projects. A variety of terms can be used to highlight the work of particular librarians: creator and contributor; curator; messenger and liaison; educator; mediator and interpreter; host; partner; innovator; hybrid scholar; advocate; and consultant.[19] There is no single or universal definition of a typical DH librarian. In a survey exploring how librarians working in digital humanities came to their roles, Molly Dahl Poremski found that many (around 78 percent) had their DH duties added to their existing job descriptions.[20] Librarians that assume these extra duties often continue to have significant portions of their position devoted to reference, instruction, or collection management. Further, 90 percent of DH librarians surveyed learned DH skills on the job.[21]

Subject Specialist Liaisons

As we observe in greater detail below, subject librarians can make important contributions to DH projects through their work in engagement, collection development, and instruction. Additionally, the knowledge and skills possessed by subject area librarians can be of great value in developing DH projects. For example, the familiarity of the subject liaison with specialized and often obscure bibliographies can assist in creating comprehensive or intentionally selective corpora to be used in textual analysis. Reference interview methods can be used in conversation with researchers, as a way of refining and scoping research questions. Finally, the research skills of subject librarians can assist in creating comprehensive literature reviews of previous scholarly research, documenting past approaches to the topic now to be addressed with digital tools and methods.

Librarians in Technology Roles/Technologists

"Digital librarians" are a hybrid of technologist and librarian, with a specific proficiency in developing online collections and other repositories. Those with technology expertise are up to date on appropriate software, provide metadata and bibliographic control, create the user interfaces, maintain server space, and work with issues of access and preservation.[22] This category may include staff from the university IT department, who are more familiar with information systems across the university but typically much less familiar with humanities research and librarianship. In some cases, those in

a technology role will play a consulting role, assisting researchers in selecting and using the appropriate software.

The Role of the Subject Librarian in Supporting DH Research and Creativity

Subject liaison positions vary greatly. At some libraries, subject area librarians provide the primary or only library support available for digital humanities research at their institutions. In these cases, providing support is a core professional responsibility, requiring these librarians to develop skill sets similar to those cultivated by IT specialists, faculty, and others affiliated with library-related or non-library-related centers for digital humanities, whether these operate under the "lab" or "service" models.

Subject area librarians whose job descriptions do not explicitly include DH responsibilities can also make important contributions to digital humanities projects, whether in connection with a digital humanities center or independent of such institutional support. Each of the three areas most commonly associated with subject liaison librarians—engagement (or outreach), instruction, and collection development—present subject liaisons with opportunities for supporting digital humanities research and creativity. Moreover, the overlap of these three areas underscores the potential value of the work of liaison librarians to digital humanities research.

Engagement

Increasingly, subject liaison librarians are asked to view engagement as their most central responsibility, as libraries move toward emphasizing "deeper engagement across campus" and engage in relationship-building with both academic departments and other centers of research and instruction support.[23] Subject area librarians who embrace this role of fostering collaboration can provide important support for both small and large-scale DH projects, by helping to organize DH-related events and connecting potential DH researchers with support resources.[24]

Local forums and events organized by subject liaisons can bring together researchers from across the disciplines to the library to share findings, learn about new tools and methods, and find new partners in collaboration. For example, in 2019, subject liaison librarians at Miami University formed a planning group with Special Collections, the Center for Digital Scholarship, and the Library Systems Department to organize and host a local digital

humanities forum, which was modeled on a similar event held at the University of North Carolina at Charlotte. Originally scheduled as a half-day event involving three one-hour sessions (panel presentations on DH in research, DH and instruction, and a "hands-on" DH tools workshop) plus breakfast and lunch, this event was delayed and scaled down due to the COVID-19 outbreak in 2020 (losing the meals and the hands-on workshop). However, the online event (with two panel sessions and a 25-minute break for introductions and open discussion) was well-attended and received, providing a chance to meet new partners in collaboration.

While engagement is often associated with communicating with campus groups and organizing events, it also pertains to providing support for DH projects large and small, from initial conception and planning, to implementation, to post-project assessment, distribution, and preservation (a framework borrowed from the broader project management literature and often applied to DH projects).[25] DH projects take many shapes[26] and exist at scales that range from course papers and projects undertaken by individual undergraduates to platforms and projects developed across years or decades by international consortia.

At the conceptualization and planning stages of a project, liaison librarians often join projects proposed by researchers; however, librarians may also be involved at the project ideation stage, especially in cases where librarians work regularly with faculty, or serve as embedded instructors in courses that involve DH-related learning. In other cases, liaison librarians may be the primary initiators of DH research projects. However a DH project is proposed, the familiarity of subject liaisons with the resources and personnel available inside and outside libraries can help connect researchers with project elements they may not even have considered, such as digitization, archival resources, metadata, accessibility, grant-writing, copyright, and digital preservation, in addition to visualization and analytic tools.

As DH projects move toward completion, subject liaisons can guide project teams to digital preservation resources as appropriate, including open access university repositories and copyright clearances when required. Liaison librarians are usually familiar with the benefits of institutional repositories, which can provide researchers with discoverability through library systems and public search engines and offer usage metrics that can help researchers make a case for the scholarly nature of their work, as opposed to materials published on free or paid nonacademic digital platforms.[27] Because liaisons have a vantage point closely aligned with the colleagues they support, they can be instrumental in ensuring that the diverse sets of

functions provided across (and outside of) libraries come together to produce successful products.

Collection Development

At most academic libraries, collection development remains a core responsibility of subject liaisons, even as academic libraries have increasingly emphasized the importance of functional librarianship and a shift toward providing experiences.[28] However, the collection development activities of subject liaisons have tended to shift away from the acquisition or bibliographic description of individual titles and toward the evaluation and configuration of digital platforms and access models, as well as involvement in making materials available through institutional repositories and other open formats.[29] This shift in collection development has brought the subject librarian in contact with digital platforms, whether through an awareness of data in collection development decisions, a familiarity with collections analytics, or acquaintance with the DH capabilities that are incorporated in platforms and products evaluated by those developing library collections.

Subject liaisons' role in acquiring databases can make them important participants in gaining access to raw materials and databases in ways that permit and facilitate DH research. Such assistance in selecting databases and negotiating terms of data access is essential, since "[w]hile tools for mining and computational analysis of text datasets have proliferated, the texts themselves are frequently protected by copyright or other intellectual property (IP) rights, or subject to license agreements that limit their access and use."[30] While the complexities involved in data research using library collections are challenging, subject liaisons, who understand the research needs of faculty and are used to working with vendors, are well-positioned to coordinate elements of library support for DH projects, and to navigate legal and technical issues that can arise in connection with the application of digital methodologies to data collections, especially to copyrighted materials.[31]

In addition to providing liaison assistance with the process of acquiring data (and the right to analyze it at scale), subject librarians are also foundational to the DH ecosystem in their traditional role of materials selectors, since most databases are acquired prior to a request associated with a particular project. This provides subject liaisons with an important opportunity to create inclusive and diverse collections, by seeking out and acquiring the raw materials for DH projects that include datasets from voices traditionally excluded and overlooked. The acquisition of such data resources can help

overcome the "Shakespeare problem" or "canon problem" that has characterized much DH work,[32] and to center the experiences of groups and individuals traditionally marginalized by academic research. Thus, the work of collection development is essential to the progressive development of DH research along anticolonial and antiracist lines.

The work performed by subject liaisons in acquiring and selecting digital collections makes them valuable partners in supporting DH research. Similarly, their work in collection management often marks subject liaisons as practitioners of DH, even among subject liaisons who might find this fact surprising. While the idea of "distant reading" thousands or millions of texts was a novel development in literary theory,[33] the practice of evaluating thousands or millions of texts for their relevance and utility for particular interpretive communities has been the daily trade of library workers at least since the mid-nineteenth century.[34] Using circulation data to determine which items, serials, or information resources to retain is itself a form of distant reading. Subject librarians may differ in their approaches to collection management and their levels of comfort with analytical tools, such as GreenGlass or Tableau, which make it possible to describe and visualize library collection usage in increasingly granular and visually appealing ways. The use of such tools is interpretive and involves judgments about users and texts similar to those involved in many text-based DH projects. Many subject-area librarians are well on the way to becoming accidental digital humanists in the course of their collection assessment and development duties.

Instruction

In a 2019 survey, instruction was the responsibility most frequently listed by subject librarians as a primary job duty, a result that confirmed the results of a 2015 survey of ARL liaison librarians in which 1,570 respondents listed instruction as a primary professional responsibility, while just 123 indicated that it was not.[35] Liaison librarians hold a wide range of teaching responsibilities, from the creation of online learning objects such as research guides and instructional modules for online learning management systems to the provision of one-shot and embedded instruction, the development and deployment of full-credit courses, and the supervision of graduate research.

Across this range of instructional activity, subject liaisons in the humanities encounter many opportunities to promote DH resources provided by the library or other campus units and are well-positioned to invite students and faculty into the mix of digital scholarship on campus and beyond. Because

they often have access to faculty and administrators, liaison librarians are also well-positioned to contribute to conversations regarding library outreach for DH instruction, which may take the form of DH workshops, short-form classes, THATCamps, data sprints, badging initiatives, embedded instruction, or library-offered for-credit courses. Liaison librarians can also participate or help to organize DH discussion or skills groups within the library itself, as a way of cultivating skills among librarians that can be shared with the wider communities within or outside of the university.

In addition to helping plan and promote DH resources, subject liaisons can themselves participate in the provision of DH instruction. For some liaisons, instruction related to DH methodologies, tools, and issues is a primary responsibility; for others, DH occupies a much smaller portion of their instructional charge. However, active library DH programming can provide professional development that can assist librarians interested in joining the DH mix.

Another relevant instructional trend (analogous to the trend towards enhanced analytics in collection development) is the incorporation of machine learning algorithms into commercial information discovery platforms. The everyday tools used by subject area librarians in teaching information literacy and research skills are already providing analytical tools that make the platforms DH-capable. While machine learning algorithms have long served to make judgments about the "relevance" of search results returned from search queries, commercial discovery platforms are increasingly experimenting with scatterplots and other novel ways of visually displaying data. And while (as mentioned above) liaison librarians should be involved in the creation of vendor agreements regarding the consumptive and non-consumptive use of platform data, it is also the case that many data vendors are taking steps to incorporate their own data tools into discovery platforms. For example, content databases such as HathiTrust, Gale, and JSTOR/Ithaka all incorporate analytical platforms (in this case HathiTrust Research Center, Gale Digital Scholar Lab, and ITHAKA Constellate, respectively).

While the future of individual analytical tools and platforms is unpredictable, it seems likely that the availability of platform-based analytical tools will increase as the functionality of putting items into folders will develop into creating larger sharable collections and the ability to run a variety of analytical tools upon those collections. Thus, even one-shot instruction sessions can provide robust opportunities for DH instruction, whether in connection with information literacy instruction or not. As more robust tools make DH research easier and more ubiquitous, the technical obstacles

that make DH instruction difficult will continue to become smaller, in ways that will provide new instructional opportunities for subject area liaisons.

Conclusion

The involvement of subject librarians in DH projects can confer a range of benefits, for small projects and large ones and across different institutional models of DH support. In addition to building inclusive and DH-ready collections, subject liaisons can use their disciplinary knowledge and long-standing relationships with researchers interested in library DH resources and partnerships to bring together the disparate elements required to conceptualize, undertake, and preserve DH projects (and also distinguish projects and proposals likely to benefit from library support from those that will not). The work of subject librarians in engaging researchers, and providing research and creative instruction across the disciplines, makes them potential envoys of DH methodologies (see Appendix 6A for some practical examples of actions subject area liaisons can take to support DH).

Over the longer term, as digital analytical tools become more ubiquitous and increasingly accessible to those who are less highly trained in specialized tools and statistical methods, the role of subject liaisons in supporting what has been called "small DH"— a term that refers to DH work involving "in-class pedagogical projects or enhancements using out-of-the box tools"— is likely to increase, as the use of visualization and machine learning in making humanistic arguments becomes increasingly routine (and, perhaps, less recognizably tied to a notion of "digital humanities" distinct from "the humanities" more broadly).[36] As forms of DH-related research, instruction, and creativity continue to develop, the role of the subject area librarian—especially where these librarians are supported in developing their DH skill sets and viewed as key project partners—provide academic libraries with important elements to add to the mix in the successful completion of DH projects, whether completed inside or beyond the institutional boundaries of the academic library.

Appendix 6A. Practical Suggestions for Subject Librarians

A subject librarian can have an active role in the DH environment on their campus. Below are some examples of activities that subject liaison librarians can take to promote DH scholarship and research.

Low commitment	1. Connect with graduate students in humanities programs, especially those who have not yet begun the thesis or dissertation process. 2. Learn the basics of your institution's technological infrastructure and environment. 3. Learn about preservation formats and standards. 4. Learn about alt-metrics and alternative ways to measure the impact of digital scholarship. 5. Explore successful digital humanities projects. 6. Learn what analytical tools are available in the databases in the library's collection.
Moderate commitment	7. Host a symposium on digital humanities and invite external participants (including faculty on campus, faculty already engaged in digital humanities scholarship, technologists, and librarians). 8. Work with faculty and undergraduate classes to design an assignment using a digital humanities tool. 9. Create an internal library interest group to discuss DH issues and to share and practice DH tools and methods. 10. Provide workshops for faculty on digital humanities tools and/or developments in scholarship. 11. Provide training for technologists in subject background for projects. 12. Seek free training on digital humanities tools provided by developers. 13. Work with technologists or online tools such as Scratch, Code School, or Code.org to learn the basics of coding. 14. Evaluate the collection to identify excluded and underrepresented voices.
Intensive commitment	15. Initiate a new digital humanities project using the library's unique subject collections.

Notes

1. Tim Bryson, Mariam Posner, Alain St. Pierre, and Stewart Varner, "SPEC Kit 326: Digital Humanities," Association of Research Libraries, November 2011:11. https://doi.org/10.29242/spec.326.
2. Emma Annette Wilson, *Digital Humanities for Librarians*, Rowman & Littlefield, 2020, xiv.
3. "Liaison, n.," in *OED Online*, Oxford University Press, accessed April 15, 2021. http://www.oed.com/view/Entry/107808.
4. John Dryden, Edward Niles Hooker, and H. T. Swedenberg, *The Works of John Dryden*, Vol. 17, University of California Press, 1956, 17:19.
5. R. David Lankes, *The New Librarianship Field Guide*, illustrated edition, Cambridge, Massachusetts: The MIT Press, 2016.
6. Maron and Pickle, "Sustaining the Digital Humanities: Host Institution Support beyond the Start-Up Phase," Ithaka S&R, June 18, 2014.
7. Ibid., 23.
8. Ibid., 24.
9. Ibid., 24.
10. Ibid.
11. Matthew Hannah, Erla P. Heyns, and Rikk Mulligan, "Inclusive Infrastructure: Digital Scholarship Centers and the Academic Library Liaison," *portal: Libraries and the Academy* 20, no. 4 (2020), 699.
12. Hannah, Heyns, and Mulligan, "Inclusive Infrastructure," 699.
13. Mary M. Case, "Partners in Knowledge Creation: An Expanded Role for Research Libraries in the Digital Future," *Journal of Library Administration* 48, no. 2 (2008), 142.
14. Harriett E. Green, "Facilitating Communities of Practice in Digital Humanities: Librarian Collaborations for Research and Training in Text Encoding," *The Library Quarterly* 84, no. 2 (April 1, 2014), https://doi.org/10.1086/675332.
15. Craig Harkema and Brent Nelson, "Scholar-Librarian Collaboration in the Publication of Scholarly Materials," *Collaborative Librarianship* 5, no. 3 (July 2013).
16. Chris Alen Sula, "Digital Humanities and Libraries: A Conceptual Model," *Journal of Library Administration* 53, no. 1 (January 2013).
17. We are using the term technologists to focus on librarians and library staff with a technology expertise, but this term could also include IT staff external to the library.
18. Harkema and Nelson, "Scholar-Librarian Collaboration in the Publication of Scholarly Materials."
19. Ying Zhang, Shu Liu, and Emilee Mathews, "Convergence of Digital Humanities and Digital Libraries," *Library Management* 36, no. 4/5 (June 8, 2015).
20. Molly Dahl Poremski, "Evaluating the Landscape of Digital Humanities Librarianship," *College & Undergraduate Libraries* 24, no. 2–4 (October 2, 2017).
21. Poremski, "Evaluating the Landscape."
22. Bethany Nowviskie, "Skunks in the Library: A Path to Production for Scholarly R&D," *Journal of Library Administration* 53, no. 1 (January 2013).
23. Antje Mays, "Dangerous Liaisons: Brainstorming the 21st Century Academic Liaison," *Proceedings of the 2018 Charleston Library Conference* 2018 (October 1, 2019).
24. Micah Vandegrift and Stewart Varner, "Evolving in Common: Creating Mutually Supportive Relationships Between Libraries and the Digital Humanities," *Journal of Library Administration* 53, no. 1 (January 2013).

25. Brett D. Currier, Rafia Mirza, and Jeff Downing, "They Think All of This Is New: Leveraging Librarians' Project Management Skills for the Digital Humanities," *College & Undergraduate Libraries* 24, no. 2–4 (October 2, 2017); Burdick et al., *Digital Humanities*; Khang and Moe, "Success Criteria and Factors for International Development Projects."
26. Alex H. Poole, "The Conceptual Ecology of Digital Humanities" *Journal of Documentation* 73, no. 1 (January 1, 2017); Wang, "Distribution Features and Intellectual Structures of Digital Humanities."
27. Stacy Konkiel and Dave Scherer, "New Opportunities for Repositories in the Age of Altmetrics," *Bulletin of the American Society for Information Science and Technology* 39, no. 4 (2013).
28. Catherine Hoodless and Stephen Pinfield, "Subject vs. Functional: Should Subject Librarians Be Replaced by Functional Specialists in Academic Libraries?" *Journal of Librarianship and Information Science* 50, no. 4 (2018): 345–60.
29. Michael Levine-Clark, "Imagining the Future Academic Library Collection," *Collection Management* 44, no. 2–4 (July 3, 2019).
30. Megan Senseney, et al., "Data Mining Research with In-Copyright and Use-Limited Text Datasets: Preliminary Findings from a Systematic Literature Review and Stakeholder Interviews," *International Journal of Digital Curation* 13, no. 1 (December 31, 2018): 184.
31. For a useful example of a liaison role in such projects, see Tara Baillargeon, Eric Kowalik, and Jennifer M. Cook, "Collaborative Digital Research: Case Study of Text Mining a Corpus of Academic Journals," *New Review of Academic Librarianship*, September 21, 2020, 1–13.
32. Laura Estill, "Digital Humanities' Shakespeare Problem," *Humanities* 8 (March 4, 2019), 45.
33. Franco Moretti, "Conjectures on World Literature," *New Left Review*, no. 1 (February 2000).
34. Geoffrey Forster and Alan Bell, "The Subscription Libraries and Their Members," in *The Cambridge History of Libraries in Britain and Ireland*, ed. Alistair Black and Peter Hoare (Cambridge University Press, 2006), https://doi.org/10.1017/CHOL9780521780971.014.
35. Nardine, Jennifer, "The State of Academic Liaison Librarian Burnout in ARL Libraries in the United States," *College & Research Libraries* 80, no. 4 (2019)."
36. Laura Braunstein, (@laurabrarian), "@miriamkp 'Small DH'—in-Class Pedagogical Projects or Enhancements to Bespoke Library Digitization Using out-of-the-Box Tools—Needs Recognition and Support, & Can Be a Low-Investment Way of Widening the Circle to Involve More People & Break down Barriers." X (formerly Twitter), March 14, 2018, https://twitter.com/laurabrarian/status/973930607481884673.

Bibliography

Baillargeon, Tara, Eric Kowalik, and Jennifer M. Cook. "Collaborative Digital Research: Case Study of Text Mining a Corpus of Academic Journals." *New Review of Academic Librarianship*, September 21, 2020, 1–13. https://doi.org/10.1080/13614533.2020.1819352.

Bryson, Tim, Miriam Posner, Alain St. Pierre, and Stewart Varner. "SPEC Kit 326: Digital Humanities." Association of Research Libraries, November 2011. https://doi.org/10.29242/spec.326.

Burdick, Anne, Johanna Drucker, Peter Lunenfeld, Todd Presner, and Jeffrey Schnapp. *Digital Humanities*. MIT Press, 2012.

Case, Mary M. "Partners in Knowledge Creation: An Expanded Role for Research Libraries in the Digital Future." *Journal of Library Administration* 48, no. 2 (2008): 141–56. https://doi.org/10.1080/01930820802231336.

Currier, Brett D., Rafia Mirza, and Jeff Downing. "They Think All of This Is New: Leveraging Librarians' Project Management Skills for the Digital Humanities." *College & Undergraduate Libraries* 24, no. 2–4 (October 2, 2017): 270–89. https://doi.org/10.1080/10691316.2017.1347541.

Dryden, John, Edward Niles Hooker, and H. T. Swedenberg. *The Works of John Dryden*. Vol. 17. University of California Press, 1956.

Estill, Laura. "Digital Humanities' Shakespeare Problem." *Humanities* 8 (March 4, 2019): 45. https://doi.org/10.3390/h8010045.

Forster, Geoffrey, and Alan Bell. "The Subscription Libraries and Their Members." In *The Cambridge History of Libraries in Britain and Ireland*, edited by Alistair Black and Peter Hoare. Cambridge University Press, 2006. https://doi.org/10.1017/CHOL9780521780971.014.

Green, Harriett E. "Facilitating Communities of Practice in Digital Humanities: Librarian Collaborations for Research and Training in Text Encoding." *The Library Quarterly* 84, no. 2 (April 1, 2014): 219–34. https://doi.org/10.1086/675332.

Hannah, Matthew, Erla P. Heyns, and Rikk Mulligan. "Inclusive Infrastructure: Digital Scholarship Centers and the Academic Library Liaison." *portal: Libraries and the Academy* 20, no. 4 (2020): 693–714. https://doi.org/10.1353/pla.2020.0033.

Harkema, Craig, and Brent Nelson. "Scholar-Librarian Collaboration in the Publication of Scholarly Materials." *Collaborative Librarianship* 5, no. 3 (July 2013): 197–207.

Hoodless, Catherine, and Stephen Pinfield. "Subject vs. Functional: Should Subject Librarians Be Replaced by Functional Specialists in Academic Libraries?" *Journal of Librarianship and Information Science* 50, no. 4 (2018): 345–60.

Khang, Do Ba, and Tun Lin Moe. "Success Criteria and Factors for International Development Projects: A Life-Cycle-Based Framework." *Project Management Journal* 39, no. 1 (March 1, 2008): 72–84. https://doi.org/10.1002/pmj.20034.

Konkiel, Stacy, and Dave Scherer. "New Opportunities for Repositories in the Age of Altmetrics." *Bulletin of the American Society for Information Science and Technology* 39, no. 4 (2013): 22–26. https://doi.org/10.1002/bult.2013.1720390408.

Lankes, R. David. *The New Librarianship Field Guide*. Illustrated edition. Cambridge, Massachusetts; London, England: The MIT Press, 2016.

Levine-Clark, Michael. "Imagining the Future Academic Library Collection." *Collection Management* 44, no. 2–4 (July 3, 2019): 87–94. https://doi.org/10.1080/01462679.2019.1610680.

"Liaison, n." In *OED Online*. Oxford University Press. Accessed April 15, 2021. http://www.oed.com/view/Entry/107808.

Maron, Nancy, and Sarah Pickle. "Sustaining the Digital Humanities: Host Institution Support beyond the Start-Up Phase." Ithaka S&R, June 18, 2014. http://www.sr.ithaka.org/sites/default/files/SR_Supporting_Digital_Humanities_20140618f.pdf.

Mays, Antje. "Dangerous Liaisons: Brainstorming the 21st Century Academic Liaison." *Proceedings of the 2018 Charleston Library Conference* 2018 (October 1, 2019): 220–32. https://doi.org/10.5703/1288284317044.

Moretti, Franco. "Conjectures on World Literature." *New Left Review*, no. I (February 2000). https://newleftreview.org/issues/ii1/articles/franco-moretti-conjectures-on-world-literature.

Nardine, Jennifer. "The State of Academic Liaison Librarian Burnout in ARL Libraries in the United States." *College & Research Libraries* 80, no. 4 (2019). https://doi.org/10.5860/crl.80.4.508.

Nowviskie, Bethany. "Skunks in the Library: A Path to Production for Scholarly R&D." *Journal of Library Administration* 53, no. 1 (January 2013): 53–66. https://doi.org/10.1080/01930826.2013.756698.

Poole, Alex H. "The Conceptual Ecology of Digital Humanities." *Journal of Documentation* 73, no. 1 (January 1, 2017): 91–122. https://doi.org/10.1108/JD-05-2016-0065.

Poremski, Molly Dahl. "Evaluating the Landscape of Digital Humanities Librarianship." *College & Undergraduate Libraries* 24, no. 2–4 (October 2, 2017): 140–54. https://doi.org/10.1080/10691316.2017.1325721.

Senseney, Megan, Eleanor Dickson, Beth Namachchivaya, and Bertram Ludäscher. "Data Mining Research with In-Copyright and Use-Limited Text Datasets: Preliminary Findings from a Systematic Literature Review and Stakeholder Interviews." *International Journal of Digital Curation* 13, no. 1 (December 31, 2018): 183–94. https://doi.org/10.2218/ijdc.v13i1.620.

Sula, Chris Alen. "Digital Humanities and Libraries: A Conceptual Model." *Journal of Library Administration* 53, no. 1 (January 2013): 10–26. https://doi.org/10.1080/01930826.2013.756680.

Vandegrift, Micah, and Stewart Varner. "Evolving in Common: Creating Mutually Supportive Relationships Between Libraries and the Digital Humanities." *Journal of Library Administration* 53, no. 1 (January 2013): 67–78. https://doi.org/10.1080/01930826.2013.756699.

Wang, Qing. "Distribution Features and Intellectual Structures of Digital Humanities: A Bibliometric Analysis." *Journal of Documentation* 74, no. 1 (January 1, 2018): 223–46. https://doi.org/10.1108/JD-05-2017-0076.

Wilson, Emma Annette. *Digital Humanities for Librarians*. Rowman & Littlefield, 2020.

Zhang, Ying, Shu Liu, and Emilee Mathews. "Convergence of Digital Humanities and Digital Libraries." *Library Management* 36, no. 4/5 (June 8, 2015): 362–77. https://doi.org/10.1108/LM-09-2014-0116.

CHAPTER SEVEN

Collaboration and CoTeaching:

Librarians Teaching Digital Humanities in the Classroom

Brian Rosenblum, Frances Devlin, Tami Albin, and Wade Garrison

Introduction: Converge, Merge, DH Is the Word

Digital humanities (DH) as an area of engagement with students, staff, and teaching faculty has been rapidly evolving at the University of Kansas Libraries (KU Libraries) over the past several years. As the popularity of DH tools, platforms, and methodologies has increased, so has the demand to support and engage teaching faculty with incorporating DH in their courses and with their own research interests. Many academic libraries, including KU Libraries, are both adjusting to and leading this shift, figuring out ways to support digital scholarship for research and teaching, while at the same time gently delineating our roles, responsibilities, and limitations.

KU Libraries, like many other Research I academic libraries across the United States and Canada, have gone through a significant reorganization.* From 2011 to spring 2013, KU Libraries implemented a process that not only envisioned a new overarching organizational structure, but also created a user-focused model to support faculty research and teaching.[1] The Research and Learning Division created through this process includes four centers

* According to the library website, KU Libraries is "one of the top 50 libraries in the Association of Research Libraries" with "more than 4.4 million print volumes" across seven libraries on the Lawrence campus (University of Kansas, "At a Glance," KU Libraries website, accessed September 21, 2014, http://lib.ku.edu/about).

CC BY

based on user groups (Faculty, Graduate, Undergraduate, and Community), populated with librarians and staff to work with those specific constituents across the KU community. The librarians who make up these centers were drawn from previous roles focusing on digital scholarship, instruction services, and traditional liaison roles. Recognizing that the activities of these librarians overlapped in many ways that were not fully being utilized, the new Research and Learning Division has helped to merge these roles. The reorganization has meant that some librarians, placed in new roles, have had a significant learning curve in their new areas of responsibility, but this has also presented opportunities to gain new knowledge and skills and to create new synergies by working with colleagues with whom they had not worked extensively prior to the reorganization.

This chapter will describe three examples of efforts by librarians with subject, instruction, and digital scholarship expertise to provide digital humanities instruction and training to students and faculty and will look at how these efforts relate to our previous and evolving roles within the library. We will also provide concrete examples of in-class assignments, describe what worked well and what could be improved, and discuss some possible ways that we ourselves might develop the knowledge and skills needed to engage in this kind of work. We hope that these examples and observations can serve as models, starting points, or inspiration for subject specialists to both learn more about digital humanities and develop their own courses, assignments, and activities.

Example 1: An Introductory Digital Humanities Assignment

Our first example is an introductory-level digital humanities lecture and assignment that was delivered as part of an Introduction to Graduate Studies research methods seminar. In Spring 2012, the Director of Graduate Studies and associate professor in the KU French department and the subject librarian for French literature (Devlin, one of the coauthors), radically restructured the department's graduate research methods class. Devlin was embedded as the subject librarian in the class and attended all sessions. Rather than the typical one-off library session, overviews of relevant sources and research strategies were integrated throughout the class at the point of need. Throughout the semester, other librarians were invited into the class to present on topics such as copyright, scholarly communications, and working with special and rare collections. The course included the elements of a traditional bibliography

and research methods class but was modified to also focus on developing practical, professionally useful skills and on an introduction to alternative academic careers. These practical skills included how to create a web-based professional portfolio with an academic curriculum vitae or a professional resume; how to write blogs on higher education issues and literary theory; how to produce teaching portfolio materials; and, of particular relevance to this chapter, how to carry out a digital humanities project. While the professor did not have a background in digital humanities, he was cognizant of the importance for humanities students to learn more about this growing area of research. The class was offered a second time in the Fall 2013 semester and was expanded to include graduate students from the Slavic and German departments to increase the class numbers and because many of the topics covered were of common interest to all. By collaborating with these other two language departments, the Introduction to Graduate Studies class can now be offered annually, rather than every two years as it had been in the past.

Two class sessions were allocated in the syllabus to the digital humanities component. One of the coauthors (Rosenblum), who had experience developing and supporting digital projects but not as much experience in classroom DH instruction, was asked to develop an assignment that would be suitable for introductory-level work, that could be described and accomplished within two class sessions, and that would still provide students with an effective, hands-on learning experience. After some preliminary research, he adapted and modified an assignment from Lauren Klein's Digital Humanities class at the Georgia Institute of Technology.[2] The first class session, led by Rosenblum, was devoted to an introductory lecture on digital humanities and included some suggested readings, examples of digital humanities projects, and pointers to resources, tools, publications, and organizations that the graduate students might find useful in learning more about digital humanities. There was also a very brief demonstration of Voyant,[3] an easy-to-use Web-based text-analysis application, and several other tools. The students were then assigned a reading, "The Hermeneutics of Screwing Around; or What You Do with a Million Books" by Stephen Ramsay,[4] and were asked to use a digital tool to apply the methodology of "screwing around" to a text of their choice, write a short blog post to describe their experience, and prepare for a short class presentation. The text of the assignment, adapted from Klein's original, is in appendix 7A at the end of this chapter.

Two open sessions were scheduled outside of class time over the next several weeks for students to get individual help with selecting and working with their chosen tool. More than half the students attended one of these

open sessions. Rosenblum and the digital humanities librarian (Garrison) provided guidance in thinking about possible uses for the tools, showing more hands-on demos of the tools and introducing basic concepts such as removing stop words and the difference between text editors and word processors. The intent of the sessions was to simply give the students some ideas and enough knowledge to get started but not determine their research question for them. Rather, the assignment was intended to spur on learning by doing through a combination of trial and error and critical thinking, in the manner that much digital humanities work gets done.

Most of the students selected Voyant to explore a literary text, probably because it was the application shown most extensively in class and it includes a number of different types of visualizations and tools within the application. Other tools used by students included Juxta, Scalar, and Poem Viewer.[5]

One student, already proficient in programming and text analysis, developed his own Python scripts to support his analysis of the use of diminutives in Russian and other Slavic languages. The students wrote about their findings on their class blog, and during the second digital humanities class session later in the semester they gave a three-minute presentation on their findings to the rest of the class. The blogs were thoughtful and presentations were successfully delivered in a fun and engaging class session. The professor leading the class was impressed with the enthusiasm and engagement of the students in exploring a new digital tool.

The students used the tools for a variety of explorations, from looking at word frequencies and usage patterns (there were many word clouds) to looking at character relationships and networks. One student used Scalar to begin work on a scholarly, multimedia edition of a nineteenth-century novel. There were several instances of more than one student using the same tool and the same text. In these cases it was interesting to see the very different thought processes they used, the different questions they asked about the texts, and the different results they obtained. Mostly the assignment was a chance for the students to get some hands-on experience working with text and to get used to the idea of experimenting with tools that are often in a constant state of development. The students in the class liked the assignment and provided positive feedback at the end of the course. Some expressed a desire for even more digital humanities!

The development and incorporation of a digital humanities assignment into the Introduction to Graduate Studies class not only introduced these students to research in the digital humanities, but also engaged them by encouraging them to "play around" with a new tool. Additionally, it was a

successful collaboration between a faculty member, a subject librarian, and a digital humanities specialist that supported faculty and graduate students in new ways of learning in the classroom and expanded their knowledge of humanistic research. The class also resulted in a new opportunity for the student noted above who was proficient in Python. This student later became the graduate student representative on the advisory board of KU's Institute for Digital Research in the Humanities (IDRH, described below) and taught an introduction to text mining workshop at IDRH's workshop series the semester following the class. This opportunity and connection probably would have gone unrealized if this collaborative effort at bringing digital humanities into the classroom had not happened. As librarians' traditional roles in teaching continue to evolve, these kinds of partnerships will be essential to leveraging librarians' expertise to offer new services and work collaboratively with faculty to integrate digital humanities into the classroom. The Introduction to Graduate Studies class, including the same digital humanities assignment, will be offered again in the Fall 2014 semester.

Example 2: A Semester-Long Collaborative Digital Project

The Center for Faculty/Staff Initiatives and Engagement came together as a unit in May 2013, comprising staff with digital scholarship, instruction, and liaison expertise. In June 2013, three librarians from this center (Rosenblum, Garrison, and Albin) began a collaboration with a religious studies professor on a semester-long assignment for his graduate-level course on the archaeological site of the mystical cult Megaloi Theoi, located on the island of Samothrace in the Aegean Sea. Rosenblum and Garrison had previous DH experience and knowledge of various tools, while Albin, who had minimal DH knowledge, had expertise in pedagogy and research instruction. As in the first example above, the professor did not have any DH experience. He had an extensive collection of personal photos from Samothrace that he wanted to incorporate into his upcoming fall seminar.

The first step in this potential collaboration was to have a couple of very casual, noncommittal, low-stress conversations with the professor about what he might be envisioning for his upcoming course. In these conversations, librarians discussed the 818 photos, what to teach, how to teach, learning outcomes, level of librarian involvement, and what DH tools or platform to use. Each meeting was exploratory and somewhat awkward and involved a considerable amount of brainstorming. The unfocused nature of

the conversations was to be expected. Librarians were working with a faculty member who was new to digital humanities, they were all in a newly formed faculty center and had to figure out how we worked together, and lastly, they were still negotiating how to define new roles and responsibilities. However, even with the conversations going in multiple directions, it was important to separately and then collaboratively envision a range of possibilities for the course. It was agreed fairly early on that this collaboration needed to be flexible and open, yet at the same time still have some structure built in for hands-on, librarian-led instruction sessions. Since this was a very early attempt by librarians in the faculty center to introduce and instruct graduate students and faculty in a digital project over a semester, it was imperative that adaptations were made as needed.

The librarians established that Omeka would be an appropriate platform to use.[6] It is designed to accommodate a range of items including photographs, is user-friendly and well-suited for collaborative work, and is free. Selecting a platform brought to light the potential time and labor involved in preparing, modifying, and maintaining the software. Rosenblum and Garrison had experience with Omeka, but Albin, who would be designing and teaching Omeka to the students, didn't even know how to properly pronounce the name of the platform.

When they contacted KU Information Technology about server space, the librarians discovered that IT didn't have a server environment with the correct specifications to run Omeka, so it was installed on an external server; the Institute for Research in the Digital Humanities (IDRH) paid a minimal monthly fee to house it there. An Omeka site was set up for the course and the entire collection of images, which had no associated metadata other than the image filename, was imported into the system. A sandbox site was also created allowing Albin to learn how Omeka worked, how to import images, how to create metadata using Dublin Core, how to display images via themes and exhibits, how to use plugins, and on occasion, how to break Omeka, which Rosenblum would then fix. Through experimenting and breaking the sandbox, Albin was able to conceptualize the types of handouts the students might need to understand, navigate, and build online open-access exhibits.

Meetings with the professor progressed over the summer. He decided that his students would use the photographs as an archive through which to theorize the role and significance of the archeological site. Each student would select a building and create an exhibit based on the photos and original text. Then as a class, they would collaboratively write an introductory page introducing the project and the exhibits. It was decided that a minimum of

three hands-on training sessions during class time (2½ hours each) would be spread out over the semester, with the possibility of a fourth session closer to the end of the semester. The librarians would also meet with students one-on-one as needed.

The first two instruction sessions were designed to introduce students to Omeka, including creating collections and exhibits, the Semantic Web, and simple Dublin Core. While it was not a goal to turn the students into mini metadata librarians, the librarians felt that it was important that students grasp the significance and differences between tagging and more controlled vocabularies. Since the students were working on an archaeological site together, they needed to consider their potential audiences and whether or not they needed to include any discipline-specific language. To get their brains thinking about terms, words, descriptors, and tagging versus metadata, they were given the assignment of looking at photos on Flickr, specifically photos of Star Wars action figures with chipmunks and buildings from the 1983 World's Columbian Exposition,* individually creating tags and metadata for specific photos based only on the images and then together as a class discussing the terms they chose for tagging and the terms they chose for metadata.

In the third instruction session, students began building their collections from the 818 items, creating rudimentary layouts for exhibits, and brainstorming on metadata. The only metadata imported with the images when they were uploaded were the file names created by the professor. After the third session, librarians worked with students one-on-one as needed. They kept in touch with the professor and toward the end of the semester held a fourth and final session. Questions were sent ahead of time, allowing the librarians to do research in preparation for the class. This time around the students' questions were much more connected to the overall aesthetic of the project website, with requests to modify the layout and look and feel of the Omeka theme. The professor, not fully understanding the expense of customization of a corporate site, also asked to consider creating a theme

* Images used for assignment: Chris McVeigh, "Space Cowboy," photograph taken September 21, 2008, Chris McVeigh, Chipmunk Adventures Album on Flickr, https://www.flickr.com/photos/powerpig/2878681351; "South Portal of Art Building," photograph, from *The Columbian Gallery: A Portfolio of Photographs from the World's Fair* (Chicago: Werner Company, 1894), in World's Columbian Exposition Collection at The Field Museum, GN90799d_CG_071w, posted to The Field Museum Library's Flickr photostream July 26, 2005, https://www.flickr.com/photos/field_museum_library/3410234992/in/set-72157616234589478.

that resembled the Waldorf Astoria's Omeka site. (The heavily customized theme the professor was referring to has since changed.) However, because only Rosenblum has some basic knowledge of the PHP and CSS necessary to modify themes and templates in Omeka, and because making such modifications creates challenges for the long-term maintenance of sites, the librarians could not accommodate those requests, especially for what was a pilot project for a class. That meeting and two subsequent meetings with the professor and his teaching assistant (TA) were to some degree frustrating for all parties.

During the final two meetings, the professor and his TA, who was very proficient with WordPress, suggested that the librarians create a WordPress site for the class and transfer all of the content from the Omeka site. There was a discussion about the pros and cons of using WordPress, with much of the conversation revolving around whether the professor and TA wanted to emphasize the final text and content of the exhibitions that the students created or whether they wanted to think of the project as an ongoing curation of a collection of several hundred individual images. (It turned out to be the former.) Mostly, however, the conversation centered on the issues of labor, commitment, and sustainability. For example, if the students and professor wanted to use WordPress as a platform, largely because of the graduate student's familiarity with the software, they would need to think about how to maintain the site in the future after the student has graduated and moved on. The libraries would not be able to support a WordPress site, not for technical reasons, but because of time and resources.

How the project concluded at the end of the semester was not necessarily a big surprise. Even though there may have been some frustration and disappointment, all parties involved, whether or not they recognize it, gained a considerable amount of knowledge about developing and implementing a digital project. What started off as a smallish undertaking—guiding students and the professor in the creation of collections and exhibits for possible public use—morphed into graduate students recognizing that their work, if it went public, would be used by people all over the world. In that sense, it's understandable that their concerns surrounding the aesthetics of the project site would increase towards the end of the semester and get pushed to the forefront. During their initial introduction to Omeka, the Semantic Web, and Dublin Core, they were not as invested as to how the site looked. Everybody was using a different theme for their exhibit, but by the end of the semester they saw the need for an overall cohesive representation of the site and had a desire for a bit more flashiness than what the default Omeka themes or the librarians could offer. Regardless of the issue of flash or fancy, Omeka still

proved to be an excellent tool for teaching students about the practical and theoretical issues involved in creating digital projects.

For the librarians, the experience of working with a faculty member and the students on assignments and courses with a DH focus was extremely beneficial. It has given us a better understanding for working and negotiating in future collaborations on campus. It taught us that parameters and common understandings of roles and responsibilities need to be negotiated and constantly reiterated, regardless of the assignment, project, or course redesign. Collaborating and partnering with faculty ensures that both the professor and students understand the complexity of DH (pros and cons, benefits, and struggles).

Example 3: Teaching the Teacher: Course Development Grants and Workshops

Our final example involves librarians not directly involved in the classroom, but helping build digital humanities capacity at the university through an incentive program designed to encourage professors to add digital humanities material to their courses. As we have seen in the examples described above, faculty often don't have the expertise to teach digital humanities to their students, even as they recognize the importance of introducing DH concepts and topics into their classroom. In addition, as we have also seen, it can be time-consuming for librarians to take on this role, especially when we are still developing our own expertise in this area. Librarians doing in-class DH instruction in this way would not be scalable if the demand significantly increases. In the long run, we may be able to better leverage our limited resources by sharing and repurposing our work and experiences in digital humanities pedagogy and by spreading DH knowledge and capacity more widely among others. The course development grant initiative administered by KU's Institute for Digital Research in the Humanities (IDRH) is one example of a small effort in this direction, and it is increasingly involving librarian expertise.*

IDRH was founded in 2010 to provide resources and training in the practices and tools of the digital humanities for the KU community and is itself an example of a strong collaborative initiative between the libraries and the campus community. The institute is supported through a partnership

* For more information, please see the Institute for Digital Research in the Humanities website, accessed September 21, 2014, http://idrh.ku.edu.

between the KU Libraries, the Hall Center for the Humanities, and the College of Liberal Arts and Sciences (CLAS) and is administered jointly by a CLAS faculty codirector (Dwyer) and a KU Libraries codirector (Rosenblum). IDRH's primary programs include a digital humanities conference held every September, a monthly seminar series held at the Hall Center for the Humanities, a digital humanities seed grant program intended to help faculty pilot new digital projects, and a regular series of hands-on workshops on digital tools and methods. In developing programs to support digital humanities research at KU, IDRH realized that education and training was a major factor in generating viable project proposals, and we have begun to explore ways to increase DH expertise among both faculty and students on campus. The Course Development Grant program is one effort towards this end.

The Course Development Grant program is intended to help develop an interdisciplinary palette of courses in digital humanities at KU.* IDRH offers a $1,000 stipend to tenured and tenure-track faculty who develop a new course in the digital humanities. The guidelines state that the course may be in any humanities or closely related discipline and may cover specific topics within a discipline (e.g., nineteenth-century English literature), as long as at least 50 percent of the course content covers DH skills, methods, and tools. The program can be used to develop undergraduate or graduate courses and favors proposals that attract students from a variety of departments and disciplines and that use open-source, nonproprietary, cross-platform tools. All proposed courses must be taught on the Lawrence or Edwards campus within three semesters of receiving funding, and participants are asked to submit a syllabus of the new course after it is offered.

Applicants submit a short two-to-three-page proposal in late spring outlining the proposed course, potential assignments, the frequency with which it will be offered, target student audience, and "the potential impact you expect the course to have on KU's digital humanities profile." Faculty from all humanities and related disciplines are invited to submit proposals. The submissions are reviewed by a small committee of librarians and faculty from IDRH community, including grant recipients from previous years. (See appendix 7B for a copy of the grant guidelines.)

* Information on the Course Development Program can be found at "Course Development Grants," Institute for Digital Research in the Humanities website, accessed February 5, 2015, http://idrh.ku.edu/course-development-grants.

While it was hoped that the program would lead to the development of a university-wide general introduction to DH course, that has not happened yet (in large part because there is not a clear departmental home for such a course, which would be highly interdisciplinary in both content and participants). Nevertheless, in three years the program has helped develop or revise seven courses across a range of disciplines:

- Introduction to Graduate Studies (French and Italian, Slavic, German combined), awarded 2014
- American Literature I (English), awarded 2014
- The Digital World of Louise Erdrich (English), awarded 2013
- Advanced German I (German), awarded 2013
- Manzoni in the Digital Age (French and Italian), awarded 2012
- Infomania (Journalism), awarded 2012
- The Digital Shakespeare (English), awarded 2012

The courses incorporate a range of digital humanities material and assignments, from the "screwing around" assignment discussed above, to introductory mapping and visualization exercises, to more in-depth analysis and critique of digital humanities projects and tools. The Introduction to Graduate Studies course discussed as example 2 above is one of our latest grant recipients, and that course is set to expand its digital humanities content in the coming years. (We plan to collect the syllabi from these courses from the instructors when they are available and make the available online and eventually make them available online.)

Grant recipients participate in a sixty-to-ninety-minute workshop session in late spring with IDRH staff, previous course development grant recipients, and, increasingly, librarians. So far, over the course of the three years the program has been in place, we have had three former subject liaisons participate in the workshop sessions. The grantees discuss intended learning outcomes and assignments for each course, and other participants, including the librarians, offer suggestions and share experiences from previous work in the classroom. These workshop sessions, while short, have proven to be stimulating and productive and a useful way for both instructors and librarians to become familiar with new pedagogical ideas and to improve and guide their course planning.

It is too soon to say how successful the Course Development Grant program has been in developing a palette of courses with significant DH content. The courses take time to develop and get into the university's course catalog. A couple of the courses have already been offered once, and others

are still in development and will be offered for the first time in 2014 or 2015. While the expectation is that instructors will continue to develop and improve the DH aspects of the classes over time, there is no guarantee that this will happen. So the long-term outcomes of this program remain to be seen.

However, there have been several clear and immediate short-term benefits. The program has been effective in tapping into the existing interest in digital humanities instruction on the part of faculty and generating new interest. It is bringing faculty without DH experience into the conversation and providing a small forum for discussing, planning, and generating assignments that can be tested out in classroom. It provides some guidance for faculty new to digital humanities. And, through the vehicle of including previous year's recipients in the workshop sessions, it has begun to generate a small community of instructors and librarians to communicate and share ideas, providing another forum for librarians and faculty to partner. In short, for a relatively low cost, it has provided a way to start and maintain conversations with faculty about digital humanities instruction, resulting in some concrete activities and instruction in the classroom.

The next steps for IDRH include getting richer feedback from instructors after they have taught their new courses and gathering course materials and outcomes (syllabi, assignments, student work, and feedback) in order to create a repository of materials that can be shared and repurposed for other instructors at KU and beyond. IDRH also intends to work with other campus units, such as KU's Center for Teaching Excellence and the Honors Program, to continue to develop and expand such efforts. Finally, there is an opportunity to include a wider range of library staff in the program, especially in helping faculty craft instruction proposals, in reviewing proposals, and in participating in the workshop sessions.

Conclusion

Digital humanities is an area of scholarship that provides rich opportunities for engagement between librarians, faculty, and students. In the examples described above, all parties involved (the students, the faculty, and the various librarians with different areas of specialization) learned new skills and gained new DH knowledge that will inform and improve the way librarians engage with future faculty and student partnerships and with each other. Each party brings their own expertise and values together, ensuring stronger partnerships, more trust in collaborations, and a willingness to stretch their

understanding of digital humanities. Librarians played a key role in conceiving, developing, and carrying out the in-class sessions and other initiatives described above.

In fact, librarians are essential to digital humanities development in the classroom for several reasons and are better positioned than many in the university to collaborate and lead the way in digital humanities instruction and engagement. The interdisciplinary nature of DH, with its focus on emerging tools and methodologies that span disciplines, means that DH expertise can't reside within a single department or school. Librarians are well situated to step in here with their own interdisciplinary expertise and connections across campus. In addition, DH's use of digital collections (whether a researcher's own private collection or materials provided by cultural heritage institutions) and the data-driven nature of DH—its engagement with issues such publishing and dissemination of knowledge, copyright and intellectual property, file formats, metadata and preservation, and managing and structuring data—are a natural alignment with the goals, activities, and professional expertise of librarians. In addition, while librarians may still be somewhat uncomfortable with our own knowledge of DH skills and methods, research faculty, as we have seen in our examples above, often have even less experience in digital scholarship and welcome guidance from librarians.

The examples in this chapter show additional reasons that librarians should not be seen as just service providers, but recognized as partners in aiding students and faculty with skill development as well as project development. Digital projects are complex endeavors that require all parties to have a mutual understanding of desired outcomes and each party's roles and responsibilities and to know what can be reasonably accomplished in any given circumstance. Even in smaller-scale initiatives, it is usually not a matter of a librarian providing a one-off class session to meet a request for a presentation or assignment in DH. Instead, librarians can start and keep alive ongoing conversation about tools, methods, learning outcomes, and collaborations that are at the heart of digital scholarship.

At KU Libraries, we hope to further develop our own expertise by implementing an internal professional development program to develop expertise in several "tracks" such as digital scholarship, teaching and learning, data, and scholarly communication. Our newly created Research and Learning Division, with its merging of traditional librarian roles, provides an opportunity for cross-training and professional development by having staff share their expertise with each other in a coordinated internal training program. The intent of this program is to give librarians with subject and instruction

experience an opportunity to strengthen their knowledge of digital humanities and for digital scholarship and data librarians to learn about creating effective learning outcomes and the learning styles of different communities. Also, training our staff to train each other and to work on collaborative digital projects will enhance our own skills and expertise and enable us to be better teachers and collaborators with faculty and students. At the same time, we want to take heed of Trevor Muñoz's framing of digital humanities in libraries: "Digital humanities in libraries isn't a service and libraries will be more successful at generating engagement with digital humanities if they focus on helping librarians lead their own DH initiatives and projects."[7] Towards that end, we plan to explore project-based programs that bring together small groups of librarians from around the library to collaborate on a small digital projects, providing an opportunity for deeper learning than can be provided in a workshop.

There are many indications that digital humanities is becoming an important area of knowledge for campus educators and one that librarians will need to be familiar with and integrate with the ACRL Framework[8] and other literacies in our work to develop assignments of varying lengths for both undergraduate and graduate students. KU Libraries have had increasing conversations with individual faculty, as well as with department heads and teaching and research units across campus—such as the Center for Teaching Excellence, the undergraduate honors program, and the Center for Undergraduate Research—about integrating digital humanities into their activities. Finally, there is a growing volume of literature devoted to digital humanities pedagogy in a variety of venues ranging from published monographs (such as this volume) and journals, to conference presentations, to blog posts and online forums. We are monitoring these sources and looking at ways to bring their knowledge into our conversations with campus partners, into the classroom, and into our own instruction and research activities.[9]

Through the multiple approaches to developing digital humanities knowledge, we are preparing ourselves to meet the evolving requirements of effective, engaged library service.

Appendix 7A. Digital Humanities Assignment

Introduction to Graduate Studies (Fall 2014)

This assignment adapted from Lauren Klein, Georgia Tech University: [site discontinued]

(1) Please read "The Hermeneutics of Screwing Around; or What You Do with a Million Books" by Stephen Ramsay: http://www.playingwithhistory.com/wp-content/uploads/2010/04/hermeneutics.pdf

In "The Hermeneutics of Screwing Around; or What You Do with a Million Books" Stephen Ramsay contrasts the controlled, ordered nature of conventional search, like Google, with a research methodology characterized by "surfing and stumbling," otherwise known as "screwing around." The result of such a research methodology, he says, can be revelatory. He asks: "Could we imagine a world in which 'Here is an ordered list of the books you should read,' gives way to, 'Here is what I found. What did you find?'"

Inspired by Ramsay's provocation, your assignment is to do just that—to use one of the tools below to apply the methodology of "screwing around" to a text of your choice. After "surfing and stumbling" through the tools and texts, you should **(2) craft a blog post** that includes:

- (a) A screen capture of the best (or most interesting) instance(s) of what it was that you found; and
- (b) A short account of 300–500 words that explains what it was that you found, why you think it's the best (or the most interesting), and what other questions you have that remain. You may also include a critical assessment of your experience learning and using the tool. For example, what were the challenges (technical and/or intellectual) of using it; how might this tool be beneficial (or not) to your research; would a close reading approach be better for this question?
- *Note: it is also okay if you find nothing significant. That is often the case, especially when learning a new tool. You can still make it "interesting" by including both (a) and (b) above in the blog post, addressing the challenges or difficulties you encountered, and explaining why you think you found nothing significant.*

Tools

Choose one of the following tools, read through the documentation, upload a text (or multiple texts), and see what you find.

VOYANT: http://voyant-tools.org/
documentation: http://docs.voyant-tools.org/
full list of Voyant tools: http://docs.voyant-tools.org/tools/ [site discontinued]

LEXOS: http://lexos.wheatoncollege.edu/
documentation: http://wheatoncollege.edu/lexomics/

JUXTA: http://juxtacommons.org/ [site discontinued]
documentation: http://juxtacommons.org/guide [site discontinued]

PAPER MACHINES (ZOTERO PLUGIN): http://papermachines.org/
documentation: http://papermachines.org/?page_id=30

POEM VIEWER: http://ovii.oerc.ox.ac.uk/PoemVis/

CORPUS.BYU.EDU: http://corpus.byu.edu/

BOOKWORM: http://bookworm.culturomics.org/ (you will need to register)

NAMED ENTITY RECOGNIZER (NER): http://nlp.stanford.edu:8080/ner or http://cogcomp.cs.illinois.edu/page/software_view/4 (you will need to download and install the application)

TEMPORAL TAGGER (SUTime): http://nlp.stanford.edu:8080/sutime (you will need to download and install software)

Texts

You can obtain full-text versions of many of the novels you are reading in class at http://gutenberg.org or http://archive.org (Note, due to copyright restrictions, translations may be older, different versions than the version you are reading for class.)

With the permission of the class professors, you may choose another text or set of texts for analysis.

*Note: If you are not already familiar with the differences between a **text editor** (e.g. TextWrangler) and a **word processor** (e.g. Microsoft Word) please see:* http://chronicle.com/blogs/profhacker/writing-power-tools-text-editors/38940

A text editor is far better tool for preparing and manipulating texts for further analysis.

Appendix 7B. Course Development Grants

http://idrh.ku.edu/course-development-grants

As part of an effort to develop an interdisciplinary palette of courses in digital humanities at KU, the Institute for Digital Research in the Humanities is offering a $1000 stipend to tenured and tenure-track faculty who develop a new course in the digital humanities.

Priority will be assigned to proposals that meet following criteria and topical foci:

- Undergraduate courses, or Undergraduate/Graduate courses
- Courses attracting students from a variety of departments and disciplines
- Courses that use open-source, non-proprietary, cross-platform tools
- Methods that can be applied to a variety of humanities disciplines

Suggested Topics

- A (general) introduction to the Digital Humanities (high priority)
- Scripting and coding
- Markup languages for humanists (XML, TEI)
- Visual representation of data
- The creation of corpora and/or use of existing corpora
- Analyzing and presenting audiovisual sources
- The ethics of data access and privacy
- Social impacts of new media
- Visual and textual models of epistemology
- Cyberinfrastructure and the humanities
- Collaborative research methods in the humanities

The course may be in any humanities or closely-related discipline, and may cover specific topics within a discipline (e.g. 19th c. English literature), as long as at least 50% of the course content covers DH skills, methods, and tools. All proposed courses must be taught on the Lawrence or Edwards campus within 3 semesters of receiving funding. We expect to make up to three awards.

All applicants who are selected for the program will be asked to participate in a one hour Digital Humanities curriculum workshop in late April or early May (TBA).

How to Apply: Interested participants are invited to submit a short proposal (two to three pages, double-spaced) that includes: (1) a narrative description of the new course, including a list of; (2) the course title and a (possible) course number; and, (3) a discussion of the potential impact you expect the course to have on KU's digital humanities profile. The proposal should indicate whether or not the course is undergraduate or graduate, the expected enrollment, whether or not it is intended as a principal course, and the frequency with which it will be offered. It is expected that the course will be offered sometime during the next three semesters and that it will be offered at least three times within the next six-year period. Faculty from all humanities and related disciplines are invited to submit proposals.

Additional Guidelines

- **The deadline for proposals is Monday, April 28, 2014. All proposals should be submitted to -------- no later than 5:00 p.m. Central Time on that day.**
- The department chair and/or dean, as appropriate, must endorse all proposals.
- Successful applicants will be notified on or before Friday, May 9.
- The workshop for program participants will be scheduled for May, 2014. Date and time to be announced. Attendance by participants in the full program of the workshop is required in order to receive the summer stipend.
- Participants will be asked to submit to IDRH a syllabus of the new course after it is offered.
- For more information, please contact Arienne Dwyer or Brian Rosenblum, co-directors, IDRH, ------- or ----@--------.

Notes

1. Erin Ellis et al., "Positioning Academic Libraries For the Future: A Process and Strategy for Organizational Transformation" (paper presented at 35th International Association of Scientific and Technological University Libraries Conference, Aalto University, Espoo, Finland, June 2–5, 2014), http://hdl.handle.net/1808/14141; see also Janice M. Jaguszewski and Karen Williams, *New Roles for New Times: Transforming Liaison Roles in Research Libraries* (Washington, DC: Association of Research Libraries, August 2013), http://www.arl.org/storage/documents/publications/nrnt-liaison-roles-revised.pdf

2. Lauren Klein, "LCC 3843: Digital Humanities—Fourth Online Assignment: Sherlock Holmes Text Analysis," 2012, http://lkleincourses.lmc.gatech.edu/dh12/files/2011/12/oa4.pdf.
3. Stéfan Sinclair and Geoffrey Rockwell, Voyant Tools website, accessed September 21, 2014, http://voyant-tools.org.
4. Stephen Ramsay, "The Hermeneutics of Screwing Around; or What You Do with a Million Books," in *Pastplay: Teaching and Learning History with Technology*, ed. Kevin Kee (Ann Arbor, MI: University of Michigan Press, 2014), http://quod.lib.umich.edu/d/dh/12544152.0001.001/1:5/--pastplay-teaching-and-learning-history-with-technology?g=dculture;rgn=div1;view=fulltext;xc=1#5.1.
5. University of Virginia, Juxta Commons website, accessed September 21, 2014, www.juxtacommons.org [site discontinued]; Alliance for Networking Visual Culture, Scalar website, accessed September 21, 2014, http://scalar.usc.edu; Oxford e-Research Centre, University of Oxford, "About" page, Poem Viewer website, accessed September 21, 2014, http://ovii.oerc.ox.ac.uk/PoemVis/about.html.
6. Roy Rosenzweig Center for History and New Media, Omeka website, George Mason University, accessed September 21, 2014, http://omeka.org.
7. Trevor Muñoz, "Digital Humanities in the Library Isn't a Service," *Trevor Muñoz* (blog), August 19, 2012, http://trevormunoz.com/notebook/2012/08/19/doing-dh-in-the-library.html.
8. Association of College and Research Libraries, "Framework for Information Literacy for Higher Education," adopted by the ACRL Board, January 11, 2016, https://www.ala.org/acrl/standards/ilframework.
9. Please see the following sources: Association for Computers in the Humanities, "How Do We Introduce Undergraduates to the Digital Humanities?" Digital Humanities Questions and Answers Thread, accessed September 23, 2014, http://digitalhumanities.org/answers/topic/how-do-we-introduce-undergraduates-to-the-digital-humanities; Brett D. Hirsch, ed., *Digital Humanities Pedagogy: Practices, Principles and Politics* (Cambridge: Open Book Publishers, December 2012), https://www.openbookpublishers.com/books/10.11647/obp.0024; Christopher Blackwell and Thomas Martin, "Technology, Collaboration, and Undergraduate Research," *Digital Humanities Quarterly* 3, no. 1 (2009), www.digitalhumanities.org/dhq/vol/3/1/000024/000024.html; College English Association, "Digital Humanities Pedagogy," special issue, *CEA Critic* 76, no. 2 (July 2014), http://muse.jhu.edu/journals/cea_critic/toc/cea.76.2.html.

Bibliography

Alliance for Networking Visual Culture. Scalar website. Accessed September 21, 2014. http://scalar.usc.edu.

Association for Computers in the Humanities. "How Do We Introduce Undergraduates to the Digital Humanities?" Digital Humanities Questions and Answers Thread. Accessed September 23, 2014. http://digitalhumanities.org/answers/topic/how-do-we-introduce-undergraduates-to-the-digital-humanities

Association of College and Research Libraries, "Framework for Information Literacy for Higher Education," adopted by the ACRL Board, January 11, 2016, https://www.ala.org/acrl/standards/ilframework.

Blackwell, Christopher, and Thomas Martin. "Technology, Collaboration, and Undergraduate Research." *Digital Humanities Quarterly* 3, no. 1 (2009). www.digitalhumanities.org/dhq/vol/3/1/000024/000024.html.

College English Association. "Digital Humanities Pedagogy." Special issue, *CEA Critic* 76, no. 2 (July 2014). http://muse.jhu.edu/journals/cea_critic/toc/cea.76.2.html.

Ellis, Erin L., Brian Rosenblum, John M. Stratton, and Kathleen Ames-Stratton. "Positioning Academic Libraries for the Future: A Process and Strategy for Organizational Transformation." Paper presented at the 35th International Association of Scientific and Technological University Libraries Conference, Aalto University, Espoo, Finland, June 2–5, 2014. http://hdl.handle.net/1808/14141.

Hirsch, Brett D., ed. *Digital Humanities Pedagogy: Practices, Principles and Politics.* Cambridge: Open Book Publishers, December 2012. www.openbookpublishers.com/product/161/digital-humanities-pedagogy--practices--principles-and-politics.

Institute for Digital Research in the Humanities. "Course Development Grants." Institute for Digital Research in the Humanities website. Accessed February 5, 2015. http://idrh.ku.edu/course-development-grants.

Institute for Digital Research in the Humanities website. Accessed September 21, 2014. http://idrh.ku.edu.

Jaguszewski, Janice M., and Karen Williams. *New Roles for New Times: Transforming Liaison Roles in Research Libraries*. Washington, DC: Association of Research Libraries, August 2013. http://www.arl.org/storage/documents/publications/nrnt-liaison-roles-revised.pdf.

Klein, Lauren. "LCC 3843: Digital Humanities—Fourth Online Assignment: Sherlock Holmes Text Analysis." 2012. http://lkleincourses.lmc.gatech.edu/dh12/files/2011/12/oa4.pdf.

Muñoz, Trevor. "Digital Humanities in the Library Isn't a Service." *Trevor Muñoz* (blog), August 19, 2012. http://trevormunoz.com/notebook/2012/08/19/doing-dh-in-the-library.html.

Oxford e-Research Centre. "About" page. Poem Viewer website. University of Oxford. Accessed September 21, 2014. http://ovii.oerc.ox.ac.uk/PoemVis/about.html.

Ramsay, Stephen. "The Hermeneutics of Screwing Around; or What You Do with a Million Books." In Pastplay*: Teaching and Learning History with Technology*, edited by Kevin Kee, 111-172. Ann Arbor, MI: University of Michigan Press, 2014. http://quod.lib.umich.edu/d/dh/12544152.0001.001/1:5/--pastplay-teaching-and-learning-history-with-technology?g=dculture;rgn=div1;view=fulltext;xc=1#5.1.

Roy Rosenzweig Center for History and New Media. Omeka website. George Mason University. Accessed September 21, 2014. http://omeka.org.

Sinclair, Stéfan, and Geoffrey Rockwell. Voyant Tools website. Accessed September 21, 2014. http://voyant-tools.org.

University of Kansas. "At a Glance." KU Libraries website. Accessed September 21, 2014. http://lib.ku.edu/about.

University of Virginia. Juxta Commons website. Accessed September 21, 2014. www.juxtacommons.org [site discontinued]

CHAPTER EIGHT

Construction and Disruption:

Building Communities of Practice, Queering Subject Liaisons

Caro Pinto

> *In Germany today, not only does the monument vanish, but so too do the traditional notions of the monument's performance. How better to remember forever a vanished people than by the perpetually unfinished, ever-vanishing monument?*
>
> *—James E. Young*[1]

SOCIETIES BUILD monuments to remember. Monuments rise to mourn the dead, to commemorate victory in war, and to reflect on injustice. Monuments are meant to influence future behaviors: to wage peace or to bend societies toward justice. Libraries are also memorial environments: many are imposing, glorious brick structures that emulate cathedrals, holding history within their walls. Culturally, we read architectural elements like brick and paned glass as the physical manifestation of the library. At Mount Holyoke, many students remark that our Reading Room *feels* like a library with its wood beams and bookshelves that invite them to contemplate and study.

While monuments were designed to facilitate collective memory, some enable collective forgetting, like the counter-monuments built in reaction

CC BY

to the bloody wars of the twentieth century.* Perhaps the library building's suggestive power has also run its course; libraries are being renovated to facilitate new types of inquiry and research, while notions of "virtual libraries" and "libraries without walls" dominate library design discourse.† Monuments and libraries are physical structures but also sets of ideas. Certainly, the work of digital humanities in libraries is not exactly parallel to how monuments instruct us to remember a vanished people or incidents of horrible violence, but memory studies holds lessons for digital humanists. "Doing digital humanities" is a process; it is a set of practices requiring deep engagement with computing, technology, and critical reflection in order to create strategies for positive library futures and engagement with teaching, learning, and scholarship on our campuses and in public discourse.

The Digital Will Not Save You

Literary scholar Murray Roston taught his students at UCLA "every generation faces a system of inherited assumptions and urgent concerns."[2] This generation of librarians faces a system of inherited concerns about the relevance of libraries in the digital era, as well as urgent calls for new training and preparation for a rapidly changing job market. How do we stay relevant in a shifting landscape that assumes that to hold steady is anathema to our profession? We celebrate change agents and push for disruption. *New Yorker* writer Jill Lepore reflected on the current embrace of "disruption" as the solution to a variety of challenges: "[E]veryone is either disrupting or being disrupted…. This fall, the University of Southern California is opening a new program: 'The degree is in disruption,' the university announced. 'Disrupt or be disrupted,' the venture capitalist Josh Linkner warns in a new book, '…mean[s] that the time has come to panic as you've never panicked before.'"[3]

---

* James Young's 1992 article in *Critical Inquiry* defined counter-monuments as "brazen, painfully self conscious memorial spaces conceived to challenge the very premises of their being" (James E. Young, "The Counter-Monument: Memory against Itself in Germany Today," *Critical Inquiry* 18, no. 2 [January 1, 1992]: 271). Built to critique the traditional monuments built after World War I that failed to prevent the violence and mass murder of the Second World War, counter-monuments demand that memorial practices extend beyond the construction of a monument.

† Hannah Bennett reflected on how perceptions of libraries have shifted from physical entities into ideas in *Art Documentation* in 2013 (Hannah Bennett, "The Psyche of the Library: Physical Space and the Research Paradigm," *Art Documentation* 32, no. 2 [Fall 2013]: 174–85, www.jstor.org/stable/10.1086/673511).

Academic library discourse also incorporates the language of disruption. The newsletter *Keeping Up With…*, published by the Association of College and Research Libraries, echoes these concerns about relevance, covering topics such as big data, patron-driven acquisitions, and digital humanities.‡ Indeed, creative destruction drives many of the changes seen in library hiring, retention, space planning, and collections. The tone of these changes ranges from gentle to corrosive.[4] The discourse of creative disruption sometimes suggests that librarians are toiling at empty reference desks in bookless libraries devoid of people. Such libraries are empty, meaningless monuments to knowledge.

The question of disruption and crisis extends to the future of the humanities. Who will save the humanities? "The humanities are in crisis again, or still. But there is one big exception: digital humanities, which is a growth industry," wrote Adam Kirsch in *The Atlantic*.[5] Are the digital humanities the disruptor-savior for traditional libraries? If so, how should librarians at smaller, teaching-oriented institutions develop digital humanities programs and services? At Mount Holyoke College, we took inspiration from a symposium that started conversations about how to build digital humanities communities of practice. The event, held in May 2013, was organized under the auspices of the Five College Consortium, a network of institutions in western Massachusetts: Amherst College, Hampshire College, Mount Holyoke College, Smith College, and the University of Massachusetts Amherst.[6] The day taught us about the importance of flexibility and the value of archival materials, collaboration, and administrative support. While the symposium aimed to address these questions at a consortial level, individual institutional participants pondered how to enact local changes. Our cross-functional group at Mount Holyoke College, now called RAD (Research and Instructional Support, Archives and Special Collections, and Digital Assets and Preservation), coalesced organically outside of traditional organizational lines. RAD actively engages with projects that meet college learning goals and objectives, especially those that speak to the intersection of technology and the traditional liberal arts. We do not interact in compartmentalized

‡ The Association of College and Research Libraries describes the *Keeping Up With…* newsletter series as "an online current awareness publication from the Association of College and Research Libraries (ACRL) featuring concise briefs on trends in academic librarianship and higher education. Each edition focuses on a single issue including an introduction to the topic and summaries of key points, including implications for academic libraries" (ACRL, "Keeping Up With…," accessed October 25, 2014, www.ala.org/acrl/publications/keeping_up_with). For more information, visit the website.

spaces, but over coffee in the library atrium, in a classroom trying to boot a circa 2000 iMac, or in a windowless conference room nudging cascading style sheets toward their rightful place in a digital exhibition. Our community of practice at Mount Holyoke was not built from the top down; it is project-based and student-focused, blessed by our administration to support curricular engagement and to ensure that our graduates are ready for the disruption-happy world they are inheriting. In the words of Johanna Drucker, "the *next* university …is a fully integrated and distributed platform" evolved from "monastic centers" and laboratory cum industrial incubator.[7] Guerilla digital humanities groups, task forces, projects, and teams are crucial steps toward realizing the "*next* university."

It Takes a Village to Build a Digital Project

Many humanists worship at the altar of the individual creative genius, believing that their best work is completed without the assistance of anyone else. Indeed, as journalist Joshua Wolf Shenk argues, "the idea of the solitary creator is such a common feature of our cultural landscape (as with Newton and the falling apple) that we easily forget it's an idea in the first place."[8] Indeed, many humanities projects—particularly live or recorded music or theatrical productions—require the talents of many individuals. Digital humanities projects also shatter the myth of the solitary genius in that they depend upon the sustained collaboration of multiple people with diverse expertise. They are often the direct result of sustained collaborative vision, resource development, and time management. It takes a village to build a digital project. Projects like One Week | One Tool illustrate how digital projects are impossible to build without robust teams, support for those teams, and teamwork to collaborate effectively. Indeed, One Week | One Tool is a self-described barn raising.* These communities organize themselves to construct the technical foundations, critical content, and public discourse that propels and sustains their projects. These dynamic communities of practice evolve to meet the demands of the present, disappearing and reappearing to meet new challenges.

* The website of the One Week | One Tool project relates its mission to barn raisings: "For centuries rural communities throughout the United States have come together for 'barn raisings' when one of their number required the diverse set of skills and enormous effort required to build a barn—skills and effort no one member of the community alone could possess" (One Week | One Tool homepage, accessed October 25, 2014, http://oneweekonetool.org [site discontinued]).

The humanities may not have been originally imagined to be group endeavors, but the work of the digital demands a sum greater than one part. Faculty, librarians, archivists, and technologists must collaborate.[9] Libraries must also find ways to better facilitate cross-department, cross-functional organization in addition to forging new types of relationships with faculty. Some organizations are evolving to meet this challenge. Research universities like the University of Florida and Michigan State have reorganized themselves to better facilitate digital projects. As Laurie Taylor and Blake Landor note of Florida: "The Digital Humanities Library Group was created without a specific charge other than to address/discuss issues in digital humanities and to schedule training in support of the group's members. While the formation of the group was approved by Library Administration …it is very much a grassroots cohort of primarily Subject or Liaison Librarians brought together by a common interest."[10]

Thomas Padilla described the Michigan State University Digital Scholarship Collaborative, noting that "Direct ties to subject areas combined with appointments to other parts of the library strengthens our ability to be effective with respect to collection preparation to support computational analysis, communicating principles of data curation, and staying informed about needs of the campus community."[11] Indeed, these examples of cross-department, cross-functional organization are on the rise within libraries. They are working to meet evolving scholarly needs and to enlist many different library constituents to support digital humanities initiatives. This kind of nimble group structure engages digital technology to facilitate new directions in the humanities.

Digital humanities centers at research universities employ faculty and staff that can scale up to meet complex demands and have a population of graduate students to help with the challenging work of enacting these changes. What does digital humanities look like on a smaller scale at a teaching institution? How can smaller colleges do digital humanities? According to William Pannapacker of Hope College, these institutions hold special advantages: "Because of their teaching focus …faculty members are more likely to be able to experiment with projects that may not lead to traditional scholarly publications. Some liberal arts colleges even have a culture of faculty-student collaborative research, which translates perfectly into the project-building methods of the digital humanities."[12] Small liberal arts colleges may lack larger budgets, flashy centers, and an army of graduate students, but they do have agile pathways for incorporating digital humanities into their curricula and job descriptions. In many ways, liberal arts colleges are ripe for DH

innovations in staffing, collaborations, and the primacy of student-directed work.

If You Fund It, They Will Come

The Andrew W. Mellon Foundation has funded several liberal arts college digital humanities initiatives, including the Tri-Co Digital Humanities Initiative (Tri-Co DH), Five Colleges, Incorporated (5CollDH), and Hamilton College. In the Five Colleges, Mellon's funding fosters new collaborations and encouragement for faculty to incorporate the digital into the curriculum. As the website states, "Five College Digital Humanities provides grants and training to, as well as encourages, faculty members within the consortium to incorporate digital technologies into humanities studies and student research. From 2011 to 2015, 5CollDH will fund faculty and staff through our collaborative grants program, and groups of students through our student fellowship program."[13] One of the central questions for librarians in the consortium has been this: How do we participate? How can librarians be change makers and active agents in these digital projects? In the Five Colleges, librarians wanted to make substantial contributions to this growing endeavor. To integrate the "digital" into the curriculum would require sustained collaboration between librarians, technologists, and faculty. Librarians wondered how to begin tackling this task across the campuses to transform the digital humanities from an abstraction into an actionable set of practices.

The libraries in the Five College Consortium are autonomous, but collaborate in meaningful ways with a shared integrated library system, a shared depository, and reciprocal lending among all five campuses. The libraries also collaborate administratively with committees, task forces, and working groups. One such committee was the Digital Environment Development and Coordinating Committee (DEDCC), which scanned the horizon to help librarians understand future challenges and opportunities. During the 2012–13 school year, DEDCC engaged in exploratory conversations about how to collaborate with faculty and students to "do" digital humanities in the Five Colleges. The group decided to sponsor a symposium whose goal was "to provide a starting place for librarians and IT staff at the Five Colleges to learn about and explore research and scholarship in the Digital Humanities in liberal arts settings. The symposium included a panel followed by breakout sessions to explore topics around the broad question *'What does it take to become an effective digital humanities community of practice in the Five Colleges?'*"[14]

Imagining a Community of Practice

The DEDCC librarians solicited proposals nationally to help the group reflect on what it would take to become an effective community of practice. The committee selected speakers to represent a variety of perspectives: Joanne Schneider, library director at Colgate University; Laura McGrane, a faculty member at Haverford College; Jen Rajchel, a post-baccalaureate resident for the Tri-Co DH Initiative (at Haverford, Swarthmore, and Bryn Mawr Colleges); Laurie Allen at Haverford, who had transformed from a traditional reference and instruction librarian into Digital Scholarship Coordinator; Alston Brake, the very first Digital Scholarship Librarian at Washington and Lee University; and Brandon Bucy, an Academic Technologist at Washington and Lee. Each speaker saw changes in their titles and responsibilities to meet new challenges presented by the digital humanities. The panels were excellent object lessons in how to transform libraries to meet the evolving needs of digital scholarship in small organizations. Often, these transformations were accomplished without adding new positions, which is always a challenge in waging organizational change.

The 5CollDH event was a one-day symposium that began with the usual mix of coffee and mingling followed by presentations from invited speakers and discussions of digital scholarship. Following lunch, the group of archivists, librarians, instructional technologists, and administrators broke out for an "unconference" of active conversations, brainstorming, and small-group exercises. What might work at Colgate or Haverford or Washington and Lee would not necessarily work at Smith, Mount Holyoke, Hampshire, Amherst, or the University of Massachusetts, so the afternoon's work session was intended to envision DH in the local context. Armed with a range of brightly colored sticky notes, pens, and enthusiasm, participants went to work brainstorming about how to facilitate a DH community of practice in the Five Colleges.

Active facilitation of these conversations was key to identifying themes and ideas that resonated across and within libraries and units. In unconference mode, participants voted on the top priorities within their groups. Themes included budgets, "special collections," "updated position descriptions," and "compelling vision." Soon, participants covered the white butcher paper with many brightly colored sticky notes. Following the conclusion of that exercise, facilitators began grouping the Post-it notes together in themes: relationships, communication, funding, skills, definitions of DH, and creating an inventory of 5CollDH projects. Groups gathered around butcher

paper to meditate on these themes. Eventually, facilitators provided orange dots for participants to use in voting on common themes that emerged to help work toward the goal of defining our community of practice. Themes emerged, such as "who else is working on an area of interest," "who is this for," "common set of principles." Once the votes were tallied, the facilitators compiled the results to act as action steps for the committee to consider for future programming. While the consortial perspective was critical for the success of the event, the local context was equally critical for our day-to-day work. We also considered how we would do digital humanities at our individual campuses. The facilitated discussions were excellent, but raised additional questions: How could we bring these conversations into our local institutional contexts? How should we create working groups on our campuses and beyond to integrate this work into our existing practices?

An opportunity to do digital humanities at Mount Holyoke College soon presented itself. The professor of the first-year connections course, a one-credit course designed to acclimate first-year students to college learning, approached the college archivist about an assignment focusing on college traditions. The objective was to inculcate first-year students with a sense of Mount Holyoke history and to allow them to learn about the traditions that make Mount Holyoke College unique. We eventually decided to create a digital exhibit that incoming students and their families could view and enjoy before they arrived on campus. The exhibit would also serve the library as a large-scale teaching tool, engaging a variety of students over the long term. And so began the first iteration of RAD: Research and Instructional Support, Archives and Special Collections, and Digital Assets and Preservation Services.

A Cross-Functional Community at Mount Holyoke

One subject liaison, one metadata librarian, or one archivist cannot play hero to the digital humanities endeavor. In our case at Mount Holyoke, we recognized that our cohort would take the form of working groups formed under the charge of producing certain projects. Our cohort coalesced in an informal but powerful way. In one example, we built an Omeka site for the first-year seminars program to empower new students to transition to college and cultivate resilience in a new community. The Omeka site would not have been possible without the collaborative efforts of everyone involved in the project, especially the students who were at the center of the research, development, and deployment stages.

Our cross-functional work is not a monument to digital scholarship or to collaboration; we fully expect that our working group will evolve to meet new challenges and reflect constructively on past work. We organized new configurations of people to attend to new tasks, but we recognized the impermanence of these configurations and our need to be open to shifting responsibilities. Positions in libraries cannot be monuments to the past or to the already completed work of other people. Libraries themselves cannot simply be static monuments to past work; we cannot rely on grand architecture to remind our funders and our constituents to care about the future of libraries in higher education. While we must remember our past to understand our present and plan for the future, we need to be nimble and sensitive to local organizational contexts.

Cross-Functional Individuals: Queering the Subject Librarian

If this generation of librarians must choose between disrupting and being disrupted, there are exciting possibilities ahead for us. Today's information professionals can disrupt the information professional binary that divides librarians and archivists, subject liaisons and metadata librarians. These binaries obscure the remarkable commonalities and complementary skill sets that make academic library communities stronger, nimbler, and ready for future challenges. Successful digital humanities work depends upon effective cross-functional collaboration. I readily and enthusiastically identify as a *librarchivist*; a franken-professional who is part archivist and special collections professional and part librarian, heir of information, collection development, instruction, and liaison to faculty, curriculum, and student research needs. While my professional identity as a "specialized generalist" empowers me to participate in a range of conversations and projects, I recognize that I need the expertise and skills of my community of allies and collaborators. Ideas that can shape our scholarly future can come from any of us. Can they come from you? We should disrupt toward solidarity and innovate toward communities of practice in the digital humanities. These communities have the potential for positive, lasting disruption in the academy, as Roxanne Shirazi has argued: "Let's join our colleagues who are struggling with the narrow system of rewards that favors individual research over (collaborative) service work. The same system in which women, people of color, and queer scholars disproportionately shoulder the burden of committee work, community building, and 'service work' that reproduces the academy."[15] If

the library of the past was a monument to the past, perhaps digital humanities can create a counter-monument, disrupting the landscape towards a new future.

Notes

1. James E. Young, "The Counter-Monument: Memory against Itself in Germany Today," *Critical Inquiry* 18, no. 2 (January 1, 1992): 271.
2. Murray Roston, quoted in Natasha Vargas-Cooper, *Mad Men Unbuttoned: A Romp through 1960s America* (New York: Collins Design, 2010), xii.
3. Jill Lepore, "The Disruption Machine," *New Yorker*, June 30, 2014, accessed February 4, 2015, http://www.newyorker.com/magazine/2014/06/23/the-disruption-machine.
4. Caro Pinto, "Creative Destruction in Libraries: Designing Our Future," *In the Library with the Lead Pipe* (blog), November 20, 2013, www.inthelibrarywiththeleadpipe.org/2013/creative-destruction-in-libraries-designing-our-future.
5. Adam Kirsch, "Technology Is Taking Over English Departments: The False Promise of the Digital Humanities," *New Republic*, May 2, 2014, www.newrepublic.com/article/117428/limits-digital-humanities-adam-kirsch.
6. Five College Consortium home page, accessed October 25, 2014, https://www.fivecolleges.edu.
7. Johanna Drucker, "The University as a Fully Integrated and Distributed Platform: A Vision," *portal: Libraries and the Academy* 14, no. 3 (2014): 326.
8. Joshua Wolf Shenk, "The End of 'Genius,'" Sunday Review, *New York Times*, July 19, 2014, www.nytimes.com/2014/07/20/opinion/sunday/the-end-of-genius.html.
9. Caro Pinto, "Digital Humanities Is a Team Sport: Thoughts on #DHTNG," *dh + lib* (blog), April 10, 2013, http://acrl.ala.org/dh/2013/04/10/digital-humanities-is-a-team-sport-thoughts-on-dhtng.
10. Laurie Taylor and Blake Landor, "Intertwingularity with Digital Humanities at the University of Florida," *dh + lib* (blog), July 23, 2014, http://acrl.ala.org/dh/2014/07/23/intertwingularity-digital-humanities-university-florida.
11. Thomas Padilla, "Digital Humanities Librarianship: Year 1," *Thomas Padilla* (blog), July 13, 2014, www.thomaspadilla.org/2014/07/13/digital-humanities-librarianship-year-one.
12. William Pannapacker, "Stop Calling It 'Digital Humanities,'" *Chronicle of Higher Education*, February 18, 2013, http://chronicle.com/article/Stop-Calling-It-Digital/137325.
13. Five College Digital Humanities, "About" page, accessed July 30, 2014, http://5colldh.org/about.
14. Digital Humanities for Liberal Arts Colleges Symposium home page, accessed July 30, 2014, https://sites.google.com/a/mtholyoke.edu/digital-humanities-for-liberal-arts-colleges-symposium.
15. Roxanne Shirazi, "Reproducing the Academy: Librarians and the Question of Service in the Digital Humanities," *Roxanne Shirazi* (blog), July 15, 2014, http://roxanneshirazi.com/2014/07/15/reproducing-the-academy-librarians-and-the-question-of-service-in-the-digital-humanities.

Bibliography

Association of College and Research Libraries. "Keeping Up With…." Accessed October 25, 2014. www.ala.org/acrl/publications/keeping_up_with.

Bennett, Hannah. "The Psyche of the Library: Physical Space and the Research Paradigm." *Art Documentation* 32, no. 2 (Fall 2013): 174–85. www.jstor.org/stable/10.1086/673511.

Digital Humanities for Liberal Arts Colleges Symposium home page. Accessed July 30, 2014. https://sites.google.com/a/mtholyoke.edu/digital-humanities-for-liberal-arts-colleges-symposium

Drucker, Johanna. "The University as a Fully Integrated and Distributed Platform: A Vision." *portal: Libraries and the Academy* 14, no. 3 (2014): 325–28.

Five College Consortium home page. Accessed October 25, 2014. https://www.fivecolleges.edu.

Five College Digital Humanities. "About" page. Accessed July 30, 2014. http://5colldh.org/about.

Kirsch, Adam. "Technology Is Taking Over English Departments: The False Promise of the Digital Humanities." *New Republic*, May 2, 2014. www.newrepublic.com/article/117428/limits-digital-humanities-adam-kirsch.

Lepore, Jill. "The Disruption Machine." *New Yorker*, June 30, 2014, , http://www.newyorker.com/magazine/2014/06/23/the-disruption-machine.

One Week | One Tool homepage. Accessed October 25, 2014. http://oneweekonetool.org [site discontinued].

Padilla, Thomas. "Digital Humanities Librarianship: Year 1." *Thomas Padilla* (blog), July 13, 2014. www.thomaspadilla.org/2014/07/13/digital-humanities-librarianship-year-one.

Pannapacker, William. "Stop Calling It 'Digital Humanities.'" *Chronicle of Higher Education*, February 18, 2013. http://chronicle.com/article/Stop-Calling-It-Digital/137325.

Pinto, Caro. "Creative Destruction in Libraries: Designing Our Future." *In the Library with the Lead Pipe* (blog), November 20, 2013. www.inthelibrarywiththeleadpipe.org/2013/creative-destruction-in-libraries-designing-our-future.

———. "Digital Humanities Is a Team Sport: Thoughts on #DHTNG." *dh + lib* (blog), April 10, 2013. http://acrl.ala.org/dh/2013/04/10/digital-humanities-is-a-team-sport-thoughts-on-dhtng.

Shenk, Joshua Wolf. "The End of 'Genius.'" Sunday Review, *New York Times*, July 19, 2014. www.nytimes.com/2014/07/20/opinion/sunday/the-end-of-genius.html.

Shirazi, Roxanne. "Reproducing the Academy: Librarians and the Question of Service in the Digital Humanities." *Roxanne Shirazi* (blog), July 15, 2014. http://roxanneshirazi.com/2014/07/15/reproducing-the-academy-librarians-and-the-question-of-service-in-the-digital-humanities.

Taylor, Laurie, and Blake Landor. "Intertwingularity with Digital Humanities at the University of Florida." *dh + lib* (blog), July 23, 2014. http://acrl.ala.org/dh/2014/07/23/intertwingularity-digital-humanities-university-florida.

Vargas-Cooper, Natasha. *Mad Men Unbuttoned: A Romp through 1960s America*. New York: Collins Design, 2010.

Young, James E. "The Counter-Monument: Memory against Itself in Germany Today." *Critical Inquiry* 18, no. 2 (January 1, 1992): 267–96.

CHAPTER NINE

A Checklist for Digital Humanities Scholarship

Elizabeth Lorang and Kathleen A. Johnson*

WITH VARIED training, experience, and interest, and working at widely different colleges and universities, subject librarians may be well prepared or utterly at sea when students and faculty ask for help with creating digital humanities scholarship. But as subject specialists increasingly connect patrons to resources both within and outside the library for developing digital humanities scholarship or collaborate in creating projects, they must have a core of knowledge and practical resources for working with their researchers. To help develop this core of knowledge, this chapter addresses foundational issues that arise in the process of creating digital humanities scholarship and offers some practical guidance. Specifically, this chapter provides a checklist for managing crucial points throughout the scholarship life cycle—whether the project is an undergraduate's class activity, a graduate student's first foray into digital scholarship, or a new or experienced faculty member's enterprise, and whether or not one's institution has an established support structure for digital scholarship. This chapter has emerged in part from discussions at a range of professional conferences and meetings, in which subject librarians have expressed significant concern about not knowing where to begin when researchers ask for assistance in their digital humanities research and scholarship. The goal of this chapter is to provide subject librarians a compact, streamlined resource to consult when approached by researchers whose scholarly

* We would like to thank Katherine Walter, Chair of Digital Initiatives and Special Collections and Co-Director of the Center for Digital Research in the Humanities at University of Nebraska–Lincoln, and the fellows of the UNL Digital Scholarship Incubator—Rebecca Ankenbrand, Geraldine Dobos, Kevin McMullen, and Brian Sarnacki—for their input on this essay and checklist.

CC BY

questions require an engagement with digital humanities methods. This checklist does not endeavor to be a detailed how-to, but rather to develop a framework within which subject specialists can work with researchers.

Why create a checklist for digital humanities scholarship, and what is the place of a checklist in an enterprise that depends so heavily on innovation and creativity? In developing a checklist for digital humanities scholarship, we draw on Dr. Atul Gawande's *Checklist Manifesto*. Gawande is a surgeon who was asked by the World Health Organization to help develop an approach to reduce surgical errors.[1] To learn more about how other professions avoid errors, he consulted with construction engineers and aviation specialists. *The Checklist Manifesto* details the process by which Gawande came to value and then develop a checklist to improve patient outcomes worldwide in surgery and to advocate for the use of checklists in any complex endeavor. "[T]he reliable management of complexity…," Gawande writes, "requires balancing a number of virtues: freedom and discipline, craft and protocol, specialized ability and group collaboration."[2] He goes on to explain how checklists can help manage complexity and strike a balance between the sometimes competing virtues (e.g., freedom and discipline): "They supply a set of checks to ensure the stupid but critical stuff is not overlooked, and they supply another set of checks to ensure people talk and coordinate and accept responsibility while nonetheless being left the power to manage the nuances and unpredictabilities the best they know how."[3] Following Gawande's model, this chapter aims to help manage the complexities of digital humanities scholarship to ensure that critical steps are not overlooked. In documenting the critical steps that cut across projects, the checklist also enables scholars to focus their energy on creativity, craft, and imagination—other key features of successful scholarship.

The umbrella term *digital humanities* covers a wide variety of approaches and types of projects, and digital humanities' definition as a field, a set of methodologies, a community of practice, a movement, or something else altogether remains an ongoing point of discussion.* Digital humanities writ large includes research and pedagogical activities, as well as public programming, and the boundaries between these activities are fluid. While subject librarians should be aware of the unfixed nature of definitions for digital humanities, the indefinite nature of the terminology ultimately is not crucial to the work of this chapter. This chapter focuses on scholarly projects in the humanities that further conversations in the researchers' fields of study and

* For a sampling of pieces that take up defining digital humanities and digital scholarship, see "Readings on Defining Digital Humanities" at the end of this chapter.

for which digital methodologies are a critical component of research, analysis, or dissemination.

In constructing a checklist for digital humanities scholarship, our goal has been to focus rather than to exclude. Our checklist endeavors to fit a wide range of digital humanities scholarship, from Web-based projects to large-scale data mining. While the sources, methods, and products of these scholarly endeavors will vary, certain key activities must be accomplished for successful digital humanities scholarship across the spectrum.† To that end, we are interested not in what is the same or what is different in the digital or in claiming any kind of shift—epistemological, paradigmatic, or otherwise—but rather in documenting what is necessary for creating successful digital humanities scholarship in a very practical sense.

Background: Lessons on Digital Humanities

The University of Nebraska–Lincoln (UNL) is a land-grant research university of 25,000 students with very high research activity and is a member of the Committee on Institutional Cooperation (CIC).[4] Before getting to the checklist proper, we want to reflect on a few lessons learned at UNL regarding digital humanities work over the past fifteen-plus years. These lessons continue to guide the efforts of the UNL Libraries and the Center for Digital Research in the Humanities to develop digital humanities scholarship in the present, and they form a backbone for the checklist that follows. Although these lessons come out of our specific institutional context, they apply broadly to the creation of digital humanities scholarship.

Lesson One

Space and equipment are not the only—nor even the most important—requirements for successful digital scholarship. Developing a knowledge base is far more crucial than physical space or technical infrastructure. Although UNL Libraries set up an E-Text Center with new desktop computers and some cutting-edge digitization equipment in the later 1990s, the library staff as well as the humanities and social science researchers had not yet developed a solid understanding of what digital scholarship might entail. In the years since, as UNL founded and developed the Center for Digital Research in the Humanities (CDRH) as a joint initiative of the University Libraries

† For a selection of pieces that discuss the work of building in digital humanities, see "Readings on Doing Digital Humanities" at the end of this chapter.

and College of Arts and Sciences, library leaders and others learned from the early experience of the E-Text Center. The E-Text Center was incorporated into the CDRH with a far more robust vision of the potential for digital humanities research and scholarship and with the necessary intellectual frameworks and staff in place. Now nearly two decades and many projects later, the CDRH launched a pilot program called the Digital Scholarship Incubator in spring 2014. Although a small, designated space with basic computing equipment is part of the Incubator, its most important characteristic is knowledge exchange: graduate student fellows agree to develop their projects in the co-working space, so that they can take advantage of the group's collective knowledge. In addition, the Incubator director works to connect the students with individuals inside the library and elsewhere on campus, as well as with information resources. The emphasis in this space is not on high or bleeding-edge technology. Rather, the space is about collective knowledge and relationships. This first lesson is significant for subject librarians because they may see the first move within institutions, or the desires of their patrons, to be for space and equipment; what is most crucial, however, is connecting individuals or groups with information, serving as a liaison for users both within the library and between the library and other campus entities, and connecting people with resources and with one another.

Lesson Two

As with scholarship in any form or medium, the researcher must be able to articulate the scholarly conversation in which her project participates, and the project should create new knowledge. Some digital projects are routine digitization endeavors: they create digital resources that are fundamentally about access to materials. These more routine digitization initiatives can usefully be compared with the electronic scholarly editions published in the open-access journal *Scholarly Editing* or the multimodal scholarship of the journal *Vectors*.[5] All of these projects involve digitization at some level, but the difference is not simply one of scale, the large-scale versus the "boutique."

In the case of editions in *Scholarly Editing*, the digitization of a text through page images and transcription is often one component of the work. In proposing an edition for *Scholarly Editing*, however, the editor must craft both a statement of significance for the edition as well as a statement of editorial approach—an editorial policy and editorial philosophy. Thus, in his "Introduction to '*Avisos a pretendientes para Indias*,'" Clayton McCarl offers a lengthy discussion of the editorial philosophy underlying the preparation

of the edition, and he connects that philosophy of the edition to all of its components, including transcription, encoding, annotation, and web display. He writes, in part, "A guiding principle behind this project has been a desire to exploit the possibilities of TEI XML to document and reveal the various levels of editorial and translational decisions involved."[6] Likewise, in the Fall 2013 issue of *Vectors*, contributor Emily Thompson, creator of "The Roaring 'Twenties: An Interactive Exploration of the Historical Soundscape of New York City," wrote to introduce her project, "The aim here is not just to present sonic content, but to evoke the original contexts of those sounds, to help us better understand that context as well as the sounds themselves. The goal is to recover the meaning of sound, to undertake a historicized mode of listening that tunes our modern ears to the pitch of the past. Simply clicking a 'play' button will not do."[7] Digital humanities scholarship requires an intellectual engagement with how the methods, form, and medium of the scholarship further the argument the scholar seeks to advance. As Thompson's description of her work makes clear, digital humanities scholarship may grow out of digitization—the digitization of a corpus of texts or images, for example, may lead to research questions or allow one to make the critical point. Especially when working with undergraduate students, it is valuable for a subject librarian to help them understand the difference between routine digitization and the development of a project that seeks to address a research question.

Lesson Three

While nearly all scholarship in any form requires collaboration, a remarkable feature of digital humanities scholarship is the way it foregrounds collaboration, often through the formation of interdisciplinary research groups. Such a group might include scholars from one or more disciplines, graduate or undergraduate students, librarians, and technologists. Various articles have commented on the ways in which such collaborations may blur the roles of the participants and how collaborative engagement in a research question or project may result in more satisfying results. Any such endeavor requires commitment of team members to listen to one another's opinions and to value the different skills and knowledge of the individuals within the group. It also requires that the project director recognize when the group's collective opinions should supersede personal inclinations. This can actually be crucial to the research.*

* For a selection of pieces that deal with collaboration in digital humanities, see "Readings on Collaborating in Digital Humanities" at the end of this chapter.

Subject Librarians and the Digital Humanities

As a subject librarian, you already have many of the tools necessary, and you know more than you think you do, to be able to assist researchers in the development of digital humanities scholarship. Knowing what is happening in the departments you serve as well as knowing your institution's collections remains critical and will allow you to be proactive in your support of digital humanities scholarship. Being aware of the interests and strengths of researchers in your departments allows you to connect them with archival or special collections from which digital humanities scholarship might emerge. It also puts the subject librarian in a position to advocate within the library and elsewhere for resources to meet researchers' needs, whether content and materials or technology. Developing and maintaining relationships with vendors also remains important. For data-mining projects, researchers sometimes need access to the underlying data of a service rather than access via the product's typical web interface. The subject librarian may serve as a liaison between the researcher and the vendor, potentially working also with acquisitions staff or legal departments. In this scenario as well, the subject librarian serves as an advocate for the information needs of the researcher. Subject librarians might also raise matters of copyright, preservation, and archiving of the project or of elements of data in the project, as well as offering traditional reference assistance and alerting people to the existence of relevant resources outside of one's home library. Beyond these specific kinds of assistance, by drawing on the checklist in this chapter, you will provide your faculty and students with a larger framework to think through the entire process and to see how your knowledge and skills fit into the big picture.

With a major digital humanities unit already in place at our institution, subject librarians have played many roles in the development of various projects. In several cases, subject librarians have served as the project director or principal investigator and have guided the intellectual pursuits and outcomes of their projects.* More frequently, however, UNL subject librarians have served in supporting roles: they have helped with early digital humanities project development; served on a joint library and college advisory committee

* Two examples of digital humanities projects created by UNL librarians are the *American Indian Treaties Portal,* headed by UNL Documents Librarian Professor Charles Bernholz (University of Nebraska–Lincoln website, accessed July 21, 2014, http://treatiesportal.unl.edu), and Architecture Librarian Professor Kay Logan-Peters's *Architectural Tour of Historic UNL (see* "About the Project," accessed July 21, 2014, https://historicbuildings.unl.edu/about.html?csrt=16247912834438985941).

centered on digital humanities, alerted faculty members to the existence of relevant collections within a larger special collection, purchased microfilms needed for a digital project, connected faculty members to resource people who could provide the underlying data for data mining, encouraged graduate students and faculty to follow standards and best practices, developed metadata frameworks according to community standards, and helped to assess the usability and enhancement of completed projects. At institutions without an established digital humanities program, the subject librarian might well find himself or herself called to provide more extensive assistance. Regardless of the level of involvement subject librarians may have in digital humanities at different institutions, as the most successful subject librarians do, continue to perform environmental scans and develop habits of mind for working with your researchers in their specific areas. The following annotated checklist articulates many of the habits of mind a subject librarian needs to use in working with any digital humanities project.

Annotated Checklist

As noted above, the goal of this checklist is to provide a framework within which subject specialists can work with researchers. This checklist is not intended to capture every contingency or to explain how to perform each step, but to raise foundational issues in creating successful digital humanities scholarship. This document is designed for subject specialists to use as they consult with the researchers they assist in connecting scholars with information, opportunities, and possibilities, but also as they develop their own scholarly projects. Ultimately, it is the scholar, or a designated member of the project team, who is responsible for completing each of the items in the checklist. Thus, the "you" referenced in the checklist items refers to the researcher or scholar.

- ☐ **Articulate the project's research question and the scholarly conversation being entered.**

  The boundaries between "routine" digitization and scholarship are not hard and fast, of course, but within a scholarly project it is fundamental from the outset that the scholar be able to articulate the research question and the scholarly conversations in which the project participates. Over time, the argument or thesis should develop.

☐ **Research other projects with similar research aims or methodologies.**

This step is analogous to performing a literature review before starting research and writing. Doing so ensures you do not pour your heart and soul into something someone has already done, and it may suggest possible collaborators.

☐ **Define the scope of the project.**

Set out the boundaries for the project, including the beginning and ending of the project, what it will accomplish, and the primary audience. If you fail to define a project, you risk struggling with a scope that continually changes and, consequently, never finishing the project.

You need a base project to build on first. It is best to keep track of good ideas, but to stick to the boundaries. At the same time, be prepared to evaluate progress toward your stated goal. Before revising the scope of the project, carefully consider the consequences of scope change, and remember that resources—including time—are finite. If the scope must change for a legitimate reason, reassess whether the project can be completed with the resources you have or can draw on.

☐ **Evaluate needed and available resources.**

Determine what resources are necessary to complete the project (hardware, software, content, expertise). Perform this evaluation early to get a handle on what it will take to complete the project. Assess what resources are available at your institution, both within and outside the library, as well as in the community. If you do not have access to all of the necessary resources, how might the problem be solved (changing the project scope, partnering with another institution or bringing additional collaborators on board, employing a different technological solution)? Bear in mind that sometimes a lack of resources leads to creative solutions.

The resource evaluation may indicate that funding (whether internal or external) is needed in order to complete the project work. If the project requires funding, at what stage is funding needed and

at what level? Where might this funding be secured? If you think the project needs funding, does that suggest a need to reconceptualize the project to complete it with resources at hand? If you need outside funding, what does that mean for the timeline of the project? Is it more important to get something done now, or can you wait for grant funding that may or may not come through? If funding is necessary, factor in the time involved to research funding opportunities, write proposals, and wait on funding decisions.

☐ **Identify project participants.**

Digital humanities projects regularly involve a range of participants, including scholars and librarians as well as students, information technology professionals, and people from outside of the academic community. Assess early in the project who these participants should be. You cannot develop your project further without the right team. Have a frank discussion about what it means to have each person involved in the project. Include in this discussion participant roles and how people will be credited and acknowledged.[8]

☐ **Develop a communication plan.**

A communication plan articulates the etiquette for how the team will communicate with one another and externally about the project. Within the team, what is a reasonable time to wait for an answer from another project participant? The communication plan may state that a lack of response is agreement and that the person asking for input will then be able to take action.

The communication plan should spell out not only how creators of the project will communicate, but also how inquiries will be handled once the project or project results are public. What is an acceptable time frame to respond? How will you document your response? Will other project members need to receive a copy of the response?

☐ **Investigate issues regarding intellectual property.**

There are a host of issues to consider regarding intellectual property and digital humanities scholarship. Before proceeding with the project, have clarity about copyright and fair use for any intellectual property used by the project. Always articulate why you have the

right to use content, even if the justification seems obvious. For intellectual property created for the project, investigate issues relating to development of potentially marketable products (such as software). Who—an individual, the team, the institution—will own the scholarship and any products? Under what terms will the project's intellectual property be made available? Document all of the decisions made about intellectual property both used by and created for the project.

☐ **Develop a project work plan.**

Identify the order of activities and the critical paths. What must happen before or after something else? What things can be done concurrently? Break the project into smaller pieces and establish milestones and benchmarks for each of these pieces as well as for the overall project. Set deadlines for the goals and monitor progress in order to avoid scope creep and project drift.

The work plan should also include a budget. You may not have a pool of money with which to work, but participants' time should be included in the budget. Remember, no one has limitless time to work on a project. Treating time as money will help to evaluate whether the goals and timeline are reasonable.

☐ **Develop a data management plan.**

Determine what project materials will need to be maintained beyond the research stage, by whom, and for how long. Project data might include source code, incoming data being processed, data results of running software, and analyses. Not everything has to be saved or maintained, but for parts of the project for which long-term use is possible, what must you do to assure that others will be able to access and use those materials? Start planning for data management at the beginning of the project because your choices will affect what data will be retained and how the data can be managed. Funding agencies increasingly require thinking about these issues before they will provide support.*

* For examples of data management plans submitted for NEH-funded digital humanities projects, see National Endowment for the Humanities, "Digital Humanities Start-up Grants," accessed July 15, 2014, www.neh.gov/grants/odh/digital-humanities-start-grants [site discontinued].

☐ **Assess project development.**

Regularly assess whether actual project work matches with the scope, timeline, goals, and budget articulated in the project initiation phase. Respond to changes as necessary. Repeat this step throughout the project work stage.

Stay alert to mission creep. Work plans are projections. You may not have the timeline right for some elements. You may have to take longer on one aspect than planned but still have a fixed deadline. What do you do to complete the work in light of change in available time? Figure out how you can respond to unexpected roadblocks.

☐ **Document project work.**

Document work throughout the entire project. With multiple people working on most digital humanities projects, turnover happens. Without documentation, a new person coming into a crucial role might have to start fresh. He or she may have to piece together the past and may make wrong assumptions; moreover, it is a waste of time to have to recreate and backtrack. Even without staff turnover, with any digital humanities project, you might step away for a few weeks to work on other responsibilities; upon return, it is hard to remember even a short time later what you did or how you did it.

Later, writing up project results also requires good documentation, and you will need detailed, sound documentation for peer review. Document both the humanities content and that you have used best practices for your field.

☐ **Seek evaluation and peer review.**

At the outset of the research endeavor, members of the project team should discuss requirements and options for evaluation of their scholarship. A host of issues, including status and rank of the scholars, requirements of sponsoring institutions and organizations, team members' ideals, and the project's purpose, may affect decisions about what form and level of evaluation is necessary. While all scholarship benefits from evaluation, for faculty seeking tenure or promotion, evaluation and peer review are especially crucial.

Time-tested methods of peer review remain viable for some forms of digital humanities scholarship: data and results of a data-mining project, for example, may be presented in a peer-reviewed journal article or in an essay collection or monograph from an academic publisher. For other scholarly products, such as websites, databases, and software code, structures for evaluation and peer review in the humanities may be less fixed.[9] Professional organizations and less formal communities of practice can offer valuable information and resources as you consider your options for evaluation.

At the conclusion of the project, follow up on the evaluation plan.

☐ **Disseminate results.**

Project results may culminate in an article, book, series of blog posts, website, source code, or input and output data, in a combination of these elements, or in completely different forms. Depending on the final forms, strategies for dissemination and promotion may be well established, such as with a book, or require more legwork on the part of the team, such as with project data made available through a data repository.

Beyond getting the results in their final form and making them available, explore possibilities for increasing the findability of your scholarship. Scholars should also work with librarians on project metadata since having good metadata is crucial to findability. Work with librarians at your institution to have websites cataloged and submitted to WorldCat, for example. Similarly, work with your institutional repository and data management librarians, if available. Promote and share project results on Listservs and social media and at conferences, as appropriate. University communications offices can also help to publicize projects in local, national, and international media.

☐ **Perform project closedown procedures.**

Taking the time to formally conclude a project is a crucial final step. Closedown procedures include completing any remaining documentation as well as following through on the data management plan. In addition, the project closedown phase is an opportunity

to thank everyone who has contributed to the project. Even—or especially—for very large research initiatives that might extend over many years, conceptualizing the initiative as a sequence of projects, each of which can be initiated, completed, and closed down, is important to long-term success.

Conclusion

With this annotated checklist, our aim has been to provide practical guidance for developing digital humanities scholarship at all scales. We hope that readers will use this checklist as a foundation to build on. We also hope that subject specialists involved in advising about or creating digital humanities scholarship will draw on other chapters in this collection, which offer a variety of case studies illustrating the development of digital humanities projects within the library and partnerships among a variety of participants. Those chapters might usefully be read as illustrations to many of the key ideas set out in the checklist.

Suggested Readings

Readings on Defining Digital Humanities

For a sampling of pieces that take up defining digital humanities and digital scholarship, see Matthew K. Gold, ed., *Debates in the Digital Humanities* (Minneapolis: University of Minnesota Press, 2012), open-access edition: http://dhdebates.gc.cuny.edu; Clifford A. Lynch, "The 'Digital' Scholarship Disconnect," *EDUCAUSE Review* 49, no. 3 (May/June 2014), https://er.educause.edu/articles/2014/5/the-digital-scholarship-disconnect; Bethany Nowviskie, *Bethany Nowviskie* (blog), accessed July 15, 2014, http://nowviskie.org; Ted Underwood, "You Can't Govern Reception," *The Stone and the Shell* (blog), May 3, 2014, http://tedunderwood.com/2014/05/03/you-cant-govern-reception.

Readings on Doing Digital Humanities

This chapter and its checklist participate in developing discussions about documenting a set of practices for "doing" digital humanities, however one may define digital humanities and whatever form that doing takes. See, for example, Anne Burdick et al., *Digital_Humanities* (Cambridge, MA: MIT Press, 2012), https://direct.mit.edu/books/oa-monograph/5346/

Digital-Humanities, which includes the chapters "Questions and Answers" (with the sections "Digital Humanities Fundamentals" [pp. 122–23], "The Project as Basic Unit" [pp. 124–25], and "Institutions and Pragmatics" [pp. 126–27]) and "Specifications" (with the sections "How to Evaluate Digital Scholarship" [pp. 128–29], "Project-Based Scholarship" [pp. 130–31], "Core Competencies in Processes and Methods" [pp. 132–33],"Learning Outcomes for the Digital Humanities" [p. 134], and "Creating Advocacy" [p. 135]). Similarly, the Fall 2012 issue of *Journal of Digital Humanities* (Daniel J. Cohen and Joan Fragaszy Troyano, eds., *Journal of Digital Humanities* 1, no. 4, http://journalofdigitalhumanities.org/1-4) is dedicated to "Closing the Evaluation Gap." Examples of webpages, blog posts, and other web resources concerned with practices for creating, sustaining, and maintaining digital projects include Association for Computers and the Humanities, "Digital Humanities Questions and Answers," accessed July 15, 2014, http://digitalhumanities.org/answers; LAIRAH (Log Analysis of Internet Resources in the Arts and Humanities), "The LAIRAH Digital Humanities Checklist," University College London website, last updated July 4, 2013, www.ucl.ac.uk/infostudies/research/circah/lairah/features; 4Humanities, "Check IT Out!" accessed July 15, 2014, http://4humanities.org/check-it-out; Paige Morgan, "How to Get a Digital Humanities Project off the Ground," personal website, June 5, 2014, www.paigemorgan.net/how-to-get-a-digital-humanities-project-off-the-ground; and Cheryl Klimaszweski, "Digital Project Checklist," last modified November 23, 2011, https://docs.google.com/document/d/1dnKMwiUR-evkcNQDEbJMMBrhIICVH_J14ku5VsmCKog/edit [site discontinued]. Likewise, Rebecca Frost Davis has created a "Process Checklist for Integrating Digital Humanities Projects in Courses" (September 13, 2012, http://rebeccafrostdavis.wordpress.com/2012/09/13/process-checklist-for-integrating-digital-humanities-projects-into-courses). The resources included here are not intended to comprise a comprehensive list, but to give a sense of what is out there and where the present checklist fits in.

Readings on Collaborating in Digital Humanities

Some readings that deal with collaboration in digital humanities projects include Amy E. Earhart, "Challenging Gaps: Redesigning Collaboration in the Digital Humanities," in *The American Literature Scholar in the* Digital *Age*, ed. Andrew Jewell and Amy E. Earhart (Ann Arbor: University of Michigan Press, 2011), 27–43; Julia Flanders, "Time, Labor, and 'Alternate Careers' in Digital Humanities Knowledge Work," in *Debates in the Digital*

Humanities, ed. Matthew K. Gold (Minneapolis: University of Minnesota Press, 2012), open-access edition: http://dhdebates.gc.cuny.edu; Bethany Nowviskie, "Evaluating Collaborative Digital Scholarship (or, Where Credit Is Due)," *Journal of Digital Humanities* 1, no. 4 (2012), http://journalofdigitalhumanities.org/1-4/evaluating-collaborative-digital-scholarship-by-bethany-nowviskie; Lynne Siemens, "'It's a Team If You Use "Reply All"': An Exploration of Research Teams in Digital Humanities Environments," *Literary and Linguistic Computing* 24, no. 2 (2009): 225–33, doi:10.1093/llc/fqp009; and Lynn Siemens et al., "A Tale of Two Cities: Implications of the Similarities and Differences in Collaborative Approaches within the Digital Libraries and Digital Humanities Communities," *Literary and Linguistic Computing* 26, no. 3 (2011): 335–48, doi:10.1093/llc/fqr028. See also Gawande, *Checklist Manifesto*, 46, 68, 70.

Checklist

- ☐ **Articulate the project's research question and the scholarly conversation being entered.**
- ☐ **Research other projects with similar research aims or methodologies.**
- ☐ **Define the scope of the project.**
- ☐ **Evaluate needed and available resources.**
- ☐ **Identify project participants.**
- ☐ **Develop a communication plan.**
- ☐ **Investigate issues regarding intellectual property.**
- ☐ **Develop a project work plan.**
- ☐ **Develop a data management plan.**
- ☐ **Assess project development.**
- ☐ **Document project work.**
- ☐ **Seek evaluation and peer review.**
- ☐ **Disseminate results.**
- ☐ **Perform project closedown procedures.**

Notes

1. Atul Gawande, *The Checklist Manifesto: How to Get Things Right* (New York: Metropolitan Books, 2010), 86–87.

2. Ibid., 79.
3. Ibid.
4. University of Nebraska–Lincoln, "About UNL," accessed July 15, 2015, www.unl.edu/aboutunl; University of Nebraska–Lincoln, "Biggest Freshman Class in 35 Years Pushes UNL's Enrollment Past 25,000," news release, September 8, 2014, https://news.unl.edu/newsrooms/today/article/biggest-freshman-class-in-35-years-pushes-enrollment-past-25k/; University of Nebraska–Lincoln, "The Big Ten Conference, CIC and APLU," accessed July 15, 2014, www.unl.edu/aboutunl/big10.shtml.
5. Amanda Gailey and Andrew Jewell, eds., *Scholarly Editing: The Annual of the Association for Documentary Editing*, accessed July 15, 2014, www.scholarlyediting.org; *Vectors Journal: Introduction,* accessed July 15, 2014, http://vectors.usc.edu/journal/index.php?page=Introduction.
6. Clayton McCarl, "Introduction to '*Avisos a pretendientes para Indias*,'" *Scholarly Editing: The Annual of the Association for Documentary Editing* 35 (2014), www.scholarlyediting.org/2014/editions/intro.avisos.html.
7. Emily Thompson, "Author's Statement," The Roaring 'Twenties: An Interactive Exploration of the Historical Soundscape of New York City, *Vectors* 4, no. 1 (Fall 2013), http://vectors.usc.edu/projects/index.php?project=98&thread=AuthorsStatement.
8. Adam Crymble and Julia Flanders, "FairCite," *Digital Humanities Quarterly* 7, no. 2 (2013), www.digitalhumanities.org/dhq/vol/7/2/000164/000164.html; Anne Burdick, Johanna Drucker, Peter Lunenfeld, Todd Presner, and Jeffrey Schnapp, *Digital_Humanities* (Cambridge, MA: MIT Press, 2012), 15, https://direct.mit.edu/books/oa-monograph/5346/Digital-Humanities.
9. Daniel J. Cohen and Joan Fragaszy Troyano, eds., *Journal of Digital Humanities* 1, no. 4 (Fall 2012), http://journalofdigitalhumanities.org/1-4.

Bibliography

ACH (Association for Computers and the Humanities). "Digital Humanities Questions and Answers." Accessed July 15, 2014. http://digitalhumanities.org/answers.

Bernholz, Charles. *American Indian Treaties Portal.* University of Nebraska–Lincoln website. Accessed July 21, 2014. http://treatiesportal.unl.edu.

Burdick, Anne, Johanna Drucker, Peter Lunenfeld, Todd Presner, and Jeffrey Schnapp. *Digital_Humanities.* Cambridge, MA: MIT Press, 2012. https://direct.mit.edu/books/oa-monograph/5346/Digital-Humanities.

Cohen, Daniel J., and Joan Fragaszy Troyano, eds. *Journal of Digital Humanities* 1, no. 4 (Fall 2012). http://journalofdigitalhumanities.org/1-4.

Crymble, Adam, and Julia Flanders. "FairCite." *Digital Humanities Quarterly* 7, no. 2 (2013). www.digitalhumanities.org/dhq/vol/7/2/000164/000164.html.

Davis, Rebecca Frost. "Process Checklist for Integrating Digital Humanities Projects into Courses." September 13, 2012. http://rebeccafrostdavis.wordpress.com/2012/09/13/process-checklist-for-integrating-digital-humanities-projects-into-courses.

Earhart, Amy E. "Challenging Gaps: Redesigning Collaboration in the Digital Humanities." In *The American Literature Scholar in the Digital Age*, edited by Andrew Jewell and Amy E. Earhart, 27–43. Ann Arbor: University of Michigan Press, 2011.

Flanders, Julia. "Time, Labor, and 'Alternate Careers' in Digital Humanities Knowledge Work." In *Debates in the Digital Humanities*, edited by Matthew K. Gold. Minneapolis: University of Minnesota Press, 2012. Open-access edition: http://dhdebates.gc.cuny.edu.

4Humanities. "Check IT Out!" 4Humanities website. Accessed July 15, 2014. http://4humanities.org/check-it-out.

Gailey, Amanda, and Andrew Jewell, eds. *Scholarly Editing: The Annual of the Association for Documentary Editing*. Accessed July 15, 2014. www.scholarlyediting.org.

Gawande, Atul. *The Checklist Manifesto: How to Get Things Right.* New York: Metropolitan Books, 2010.

Gold, Matthew K., ed. *Debates in the Digital Humanities.* Minneapolis: University of Minnesota Press, 2012. Open-access edition: http://dhdebates.gc.cuny.edu.

Klimaszweski, Cheryl. "Digital Project Checklist." Last modified November 23, 2011. https://docs.google.com/document/d/1dnKMwiUR-evkcNQDEbJMMBrhIICVH_J14ku5VsmCKog/edit [site discontinued].

LAIRAH (Log Analysis of Internet Resources in the Arts and Humanities). "The LAIRAH Digital Humanities Checklist." University College London website. Last updated July 4, 2013. www.ucl.ac.uk/infostudies/research/circah/lairah/features.

Logan-Peters, Kay. "About the Project." *An Architectural Tour of Historic UNL website.* Accessed July 21, 2014. https://historicbuildings.unl.edu/about.html?csrt=16247912834438985941.

Lynch, Clifford A. "The 'Digital' Scholarship Disconnect." *EDUCAUSE Review* 49, no. 3 (May/June 2014). https://er.educause.edu/articles/2014/5/the-digital-scholarship-disconnect.

McCarl, Clayton. "Introduction to '*Avisos a pretendientes para Indias*.'" *Scholarly Editing: The Annual of the Association for Documentary Editing* 35 (2014). www.scholarlyediting.org/2014/editions/intro.avisos.html.

Morgan, Paige. "How to Get a Digital Humanities Project off the Ground." Personal website. June 5, 2014. www.paigemorgan.net/how-to-get-a-digital-humanities-project-off-the-ground.

NEH (National Endowment for the Humanities). "Digital Humanities Start-up Grants." Accessed July 15, 2014. www.neh.gov/grants/odh/digital-humanities-start-grants [site discontinued].

Nowviskie, Bethany. *Bethany Nowviskie* (blog). Accessed July 15, 2014. http://nowviskie.org.

———. "Evaluating Collaborative Digital Scholarship (or, Where Credit Is Due)." *Journal of Digital Humanities* 1, no. 4 (2012). http://journalofdigitalhumanities.org/1-4/evaluating-collaborative-digital-scholarship-by-bethany-nowviskie.

Siemens, Lynne. "'It's a Team If You Use "Reply All"': An Exploration of Research Teams in Digital Humanities Environments." *Literary and Linguistic Computing* 24, no. 2 (2009): 225–33. doi:10.1093/llc/fqp009.

Siemens, Lynne, Richard Cunningham, Wendy Duff, and Claire Warwick. "A Tale of Two Cities: Implications of the Similarities and Differences in Collaborative Approaches within the Digital Libraries and Digital Humanities Communities." *Literary and Linguistic Computing* 26, no. 3 (2011): 335–48. doi:10.1093/llc/fqr028.

Thompson, Emily. "Author's Statement." The Roaring 'Twenties: An Interactive Exploration of the Historical Soundscape of New York City. *Vectors* 4, no. 1 (Fall 2013). http://vectors.usc.edu/projects/index.php?project=98&thread=AuthorsStatement.

Underwood, Ted. "You Can't Govern Reception." *The Stone and the Shell* (blog), May 3, 2014. http://tedunderwood.com/2014/05/03/you-cant-govern-reception.

UNL (University of Nebraska–Lincoln). "About UNL." Accessed July 15, 2014. www.unl.edu/aboutunl.

———. "Biggest Freshman Class in 35 Years Pushes UNL's Enrollment Past 25,000." News release, September 8, 2014. https://news.unl.edu/newsrooms/today/article/biggest-freshman-class-in-35-years-pushes-enrollment-past-25k/.

———. "The Big Ten Conference, CIC and APLU." Accessed July 15, 2014. www.unl.edu/aboutunl/big10.shtml.

Vectors Journal. Introduction. Accessed July 15, 2014. http://vectors.usc.edu/journal/index.php?page=Introduction.

CHAPTER TEN

Moderating a Meaningful DH Conversation for Graduate Students in the Humanities

Kathleen A. Langan and Ilse Schweitzer VanDonkelaar

Introduction

The nature of academic librarianship traditionally calls for librarians to serve as subject liaisons. To fulfill such a role, it is common for academic librarians to hold an additional higher degree beyond the MLS. According to a recent study, "13% are doctoral degree holders; 47% have a second masters' degree."[1] This educational background allows librarians to be professionally ambidextrous to institutions of higher education (IOHE) in ways that one does not expect from other faculty on campus. Academic librarians work fluidly across fields, departments, and units as specialists to support the academic community in a variety of capacities, such as providing research consultation, teaching research methodology, and assisting in course development. The ability to navigate between these academic spheres proves an invaluable advantage when accommodating an inherently interdisciplinary field of study like the digital humanities (DH), which relies heavily on cross-campus cooperation. Not only do subject librarians have the necessary content knowledge, they also have an additional kick of technical aptitude that positions them to mediate a DH conversation and to facilitate collaboration in otherwise disparate DH efforts across campus.

Because of its nebulous nature, DH is a nonhierarchical, integrative discipline. It is often hard to identify who might be best suited as campus liaison to a DH initiative. Some institutions are fortunate to have an appointed DH lead or team. For those institutions that do not have a formally identified

point person—as is the case at Western Michigan University (WMU)—the responsibility often falls to a librarian who acts as liaison for humanities-related subjects. As a result, these librarians are charged with teaching DH skills, pedagogy, and methodology, not only to faculty but also, and arguably more important, to graduate students. This chapter presents the importance of the subject librarian's role in developing a purposeful DH initiative devoted to the professional preparation of graduate students. It describes a case study at WMU, where momentum for a centralized DH initiative found its source and purpose in a cohort of vocal graduate students who turned to subject librarians for guidance. This unexpected collaboration culminated in much needed and realistic DH learning opportunities for graduate students, developed by subject librarians.

Situating the Conversation

WMU is a mid-sized, midwestern, doctorate-granting public university with a full-time enrollment of approximately 21,000 undergraduate and 5,000 graduate students. WMU is a tier 1 research institute with over 900 full-time board-appointed faculty. Even though individual faculty members are involved with DH projects, there is no current centralized digital humanities initiative, nor is there any formal DH curriculum.

Unlikely Initiators of the DH Conversation at WMU

In the spring of 2013, doctoral students from WMU's English department approached subject librarians to learn more about digital humanities with the strong concern that they lacked any formal exposure to or training in DH. They worried that the apparent gap in formal DH training or applied knowledge would hinder their success when applying and interviewing for academic faculty positions. DH was fast becoming a prominent skill set found in a high percentage of the faculty job descriptions posted on the Modern Language Association (MLA) Job Information List (JIL). In turning to subject librarians, they sought answers to questions such as these. How could they become more conversant in the broader academic discussion regarding DH? How could they become aware of technical, philosophical, and pedagogical issues or identify key players involved in DH initiatives? While these concerns were immediate and personal for the students involved, their questions set in motion a broader collaboration that extended across disciplinary lines at WMU and resulted in a plan to formalize the teaching

of DH at the graduate level. The question about the unstable job market also opens up the discussion of time to degree, incurred debt, and, ultimately, attrition rates. What ethical responsibility does the institution hold for these students? If students feel they are at a disadvantage when entering the job market, how do they justify pursuing a higher degree? How can different parts of IOHE help?

The National Conversation on the State of Graduate Studies in the Humanities

The concerns of WMU's graduate students are realistic, validated by the national conversation regarding the current state of graduate education in the humanities, a conversation that is happening among various constituents such as the MLA and the American Historical Association (AHA), which are framing the current state of graduate education around the future of their fields and professions. For the better part of the 2000s, potential jobs in the humanities greatly outnumbered degrees earned. In the 2008–09 cycle, however, the ratio reversed, with a significant decrease in jobs offered, a trend that continues today.[2] In the 2012–13 cycle, the number of tenure-track positions advertised in the MLA JIL fell by roughly 5 percent.[3]

Not only is the field getting smaller, the nature of the field is changing, requiring more technological expertise, as is evident in job descriptions for academic positions. In his essay "Digital Humanities and the 'Ugly Stepchildren' of American Higher Education," Luke Waltzer describes the downturn in the humanities job market and the concurrent flourishing of DH and alternative-academic (alt-ac) positions and initiatives in IOHE as well as in governmental and private funding bodies, such as the National Endowment for the Humanities' (NEH) Office of the Digital Humanities, HASTAC (Humanities, Arts, Science, and Technology Alliance and Collaboratory), Google, and the Andrew Mellon Foundation, to name a few.[4] This increase in industry support for technological approaches to humanities research and teaching is likewise reflected in academic job ads. According to the MLA Office of Research, "ads tagged with the index term 'technology and digital media' represented 19.0% of ads in the English edition (up from 7.7% in 2003–04, when this option first appeared) and 10.2% of ads in the foreign language edition (up from 5.9% in 2003–04)."[5] A simple job search on the 2013 MLA JIL with "digital humanities" in the description yielded 19 (of 259 total) results. Of these positions, several exclusively sought DH experts and included the following titles: digital humanities design consultant, director

of digital studies center, assistant professor in DH, postdoc in DH, assistant professor in emerging media. Multiple other job descriptions seeking candidates for professorships in literature and the environment, Shakespeare studies, eighteenth-century and Victorian literature, and rhetoric and communication all listed DH as a desirable subspecialty. Academic jobs now require expertise in multiple areas, alongside teaching experience and an expectation that candidates will already be published in their fields. Concurrently, many graduate students are aware that the likelihood of securing a tenure-track position is low enough that they must also market themselves to the growing nonacademic and alt-ac employers, which are recruiting students with hybrid expertise in technology and humanities.

The recent push to prepare students for alt-ac positions emerges in the midst of a conventional and pervasive academic elitism that devalues non-tenure-track faculty positions. Historically, the sentiment among academics has been that students compromise or settle for something less when they choose an alt-ac job, as these positions may be viewed as less than desirable and less prestigious than tenure-track positions. This is a demoralizing outcome for humanities graduates who find themselves without faculty positions and potentially disconnected from their departments and cohorts and who may carry feelings of being undervalued second-class citizens of academia if this prevailing mindset does not change.

By offering recommendations and best practices to universities and departments, professional associations such as the MLA and AHA proactively acknowledge the need to responsibly prepare graduate students with a meaningful and applicable education. In a 2012 interview, then MLA president Michael Bérubé is paraphrased as saying, "curriculums …should be redesigned to emphasize collaboration, but the question will be how it's valued by future employers, and by institutions themselves."[6] Bérubé is quoted as saying collaboration "runs up against the barriers of the institutional reward system," but also that "'interdisciplinarity' will play a crucial role in reforming graduate education in the humanities, in part because it will prepare graduates for a greater array of employment, both inside and outside academe."[7] Though DH is not explicitly mentioned as one of those collaborative initiatives, it is definitely understood. In fact, this collaborative, interdisciplinary approach could be seen as a clear training path toward equalizing alt-ac careers. According to Vimal Patel, Russell Berman, who led the MLA task force on higher education said, "Departments should be more clear with students from the start that tenure-track jobs are becoming harder to find …and should also explain to students what else they could

do with a language or literature Ph.D. Career options off the tenure track."[8] Berman is also quoted as saying, "the subject matter may, in fact, be far from literature …but the rich professional formation acquired during the course of doctoral study can be put to good use."[9] Students won't necessarily need to look outside of their field if we or they broaden the definition of the field itself.

This trend is happening in other humanities-related disciplines where job offers are falling and attrition is high. In 2011, the president and the executive director of the AHA made similar pleas for change in graduate education. They stated that the AHA needs "to examine the training we offer, and work out how to preserve its best traditional qualities while adding new options. If we tell new students that a history PhD opens many doors, we need to broaden the curriculum to ensure that we're telling the truth…. There's the whole exploding realm of digital history and humanities, and the range of skills required to practice them."[10] Moreover, as Waltzer points out, redefining graduate education in the humanities to include training in digital tools and creation is specifically suited for and already being undertaken by students: "Graduate students and junior scholars are more confidently embracing what digital tools can mean for their work and are more likely than their predecessors to imagine a career path that revolves around their identities as digital humanists."[11]

The MLA has called for academic programs to embrace technology when envisioning a new graduate curriculum. In Flaherty's summary of the MLA report, she highlights that the report calls for "more technological training, and says students should be encouraged to test and develop new tools and techniques for the study of literature and languages."[12] The MLA executive council also recognizes the benefit in developing interdisciplinary connections with others on campus.

However, in a June 2014 response to the MLA report, ten humanities scholars pointed out that the new career training suggested by the report "places increased burdens on graduate program faculty (directors, in particular). The report somehow expects faculty to provide training for students in areas where faculty themselves may not be adequately trained" including, as the authors of this response point out, "significant training in new digital methodologies."[13] Library faculty, particularly those fluent in digital research methods, seem to be the ideal candidates to supplement traditional humanities training. In the article "Who Prepares Humanities Ph.D.'s for a Nonacademic Search?" Wood and Gurwitz enumerate several potential campus members to train doctoral humanities students, including advisors,

the department itself, and campus career services.[14] Surprisingly, nonacademic unit services such as libraries were noticeably absent from the article, odd because academic librarians represent very successful role models for nontraditional academic career paths.

WMU's Annual Enrollment and Degrees Conferred

WMU offers three doctoral programs traditionally defined under the humanities: English, Spanish, and comparative religion, and one interdisciplinary doctoral program in history. Add to this the average head count in traditional humanities-focused master's programs in communication, English, Spanish, comparative religion, and philosophy and the numbers rise quickly. There are also two interdisciplinary master's programs in history and medieval studies.

There are, on average, 115 students annually completing a higher degree in a humanities-related program from WMU and entering the workforce. This number justifies the need for institutional responsibility to provide viable and relevant professional development to graduate students. Attrition is a real concern for departments, and students leave programs for many reasons. In 2013, a national survey of graduate students in history revealed that "students did not feel they had been adequately prepared for the nonacademic job market by either their departments or their universities."[15]

The Emerging DH Conversation at WMU

Turning to their own faculty in English, Spanish, or history to learn about DH would have seemed the obvious choice for WMU graduate students, but the students recognized that librarians have obvious technical advantages and an institutional familiarity and flexibility necessary to start a successful conversation. Subject librarians developed a series of events to educate graduate students broadly on DH and to bring together faculty members and students working at various levels of familiarity with DH. Each activity or event was born from the previous one; the initiative was very organic in nature. Response to and attendance at all events were overall positive and encouraging, validating a much needed conversation.

Event One

The first event was a one-shot session folded into a spring 2013 Graduate Student Research Fair, planned and sponsored by University Libraries, in which graduate students were introduced to the library's digital resources.

This session was a small DH discussion group, organized by our Humanities Librarian and Head of Special Collections, envisioned as a "meat and potatoes" introduction to key terms in DH, as well as an exploratory talk to determine how much interest graduate students had in digital studies. From this initial plea, students expressed a need for more in-depth instruction.

Event Two

To answer this request, librarians put together a summer seminar series on DH. Not limited to graduate students, this summer series had a goal to garner exposure and publicity for extant DH projects and resources and to begin a broader conversation with faculty, students, staff, and administrators interested in DH. In the four-session series, invited speakers included faculty and staff from subject departments and technical units, such as the libraries' digitization center, and covered technical aspects of DH or content-driven projects. In the course of this summer series, it became apparent that each of the faculty presenters had worked independently and with relatively little institutional support to develop his or her DH project. Attendees, including the Director of the University Center for the Humanities, suggested that we should continue this conversation in a more public venue to raise awareness of our faculty's current work in DH as well as to determine how the university might better support interdepartmental DH ventures.

Event Three

Interest in DH emerged from across campus, and a third wave of events was held the following semester for administrators, faculty, and students in the form of a working group sponsored by the WMU Center for the Humanities. For this interdisciplinary working group, the goal was to identify expertise and infrastructure to support a centralized DH initiative, keeping graduate students' needs central to the mission. More important, the subject librarian worked in close collaboration with a graduate student from the English department to plan each group meeting's agenda and to move this initiative forward.

The interdisciplinary working group in fall 2013 was funded by the WMU University Center for the Humanities and was open to the entire community with a guided conversation in hopes of answering one key question: Is there a need or a desire for a centralized DH initiative at WMU? Because the conversation was allowed to develop organically, the focus moved away from this initial question and toward "How can we best serve our students

and faculty, given the resources we have?" Each of the four monthly meetings addressed a different topic of DH at WMU:

1. Identifying and showcasing extant projects and experts. Can we build a knowledge base from their experiences?
2. Navigating the digital diaspora. Identifying technical and financial resources and ways to collaborate.
3. Implications for graduate students. Institutional responsibility for professional development for future leaders in the field.
4. How do we use this new information? What projects or outcomes are feasible?

Over the course of this multi-month discussion, participants kept circling back to the same issues: a need to determine what resources for building and maintaining DH projects were already available to the WMU community and where these resources could be found. One participant in the first working group meeting noted that the very discussion demonstrated a need to have not only a central person who would deal with all DH-related requests on campus, but also a central process or a way of communicating so that faculty, staff, and students would know who's doing what and where and with whom (as relates to digital projects), and what resources they might need. At the conclusion of the first meeting, participants agreed to work together to create a Google document in which the working group could compose a mission statement for the digital humanities at WMU, using other centers and initiatives as exemplars. At the second meeting, "Navigating the Digital Diaspora," as the conversation increasingly turned to the idea of locating a DH center on campus, one administrator observed that there was too much concern about where our initiative would be housed, when the group should be focusing on getting people together and talking to each other. The key necessity at this point was to streamline the process of organizing DH projects and getting those projects underway, and the "center" would follow. Another participant suggested a new way to understand the concept of a "DH center"—that is, to regard the "center" as a point person (or people) and to establish an advisory board to lead and coordinate tutorials, workshops, and mentoring on DH projects. These points of conversation evolved into larger goals of constructing a workflow document that would streamline the process of locating support, funding, and resources at WMU for faculty pursuing DH research, as well as developing a means of surveying or assessing what kinds of independent digital projects were already underway on campus.

Attendance at each working group meeting varied depending on the topic, though there was a good representation of different constituencies (faculty, administrators, staff, and students) at all meetings. At the end of the semester, it was decided to continue to meet bimonthly with a core group of volunteers during the spring of 2014. Goals included completing projects suggested during fall meetings, including the workflow document, as well as planning a THATCamp (The Humanities and Technology Camp) for spring 2014, and a future graduate-level course in digital humanities.

Outcomes Born from the Conversation

There have been many tangible and intangible outcomes as a direct result of this year-long conversation, including short-, medium-, and long-term.

Short-term outcomes include developing a self-help digital workflow guide, available on WMU Libraries online research guides for faculty, staff, and students. Tracing the process of a typical humanities project from its genesis to publication, the workflow guide prompts the user to conceptualize the design and requirements of a DH project in terms of purpose, audience, sustainability, technical and financial requirements, upkeep and updating, copyright requirements, and necessary collaborators. After posing these questions, the workflow guide suggests resources available at WMU and beyond to support the researcher as the project moves forward. It includes contact information and information regarding considerations to anticipate during the development and creating of a DH project. The document also connects people, offices, and resources for teaching with established DH projects. This is a living document and will require continuous updating as resources change.

Medium-term outcomes include the development and hosting of a THATCamp at WMU in May 2014. WMU is fortunate to be the host of the annual world-renowned International Congress on Medieval Studies (ICMS). Each year, thousands of medievalists (ranging from undergraduates to emeriti to independent scholars) from around the world descend on Kalamazoo, Michigan, from Wednesday through Sunday in early May to share research and to network. This year, the authors attached a THATCamp to the ICMS with the hope of continuing it annually. For this first venture, we invited a keynote speaker from WMU to discuss game theory in the context of teaching and researching in the humanities. The camp was advertised to the traditional ICMS clientele as well as WMU faculty, staff, and graduate students, using primarily social media outlets and university

public relations venues. In total, there were about twenty campers, including a number of medievalists from around the country, several subject librarians among them. Half of its attendees were graduate students from various programs at WMU. Largely an experiment, this was regarded as an opportunity to host interested and curious parties from on and off campus to talk informally about different aspects of teaching humanities and technology.

Long-term goals include developing courses for a more formalized DH education, including introductory graduate seminars with a potential for more advanced courses in the future. Teaching is the natural conclusion to better preparing students for the job market, whether they pursue traditional faculty jobs or alt-ac careers. An ever-increasing number of individual DH-related courses and academic programs in DH are appearing across different institutions both nationally and internationally. However, it is difficult to get such efforts started, as some faculty would prefer to dedicate their time to their own research instead of teaching or developing new coursework, even though they recognize the importance of teaching.[16] With little support for their own DH research projects, it is not surprising that few faculty members are qualified or willing to guide graduate students through the new DH quagmire. With faculty perhaps too encumbered with the hassles of developing DH projects to commit to teaching introductory seminars in DH, subject librarians seem a more efficient choice to lead DH initiatives on campus, as they are not as ensconced in one discipline and constantly engage with organizational resources and technology-based infrastructure. The interdisciplinary nature of academic librarianship puts subject librarians in an ideal position to foster collaboration among key players on campus, including graduate students. Yet it may be hard to drum up support to pull librarians from traditional duties and allow them to teach a DH graduate course: Warwick calls it a "circular problem."[17] In paraphrasing Terras, Warwick agrees that "the lack of teaching programmes may be partially due to the fact that it has taken some time for digital humanities to be accepted as a legitimate discipline."[18]

Other broader long-term goals include the possibility of creating an undergraduate minor and a graduate certificate, programs that would take advantage of existing courses in the WMU catalog. We will also explore marketing DH courses and programs to students from non-humanities departments, such as graphic design, geographic information systems (GIS), and computer sciences.

Developing an Academic Presence at the Graduate Level

The most important development of this collaboration has been creating a long term-goal and formalizing graduate instruction in DH with the creation of a for-credit DH seminar for graduate students. Though developed for the immediate and practical benefit of students, this seminar will also (hopefully) bring visibility and relevance to the DH research being performed independently by WMU faculty; as Warwick has it, "teaching is important to digital humanities in many ways: it helps to train the next generation of digital humanities scholars and practitioners, of course, but it also gives the subject a sense of stability in institutional terms."[19] At WMU, graduate students were very vocal in their support of formal instruction. In response, our intention is to offer this one seminar as a test run and then submit an application to the graduate curriculum faculty senate committee to add it to the course catalog. In the process of developing a digital humanities course, however, institutional administrative barriers due to accreditation, funding, and other factors hindered the creation of a truly interdisciplinary initiative despite broad interest across departments, units, and colleges. Before any content, syllabus, or pedagogical approach could be developed, several administrative necessities needed to be addressed.

The first issue in developing a graduate course in DH was finding a philosophically appropriate academic home. Since the libraries are not an academic unit, they are unable to house a course. A DH course attached to one particular academic department contradicted the interdisciplinary spirit of the field. It was decided that the WMU Graduate College, the overarching unit providing support and advocacy for graduate students (not to mention granting their degrees), would be the best academic home for this course. The next hurdle was finding an instructor of record who met the Graduate College's rigorous standards for instruction. For all the suitability and strengths of the subject librarian to serve as a DH point person on campus, administrative issues emerge when the librarian is called upon to teach. At WMU, for example, since University Libraries are not an academic unit, they cannot nominate someone to teach at the Graduate College; the nomination has to be from a doctoral-granting department. (The WMU English department was willing to nominate the librarian as the instructor of record.) Though this issue has been resolved, other issues emerge for the librarian who does not serve as instructor on a regular basis: Does the librarian receive

release time and fold this course in with her other normal duties, or does she teach this course as an overload? Another unresolved issue is to decide between a face-to-face course, an online courses, or a hybrid course. Choosing e-learning, while eminently suitable to a course on digital tools, methods, and texts, brings on other sets of issues, as those courses are set up and paid for by a different nonacademic unit at WMU. Ultimately, answers to these questions will come largely from the college that ends up housing this course; the librarian as instructor must work with administrators to map out this new instructional territory.

Intangible Outcomes

If anything, being involved in the planning process proves to the graduate students that one does not need to be so leery of DH, that the field is neither impenetrable nor mysterious. Handing over responsibilities to the graduate students as initiators of a campus-wide conversation as well as directly involving them in organizing events that require pulling people in from all levels of the university—upper administration to faculty to students to staff—is invaluable. The exposure to the policies and procedures of these sorts of events have the unintended benefit of showing graduate students what will be expected of them for the service portion of the tenure process. So much emphasis is placed on teaching and scholarship, yet it is events like this that move universities forward, foster collaboration, and initiate change.

The graduate students who attended the aforementioned DH events were exposed to upper-level administrative discussions that revolved around campus planning, academic units, and how the greater academic entity functions. As part and parcel of this, graduate students (especially those who were involved from the beginning of this venture) were also exposed to the difficulties of mobilizing disparate offices and individuals across a university to move toward a common goal. Chief among these challenges was the issue of funding, as again and again we were told that, while offices and administrators were (in their own words), philosophically, intellectually, and spiritually supportive of our goals, the realities of an ever-shrinking university budget made establishing a "Center for DH" a near-impossibility. Additional challenges became obvious when faculty and librarians talked about difficulties they had experienced when trying to get all of their ducks in a row to apply for funding for DH projects. Finally, graduate students witnessed a certain level of cooperation fatigue when, after several meetings, attendance began to lag and the working group self-selected itself down to a core group of truly

motivated members. With this reduction in participation, and with more responsibilities for planning, organizing, and developing content falling to the subject librarian and a graduate student from the English Department, it became apparent the degree to which these kinds of ventures require a constellation of elements to keep themselves afloat: dynamic and innovative leaders, continued funding, interested and involved stakeholders from across the campus, a clear vision for future work and goals, and members willing to volunteer time and effort to contribute to group projects.

As the year waned, the organizers saw a drop-off in a number of these elements (not to mention experiencing their own fatigue at having to manage this group and its projects on top of other full-time job requirements). The challenge of seeing the various short- and long-term goals through to completion at the end of the academic year made clear the requirement for both the energy (and job-search-related anxiety) of graduate students to push us forward, as well as the knowledge of how to navigate convoluted university systems and communications that comes with experience in faculty or library roles.

Recommendations

As we move forward with our plans for the graduate-level DH course and continue the discussion of the place of DH at WMU, we offer this advice to universities, librarians, and students who find themselves in positions similar to ours:

Think interdisciplinarily. If one unit, office, or department says no, look around you. There is a solution (or a home!) somewhere. As subject librarians, realize that institutional and administrative barriers may prevent you from developing or teaching a course. You may need to work closely with colleagues in other departments or units to plead your case.

Clarify professional duties. It is worth mentioning that there was institutional misunderstanding of the definition of information literacy. Administrators questioned why a subject librarian would want to teach a course in digital humanities because it is not true information literacy. Even though other teaching librarians understood the impact of teaching a semester-long graduate course on digital humanities, subject librarians may find that they need to justify themselves. In reality, teaching a semester-long graduate course in digital humanities to support graduate students in being better prepared is as important, if not more, than teaching one-shot information literacy to first-year writing students. It is truly demand-driven service.

There are tangible outcomes that can be assessed to measure programmatic effectiveness. It could potentially attract more graduate students, thereby generating more revenue for departments or units. Finally, it increases the visibility of the university libraries as providing relevant academic support as well as the academic value of subject librarians.

Time. This process has thus far taken fifteen months, with another academic year projected to work through the various administrative hurdles to launch either new programs or courses. It took a lot of public events and discussions to identify what our unique institutional need actually was. Rather than going into the conversation with an already-made decision, it is recommended that you listen to all participants. It is surprising to see who emerges as the most organized in vision and motivation—in this case, the graduate students. The audience will, of course, vary from institution to institution.

The New Academic Model

The new reality for the humanities will likely see more complex models of collaboration; as Reid has it, "the default mode for humanities academic labor has been for a professor to work independently.… As the humanities shift into a digital assemblage, however, these practices are changing, and this is already apparent in digital humanities fields, where research indicates a growing amount of collaboration."[20] Reid is partly correct, yet this observation leaves out the librarians who work with faculty to produce this research. As the humanities take on more digital tools and forms, the scholar/subject librarian relationship, which previously produced isolated instances of collaboration for independent scholarship and research, will shift to more multiphasic, team-based initiatives, setting many other changes in motion: changes in job responsibilities, graduate education, and subject librarian responsibilities. As we add graduate students to this equation, more complex research projects provide moments when students can be trained collaboratively by subject librarians and subject faculty, working as part of a larger team with more diverse expertise. These collaborations also provide recruitment and mentorship opportunities for librarians as they work together with graduate students in a newly emerging academic model where technology, humanities, research, and teaching intersect. This intersection, DH librarianship, is fast becoming a hybrid discipline of its own, defusing the ac/alt-ac debate.

Adding a technological component to the study of the humanities has created a philosophical schism between the old and new guard; unfortunately, the "digital" *can* drive people, resources, and ideas apart. And yet, with this ideological change, job expectations are evolving to a point where new scholars and freshly minted PhDs are expected to be conversant in the new philosophies and methodologies of digital research, thus rectifying the schism and uniting old and new approaches to pedagogy and research. This is quite a lot to expect of graduate students, and certainly necessitates re-envisioning humanities graduate education. In fact, the new vision of graduate education modernizes and renews the role and function of the subject librarian in higher education, whose traditional role on campus is threatened, because of other reasons, by obsolescence. By taking on the responsibility of teaching DH, subject librarians are teaching to a new vision of information literacy, that of multiliteracy, which encompasses digital and media literacy, and serving as academic and professional mentors to graduate students who might not have previously seen them in that light.

IOHE hold an ethical responsibility to graduate students and their professional development. Following the examples set forth by the MLA and AHA, the Association of College and Research Libraries (ACRL) is well positioned to provide organizational support to subject librarians as integrative to graduate education in the humanities and for preparing graduate students to be viable in the field of digital academia. By developing DH-related seminars, working groups, and courses, subject librarians foster the otherwise orphaned academic discipline of DH and are pivotal in uniting disparate DH efforts scattered across departments, people, and pedagogies. At WMU, subject librarians provided more than just basic instruction on DH theories and applications for graduate students. They established a working coalition and institutionalized the teaching of DH, ensuring that the "digital" becomes a part of the WMU humanities legacy.

Notes

1. Shin Freedman, "Questions about Academic Librarians: Factors Influencing Our Academic Identity" (presentation, Charleston Conference: Issues in Book and Serial Acquisition, Charleston, SC, November 6–9, 2013), 320, doi:10.5703/1288284315280.
2. Modern Language Association Office of Research, *Report on the MLA* Job Information List, *2012–13* (New York: Modern Language Association, October 2013), www.mla.org/pdf/rptjil12_13web.pdf; David Laurence, "Our PhD Employment Problem, Part 1," figure 1, *The Trend* (blog), February 26, 2014, http://mlaresearch.commons.mla.org/2014/02/26/our-phd-employment-problem.

3. Luke Waltzer, "Digital Humanities and the 'Ugly Stepchildren' of American Higher Education," in *Debates in the Digital Humanities*, ed. Matthew K. Gold (Minneapolis: University of Minnesota Press, 2012), 335–37.
4. Ibid.
5. MLA Office of Research, *Report on the MLA* Job Information List, 3.
6. Michael Bérubé, paraphrased in Colleen Flaherty, "Fixing Humanities Grad Programs," *Inside Higher Ed*, December 7, 2012, para. 8, www.insidehighered.com/news/2012/12/07/mla-president-says-reforming-graduate-education-humanities-requires-hard-decisions#ixzz33E2CqU9B.
7. Ibid., para. 8, 11.
8. Russell Burman, quoted in Vimal Patel, "Ph.D. Programs Should Change but Not Shrink, MLA Says," *Chronicle of Higher Education*, May 28, 2014, para 7, http://chronicle.com/article/PhD-Programs-Should-Change/146809.
9. Ibid., para. 8.
10. Anthony T. Grafton and James Grossman, "No More Plan B: A Very Modest Proposal for Graduate Programs in History," *Perspectives on History: A News Magazine of the American Historical Association* 49, no. 7 (October 2011), para. 11, www.historians.org/publications-and-directories/perspectives-on-history/october-2011/no-more-plan-b.
11. Waltzer, "Digital Humanities and the 'Ugly Stepchildren,'" 339.
12. Colleen Flaherty, "5-Year Plan," *Inside Higher Ed*, May 28, 2014, www.insidehighered.com/news/2014/05/28/mla-report-calls-phd-program-reform-including-cutting-time-degree; see also Modern Language Association, *Report on the MLA Task Force on Doctoral Study in Modern Language and Literature* (New York: Modern Language Association, May 2014), www.mla.org/pdf/taskforcedocstudy2014.pdf.
13. 10 Humanities Scholars, "Don't Capitulate. Advocate," *Inside Higher Ed*, June 24, 2014, para. 8, www.insidehighered.com/views/2014/06/24/essay-critiques-mla-report-graduate-education#ixzz35flA96jv.
14. L. Maren Wood and Beatrice Gurwitz, "Who Prepares Humanities Ph.D.'s for a Nonacademic Search?" *Chronicle of Higher Education*, July 15, 2013, para. 9–20, http://chronicle.com/article/Who-Prepares-Humanities/140253.
15. Ibid., para. 8.
16. Claire Warwick, "Institutional Models for Digital Humanities," in *Digital Humanities in Practice*, ed. Claire Warwick, Melissa Terras, and Julianne Nyhan (London: Facet Publishing, 2012), 208–9.
17. Ibid., 208.
18. Ibid.; see also Melissa Terras, "Disciplined: Using Educational Studies to Analyse 'Humanities Computing,'" *Literary and Linguistic Computing* 21, no. 2 (2006): 229–246, doi:10.1093/llc/fql022.
19. Warwick, "Institutional Models," 209.
20. Alexander Reid, "Graduate Education and the Ethics of the Digital Humanities," in *Debates in the Digital Humanities*, ed. Mathew K. Gold (Minneapolis: University of Minnesota Press, 2012), 356.

Bibliography

Flaherty, Colleen. "5-Year Plan." *Inside Higher Ed*, May 28, 2014. www.insidehighered.com/news/2014/05/28/mla-report-calls-phd-program-reform-including-cutting-time-degree.

———. "Fixing Humanities Grad Programs." *Inside Higher Ed*, December 7, 2012. www.insidehighered.com/news/2012/12/07/mla-president-says-reforming-graduate-education-humanities-requires-hard-decisions#ixzz33E2CqU9B.

Freedman, Shin. "Questions about Academic Librarians: Factors Influencing Our Academic Identity." Presentation, Charleston Conference: Issues in Book and Serial Acquisition, Charleston, SC, November 6–9, 2013. doi:10.5703/1288284315280.

Grafton, Anthony T., and James Grossman. "No More Plan B: A Very Modest Proposal for Graduate Programs in History." *Perspectives on History: A News Magazine of the American Historical Association* 49, no. 7 (October 2011). www.historians.org/publications-and-directories/perspectives-on-history/october-2011/no-more-plan-b.

Laurence, David. "Our PhD Employment Problem, Part 1." *The Trend* (blog), February 26, 2014. http://mlaresearch.commons.mla.org/2014/02/26/our-phd-employment-problem.

Modern Language Association. *Report on the MLA Task Force on Doctoral Study in Modern Language and Literature.* New York: Modern Language Association, May 2014. www.mla.org/pdf/taskforcedocstudy2014.pdf.

Modern Language Association Office of Research. *Report on the MLA* Job Information List*, 2012–13*. New York: Modern Language Association, October 2013. www.mla.org/pdf/rptjil12_13web.pdf.

Patel, Vimal. "Ph.D. Programs Should Change but Not Shrink, MLA Says." *Chronicle of Higher Education,* May 28, 2014. http://chronicle.com/article/PhD-Programs-Should-Change/146809.

Reid, Alexander. "Graduate Education and the Ethics of the Digital Humanities." In *Debates in the Digital Humanities*, edited by Mathew K. Gold, 350–67. Minneapolis: University of Minnesota Press, 2012.

10 Humanities Scholars. "Don't Capitulate. Advocate." *Inside Higher Ed*, June 24, 2014. www.insidehighered.com/views/2014/06/24/essay-critiques-mla-report-graduate-education#ixzz35flA96jv.

Terras, Melissa. "Disciplined: Using Educational Studies to Analyse 'Humanities Computing.'" *Literary and Linguistic Computing* 21, no. 2 (2006): 229–246. doi:10.1093/llc/fql022.

Waltzer, Luke. "Digital Humanities and the 'Ugly Stepchildren' of American Higher Education." In *Debates in the Digital Humanities*, edited by Matthew K. Gold, 335–49. Minneapolis: University of Minnesota Press, 2012.

Warwick, Claire. "Institutional Models for Digital Humanities." In *Digital Humanities in Practice,* edited by Claire Warwick, Melissa Terras, and Julianne Nyhan, 193–216. London: Facet Publishing, 2012.

Wood, L. Maren, and Beatrice Gurwitz. "Who Prepares Humanities Ph.D.'s for a Nonacademic Search?" *Chronicle of Higher Education,* July 15, 2013. http://chronicle.com/article/Who-Prepares-Humanities/140253.

CHAPTER ELEVEN

Spaces, Skills, and Synthesis

Anu Vedantham and Dot Porter

Introduction

The term digital humanities is not universally embraced. Faculty, students, and librarians generally greet it with nonplussed expressions. What does it mean? Is there such a thing as "analog humanities"? What is included? What is excluded? Who is a digital humanist? Who is not?[1]

We face similarly confused reactions when librarians venture into new territory—fancy study spaces, makerspaces, embedded librarianship, and so on. When a library opens a loud study space, the campus reacts with "But the space has no books. Why is this space in the library? What makes this space a library anyway?" Our miles of bookcases also receive confusing feedback. Some faculty members state with pride that they never enter the library. Others become incredibly upset when books are moved to offsite storage. As we change our perceptions of physical books, we are forced to confront our definitions of how we manage our academic work.

In contrast, consider trends in industries such as banking or real estate. One expects banks and realtors to have functioning, attractive websites and to use technology to their advantage. We do not use terms such as *digital banking* or *digital real estate*. There are parallels within the university. Faculty and students in the sciences have integrated coding and technology into their work without facing terms such as *digital physics* or *digital biology*. Investing in appropriate technologies is a requirement in all fields, and the humanities are no exception.

In this chapter, we describe three learning spaces at Penn Libraries in terms of planning history, physical layout, technology capabilities, staffing,

CC BY

and programmatic decisions. We describe how support for "digital humanities" (DH) has grown deeper, more varied, better funded, and more effectively coordinated across units.

The work we describe is much greater than the efforts of the two authors. It reflects the dedication and creativity of scores of humanities scholars, librarians, instructional design staff, and administrators. We present our experiences and ideas in the larger context of Penn Libraries, attempting to distill concepts that may resonate in your campus context.

We write this chapter in our own voices. One of us works on the first floor of our main library in a busy area where scores of people pass by her office daily on campus tours. One of us works on the sixth floor of the same building with a city skyline view, a gallery, and quiet surroundings. The spaces we inhabit as individuals color our own academic work as well as the relationships we build within and beyond campus.

University Context

The University of Pennsylvania (Penn) is an Ivy League university in an urban location. Penn has twelve schools and is consistently ranked among the top ten universities in the United States. The university's responsibility-centered budgeting structure contributes to a highly decentralized campus environment. Innovations happen continuously all over campus, but sharing of ideas across units can be a challenge.

The School of Arts and Sciences houses all the humanities disciplines. The school has twenty-seven academic departments and offers fifty-four majors for over 6,300 undergraduate students as well as thirty-three doctoral and ten master's programs for more than 2,100 graduate students.[2]

Penn Libraries is seen as a neutral place that does not favor one discipline over another (we hear the phrase "The library is like Switzerland"). We are recognized as the provider of comfortable, nurturing spaces, and strong technology infrastructure; in particular, our management of the campus-wide Canvas courseware system helps us engage with teaching and learning across campus.

Librarians at Penn have staff ranking, not faculty positions, a situation that reflects and contributes to perceptions of librarians as supporters of research, rather than as full collaborators or independent researchers. In recent years, we have developed staff-led research projects to engage staff with library work and collections and to support them in personally experiencing how it feels to do research. It is a slow but worthwhile process,

and we believe it could strengthen the relationship between librarians and faculty.

DH projects are conducted at Penn in a variety of contexts, within academic departments as well as in the libraries. We offer a few examples. PennSound, supported by the School of Arts and Sciences, records poetry readings and makes those recordings available for download. The Schoenberg Institute for Manuscript Studies (SIMS) focuses on DH projects in the context of medieval manuscript collections. The Penn Humanities Forum has launched a Digital Humanities Forum (DHF),[3] which, in collaboration with Penn Libraries, has held several large symposia each year, bringing to campus internationally known DH leaders. A faculty committee is exploring the DH landscape on campus as part of strategic planning in the School of Arts and Sciences. Library conversations with this committee have helped us articulate our own capabilities, resulting in a crowdsourcing project to catalog library expertise to answer the question "Who to ask for what?"[4]

We structure this chapter in three segments to reflect the title—spaces, skills, and synthesis—beginning with our physical spaces.

Spaces

Libraries are associated primarily with a sense of place. We "go to the library," a physical building that contains resources to help us think, read, find materials, and conduct research. The appearance and functionality of the library can affect our research productivity as well as our metacognition—our perception of our own productivity. Spaces designed to encourage brainstorming and discussion need to look, feel, and function differently from those designed to support deep reading, note taking, writing, or presentation practice.

As documented by the Learning Spaces Collaboratory *Guide*, creative combinations of space and technology can catalyze scholarship.[5] Colocation of library expertise near collaboration spaces can help faculty and students draw connections between traditional library capabilities (collections, subject specialists, bibliographers, catalogers) and newer capabilities (scanning, digitization, data analysis and visualization, new media creation, web/blog design, audio/video editing). Effective space management through provision of reliable technology infrastructure can impart confidence to faculty and students taking on DH projects.

When a library space works well, it becomes a coveted destination.[6] We observe people "voting with their feet." During final exams one semester at the Weigle Information Commons, a young woman looked around and,

finding no open tables, pulled up her chair to a recycling bin and used it as a table for her laptop. She preferred to write her research paper there in the crowded open area rather than move to empty seats one floor above. When a space "feels right" for a particular purpose, people choose it again and again even when it is overflowing. Over 2,000 groups of students on average reserved space in the Weigle Information Commons each month during 2013.[7]

The Importance of Learning Space Design

Think about your local mall. If it included a long row of identical coffee shops, would you still want coffee? How do endless stacks of books affect you? Recently Penn Libraries hosted a qualitative research exhibit created by a graduate nursing class on the topic of stress.[8] Graduate students described their concerns through photographs. One graduate student reflected on the Sisyphean pessimism created by a photo of endless rows of book stacks on our fourth floor. When presented in creative ways, collections can instead inspire optimism and curiosity in learners. Bookstores, for example, choose artistic setups over book stacks, specializing in cozy corners, topical exhibits, and displays. Library spaces must prioritize the sense of place, an atmosphere that invites you to walk around and find a corner for your needs. We describe three spaces in the Penn Libraries that differ in layout, function, staffing, and programming—to support specific types of academic work.

Space Planning History

In 2006, Penn Libraries opened the Weigle Information Commons (WIC) after a five-year planning process with committees representing faculty, students, and staff. A joint funding model with the School of Arts and Sciences created a renewing connection between Penn Libraries and the humanities departments. Formal "program partner" relationships with campus organizations including the writing center, academic support services (including support for students with disabilities), public speaking center, undergraduate research opportunities center, and career services help maintain the momentum built during planning. The space aims to support small-group collaboration by undergraduate students, with a one-stop-shop approach to academic support services.

In 2013, Penn Libraries opened the Kislak Center for Special Collections, Rare Books and Manuscripts after a seven-year planning effort. The large-scale renovation of two floors created spaces for presentations, discussions,

and teaching as well as spaces for storing, cataloging, displaying, and working with library-owned materials. The space aims to encourage broad and creative use of physical materials and to serve as a state-of-the-art gathering place for campus.

In 2014, Penn Libraries opened the Collaborative Classroom after a two-year planning effort that included active collaboration with an undergraduate student group, the Student Committee on Undergraduate Education (SCUE), and visits to other universities. The project complements a campus initiative funded by a 2013 grant from the Association of American Universities (AAU).[9] The space aims to support campus interest in new pedagogical methods including "flipped classroom" instruction.

Facilities, Furniture, and Technology

WEIGLE INFORMATION COMMONS (WIC)

The WIC has 6,600 square feet of space on the first floor of the main library building. It includes twelve Data Diner Booths that look and function just like their restaurant counterparts with the addition of computers. Each seats six people, and the benches are long and wide to accommodate the occasional nap. The high bench backs create a conversational buzz while limiting eavesdropping. Mixed in with the booths are ten study rooms for private conversations and two open alcoves for large-group brainstorming. The space includes a thirty-five-seat seminar room with built-in videoconferencing and the Vitale Digital Media Lab for self-service media creation. The variety of spaces in close proximity enables powerful synthesis activities.[10] Humanities classes will start with lecture or large-group discussion in the seminar room, break up for small-group conversations in the booths, and reconvene to share conclusions.

Several rooms allow self-service high-definition video recording and videoconferencing (Skype, Google Hangouts, etc.), role playing, and live interviews. The Vitale Digital Media Lab supports self-service media creation with a variety of gadgets including a vinyl record digitizer, a slide scanner, and a large-format printer. Faculty and students can borrow video cameras, audio recorders, and portable scanners for offsite use.

KISLAK CENTER FOR SPECIAL COLLECTIONS, RARE BOOKS AND MANUSCRIPTS

The Kislak Center occupies the two top floors of the same library building. Renovations have made collections of rare materials dramatically more interesting and accessible. The Class of 1978 Pavilion is a glamorous space with

built-in videoconferencing and video-recording capability that can accommodate 140 people for lectures or be rearranged for smaller discussions. Nearby is a row of three 20-person seminar rooms that can be combined by removable partitions.

The center includes the Vitale Media Lab (Vitale II), which focuses DH programming on cultural heritage and special collections materials. The room can be reconfigured easily for groups of up to twenty people and includes a high-definition ceiling-mounted camera for enlarging manuscripts on large monitors and sharing via videoconference. Closely related to Vitale II is the Schoenberg Institute for Manuscript Studies (SIMS), which focuses on research around medieval manuscripts, both physical and digital.

Down the hall, the Rare Book Reading Room provides a central space with small pullout rooms for researchers to handle materials. The Kislak Center also includes the Goldstein Exhibit Hall, the historically important Lea Library with its collection of eighteenth-century manuscripts, the Moelis Terrace for informal study, and a balcony with a splendid view of the Philadelphia skyline.

COLLABORATIVE CLASSROOM

The Collaborative Classroom is an active learning space that includes a thirty-seat classroom for formal instruction set into a wide "porch" area where students can work informally in small groups. The classroom is located on the first floor next to WIC and a rare books exhibit area. Created in a room that formerly housed government documents, all the walls are writable surfaces. Designed for problem-solving activities, the space contains five round tables (with power and Internet wired into the centers) and one instructor station, each with a projection screen. Technology and furniture choices were made to explicitly support "flipped classroom" pedagogies.[11] An elaborate audiovisual system gives faculty members one-touch power to take or cede control over the six display screens.

COMMON ASPECTS AND A NEIGHBORHOOD

Common technology capabilities across the three spaces include movable furniture, ability to display from personal laptops, access to MacBook laptops and iPads, and an extensive inventory of educational software.

All three spaces are in the Van Pelt-Dietrich Library Center, a half-century-old building with a twenty-four-hour café on the lowest level. Together they create a powerful "neighborhood." Faculty and students can accomplish different tasks on different floors—touch a manuscript in the reading room,

take photos of a few pages in Vitale II, create a video of their conversation about the photos in a WIC study room, or annotate the photos on the writable walls in the Collaborative Classroom. Helping patrons explore, understand, and feel ownership over the capabilities available in our neighborhood of spaces is the challenge we face as librarians.

Skills

In support of learning spaces, Penn Libraries has invested in expanding skill sets of library staff through professional development of current staff and recruitment of individuals with specialized skill sets, partially through postdoctoral DH fellowships. Reflections on DH support have required a reexamination of what it means to be a librarian and who is needed on deck in a modern library. We discuss three specific roles we perceive in the relationships librarians have developed with DH researchers.

Librarian as Concierge

Successful DH projects require thoughtful planning. Eliciting the initial vision for a project requires careful conversations similar to the traditional reference interview. What do you want to accomplish? How will you recognize success? What obstacles do you perceive? How will you organize your project effectively? What skills are needed for the project to be successful? Who has those skills? It is worth mentioning that this is true whether the project instigator is student, faculty, or librarian, new to DH or seasoned.*

Subject specialists who are themselves new to or nervous with technology tools may hesitate to guide faculty choices, defaulting to a "But that's not my role" response. Having instructional design and information technology experts on staff and pairing them with subject specialists can improve library contributions in the initial stages of a DH project. Creating collaborative relationships between IT experts and subject specialists also has positive side effects as described later in the context of our Digital Scholarship Workshop series.

A few tensions arise that are worth tackling head-on. Sometimes librarians are seen as insiders, and sometimes as outsiders. The librarian can stay outside the departmental politics, helping all members of a department and not distinguishing across hierarchies of tenured/untenured, standing/

* We use the term *project instigator* to refer to the person who is leading a project, whether faculty, student, or librarian.

ad-hoc, faculty/graduate student/staff, and so on. The outsider role can be especially helpful in sharing successes and challenges across departments. If one department or research group faces difficulties with implementing a particular software program, the librarian can help share that information (while maintaining strict privacy guards) with another department that is considering a similar project. Successes can be shared through the library website and blogs, where we showcase good projects and abstract out from project details to the general functionality, helping with replication across campus.[12] Facilitating peer-to-peer sharing for faculty, such as our annual Engaging Students through Technology symposia, be highly effective.[13]

Librarians can benefit from insider knowledge if they have personal experience conducting research so that they come to the project understanding how difficult it can be to achieve productive research results. A librarian with strong relationships (think "embedded librarianship") within a department can develop intuition that helps when new DH projects emerge.

Librarian as IT Expert

To support or lead DH projects, libraries need staff with both wide and deep expertise, A project instigator may want to first look at a range of options before picking a tool set. (Should I start with a website or a multimedia-friendly database? Would a podcast work for me, or is video needed?) At this stage, assistance from the library needs to be perceived as unbiased and impartial. Project instigators do not want to be "convinced into" a particular tool, platform, or software choice, particularly if it seems like that the librarian is reverting to his or her own personal expertise ("If you have a hammer, everything looks like a nail").

During this initial stage of tool selection, we need to acknowledge that exploring a new tool can be surprisingly stressful. An interface that feels natural to a regular user can look arcane and intimidating to a newcomer. When librarians take time to prepare extensive demonstration materials in a disciplinary context, faculty are more easily able to look past user interface limitations and glimpse the capabilities of a DH tool.

Once the project instigator has chosen the tools, deep knowledge is in demand. The project instigator needs to be paired up with a librarian who knows those tools well, can point out pitfalls ahead of time, and can effectively liaise with other library staff as needed. Every DH tool has its strengths and weaknesses, and knowing details early can reduce frustration.

We have worked toward providing a "geek squad" of sorts by making the expertise of our staff more transparent for faculty and students. We have used simple tools—a publicly editable Google Doc tied to a Springshare LibGuide—to collect data from our librarians about their own expertise with DH tools. The crowdsourcing project took about four weeks. Librarians added in their names next to tools they were comfortable with. The results provide an easy place to find out whom to ask about what, and a useful listing of DH tools building on the DIRT Directory framework.[14]

Librarian as Researcher

We face ambivalence from faculty members and students on the perception of librarians as researchers. As faculty and students use more online resources, subscription databases as well as materials available openly through the Internet, librarians need to develop new skills and approaches. One approach taken at Penn Libraries is to support library staff in some departments to be DH researchers in their own right in addition to supporting faculty and student research.[15] We see advantages for librarians to stay active in personal research activities—to continue their own learning and to remain connected to the difficulties of being a learner.

We have successfully used blogging as a way to expose the scholarship activities within the library to the broader campus community. We have several blogs highlighting the work of different units, including the *Schoenberg Institute* blog, the DH tag on the general *PennWIC* blog, and the *Unique at Penn* blog.[16]

Staffing Details

Providing support for the three types of roles defined above requires library staffing that is varied and flexible. Our spaces are supported by traditionally educated librarians, instructional designers, and software geeks. In addition to permanent staff, we take advantage of multiyear fellowships, year-long graduate internships, short-term visiting scholars, and ad hoc consultants.

WIC includes three full-time and two half-time staff and up to a dozen graduate and undergraduate students. Staff expertise includes video editing (Final Cut Pro, Premiere, iMovie), graphic design (Photoshop, InDesign, Illustrator), web design (WordPress, CSS), instructional design (lesson planning, assignment design), data analysis (Excel, Microsoft Access), and so on.

The Kislak Center has many full-time and part-time staff, including curators to collect and interpret collections, catalogers to describe materials, and

staff to manage the reading rooms. SIMS has a separate research agenda, undertaken primarily by librarians and library staff. It is not a support unit for faculty and students, although faculty and students collaborate on SIMS projects. Directly involved with DH support are five full-time staff: two curators for Digital Research Services, a digital content programmer, a project manager for the Schoenberg Database of Manuscripts,[17] and a web developer. Staff expertise includes software development using programming languages, web server management, data mining, text mining, data visualization, TEI encoding, and Omeka.

The Collaborative Classroom is supported by several full-time librarians (including subject specialists and social science research specialists) and graduate students in the Research and Instructional Services department with assistance from WIC. A Teaching and Learning Fellow with a doctorate in education supports faculty who teach in the space.

Communication among the groups of staff has been crucial and an ongoing challenge. One effective mechanism is *The Thread*, a library-facing blog where staff share updates informally.[18] Another is the Public Services Forum, a monthly library-wide meeting for librarians to share accomplishments and challenges. A recent Digital Scholarship Workshop series has successfully engaged librarians and the campus community by exploring DH topics through joint presentations that include tool demonstrations as well as conversation about effective liaison librarianship.[19]

Program Support Models

Our three spaces and their staff configurations support different approaches to DH work. Collaborations in WIC typically begin with a faculty request for support for a course-related assignment. We meet with the faculty member to understand goals, course design, and expectations. We design and conduct training sessions for the students, as well as for teaching assistants.[20] We assist with setup of shared spaces (physical and online) and workflows. We hold open work sessions for students shortly before assignment deadlines to help manage last-minute stresses. We create showcases of exemplary student work.[21] Demand for course-level assistance has been steady for several years,[22] and we provide custom training on web design (WordPress, Tumblr, Google Sites), graphic design (Photoshop, PowerPoint, Instagram), and video (iMovie, Final Cut Pro, iPhoto, Canvas).

We explore new technologies, purchase sample gadgets for lending, help people feel more comfortable with new technologies, and showcase examples

that might fuel replication. We provide iPads and laptops to support ad hoc use, such as a class project involving iPads and rare books.[23] We organize workshops for general audiences and provide ad hoc consultations. In 2013, WIC conducted over 220 open workshops for over 1,700 attendees and about 80 workshops by request for 1,000 attendees.[24]

Once a semester, we hold a Gadget Day for people from around campus to share their favorite new toys. We provide access to Lynda.com for self-paced skill improvement, especially with the Adobe Creative Suite of software. Once a year, we hold an Engaging Students through Technology Symposium, which attracted over 150 faculty and graduate students last year. Our faculty advisory group helped us build a faculty development module on Nurturing Student Creativity through Video Projects.[25] We run an annual Mashup Video contest to recognize student creativity.

Collaborations in the Kislak Center often include working with special collections materials, in addition to technical training. Programming offered through Vitale II tends to be ongoing and aimed at specific audiences, rather than through repeat sessions of discrete workshops. Much programming centers on what we call Focused Labs, set times each week for faculty, students, and librarians to meet and work through specific tools and techniques. The Focused Labs for 2013–2014 included Code Academy courses in Python and HTML/CSS[26] and an Omeka users group. Although all started off strong, by the end of the spring semester attendance had decreased. We are exploring ways to strengthen Vitale II programming, encouraging librarians in other departments to hold sessions in the space. We have organized several DH-focused events including three unconferences—PhillyDH@Penn 2013 and 2014, and THATCamp Penn 2012.[27]

Since opening in February 2014, the Collaborative Classroom has hosted several full-semester courses, brainstorming sessions for academic departments, outreach events, and workshops on DH topics including social media and mind mapping. In one presentation for the Digital Humanities Forum, a faculty member presented an extensive DH project that focuses on Philadelphia history and includes mapping, GIS, video, and animation technologies.[28]

Synthesis

In this section, we explore examples of synthesizing spaces and skills. An important component of DH work is to assist librarians and staff with IT expertise in learning to articulate how what they know (creating a searchable web catalog) is helpful for a DH project (create a searchable catalog of

student-selected items from the Penn Museum). Staff with significant IT expertise may not always understand the nervousness of faculty and students who are just getting started. Subject librarians can help bridge gaps and translate jargon.

Examples

The examples below are actual projects, though some details have been stylized.

- A professor of Persian history visited the Vitale Digital Media Lab in WIC carrying an oversize battered book. She wanted to create an English translation of the book, which contained several hundred pages of Arabic text and color illustrations. We guided her through scanning and photography options available across different departments. After she had created a project plan, we trained her undergraduate students to scan and resize the original illustrations and lay out the pages with English text replacing the Arabic text. Over several weeks, her project grew to include scanning, optical character recognition, color management, graphic design, file and scan management, workflow planning, and backup procedures. She reflected that our guidance on the complexity of the process was especially helpful because she could assess feasibility and costs in terms of her time and her students' time.
- A professor of South Asian studies came to WIC with ideas for her eighty-person introductory undergraduate class. Students were preparing simplistic presentations and class time watching group presentations was minimally engaging. We explored a variety of technology options, including screen video creation (Canvas, Jing, PowerPoint), forms and polls (Google Forms, online voting systems), video editing (Final Cut Pro, iMovie, Windows Movie Maker) and video sharing (YouTube, Canvas, Blackboard). We conducted training for the professor and her four teaching assistants on how to manage the new course assignment: five-minute videos by each student that would be watched and voted on by the full class, with the winners shown at an end-of-term celebration. The process of creating an effective assignment included iterative analysis of the tools available in active conversation with the course TAs, demos in the classroom, and "tech office hours" for the students shortly before each deadline. The results are included in a class showcase page.[29] After watching the professor

present at our annual symposium,[30] several faculty members around campus decided to incorporate this simple voice-over–PowerPoint technology solution in their courses.

- Several librarians expressed concern about their own knowledge regarding popular digital tools and techniques. They created a Digital Scholarship series to bring together librarians, faculty, students, and staff for workshops primarily led by librarians. Topics included choosing the right online exhibit software, qualitative research, choosing a citation manager, working with archival materials, and an Omeka overview. The series has provided short lunchtime workshops to explain what each tool does and why it is useful so that librarians can make informed decisions about which tools they want to learn.
- The Director of the Kislak Center and an English professor co-taught a freshman seminar The World of Manuscripts (ENGL 016.304), which introduced students in the class to a wide range of manuscript and manuscript-like materials (including cuneiform tablets and letters of Mary Shelley). As their final project, the students undertook a study of Penn's Wycliffite New Testament, Ms. Codex 201,[31] each student taking responsibility for a different aspect of the book. That study culminated in a short video, made with the assistance of library staff and using the ceiling camera available in Vitale II.[32] The video is public on the SIMS YouTube channel and showcases the contributions of the students and librarians.[33]
- Not all collaborations unfold as intended. A history of art professor came to the Kislak Center interested in a website to combine artifacts and archival documents relating to the Beth Shean archaeological dig held by the Penn Museum. The semester-long project included a small grant from the Digital Humanities Forum. Library staff conducted Omeka training, and students were responsible for selecting, curating, scanning, and loading materials into Omeka. However, the students found Omeka clunky and the exhibit-building function difficult. The hosted nature of the service limited our ability to customize the experience using server back-end functionality. The students chose to move the project to Squarespace, a commercial website hosting platform, without input from library staff. Although the Squarespace website is beautiful, the cataloging and reuse capabilities possible with Omeka were lost. We are now exploring ways to possibly combine an Omeka back end with a Squarespace front end.[34]

- WIC's iPads in the Classroom program has worked well in conjunction with rare books and our stacks.[35] One English class used our iPads to explore rare materials with interactive research into authors and provenance. The professor reflected on her experiences, saying, "While I've given students similar assignments for several years, this is the first year that they've done the depth of research that I expected."[36]

These examples emphasize the importance of choosing between tools and surveying the universe of available tools. Faculty appreciate a safe space to play in, a sandbox of sorts, where trying out a new tool has a low level of risk. Once a professor has identified a tool that is a good fit, the need for hands-on training can be met in many ways, including using online tutorials from vendors. The librarian need not feel compelled to have expertise in all tools of potential interest, but the librarian does need to stay up-to-date on name recognition of the universe of currently popular tools.

Repercussions for New Media Literacies

In addition to raising standards for librarian expertise, the importance of effective synthesis of DH tools leads to higher expectations for students and faculty. Alan Dix has written about the popular YouTube video "Middle Ages Tech Support" to explain how a familiar technology, such as the book, may have stumped scholars in previous centuries.[37] Today, our expectations for technology skills increase steadily. Mastery of skills in video creation, blogging, social media use, web design, text mining, and data analysis has become an expectation rather than an aspiration. Both faculty and students can face pressures to ramp up their own digital skills,[38] sometimes on short notice. The Technology Acceptance Model (TAM) may help us anticipate areas of difficulty.[39] For example, perceptions of difficulty of use and perceptions of usefulness impact how some students acquire video-creation skills.[40] We may want to consider identity-related obstacles to the process of gaining digital skills. Explicit consideration of digital literacies and obstacles to their acquisition can help us organize how we support DH projects.

Conclusion

In this chapter, we have reflected on the evolution of three spaces at the University of Pennsylvania Libraries. We end with a few philosophical comments.

All three spaces discussed here were created after extensive visits to other campuses and in coordination with national organizations considering

learning-space design. Reviewing concrete examples from other contexts has helped us make informed choices when meeting with contractors and architects. Our staffing and programming decisions have also benefited from conversations with colleagues in other institutions.

How we manage our spaces after the ribbon cutting has been an important choice. We have taken the "better broken than dusty" attitude towards our shiny new spaces, taking care to make all of campus feel welcome and comfortable. It can be tricky to maintain high-tech spaces when they are packed with patrons who won't always behave as expected. When we hosted the forty-eight-hour Penn Apps hackathon, every bit of floor space in the Kislak Center was filled with sleep-deprived undergraduates. Opening up spaces deliberately requires taking risks, and support for risk taking (with a sense of humor) from the highest levels of library administration has been essential.

Skill sets come in and out of fashion. The human connections we make with faculty and students are as important, if not more, than the specific technical skill we bring to a consultation. The push and pull between librarian-as-support-staff and librarian-as-researcher is a real one, and each librarian finds his or her own comfortable spot. Through writing this chapter together, we explored our own perceptions of this continuum.

Writing this chapter together has also been an example of successfully crossing organizational boundaries. As we struggle through our own writing and research projects, it increases our ability to empathize with faculty and students embarking on their research efforts.

Acknowledgments

This article has benefited from the assistance of Kim Eke, Jim English, Vickie Karasic, Will Noel, Nick Okrent, Rebecca Stuhr, and many other colleagues from the University of Pennsylvania.

Notes

1. Melissa Terras, Julianne Nyhan, and Edward Vanhoutte, eds., *Defining Digital Humanities: A Reader* (Farnham, UK: Ashgate Publishing, 2013), www.ashgate.com/isbn/9781409469636 [site discontinued].
2. University of Pennsylvania, "Penn Facts," accessed February 7, 2015, www.upenn.edu/about/facts.php [site discontinued].
3. University of Pennsylvania, Penn Digital Humanities Forum website, accessed February 7, 2015, http://humanities.sas.upenn.edu/dhf.shtml.

4. See the DH Tools list at Penn Libraries, "Tools and Library Experts, in Alphabetical Order," Guides: Digital Humanities, last updated October 17, 2014, http://guides.library.upenn.edu/toolexpertise.
5. See Jeanne L. Narum, ed., *A Guide for Planning for Assessing 21st Century Spaces for 21st Century Learners* (Washington, DC: Learning Spaces Collaboratory, November 2013), www.pkallsc.org/basic-page/lsc-guide-planning-assessing-21st-century-spaces-21st-century-learners.
6. See Joan K. Lippincott, "Information Commons: Meeting Millennials' Needs," *Journal of Library Administration* 50, no. 1 (2010): 27–37.
7. See WIC annual reports at Weigle Information Commons, "Reports and Publications," Penn Libraries, University of Pennsylvania, accessed February 7, 2015, http://commons.library.upenn.edu/reports-and-publications [site discontinued].
8. See Anu Vedantham, "Pressure Release," *PennWIC* (blog), May 2, 2014, http://pennwic.wordpress.com/2014/05/02/pressure-release.
9. Grant press release article available at University of Pennsylvania, "Penn to Implement AAU Undergraduate STEM Education Initiative," news release, June 25, 2013, https://penntoday.upenn.edu/news/penn-implement-aau-undergraduate-stem-education-initiative.
10. Details are available at Anu Vedantham, "Learning Space Design," *PennWIC* (blog), July 3, 2014, http://pennwic.wordpress.com/2014/07/03/learning-space-design.
11. A photo gallery of teaching examples is available at Weigle Information Commons, "Collaborative Classroom," Flickr album, accessed February 7, 2015, https://www.flickr.com/photos/pennwic/sets/72157645502444793.
12. See, for example, our success stories category at "Success Stories," *PennWIC* (blog), last updated December 10, 2014, http://pennwic.wordpress.com/category/success-stories.
13. See details from several years at Weigle Information Commons, "Engaging Students through Technology Symposium," Penn Libraries, University of Pennsylvania, 2014, http://commons.library.upenn.edu/engaging-students-through-technology-symposium [site discontinued].
14. Penn Libraries, "Tools and Library Experts"; see DiRT Directory (formerly part of Project Bamboo) at DiRT: Digital Research Tools, accessed February 7, 2015, http://dirtdirectory.org [site discontinued].
15. See Dot Porter, "What If We Do, in Fact, Know Best? A Response to the OCLC Report on DH and Research Libraries," *dh + lib* (blog), February 12, 2014, http://acrl.ala.org/dh/2014/02/12/what-if-we-do-in-fact-know-best-a-response-to-the-oclc-report-on-dh-and-research-libraries.
16. See *Schoenberg Institute for Manuscript Studies*, Penn Libraries blogs, University of Pennsylvania, accessed February 7, 2015, http://schoenberginstitute.org; *PennWIC* (blog), posts tagged "digital humanities," accessed February 7, 2015, http://pennwic.wordpress.com/tag/digital-humanities; *Unique at Penn*, Penn Libraries blogs, University of Pennsylvania, accessed February 7, 2015, http://uniqueatpenn.wordpress.com.
17. Schoenberg Database of Manuscripts, Penn Libraries website, accessed February 7, 2015, http://dla.library.upenn.edu/dla/schoenberg.
18. Penn Libraries' staff blog is at *The Thread*, Penn Libraries blogs, University of Pennsylvania, accessed February 7, 2015, http://libthread.wordpress.com.
19. See *PennWIC* (blog), posts tagged "digital scholarship," accessed February 7, 2015, http://pennwic.wordpress.com/?s=digital+scholarship.
20. See training materials at Penn Libraries, Guides: Tutorials and Tech Guides, last updated November 14, 2014, http://guides.library.upenn.edu/tutorials.

21. See Weigle Information Commons, "Student Work Showcase," accessed February 7, 2015 http://commons.library.upenn.edu/student-work-showcase.
22. See course listing at Weigle Information Commons, "Course Usage of the Weigle Information Commons," Penn Libraries, accessed February 7, 2015, http://commons.library.upenn.edu/course-usage [site discontinued].
23. See Cathy Turner, "Using iPads in the Rare Book Room," *PennWIC* (blog), January 5, 2013, http://pennwic.wordpress.com/2013/01/05/using-ipads-in-the-rare-book-room.
24. See Weigle Information Commons, "Reports and Publications."
25. See Weigle Information Commons, "ELIXR MERLOT Faculty Development Initiative," Penn Libraries, University of Pennsylvania, accessed February 7, 2015, http://elixr.merlot.org/case-stories/teaching-strategies/nurturing-student-creativity-with-video-projects/ [site discontinued] funded through ELIXR MERLOT.
26. CodeAcademy, "Recommended for You," accessed February 7, 2015, www.codecademy.com/learn?choice=language.
27. See PhillyDH@Penn 2013 website, last updated June 27, 2013, http://penn2013.phillydh.org [site discontinued]; PhillyDH@Penn 2014 website, last updated June 20, 2014, http://penn2014.phillydh.org; THATCamp@Penn: The Humanities and Technology Camp website, last updated May 3, 2012, http://penn2012.thatcamp.org.
28. See Amy Hillier, Mapping Du Bois, *The Philadelphia Negro* website, accessed February 7, 2015, www.mappingdubois.org.
29. Lisa Mitchell's Course Showcase at Weigle Information Commons, "PowerPoint Videos—Showcase 2012," accessed February 7, 2015, http://commons.library.upenn.edu/showcase-mitchell-sast-2012
30. See Lisa Mitchell, "Screen Videos as Student Projects: Engaging Students through Technology 2011," YouTube Video, 12:59, posted by pennlibraries November 30, 2011, https://www.youtube.com/watch?v=titDzisK5FE.
31. See Penn Libraries, "Ms. Codex 201—New Testament in the Translation of John Wycliffe," Penn in Hand: Selected Manuscripts, accessed February 7, 2015, http://hdl.library.upenn.edu/1017/d/medren/1551783.
32. simsmss, "Penn Ms. Codes 201, Presented by Students in the World of Manuscripts Freshman Seminar (ENGL 016.304)," *Schoenberg Institute for Manuscript Studies* (blog), Penn Libraries blogs, January 22, 2014, http://schoenberginstitute.org/2014/01/22/penn-ms-codex-201-presented-by-students-in-the-world-of-manuscripts-freshman-seminar-engl-016-304.
33. Alexandra Pierson et al., "University of Pennsylvania Library's Ms Codex 201," YouTube video, 9:04, posted by SchoenbergInstitute December 18, 2013, https://www.youtube.com/watch?v=oOkJ5bm-jZ8.
34. Beth Shean: After Antiquity website, accessed February 7, 2015, http://beth-shean.squarespace.com.
35. See Weigle Information Commons Guides, "iPads in the Classroom," last updated October 13, 2014, http://guides.library.upenn.edu/ipad.
36. Turner, "Using iPads in the Rare Book Room."
37. Alan Dix, "Physicality and Middle Ages Tech Support," *Alan Dix* (blog), December 13, 2007, http://alandix.com/blog/2007/12/13/physicality-and-middle-ages-tech-support.
38. See, for example, a discussion of video literacy at Anu Vedantham and Marjorie Hassen, "New Media: Engaging and Educating the YouTube Generation," *Journal of Learning Spaces* 1, no. 1 (2011).

39. See initial proposal for the TAM model at Fred D. Davis, "Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology," *MIS Quarterly* 13, no. 3 (September 1989): 319–40.
40. See, for example, research on gender-related issues in video creation skills in Anu Vedantham, "Making YouTube and Facebook Videos: Gender Differences in Online Video Creation among First-Year Undergraduate Students Attending a Highly Selective Research University" (EdD diss., University of Pennsylvania, 2011).

Bibliography

Beth Shean: After Antiquity website. Accessed February 7, 2015. http://beth-shean.squarespace.com.

CodeAcademy. "Recommended for You." Accessed February 7, 2015 www.codecademy.com/learn?choice=language.

Davis, Fred D. "Perceived Usefulness, Perceived Ease of Use, and User Acceptance of Information Technology." *MIS Quarterly* 13, no. 3 (September 1989): 319–40.

DiRT: Digital Research Tools website. Accessed February 7, 2015. http://dirtdirectory.org [site discontinued].

Dix, Alan. "Physicality and Middle Ages Tech Support." *Alan Dix* (blog), December 13, 2007. http://alandix.com/blog/2007/12/13/physicality-and-middle-ages-tech-support.

Hillier, Amy. Mapping Du Bois. *The Philadelphia Negro* website. Accessed February 7, 2015. www.mappingdubois.org.

Lippincott, Joan K. "Information Commons: Meeting Millennials' Needs." *Journal of Library Administration* 50, no. 1 (2010): 27–37.

Mitchell, Lisa. "Screen Videos as Student Projects: Engaging Students through Technology 2011." YouTube video, 12:59. Posted by pennlibraries November 30, 2011. https://www.youtube.com/watch?v=titDzisK5FE.

Narum, Jeanne L., ed. *A Guide for Planning for Assessing 21st Century Spaces for 21st Century Learners*. Washington, DC: Learning Spaces Collaboratory, November 2013. www.pkallsc.org/basic-page/lsc-guide-planning-assessing-21st-century-spaces-21st-century-learners.

Penn Libraries. "Digital Humanities." Guides, Penn Libraries, University of Pennsylvania. Last updated October 17, 2014. http://guides.library.upenn.edu/toolexpertise.

———. "Ms. Codex 201—New Testament in the Translation of John Wycliffe." Penn in Hand: Selected Manuscripts, Penn Libraries, University of Pennsylvania. Accessed February 7, 2015, http://hdl.library.upenn.edu/1017/d/medren/1551783.

———. Schoenberg Database of Manuscripts. Penn Libraries, University of Pennsylvania. Accessed February 7, 2015. http://dla.library.upenn.edu/dla/schoenberg.

———. "Tutorial and Tech Guides." Penn Libraries, University of Pennsylvania, last updated November 14, 2014, http://guides.library.upenn.edu/tutorials.

PennWIC (blog). Accessed February 7, 2015. http://pennwic.wordpress.com.

PhillyDH@Penn 2013 website. Last updated June 27, 2013. http://penn2013.phillydh.org [site discontinued].

PhillyDH@Penn 2014 website. Last updated June 20, 2014. http://penn2014.phillydh.org.

Pierson, Alexandra, James Bessolo, Linda Valadez, and Patricia Kamwela. "University of Pennsylvania Library's Ms Codex 201." YouTube video, 9:04. Posted by SchoenbergInstitute December 18, 2013. https://www.youtube.com/watch?v=oOkJ5bm-jZ8.

Porter, Dot. "What If We Do, in Fact, Know Best? A Response to the OCLC Report on DH and Research Libraries." *dh + lib* (blog), February 12, 2014. http://acrl.ala.org/

dh/2014/02/12/what-if-we-do-in-fact-know-best-a-response-to-the-oclc-report-on-dh-and-research-libraries.
Schoenberg Institute for Manuscript Studies. Penn Libraries blogs, University of Pennsylvania. Accessed February 7, 2015. http://schoenberginstitute.org.
simsmss. "Penn Ms. Codes 201, Presented by Students in the World of Manuscripts Freshman Seminar (ENGL 016.304)." *Schoenberg Institute for Manuscript Studies*, Penn Libraries blogs, University of Pennsylvania, January 22, 2014. http://schoenberginstitute.org/2014/01/22/penn-ms-codex-201-presented-by-students-in-the-world-of-manuscripts-freshman-seminar-engl-016-304.
Terras, Melissa, Julianne Nyhan, and Edward Vanhoutte, eds. *Defining Digital Humanities: A Reader*. Farnham, UK: Ashgate Publishing, 2013. www.ashgate.com/isbn/9781409469636 [site discontinued].
THATCamp@Penn: The Humanities and Technology Camp website. Last updated May 3, 2012. http://penn2012.thatcamp.org.
The Thread. Penn Libraries blogs, University of Pennsylvania. Accessed February 7, 2015. http://libthread.wordpress.com.
Turner, Cathy. "Using iPads in the Rare Book Room." *PennWIC* (blog), January 5, 2013. http://pennwic.wordpress.com/2013/01/05/using-ipads-in-the-rare-book-room.
Unique at Penn. Penn Libraries blogs, University of Pennsylvania. Accessed February 7, 2015. http://uniqueatpenn.wordpress.com.
———. Penn Digital Humanities Forum website. Accessed February 7, 2015. http://humanities.sas.upenn.edu/dhf.shtml.
———. "Penn Facts." Accessed February 7, 2015. www.upenn.edu/about/facts.php [site discontinued].
———. "Penn to Implement AAU Undergraduate STEM Education Initiative." News release, June 25, 2013. https://penntoday.upenn.edu/news/penn-implement-aau-undergraduate-stem-education-initiative.
Vedantham, Anu. "Learning Space Design." *PennWIC* (blog), July 3, 2014. http://pennwic.wordpress.com/2014/07/03/learning-space-design.
———. "Making YouTube and Facebook Videos: Gender Differences in Online Video Creation among First-Year Undergraduate Students Attending a Highly Selective Research University." EdD diss., University of Pennsylvania, 2011.
———. "Pressure Release." *PennWIC* (blog), May 2, 2014. http://pennwic.wordpress.com/2014/05/02/pressure-release.
Vedantham, Anu, and Marjorie Hassen. "New Media: Engaging and Educating the YouTube Generation." *Journal of Learning Spaces* 1, no. 1 (2011).
———. "Collaborative Classroom." Flickr album. Accessed February 7, 2015. https://www.flickr.com/photos/pennwic/sets/72157645502444793.
———. "Course Usage of the Weigle Information Commons." Penn Libraries, University of Pennsylvania. Accessed February 7, 2015. http://commons.library.upenn.edu/course-usage [site discontinued].
———. "ELIXR MERLOT Faculty Development Initiative." Penn Libraries, University of Pennsylvania. Accessed February 7, 2015. http://elixr.merlot.org/case-stories/teaching-strategies/nurturing-student-creativity-with-video-projects/ [site discontinued].
———. "Engaging Students through Technology Symposium." Penn Libraries, University of Pennsylvania, 2014. http://commons.library.upenn.edu/engaging-students-through-technology-symposium [site discontinued].

———. “PowerPoint Videos—Showcase 2012.” Penn Libraries, University of Pennsylvania. Accessed February 7, 2015. http://commons.library.upenn.edu/showcase-mitchell-sast-2012.

———. “Reports and Publications.” Penn Libraries, University of Pennsylvania. Accessed February 7, 2015. http://commons.library.upenn.edu/reports-and-publications [site discontinued].

———. “Student Work Showcase.” Penn Libraries, University of Pennsylvania. Accessed February 7, 2015. http://commons.library.upenn.edu/student-work-showcase.

Weigle Information Commons Guides. “iPads in the Classroom.” Penn Libraries, University of Pennsylvania. Last updated October 13, 2014. http://guides.library.upenn.edu/ipad.

CHAPTER TWELVE

Sustaining the Digital Liberal Arts:

Institutional Challenges in Looking Beyond Grant Funding

Ginny Moran, Aisling Quigley, Brooke Schmolke, and Louann Terveer

Introduction

As *Digital Humanities in the Library* (2015) established, and this work continues to demonstrate, digital humanities (or DH) projects are inherently collaborative. While working collaboratively is rewarding, it is also challenging. Since the first edition of this book was published, there has been new and continued scholarship in librarianship and digital humanities on some of the difficult side-effects of this collaborative work, including the uneven distribution of emotional labor and an unhealthy reliance on precarious labor. Because of the interdisciplinary and hybrid nature of this work, it also gives rise to liminality and vocational awe. Each of these circumstances has the potential to contribute to burnout, turnover, and long-term project sustainability challenges.[1]

The four contributors to this piece come from an academic background in library and information science and occupy staff roles as librarians, academic technologists, and a postdoctoral fellow in a dual staff/faculty role. Each brings experience working in the liminal spaces of librarianship and academic technology in higher education settings. We share a collective desire to shift digital humanities work from being grant-funded, temporary, and project-based to institutionally integrated, supported, and sustainable through initiatives that move beyond individually cultivated relationships.

CC BY NC ND

This chapter reflects our experiences and observations as professionals working at Macalester College, a small private liberal arts college located in Saint Paul, Minnesota with an enrollment of approximately 2,000 undergraduates. The College places great emphasis on fostering transformational learning experiences among and across its campus community, affirming a commitment to "the intellectual growth of the students, staff and faculty through individual and collaborative endeavor."[2] Since 2014, the Digital Liberal Arts program, or DLA, has aspired to facilitate cross-disciplinary conversations and partnerships that contribute substantially to the scholarly and pedagogical landscape of Macalester. While this chapter focuses on activities occurring in the context of the College, the challenges and opportunities discussed are broadly applicable to other institutions incorporating a similar digital approach.

Although scholars and practitioners at Macalester have been using computational methods within various academic disciplines for some time, the college had the unique opportunity to formally dedicate time and energy to the development of a Digital Liberal Arts program when it was awarded a 2014 grant from the Andrew W. Mellon Foundation entitled "The Liberal Arts in the Digital Age."[3] The term "digital liberal arts" may be less legible or familiar to some members of the academic community than the term "digital humanities." The digital humanities (DH), stemming from humanities computing, has had decades to establish itself. DLA, meanwhile, is still a new term, one that seems to have emerged in the early 2010s, when it was attached to several Mellon grants. While DH and DLA are similar, the latter forefront the liberal arts and their emphasis on collaboration and interdisciplinarity. As William Pannapacker first suggested in 2013, and Kevin Butterfield later reiterated, DLA has the potential to embrace the entire liberal arts campus, including departments and divisions that may not be considered a branch of the "humanities."[4]

We discuss here challenges Macalester has faced in shifting from one-off project-based management and preservation work to ongoing maintainable infrastructures within the context of challenges common to other institutions: administrative support, funding, and staffing. Institutional structures in staffing as well as faculty/staff divides create challenges in terms of labor compensation, allocation of time, and academic freedom. It is also important to acknowledge the context in which this piece was written and the impact of the public health crisis on our environment. COVID-19 pandemic health measures have put additional strain on institutions for funds and on staff for time, while also limiting opportunities for relationship and capacity building.

We hope a discussion of these concerns will speak to the experience of readers in a variety of roles and institutions who are doing this work. We also hope this chapter provides some language and suggestions as a starting point for conversations about ongoing sustainability, staff longevity, and salary differentials.

The Challenge of Sustaining the Digital Liberal Arts

Despite existing in some shape or form for decades, the digital humanities and the digital liberal arts continue to evade precise definition. The US Department of Education published a definition as part of the 2020 Classification of Instructional Programs, but this does not encompass everything that may be considered DH scholarship.[5] This ambiguity presents a challenge to administrators and other institutional stakeholders and reflects the myriad ways this work is defined, developed, and supported across colleges and universities.

Grant-funding is of tremendous importance to supporting innovative programming at institutions of higher education, but grant-funded programs and individuals cannot necessarily change underlying and long-standing campus infrastructures. Macalester's current three-year grant cycle illustrates the limited and temporary human resources support given to DLA on our campus for approximately a decade. DLA leadership funded through the grant comprises a full-time postdoc and release time for a faculty director, although these roles were fused for the 21/22 academic year, as the faculty director went on sabbatical. The temporary funding of DLA personnel raises several opportunities and challenges for the DLA program in terms of long-term sustainability. As Miriam Posner and others have articulated, sustainability challenges are inherent to grant-funded programs.[6] This issue of precarious labor looms large, as has been demonstrated time and again, "it is always the skilled people and the communities they build who will sustain programs and projects."[7] Although various and important activities occur while a program is active—curricula developed, workshops facilitated, equipment acquired, websites launched, and the like—sustained support of these activities remains unlikely without dedicated staff and funding. We will discuss in more detail the impact of support from senior administrators, institutional austerity in higher education, and human resources on DLA initiatives and proceed with a discussion of our own experience at a liberal arts institution.

Support from Senior Administrators

A major consequence of any three-year grant-funded project is that, in addition to their regular job duties, staff across campus may be charged with carrying forward the institutional memory of previous grant activities and continuing to provide specialized training or consultations without fair compensation or support. The literature strongly supports the need for senior administrator buy-in to develop sustained staffing and infrastructure to continue digital humanities initiatives. The Ithaka S+R report, *Sustaining the Digital Humanities: Host Institution Support beyond the Start-Up Phase*, outlines this need as one of the top success factors for a coordinated campus strategy in supporting digital humanities work, key to developing DH as an institution-wide priority with a cohesive campus-wide plan that draws from multiple campus units and departments. In addition, the report discusses the need for senior administrators to be able to understand what DH is and what makes DH research unique. Communication should work in both directions, and it is important for senior administrators to clearly communicate back to existing and new faculty on the importance and value of DH research, especially in terms of tenure and promotion. Many of the institutions surveyed mention support for DH work existing on campuses in "nodes" or "pockets," as is the case at Macalester. Of note is the traction gained and resources secured at some liberal arts colleges by aligning DH goals with the existing institutional teaching-focused and student-centered aims and mission.[8] This gives us much food for thought as Macalester enters a phase of institution-wide strategic planning under a president who joined the college in June 2020. Now seems like an ideal opportunity to use the digital skills grown during the pandemic to continue to bolster faculty's confidence in using technology-enabled pedagogies such as those used with DH/DLA research methodologies.

Looking from the perspective of the librarians' role in DH projects, Posner calls out the need for institutional support and incentives for "DH to flourish in a sustained way." Specifically, she notes leadership buy-in for permanent staff, ongoing funding, dedicated work space, and the acceptance of failure alongside successes. In addition, Posner outlines what individual librarians need to fully support DH initiatives: training, support for librarian-conceived initiatives, time assigned as part of a position (not "add-on" duties), authority for those shepherding DH projects to gather resources and work hours from others, technical infrastructure that is flexible, incentives and recognition, and established equity in DH collaborations, especially across power structures.[9]

Others have explored specific ways in which administration can support the organization of this work across multiple individuals and campus units. Through Vinopal and McCormick's exploration of the issues of sustainability and scalability, they propose a tiered model of services to support digital scholarship. In addition to the need for selection and scoping of services and projects supported, the article discusses other challenges that are at the purview of the library and/or institutional administrative levels, such as organizing/reorganizing staffing, providing time and resources for staff professional development, funding, strategic vision, and guidance. The final challenge: "Authority and time to accomplish," is offered as a critical provision so that those that manage digital scholarship services have the authority to work toward the established goals and vision. The authors call for "change in organizational mindset and practice" so that distributed and ad-hoc work is recognized and allocated time by the administration and supervisors.[10]

INSTITUTIONAL AUSTERITY IN HIGHER EDUCATION

The economic impacts of the COVID-19 pandemic exacerbate an ongoing trend of institutional austerity in higher education taking place since the 2008 Great Recession. The last decade brought a decrease in state funding to public institutions, budget cuts, and layoffs; these measures have, in turn, resulted in larger class sizes, increased student debt, and increased inequality.[11] Although financial impacts have hit public institutions especially hard, private universities and colleges have not been immune to taking measures to balance the budget, such as hiring freezes, reducing class offerings, shutting down facilities, and eliminating lower enrollment programs—especially disciplines in the humanities such as arts and languages.[12]

Additional COVID-19 pandemic era economic impacts have manifested on campuses as budget cuts, staff dismissals, unpaid furloughs, early retirement options, hiring freezes, merit pay freezes, reduction of institutional endowments, delayed student enrollment, and losses of revenue due to students studying remotely—such as refunds of room and board and student campus fees.[13] Macalester College, as with other small liberal arts institutions in its peer group, enacted a "hiring chill" with few open or new positions being filled, early retirement packages, salary reductions for senior staff, and a temporary suspension of the college's retirement contribution to employees.

In addition to these financial impacts, the stress of quickly adjusting to address new modes of teaching and learning online was significant, especially considering how slowly higher education, in general, can be to enact change. Macalester, as a small private liberal arts college, emphasized in-person

instruction prior to being forced to make changes due to the pandemic. As private liberal arts colleges grapple with the changes in instructional delivery and student expectations, there is a palpable sense that some sort of line was crossed and that things can't go back to the way they were. There is continued speculation in higher education about whether online and in-person learning opportunities woven into course structures, such as blended and hybrid learning models, will become the new normal for all institutions, altering pressure points for budgets and funding.[14]

As the pandemic brought higher education structural inequities to the forefront, campuses that were already working toward "digital transformation" (Dx) fared better in what seemed like an almost instantaneous shift to online courses and remote work. EDUCAUSE defines digital transformation as "a series of deep and coordinated culture, workforce, and technology shifts that enable new educational and operating models and transform an institution's operations, strategic directions, and value proposition." Some shifts that EDUCAUSE points out as leading to Dx are a focus on institutional alignment to a central strategy, embedding IT Into those institutional priorities, continuous adaptation of worker roles to keep pace with evolving campus needs, and expanding the role of data and technology in not only administrative activities but also research, teaching, and learning.[15]

Austerity and emergency pandemic measures have institutions cutting budgets across the campus, including Information Technology (IT) budgets. An institution's ability to support DLA is dependent on its ability to invest in and support untried technologies and these austerity measures hamper that potential. An October 2020 poll found that 65 percent of the colleges and universities surveyed reported a median IT budget decrease of 10 percent with additional cuts anticipated.[16] These cuts come at a time when there are growing and continuously evolving campus IT needs to support core campus operations and student learning. These sometimes-competing technology needs include security, privacy, learning analytics, learning management systems (LMS), hardware repair and replacement, software, interoperability between systems, single sign on, and the list continues. This can put software purchases and support for more niche research needs low on the priority list. The one-off approach often employed with DLA activities, fitting the technology to the research need, may also be at odds with growing trends in higher education to incorporate campus-wide, cookie-cutter approaches to curriculum design by employing adaptive learning technologies and student learning analytics. Macalester has managed to continue to prioritize both—providing institutional-level solutions and matching tools to needs. Although

the technical aspects are integral to supporting DLA, especially in keeping current with the evaluation, purchase, and tech support of emerging technologies, supporting individuals that are new to these technologies and methods is equally or more time consuming.

HUMAN RESOURCES

From a human resources perspective, collaborative DH or DLA work can be difficult to accomplish in a work environment where there are many administrative structures to consider. Reporting lines, faculty or staff status, and perceptions of the value of the work itself can diminish productivity or lead to missed opportunities through lack of intentional consultation and coordination.

At Macalester, we have a highly collaborative relationship between the library and ITS staff despite separate reporting lines: ITS is newly in its own reporting line after being previously situated in the finance reporting line, while the library is within the academic line. Librarians and educational technologists both see the importance of developing student, faculty, and staff digital literacies that enable efficient and ethical use of technologies, and frequently meet both inside and outside DLA working group structures to discuss support mechanisms for projects as they develop. Individuals from ITS and the library work together to explore how campus community members build these and other meta-literacy skills in tandem with the work of the DLA team as a parallel, but related, system of support meshing well with existing library initiatives of building student information and meta-literacies. Nonetheless, librarian and educational technologist professional development is often conducted at the individual level, and competing commitments often derail the best intentions of "train the trainer" plans. This makes sustainable skill building across the whole team difficult.

At Macalester we do see some of these cultural and workforce trends starting to evolve and influence how the institution as a whole approaches its priorities—including the role of technology on campus. As our campus has returned to mostly in-person courses in the 2022 academic year, we hope that our community continues to build on the skills and digital literacies that were gained in the sudden shift to online learning.

There is evidence that faculty are becoming more comfortable with incorporating learning technologies into their curriculum. Self-proclaimed "luddites" are now able to navigate and successfully host class sessions via conferencing software, for example. Prior to the pandemic, the DLA postdoctoral fellow committed to four formal faculty consultations per academic

year. With the many changes necessitated by the public health crisis, she experienced an increase in the number of faculty consultations and class visits as more individuals took on the challenge of incorporating digital projects into their coursework and scholarship. Indeed, during the 2020/21 academic year, the DLA postdoctoral fellow consulted with approximately fifteen different faculty members and was embedded in two courses in Spring 2021 (in addition to teaching her own class). Although capacity for engaging in digital projects has decreased somewhat during the Fall 2021 semester, as faculty and staff have been endeavoring to readjust to in-person teaching during a pandemic, the demand for support of digital scholarship and digitally inflected assignments remains high.

While educational technologists are generally professional staff positions in most institutions, librarians can have a variety of job classifications. At Macalester College, librarians are classed as staff alongside academic technologists. This can mean limited support opportunities for most of the individuals that sustain DLA work on campus as grant programs frequently limit staff access to funds and instead prioritize funds going to faculty or students. The divisions between faculty and staff can also create limited relationship-building opportunities, as librarians and educational technologists aren't always in the same faculty circles on campus, such as curricular committees. In addition, librarians occupy a liminal space on campus, providing student support as well as classroom instruction and faculty support in a variety of capacities. Educational technologists at Macalester also provide student support and classroom instruction but with a larger emphasis on faculty support. In addition, educational technologists have offices dispersed across campus within the academic buildings and near the faculty they work with. Librarians are siloed within the library building. This can exacerbate the faculty/staff divide, making librarian outreach to faculty more challenging as the librarian has to insert themselves in situations that put them in proximity with faculty for project discussions and lean on outreach from the educational technologists to bring them into projects.

Another staffing challenge is that while the librarian and the educational technologist are often bringing similar levels of professional expertise to projects, librarians are often paid less on a college campus. Although in digital humanities work, the level and value of support that both librarians and technologists bring to the work is similar, average pay is $63,654 per year for Digital Initiatives Librarians and $55,883 for Research & Instruction Librarians compared to $73,191 per year for Instructional Technology Faculty Support Managers and $58,828 for Instructional Designers, according to

the 2019–20 Professionals in Higher Education Survey.[17] In addition, this salary data includes librarians who have both faculty and staff status. At Macalester, this is echoed in the salary ranges for librarians and educational technologists, with the 2021 librarian range being $51,477–$77,215 and the educational technologist range being $56,259–$84,388.

Digital humanities projects can work best when librarians are full collaborators or the researchers themselves, as Muñoz and Posner assert, but this type of research or project work is rarely built into the structure of librarian jobs.[18] This is a reminder of the differences in how faculty and staff are expected (or not) to engage in research projects, and in institutions where librarians are staff, librarian-driven research can be even more difficult. Logsdon et al. note that the librarian's breadth of experience across disciplines *is* our expertise; that rarely leaves an opportunity for in-depth research and focus.[19] Posner discusses the difficulty in creating the space for that type of research to be conducted by librarians as a lead on a project, particularly for librarians who don't have digital scholarship in their job title.[20] This distinction between faculty research expectations and the dual role of librarians and other staff in both facilitating and leading such research can result in a lot of starter projects often focused on classroom pedagogy, and few deeper projects. This focus on DLA projects related to classroom pedagogy creates a great deal of student interest in continuing this sort of work, but most grants are not set up to reward staff-driven and student-led projects.

DLA in Third Spaces

The DLA Team, created near the culmination of the first grant, continues to provide the foundation for curricular and scholarly innovation, even and especially during the COVID-19 pandemic. Particularly in the remote work environment necessitated by the public health crisis of 2020–2021, the team operated according to a "network model." This mode, according to Nancy L. Maron and Sarah Pickle, "might have a strong hub at its center …and many nodes supplying specialist support as needed."[21] At Macalester, the team is distributed across campus, incorporating Information Technology Services (including Academic Information Associates), the Library, Civic Engagement Center, geography, and the Jan Serie Center for Scholarship and Teaching.

In addition to these staff, several spaces on campus support digital liberal arts work in different ways. Spaces that serve the campus more broadly and are located within the campus library are the Digital Resource Center (DRC)

and the Idea Lab, one of which has undergone substantial changes over the past few years, the other of which is new to campus. Spaces with more limited access include the Interdisciplinary Media Lab, a video production space that is primarily used by students and faculty in the media and cultural studies department (MCST), and lab spaces in theater, art, and the sciences.

The Idea Lab, located on the second floor of the library, is managed through the Department of Entrepreneurship and Innovation in collaboration with the DRC, and is staffed by twenty student workers, and currently two full time staff. The Idea Lab provides access to a variety of tools within the space, including sewing and craft supplies, computers with specialized software, vinyl cutters and more—and regularly hosts small low-stakes student-hosted workshops along with larger workshops. This space opened following a renovation of the second floor of the library in summer 2017, although in-person activities were suspended between March 2020 and the start of the 2021/22 academic year. The space provides a location for the intersection of the digital and physical, is regularly used as part of courses in brainstorming and planning projects, and is heavily used by students.

Situated departmentally in ITS, the Digital Resource Center has evolved considerably since it was established in the 1980's as a language lab. The DRC is now a part of Academic Technology Services alongside the Academic Information Associates, and provides access to audiovisual equipment and laptop checkout, 3D printing and scanning support, poster printing, audio recording space, and flexible computer lab space with specialized software. The DRC is staffed by 30 student workers who are trained in software and hardware to support classroom projects and is managed by one full time staff member. In addition to the DRC Manager, the current DLA Director and postdoc also have offices in the DRC.

Within the past five years, the DRC space and staffing have undergone considerable changes. Since 2017, departmental reorganizations within ITS shifted the management of the space from a combined role of Academic Information Associate and DRC manager to AV event support and DRC manager, and then finally to a dedicated DRC manager role in 2019; during this same period, a total of four different individuals served in these roles, creating both instability in student worker training and support for the campus. This also presented the opportunity to reimagine the space and how it served campus. In 2019, an initial proposal titled "Reimagining the Digital Resource Center" began the process of rethinking both location and services of the space, with the library identified as a strong candidate for relocation. This initial proposal led to a larger plan and proposal focused

on reimagining the whole library space, and ideally co-locating the DRC and Idea Lab within the library. Preliminary planning for this project was completed in March 2020, just prior to the campus shutdown at Macalester due to the COVID-19 pandemic lockdown. These plans were postponed indefinitely as the scope of the pandemic grew, but space constraints in the existing DRC and the desire for stronger collaboration between the library, DLA, and DRC remained.

In January 2021, ITS, the Library, and Facilities Services began the process of planning a move of the DRC into existing spaces in the library's lower level with minimal renovation focused on increasing ventilation for 3D printing. The DRC moved in May 2021 into a restructured space and location in the library. The new space better supports the reimagined mission of the DRC to "provide students, faculty, and staff with access to expertise and resources to help them actively engage with technology as a part of their scholarly lives." The library is more centrally located at the heart of campus, and the DRC space offers one physical location that can serve as a hub for DLA on campus. The director of the library sees these shared spaces and cooperative programming as helping to create collaborations rather than having them reside within the library as "tenants." However, Macalester's DLA team has not been able to leverage these third spaces yet in a comprehensive fashion, in large part due to the pandemic.

Working in the Now, Looking to the Future

In the past several months, the Macalester DLA Team has had to adjust plans and rearrange budget lines to respond to the realities of a global pandemic. Travel was impossible, and many conferences and workshops were (and still are) held in a virtual environment, so some of our faculty and staff development funds were left unspent. These challenges did offer space for some new ideas and opportunities to emerge. For one, the DLA Team decided to implement a new Faculty Fellows initiative during the 21/22 year. The DLA Faculty Fellowship is a cohort-based opportunity designed to enhance DLA capacity in relationship to faculty scholarship and teaching. This year, the program received seven promising applications, and was able to fund a larger cohort than initially imagined. The first cohort features a range of projects, including an interactive web-based documentary, an open educational resource for figure drawing, a virtual exhibition, and a digital map featuring interviews about a regional park. Our hope is that Faculty Fellows, through exchanging ideas, resources, and questions with one another, will begin to

foster a support system that will help sustain this work moving forward. The program will also further increase the visibility of DLA on our campus and perhaps persuade stakeholders that the broader initiative is worthy of permanent funding.

The DLA Team also created and deployed workflow guidelines and a "stock agenda" for faculty consultations, partly through delving deeply into a 3D workflow plan during the Institute for Liberal Arts Digital Scholarship (ILiADS) in Summer 2021. These guidelines are intended to not only facilitate individual meetings between staff and faculty, but to provide a template that can be used and understood across disciplinary boundaries. With this template, we hope that the labor necessitated by these ambitious curricular and scholarly projects is distributed more evenly and can be replicated and iterated by individuals who may join the institution in years to come.

However, we find ourselves in a largely reactive place still, having had little time for reflection and planning, especially since March 2020. Workflow guidelines do not resolve the staffing issues and concerns about compensation, but the guidelines do mitigate some of the concerns about uneven labor distribution and ongoing sustainability of portions of the support ecosystem. While constant iteration and states of perpetual beta testing can bring forward great creativity, when done in a continually reactive mode, it only creates disjointedness. There is much still to do to nurture an ecosystem of support that sustains and acknowledges staff expertise, collaborative relationships, and innovative programming. This brings us to where we are now, with many more questions than answers.

Going forward, are there meaningful ways to incorporate other campus conversations and needs such as the relationships between DLA teaching methods and community engaged research, textbook affordability issues, or inclusivity in pedagogies and course materials? We are inspired by Caro Pinto and her assertion, "We should disrupt toward solidarity and innovate toward communities of practice in the digital humanities."[22] These connections could center DLA as a curricular nexus where students learn to think critically and communicate multi-modally, all while embracing educational practices and learning methodologies that work toward a more inclusive and equitable learning environment. For example, Michigan State University has found ways to connect multiple learning-focused initiatives in creating the Lab for the Education and Advancement in Digital Research (LEADR) which focuses on the learning objectives of information literacy, digital literacy, data literacy, and computational analysis.[23] At our institution, this type of framework could help tie together other collaborative and curriculum

support already happening at the staff level of the library and ITS; it may be a productive way to frame, model, and scope collaborative instruction and projects with campus instructors. In addition, it may help raise awareness at the administrative level to DLA successes, in terms of measurable student learning. Meanwhile, individuals doing the work could build the collaborative infrastructure and garner institutional support needed to scale and sustain DLA work. Perhaps in framing DLA within these other closely related topics and growing campus concerns, we can make the case for the digital liberal arts as a campus-wide initiative.

Postscript

This chapter, originally drafted in late 2021, addresses the challenge of precarious labor and the sustainability issues that are inherent to grant-funded programs. In the intervening time to press, the authors are happy to report that Macalester College has made an institutional commitment to DLA by creating a new and permanent position in the library. This position, DLA Librarian and Program Manager, reports to the Director of the Library and facilitates the work of the DLA Team, coordinates the Faculty Fellows program, and teaches two courses a year. In addition, the college provides support for two part-time DLA Student Assistants, who help manage the DLA Newsletter, support faculty projects, and lead student-focused workshops. Macalester's commitment has allowed DLA to continue its work with the network model described in the chapter. Also in the time since this chapter was written, Macalester not only lifted their suspension of the College's retirement contribution, but also reimbursed employee retirement accounts for any contributions lost during the suspension.

Notes

1. Several articles have been written about labor issues within the digital humanities and especially as they relate to the library, including Trevor Muñoz's "Digital Humanities in the Library Isn't a Service," *Gist*, August 19, 2012, https://gist.github.com/trevormunoz/3415438; Paige Morgan's "Not Your DH Teddy-Bear; or, Emotional Labor Is Not Going Away," *Dh+lib*, July 2016, https://acrl.ala.org/dh/2016/07/29/not-your-dh-teddy-bear/; and Roopika Risam's "Diversity Work and Digital Carework in Higher Education," *First Monday*, March 1, 2018, https://doi.org/10.5210/fm.v23i3.8241. In addition, the following have been seminal in describing burnout and vocational awe: Fobazi Ettarh's "Vocational Awe and Librarianship: The Lies We Tell Ourselves," *In The Library With The Lead Pipe*, January 10, 2018, https://www.inthelibrarywiththeleadpipe.org/2018/vocational-awe/; April Hathcock's, "Let Labor Be Labor," *At The Intersection* (blog), May 12, 2016, https://aprilhathcock.wordpress.com/2016/05/12/

let-labor-be-labor/; and Kaetrena Davis Kendrick and Ione T. Damasco's "Low Morale in Ethnic and Racial Minority Academic Librarians: An Experiential Study," *Library Trends* 68, no. 2 (2019): 174–212, https://doi.org/10.1353/lib.2019.0036.

2. "Mission and History," Macalester College, accessed December 21, 2021, https://www.macalester.edu/about/mission/.
3. "The Liberal Arts in the Digital Age: Macalester College," Andrew W. Mellon Foundation, December 4, 2014, www.mellon.org/grants/grants-database/grants/macalester-college/41400641. The first Mellon grant, managed by the Jan Serie Center for Scholarship and Teaching, supported the work of a faculty director and postdoctoral fellow for three years, and raised awareness of DLA on campus. The director and postdoc as well as librarians, campus educational technologists, and staff from the Civic Engagement Center and geography department consulted on various digital projects, and began to organize as a DLA working group. On the heels of "The Liberal Arts in the Digital Age," the College received a "Making Meaning" grant in 2018 partly to further the work of DLA. This second Mellon grant also supports a faculty director and postdoctoral fellow, in addition to faculty research and staff professional development grants. Further, the hope with this second influx of funding has been to develop a more sustainable DLA program that may persist beyond this grant cycle.
4. William Pannapacker, "Stop Calling It 'Digital Humanities,'" *Chronicle of Higher Education*, February 8, 2013, http://chronicle.com/article/Stop-Calling-ItDigital/137325.
5. "Detail for CIP Code 30.5202: Digital Humanities," CIP: The Classification of Instructional Programs, National Center for Education Statistics, last modified 2020, https://nces.ed.gov/ipeds/cipcode/cipdetail.aspx?y=56&cipid=93061.
6. Miriam Posner, "Money and Time," Miriam Posner's Blog, March 14, 2016, https://miriamposner.com/blog/money-and-time/.
7. Christina Boyles et al., "Precarious Labor and the Digital Humanities," *American Quarterly* 70, no. 3 (2018): 698, https://doi.org/doi:10.1353/aq.2018.0054.
8. Nancy L. Maron and Sarah Pickle, "Sustaining the Digital Humanities: Host Institution Support beyond the Start-Up Phase," *Ithaka S+R*, June 18, 2014, 49, https://sr.ithaka.org/wp-content/uploads/2015/08/SR_Supporting_Digital_Humanities_20140618f.pdf.
9. Miriam Posner, "No Half Measures: Overcoming Common Challenges to Doing Digital Humanities in the Library," *Journal of Library Administration* 53, no. 1 (January 1, 2013): 43–52, https://doi.org/10.1080/01930826.2013.756694.
10. Jennifer Vinopal and Monica McCormick, "Supporting Digital Scholarship in Research Libraries: Scalability and Sustainability," *Journal of Library Administration* 53, no. 1 (January 1, 2013): 15, https://doi.org/10.1080/01930826.2013.756689.
11. Alexander Fangmann, "Budget Shortfalls and Planned Cuts Threaten Higher Education Across the US," *World Socialist Web Site*, May 16, 2020, https://www.wsws.org/en/articles/2020/05/16/bud2-m16.html.
12. Amanda Achin, "Interview with Rich Levy: Fighting Austerity in Higher Education," *Educators for a Democratic Union* (blog), April 27, 2020, http://educatorsforademocraticunion.org/interview-fighting-austerity-in-higher-education.
13. Laura Czerniewicz, "The Struggle to Save and Remake Public Higher Education," *University World News* (blog), April 30, 2020, https://www.universityworldnews.com/post.php?story=20200428154746989.

14. Dian Schaffhauser, "25 Ed Tech Predictions for 2021," *Campus Technology*, 2021, https://campustechnology.com/articles/2021/01/04/25-ed-tech-predictions-for-2021.aspx.
15. D. Christopher Brooks and Mark McCormack, "Driving Digital Transformation in Higher Education," ECAR Research Report, Louisville, CO: ECAR, June 2020, https://www.educause.edu/ecar/research-publications/driving-digital-transformation-in-higher-education/2020/introduction-and-key-findings.
16. Susan Grajek, "EDUCAUSE QuickPoll Results: IT Budgets, 2020–21," *EDUCAUSE Review*, October 2, 2020, https://er.educause.edu/blogs/2020/10/educause-quickpoll-results-it-budgets-2020-21.
17. "Professional Salaries," HigherEdJobs, accessed December 18, 2021, https://www.higheredjobs.com/salary/salaryDisplay.cfm?SurveyID=54.
18. Trevor Muñoz, "Digital Humanities in the Library Isn't a Service," *Gist*, August 19, 2012, https://gist.github.com/trevormunoz/3415438; Posner, "No Half Measures."
19. Alexis Logsdon, Amy Mars, and Heather Tompkins, "Claiming Expertise from Betwixt and Between: Digital Humanities Librarians, Emotional Labor, and Genre Theory," *College & Undergraduate Libraries* 24, no. 2–4 (June 5, 2017): 155–70, https://doi.org/10.1080/10691316.2017.1326862.
20. Posner, "No Half Measures."
21. Maron and Pickle, "Sustaining the Digital Humanities," 34.
22. Caro Pinto, "Construction and Disruption: Building Communities of Practice, Queering Subject Liaisons," in *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, ed. Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb (Chicago: Association of College & Research Libraries, 2015), 49, http://www.ala.org/acrl/sites/ala.org.acrl/files/content/publications/booksanddigitalresources/digital/9780838987681_humanities_OA.pdf.
23. Brandon T. Locke, "Digital Humanities Pedagogy as Essential Liberal Education: A Framework for Curriculum Development," *Digital Humanities Quarterly* 11, no. 3 (2017), http://www.digitalhumanities.org/dhq/vol/11/3/000303/000303.html.

Bibliography

Achin, Amanda. "Interview with Rich Levy: Fighting Austerity in Higher Education." Educators for a Democratic Union (blog), April 27, 2020. http://educatorsforademocraticunion.org/interview-fighting-austerity-in-higher-education.

Andrew W. Mellon Foundation. "The Liberal Arts in the Digital Age: Macalester College." The Andrew W. Mellon Foundation, December 4, 2014. www.mellon.org/grants/grants-database/grants/macalester-college/41400641.

Boyles, Christina, Anne Cong-Huyen, Carrie Johnston, Jim McGrath, and Amanda Phillips. "Precarious Labor and the Digital Humanities." *American Quarterly* 70, no. 3 (2018): 693–700. https://doi.org/doi:10.1353/aq.2018.0054.

Brooks, D. Christopher, and Mark McCormack. "Driving Digital Transformation in Higher Education." ECAR Research Report. Louisville, CO: ECAR, June 2020. https://www.educause.edu/ecar/research-publications/driving-digital-transformation-in-higher-education/2020/introduction-and-key-findings.

Czerniewicz, Laura. "The Struggle to Save and Remake Public Higher Education." *University World News* (blog), April 30, 2020. https://www.universityworldnews.com/post.php?story=20200428154746989.

Fangmann, Alexander. "Budget Shortfalls and Planned Cuts Threaten Higher Education Across the US." World Socialist Web Site, May 16, 2020. https://www.wsws.org/en/articles/2020/05/16/bud2-m16.html.

Grajek, Susan. "EDUCAUSE QuickPoll Results: IT Budgets, 2020–21." EDUCAUSE Review, October 2, 2020. https://er.educause.edu/blogs/2020/10/educause-quickpoll-results-it-budgets-2020-21.

HigherEdJobs. "Professional Salaries." Accessed December 18, 2021. https://www.higheredjobs.com/salary/salaryDisplay.cfm?SurveyID=54.

Locke, Brandon T. "Digital Humanities Pedagogy as Essential Liberal Education: A Framework for Curriculum Development." *Digital Humanities Quarterly* 11, no. 3 (2017). http://www.digitalhumanities.org/dhq/vol/11/3/000303/000303.html.

Logsdon, Alexis, Amy Mars, and Heather Tompkins. "Claiming Expertise from Betwixt and Between: Digital Humanities Librarians, Emotional Labor, and Genre Theory." *College & Undergraduate Libraries* 24, no. 2–4 (June 5, 2017): 155–70. https://doi.org/10.1080/10691316.2017.1326862.

"Mission and History." Macalester College, accessed December 21, 2021, https://www.macalester.edu/about/mission/.

Maron, Nancy L., and Sarah Pickle. "Sustaining the Digital Humanities: Host Institution Support beyond the Start-Up Phase." *Ithaka S+R*, June 18, 2014. https://sr.ithaka.org/wp-content/uploads/2015/08/SR_Supporting_Digital_Humanities_20140618f.pdf.

Muñoz, Trevor. "Digital Humanities in the Library Isn't a Service." *Gist*, August 19, 2012. https://gist.github.com/trevormunoz/3415438.

National Center for Education Statistics. "Detail for CIP Code 30.5202: Digital Humanities." CIP: The Classification of Instructional Programs. Last modified 2020. https://nces.ed.gov/ipeds/cipcode/cipdetail.aspx?y=56&cipid=93061.

Pannapacker, William. "Stop Calling It 'Digital Humanities.'" *Chronicle of Higher Education*, February 8, 2013. http://chronicle.com/article/Stop-Calling-ItDigital/137325.

Pinto, Caro. "Construction and Disruption: Building Communities of Practice, Queering Subject Liaisons." In *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*, edited by Arianne Hartsell-Gundy, Laura Braunstein, and Liorah Golomb, 39–51. Chicago: Association of College & Research Libraries, 2015. http://www.ala.org/acrl/sites/ala.org.acrl/files/content/publications/booksanddigitalresources/digital/9780838987681_humanities_OA.pdf.

Posner, Miriam. "Money and Time." *Miriam Posner's Blog: Digital Humanities, Data, Labor, and Information* (blog), March 14, 2016. https://miriamposner.com/blog/money-and-time/.

Posner, Miriam. "No Half Measures: Overcoming Common Challenges to Doing Digital Humanities in the Library." *Journal of Library Administration* 53, no. 1 (January 1, 2013): 43–52. https://doi.org/10.1080/01930826.2013.756694.

Schaffhauser, Dian. "25 Ed Tech Predictions for 2021." Campus Technology, 2021. https://campustechnology.com/articles/2021/01/04/25-ed-tech-predictions-for-2021.aspx.

Vinopal, Jennifer, and Monica McCormick. "Supporting Digital Scholarship in Research Libraries: Scalability and Sustainability." *Journal of Library Administration* 53, no. 1 (January 1, 2013): 27–42. https://doi.org/10.1080/01930826.2013.756689.

CHAPTER THIRTEEN

The Digital Humanities Summer Scholars Program:

Opportunities for Change During the COVID-19 Pandemic and Providing a Space for Diversity, Equity, and Inclusion in the Library

Angela Perkins

Introduction

"What do you know about Digital Humanities (DH) or digital scholarship?" This is a question that I ask every time I interview undergraduate students for the Digital Humanities Summer Scholars (DHSS) program, a six-week intensive learning experience consisting of seven to eight students (termed "Summer Scholars" once they join their new cohort) focused on learning how to conduct original research within a DH theoretical framework at the Lafayette College Libraries. The DHSS program is funded by the Lafayette Libraries, and the Summer Scholars receive a stipend and coverage of the cost of their summer housing for the duration of the program.[1] Since 2018, I have been directing the DHSS program at Lafayette College, a small liberal arts college located in Easton, PA, with about 2,600 students ranked 40th among national liberal arts colleges by U.S. News & World Report.[2]

During the COVID-19 pandemic in 2020, Lafayette switched to all-virtual, remote learning in the middle of the spring semester.[3] After the College's approval of conducting summer research programs remotely, I had to pose the question to myself that I usually ask our students when I was given the option to either put the DHSS program (which was slated to run from May 19 to June 26) on hold, or continue remotely, with all the difficulties that

CC BY NC ND

would entail. With a little over a month to prepare, I decided that not only should we go ahead remotely, but that the DHSS program was uniquely positioned to succeed despite its historically in-person format, due to its inherent relationship to (and interrogation of) the digital space. This chapter will explore a) the DHSS program's pedagogical goals and how the program was conducted prior to the pandemic; b) the specific limitations that the DHSS program faced during the pandemic, including how to encourage and maintain group communication during an online program, versus a traditionally in-person experience; an inability to rely on solely synchronous instruction due to Zoom fatigue; as well as technological difficulties like unstable Wi-Fi connections off-campus; and c) how the program changed in order to achieve its goals and aims, accommodate the Summer Scholars' learning and research needs, and foster collaboration and community based in DH principles. As Simon Mahony and Elena Pierazzo declare in their book chapter "Teaching Skills or Teaching Methodology?" in the book, *Digital Humanities Pedagogy: Practices, Principles and Politics*, DH curriculum ideally should "focus on teaching students new approaches and new ways of thinking about the humanities and—in order to accomplish this with different groups of learners at disparate levels—that there is a need for teaching methodological approaches and not simply technological skills."[4] The DHSS program is designed to help Summer Scholars to "develop their collaborative and interdisciplinary skills—skills that are increasingly required within and outside of the academy. Thus there is a real need for training students in collaborative methods and reflective practices in order to build a community of learning that will lead to a community of practice."[5] Because of these values, all candidates for the DHSS program, regardless of acceptance status, are invited to join the Lafayette Libraries' Digital Humanities Community of Practice[6] so that they are able to learn about DH and digital scholarship events, and continue to nurture their interest in DH by learning from active members of the community on campus.

Additionally, being an early career librarian and the only Black woman librarian at my institution informs my direction of the DHSS program, in my interaction with students through recruitment, their acceptance into the DHSS cohort, facilitating execution of their DH projects, and beyond. One of the most important aspects of the DHSS program, and one that I know that my fellow librarians, faculty, and other DHSS collaborators are most proud of, is the diversity of the Summer Scholars' final digital research projects; the multilayered research interests embedded in the DHSS projects truly

demonstrates the Libraries' (and the College's) commitment to diversity, equity, and inclusion as a valued priority in research, teaching, and learning.

As director of the DHSS program, I try to engineer a space in which students are not only empowered by me but each other—especially each other. I am someone who interacts with them, works alongside them, and I am often learning just as much as they are. To ensure that the Summer Scholars' research journeys and subsequent digital research projects are well served, gaps between the library and faculty and academic departments are bridged through helping to connect students with those faculty, librarians, and staff who share their research interests, and/or have a facility with the kind of digital humanities and/or digital scholarship they want to pursue. It is not required for Summer Scholars to have a faculty mentor while in the program, but identifying and approaching a faculty member who can provide expert knowledge in their research topic is highly encouraged. Faculty members also provide consistent support to the DHSS program by sharing their particular DH experience with the Summer Scholars in classes and workshops, from tutorials on digital tools they specialize in using to presenting on DH research projects they have spearheaded. The participating faculty who teach during the DHSS program provide valuable insight into the intellectual reasons why a Summer Scholar may choose a certain DH methodology or digital tool for their own research, and how to get started once they choose, lowering the barrier to entry some students may feel when thinking about how to conduct research in a DH framework. Collaborating with my colleagues as they assist in instruction in workshops allows me to keep developing the program's pedagogical and philosophical foundations.

This chapter aims to be of interest to those working at the intersection of libraries and DH who would like to start these kinds of programs at their own institutions, in their own libraries, but may lack access to others who have successfully coordinated these kinds of efforts. Roopika Risam, Justin Snow, and Susan Edwards of Berry Library, Salem State University, in Salem, Massachusetts, articulate this struggle in the introduction to their article, "Building an Ethical Digital Humanities Community: Librarian, Faculty, and Student Collaboration":

> The lack of attention to undergraduate education in digital humanities is a glaring omission that has left digital humanities practitioners who work primarily with undergraduate students struggling to adapt scholarly practices envisioned for graduate students to the fundamentally different needs of undergraduate

> populations. While collections like Hirsch's Digital Humanities Pedagogy[7] include essays on undergraduate digital humanities work, and a forthcoming issue of *Digital Humanities Quarterly* will focus specifically on undergraduates, scholarship has tended to focus primarily on the needs of graduate student teaching and training. An additional gap involves conversations around pedagogical instruction; the focus, instead, is on building robust infrastructure for digital humanities training…. There are logical reasons for this: (1) the infrastructure-heavy dimensions of digital humanities; (2) the specialized nature of faculty research that offers important professional development skills for graduate students and can require graduate-level training; and (3) the worsening humanities academic job market, which has led to an ethical need for value-added skills in humanities PhD programs.[8] However, it leaves open the question of how to foster the growth of digital humanities initiatives for undergraduate students.[9]

These previously existing difficulties, combined with sudden adjustment to remote learning due to the pandemic, show that Coursera CEO Jeff Maggioncalda's thought quoted in *Inside Higher Ed* at the start of the pandemic probably resonates with many librarians seeking to establish effective undergraduate DH programs at their institutions: "With more than 400 million students disrupted due to the spread of COVID-19, we are experiencing a watershed moment for education systems around the world. It tragically illustrates the need for higher ed institutions to build a technological backbone and digital competency to weather this crisis and to enter a new era of teaching and learning in a digital world."[10]

History and Evolution of the DHSS Program

The DHSS program began as a result of an Andrew Mellon Grant received by the Lafayette Libraries in 2012, which focused on raising the profile of DH on campus. The program was initially designed as the Libraries' first undergraduate internship for students to pursue their own research under the supervision of a Council on Library and Information Resources (CLIR) postdoctoral fellow at the Libraries in spring and summer 2015.[11] Sarah Morris, former director of DHSS, describes CLIR fellow Emily McGinn's intent for the program in "The Digital Humanities Summer Scholarship: A Model For Library-Led Undergraduate Digital Scholarship":

> The program was designed for an original goal of finding a way to value independent student work and exploration. It aimed to "give students' ideas the space and support they deserve" and also to "create something where students felt free to experiment, fail, adapt, and grow" (McGinn, pers. comm.). When creating the program, the developer investigated other programs that existed at liberal arts colleges but did not find a model that fit perfectly with her vision—many of the models either had students contributing to an ongoing project or had them explore tools without ultimate application. Desiring something that allowed more student initiative, she used an existing program for student research at Lafayette, the Excel Scholar Program, as a blueprint.[12]

The EXCEL Scholars Program is an opportunity which "enables high-performing students to assist faculty members with their scholarship,"[13] and has held an important, long-standing distinction for Lafayette students as a way to engage in research on campus. It was the existence of the EXCEL program that made it possible to make the case that the DHSS program could be an option for Lafayette students to pursue original research.[14]

Upholding standards of intellectual rigor is a guiding force in design of the program. This is accomplished through emphasizing the elements of research; establishing foundations in information/digital/media literacy; encouraging experimentation with transformative digital technology in the form of DH software and tools, and/or digitization leading to access to "hidden" or underrepresented archives thereby addressing archival silences; and promoting diverse ideas, encouraging interdisciplinary, multimodal research methods and methodology.

The best practices of project management, including transparency through documentation throughout the research process, are also fundamental, allowing the Summer Scholars to adopt an agile, flexible approach to the design of their digital research projects, as well as gain transferable skills that will be beneficial to them in their careers. The above principles allowed and encouraged me to conceive of converting the DHSS program from an intimate, in-person experience to a fully remote one.

The availability of an original research program on the Lafayette College campus during the COVID-19 pandemic lockdown continued the Libraries' commitment to being a place for curious, enthusiastic students of all kinds to learn the craft of research and nurture their own scholarship aims and goals. The DHSS program recruitment period began in January 2020,

pre-pandemic, and accrued thirty-seven interviews (students are required to participate in a brief interview prior to application, and are asked to give a little information about themselves, their academic goals, their major, and their research interests and/or potential DHSS project idea), with eight Summer Scholars chosen by April 2020, out of thirty-one applicants. The Summer Scholars received a $2000 stipend, but coverage of their housing for the duration of the program was eliminated; summer housing for students on campus was almost totally curtailed because of the COVID-19 pandemic policies at that time. A one-year deferral of entrance to the program was offered to accepted students, and one accepted student did defer their participation in the program until DHSS 2021.

The DHSS program is a six-week summer program, with the ultimate goal of the completion and presentation of a final digital project based on a student's original research. Students do not currently receive academic credit for participating in the program. Lectures, workshops, and labs are led by myself and participating Lafayette College librarians (e.g. Director, Research & Instructional Services; Head of Electronic Resources; Director, Digital Scholarship Services, etc.), staff, and faculty. I approach the program's design as a community-oriented, unit-based introduction to conducting research and to DH methods and methodologies. These units include understanding the elements of research, refining the research question, building a research argument, and writing annotated bibliographies and literature reviews; evaluating digital research projects; introduction to mapping, spatial humanities, and GIS; introduction to text analysis, topic modeling, and data visualization; introduction to network analysis; and introduction to digital storytelling and digital art.

DH practitioners outside of Lafayette are invited to share insights on their work by leading workshops. While most of the guests of DHSS visited in person to conduct their workshops or talks, it was a 2019 visit held on Skype by Dr. Michael Witmore, director of the Folger Shakespeare Library in Washington, DC (a workshop regarding his use of text analysis with Shakespeare's plays as his dataset[15]) that proved to me it was possible to offer unique instruction in DH in a completely virtual learning environment.

The DHSS director and the Summer Scholars meet for the class lecture or workshop in the morning three times per week (usually MWF), and a three-hour lab two times per week (WF) in the afternoon. The initial iterations of the DHSS program met for less time; when I started directing the program, I conducted informal interviews with former Summer Scholars, mainly from the 2017 cohort, and a response that I repeatedly received was

that they wished that they had more opportunity to meet with the director and their fellow Summer Scholars as a group. These former Summer Scholars mentioned that they felt there was a time right in the middle of the six weeks when they were creating their digital projects where they felt adrift, lonely, and sometimes discouraged. My strategy to avoid these grievances and pitfalls was to increase the number of meetings to multiple times per week, a couple of times a day, with the classes and workshops being mandatory, and attendance of the three-hour labs being highly encouraged.

The labs, in my conception, serve as a period when the Summer Scholars know they can come and meet with me, the DH Teaching Fellow, and/or their fellow Summer Scholars and ask questions, have conversations pertaining to their research, or even just relax and socialize. The DH Teaching Fellow, a teaching assistant position I created in 2019 specifically for former Summer Scholars, was also another way for them to communicate any comments, questions, or issues during the course of creating their digital project during the program. I will explain more about this position in the following section regarding diversity, equity, and inclusion (DEI) in the library funded through DH research.

Most of all, the labs are arenas for working out research questions, the bedrock of each Summer Scholar's digital research project. The research question is examined continually in the program; lab time is optional since the Summer Scholars should feel free to take the time to seek the information they need, whether it be physical information in the library, or online within databases or digital archives. Many times, after the Summer Scholars find the literature they need to begin answering or speaking to their research questions, they realize the question has changed, and they need a sounding board through which they can recalibrate and get back on track again. Lab time is often repurposed for workshops when we are lucky to have a surfeit of guest talks, but generally, lab times are for the Summer Scholars to come together as a group, to build a sense of community appropriate for burgeoning practitioners of DH.

It was this need to set the stage for group collaboration which first came to mind when thinking about the obstacles that the COVID-19 pandemic lockdown would bring to conducting the DHSS program virtually. Is it possible to effectively facilitate group learning and collaboration in research in a virtual setting? To negotiate the drawbacks of nurturing group collaboration and teaching a complex and dense topic like DH research remotely, it was necessary to contemplate communication: how I was going to communicate

with the Summer Scholars, how they would communicate with me, and how they would communicate with each other as a group?

For the DHSS 2020 cohort, classes, workshops, and labs were delivered synchronously, in the interest of meeting as a group, and asking and answering questions in real time. This is especially important in the discussion-rich first two weeks of the program focused on revising the research question, and establishing the fundamental steps of the research process.[16] The required work assigned in the program—including the weekly readings associated with each thematic unit, accompanied by discussion questions which guide the synchronous group discussions; weekly journal entries which compel the Summer Scholars to not only maintain project organization and an ethical transparency about their research process, but allow each to share their respective journeys; and the writing and revising of annotated bibliographies and literature reviews—is dense and scaffolded. Each Summer Scholar absorbs the glut of information and resources provided at their own pace, learning-while-doing as they go through the steps of creating their digital research projects. The Summer Scholars are always comprised of a group of students with widely differing levels of experience with both traditional research and DH or digital scholarship; it is possible that some of the cohort have not even written a traditional research paper. Simulating the flow of an in-person class or workshop gives them opportunities to voice confusion and then benefit from their fellow Summer Scholars' insights.

Our initial mode of having classes, workshops, and labs was by videoconferencing software Google Meet. We switched to the Zoom platform after the campuswide introduction of professional Zoom accounts. With the three-hour lab, I made the commitment to keep an open room on Zoom where I was always available for questions and comments. The Summer Scholars themselves were also invited to stay in this Zoom space and work on their projects, so they were also available to their cohort. Many of the Summer Scholars did choose to stay and used the opportunity to ask questions or present challenges that they were facing with their projects to myself and the group. Breakout rooms were crucial for one-on-one conferences the Summer Scholars preferred were private.

While running the synchronous sessions, online teaching made achieving classroom flow much harder than in person. Learning how to best use the features of both platforms, especially breakout rooms, was somewhat of a learning curve for me, and technical issues such as my and some of the Summer Scholars having poor Wi-Fi connections inhibited some of the efficacy and rhythm of our meetings. Additionally, we were in different time zones,

including a group member located in England, who was joining us on Zoom quite early in the morning most meeting days. It was technical and logistical issues like these, in addition to physical, emotional, and mental health problems, that contributed to the encroaching Zoom fatigue from logging many virtual hours during the week in our classes, workshops, and labs.

The key to combating Zoom fatigue, in addition to many physical breaks in between sessions, was having multiple ways for all of us to communicate outside of that environment. Preparing for the DHSS program, I knew that a forum must be created for the Summer Scholars to be able to visit and communicate with me and each other more easily. The introduction of the Slack platform was an important new piece of forming an atmosphere where it was possible to speak with each other instantly, if necessary. The Slack app was also useful as storage for the electronic files of the weekly readings (as a library outside of the DHSS program Spaces page, where the syllabus, calendar, and schedule are located for reference. Lafayette College uses the Moodle learning management system, and Spaces is a Moodle sub-page function), as well as for submitting discussion questions based on the readings. We used Slack for the required weekly journal entries, with individual threads designated for each Summer Scholar, where it was possible to see and comment on each other's journal entries. Slack was an easy-to-use and quick tool to adopt alongside the usual means of communication, such as the DHSS Spaces page, email, and phone.

The 2020 DH Teaching Fellow also took the initiative to hold her own weekly Zoom office hours where the Summer Scholars were invited to have consultations with her regarding their research and the creation of their digital projects.

Even with all these modes of contact, emotional, mental, and physical stress was unavoidable for the Summer Scholars during the first summer of the COVID-19 pandemic. Simply communicating more, or better, was sometimes not enough to prevent work slowing down, work stopping completely, absences from class, and illness, among other personal issues.[17] It was sometimes difficult to monitor or intuit whether or not a Summer Scholar was on the right track with their research question, choice of method and methodology, and/or formatting of their research project when the Summer Scholar was offline. This was one of the main reasons that I required completion of both an annotated bibliography and a literature review, whereas in the past, I did not. Often, in person, it is possible to sense or spontaneously check in when the Summer Scholar is in obvious dismay or distress; spontaneity is less likely to occur online.

Apathy can lead to late-stage dissolution of the project, an outcome that I try to avoid at all costs in the program. The reason for this is less that I want or need them to complete a digital project, but that my true goal in the program is for the Summer Scholars to gain holistic experience in executing a DH research project in which failure is baked into the process; in fact, the research, and indeed, the digital project is never "completed" so much as always available to be thought about, revised, and updated as they continue in the quest to answer their research questions. As soon as the new DHSS cohort is assembled, I communicate this with them, to instill a shared value system, as well as build confidence that their research experience will be effective and positive by virtue of committing to weathering the ups and downs inherent within the research process. As 2015 and 2016 Summer Scholars, Ian Morse, Mila Temnyalova, and William Gordon noted in their chapter for the book, *Scholarship in the Sandbox*:

> A project is oftentimes not about the final result. The majority of research rarely makes it into publication or wider distribution. This learning—about the process, tools, the limitations, and the impossibilities—can be more valuable than an end product. Such was the case with our research as members of the Digital Humanities Summer Scholars (DHSS) program …the digital humanities (DH) is a relatively new field with a diverse community of scholars and practitioners. While there is not one single definition that encapsulates it, its spirit is best described with continual rethinking, questioning, and demonstrating through a plethora of projects and collaborations. If our own DHSS projects were judged only by our initial aims, they would be failures. Yet it was in that failure that we learned the most.[18]

Though remote instruction is sometimes onerous, the positives were numerous and carried us to the finish line of accomplishing our goals in the DHSS program that year. The Share Screen function within Zoom became a crucial tool in the teaching and learning process; other beneficial tools included being able to screenshot and easily record sessions with the participants' permission, and without ITS intervention, and have the recordings available for teaching and marketing purposes. Seeing more detailed demonstration of different digital tools by the invited guest experts, right on one's own screen, and everyone seeing the same images and process (versus in-person teaching where often the tool will be demonstrated to the class by using

a projection screen, then a break must be taken after the demonstration for each individual to try the tool at different paces on their own screens), allows a deeper understanding of what exactly the digital technology does, how it works mechanically, and how it may (or may not) relate to one's own DH project.[19]

One tutorial-based workshop that was held during the DHSS program in 2020 which flourished in the virtual environment was a presentation of Bucknell University's *Moravian Lives* DH project by the creators, Katherine Faull, Professor of Comparative and Digital Humanities, and Carrie Pirmann, Social Sciences Librarian at Bucknell's Bertrand Library.[20] In addition to explaining how the project was conceived—including describing the map interface which connects metadata about thousands of memoirs from the Moravian Church—the workshop also included a tutorial for the digital tool Transkribus, which is used to transcribe archival documents. This tutorial included not just a demonstration of Transkribus by the creators, but they also engaged the Summer Scholars in a bit of crowdsourcing by having them directly engage with the *Moravian Lives* website platform, and transcribe documents themselves using the tool. This was a unique opportunity to simultaneously understand how a DH project is created through instruction by experienced creators themselves, and actively participate in building a DH project up through learning how a digital tool impacts the research.

Another example of an effective virtual tutorial workshop in DHSS 2020 was Lafayette College Assistant Professor of International Affairs, Caleb Gallemore, and his topic modeling workshop, where he demonstrated the programming language, R. The workshop was focused on text as data, especially as a way to analyze how humans communicate meaning through socialization and discourse. Central to this idea is that there is much more text that is relevant to the project than is possible to read or account for with our own human brain.[21] The other side of text as data is the fact that unstructured data requires much cleaning and standardization because computers understand text as strings of characters like letters, numbers, symbols, etc. (also known as tokens), and these must be identical for the computer to understand they are the same. It is necessary for the researcher to clean the data so that it is comprehensible to the computer, and, in turn, useful for the researcher's analysis.[22]

Professor Gallemore showed the Scholars, through sharing his screen on Zoom, how to clean and refine a dataset using R, and use topic modelling, which, roughly defined, is to look at pairings of tokens across multiple documents to try to assess which topics they came from.[23] His workshop became

an opportunity for those Summer Scholars who were interested in using R, or a programming language like it, to connect and consult with an expert.

The Summer Scholars themselves, in each iteration of the DHSS program, are required to sign up for Tool Demo presentations about midway through, prior to when most make the final decision on what software platforms or tools that they will choose for their own digital research projects. The members of the DHSS cohort provided background information and a short demonstration of how digital tools such as Voyant, ArcGIS, R, Omeka, Scalar, and Tableau work, much like the described professional tutorials.

Lastly, the Scholars' final digital research project presentations—during the last week of the program—showcase not only the final products, but also the process of creating the digital research projects, including confirming DH methodology and tools; choosing software platform for presentation of data and data analysis; and format of presentation to the DH community. Pre-pandemic, the DHSS final presentations were held in-person for the Lafayette College community, including the Libraries and Lafayette faculty and staff. The program being entirely virtual in 2020 made it possible to hold the final presentations via Zoom, allowing for the guest list to expand beyond the Lafayette community out to members of the DH/digital scholarship community at regional institutions (i.e., Pennsylvania Consortium for the Liberal Arts and their Digital Scholarship Student Symposium; Bucknell University Digital Scholarship Conference, etc.), as well as friends and family. The final presentations were recorded within Zoom, and can now be made available to all those that could not attend the event at that time. The 2020 DHSS cohort was also invited to present their final digital projects at the Lafayette College Center for the Integration of Teaching, Learning, and Scholarship (CITLS) July 2020 virtual event, the Summer Research Colloquium.[24]

The DHSS Program: DEI in the Library through Funded DH Research

The DHSS program is able to provide a space for diversity, equity, and inclusion (DEI) to thrive within the student body and the Libraries through funded DH research, especially within the often complicated and difficult framework of the COVID-19 pandemic.

The DHSS Program, while selective, has fundamental academic and ethical values of access to information and intellectual freedom at its core in the process of recruiting and choosing the students who will make up the

new DHSS cohort. Varied and multidisciplinary research ideas—turning a critical eye toward sociopolitical and cultural events or phenomenon—are enriched by each student's choice to delve into DH research methods and methodology. In addition to the United States, a number of other countries are represented in the program, either by nationality of the Summer Scholar, or by research topic; for example, in DHSS 2020, these included Cambodia, Greece, the United Kingdom, Mongolia, Ethiopia, and Guyana. There is no requirement for a student to be majoring in any specific discipline or field, and they may be any class year as long as they are enrolled as a student at Lafayette during the summer that they wish to attend. This flexible eligibility leaves room open for surprise when it comes to the types of research questions proposed. As I told a journalist for *The Lafayette*, the student newspaper: "What I really enjoy about the program is learning that a student who has a major in one of those [STEM] fields has a really rich humanities interest underlying that, but they don't get to manifest it very often."[25] DH research project topics from the DHSS 2020 cohort included:

- Khmer classical dance in modern Cambodian pop music;
- The social geography of Lafayette College;
- Indo-Guyanese women poets;
- The "femme fatale" in film and the distortion of female criminality.[26]

My colleagues at the Lafayette Libraries and I expound on one of the above projects, *The Social Geography of Lafayette College*, in the journal article, "Using Digital Scholarship Methods and Tools to Enhance and Transform Learning during COVID-19 at a Liberal Arts College":

> An example of one of the varied, multimodal 2020 DHSS projects is *The Social Geography of Lafayette College*, which examined diversity and inclusion at the institution through analyzing data about Black spaces on campus using the theory of social geography. The scholar's dataset, consisting of historical events and locations such as buildings on campus concerning Black [students and other] students of color, was collected and compiled into an Excel spreadsheet using information from the Association of Black Collegians (ABC) Archives, housed at Lafayette Libraries' Special Collections and College Archives. After refining and analyzing the data, it was then transferred to a map created using the digital tool StoryMap JS by the Knight Lab at Northwestern University, the purpose of which was to

> understand Lafayette as a space in relationship to the Black student experience at the college.[27]

"Black Spaces at Lafayette College," the StoryMap created by Imane Halal ('23), then a first year student, expands upon the work she did as an intern on the ABC Archives project; the original concept for her project was simply a timeline, a bare bones version which she completed as a part of her work for the ABC Archives (the timeline itself was part of credit for an Africana Studies course titled *The Black Experience*). Halal describes at her project's website, "When I first began this project, I had a very different goal in mind. I thought I would be simply analyzing data about Black spaces at Lafayette… while doing my original annotated bibliography I realized I felt a disconnect to where my project, and the sources I had gathered in relation to processing the data." She goes on to detail the fact that she realized, through the theoretical frameworks of social geography and intersectional feminism, that particular spaces at Lafayette College are racialized and affect the spaces that Black and brown students are able to access and inhabit over the years.[28]

In the project's StoryMap, Halal details some of the history of the Black experience at Lafayette, including the first group of accepted Black male students (Lafayette College was incorporated through an act of the Commonwealth of Pennsylvania legislature, in 1826;[29] the 1832 class included Aaron Hoff, the first Black man to attend the College), and Black female students (the first group of Black women attended in 1970, the first year of acceptance of women students at the institution), and how this history affected the use of spaces (e.g., social events, the location of Black cultural houses, etc.) on campus.[30] The user is able to click on points on the map, exploring an interactive timeline which presents this information in the form of text and images digitized from the ABC Archives.

This DH project, along with its riveting final presentation at the virtual DHSS final presentations in June 2020, recorded via Zoom and made available publicly, spoke to the national cultural moment following the killing of George Floyd in police custody in Minneapolis, MN, on May 25, 2020,[31] and to the rising concern about racism and racial discrimination on campus.[32] Halal said of her project at her presentation, regarding the theoretical framework that she used to examine institutional racism and campus experiences of nonwhite students, "it's also important to understand what a racialized geography is and how that is experienced. In another study by [Adrian] Leftwich, he also goes on to state that racialized geography is experienced in daily routines as well, such as getting subtle stares, ambiguous comments, walking

from one class to another, eating in the cafeteria, and visiting a professor. So, it can be very overt, or also covert. And in this way, students, specifically Black students and students of color, experience a racialized campus, or a racialized geography."[33] Halal is quoted in the *Lafayette News* article, "Humanities, Digitized!" as being "not tech savvy," but is noted as having created two different sites, one focused on her research, and the other on the timeline of campus events, "changing mapping programs to best depict what she had."[34]

Greater access to scholarly information in the cultural record that is underappreciated, or may not have been made available otherwise, is an impactful result of the Summer Scholars' research in the short period of time that they work on their digital research projects. The Summer Scholars also become more deeply connected to the Libraries, not only through immersion in its vast physical and electronic resources, but also through participating in the Libraries' community through presenting their ideas, contributing to further scholarship, and even working.

The DH Teaching Fellow position is an annual opportunity for student employment, which we at the Libraries are always concerned with creating. The DH Teaching Fellow position specifically encourages a former Summer Scholar to continue their interest in DH and digital scholarship, as well as gain experience in coordinating a summer research program. The DHSS 2020 cohort was aided by my second DH Teaching Fellow, Milena Berestko ('22), an international student from Poland; as a 2019 Summer Scholar experienced in the process of the DHSS program and the execution of a successful digital research project, she was an invaluable resource to the new DHSS cohort as they embarked on their own digital research projects. As noted in *The Lafayette* student newspaper, Berestko's DHSS project, *The Mosaic of Romani World*, focused on her home country and its discriminatory relationship with the Romani people: "I see so many connections now to how so many minority groups are treated currently. And it's way easier for me to understand their treatment in the modern way when I have strong background in the Roma community," she remarked.

Enhancing the DHSS program community extended to keeping her own weekly office hours alongside mine; reviewing and giving feedback on the group's annotated bibliographies, literature reviews, and project plans; as well as creating a "How to Request a Mentor" mini-guide prior to the start of the 2020 DHSS program. The mini-guide was a wonderfully brief and well-written document based on her wish that she had approached a faculty mentor while she was a Summer Scholar; reasons cited as to why the Summer

Scholars should seriously consider approaching a mentor included help finding sources, connecting them with other experts, and narrowing the scope of their research.[35]

My own experience as a Black female professional librarian, with less than five years experience, influences my vision for the DHSS program. I am especially proud to maintain a safe and inviting space for students to further their academic and professional goals. Serving as one of the readers on his thesis committee, I helped a 2018 DH Summer Scholar, Uche Anomnachi ('19), mold his final DHSS research project, "Racial Perceptions of Anime in the United States,"[36] into his senior thesis, which subsequently earned him a place in graduate school at The University of Iowa to continue his research with a professor who shares the same interests. The DHSS program has the pleasure of having a new Rhodes Scholar in its 2019 cohort.[37] Victoria Puglia ('21) informed me that she did get the chance to talk about her experience in the DHSS program in her Rhodes interview, and why she thinks that DH is important and useful to her research. I, very recently, was asked by a Lafayette College professor to contribute my experience working with a 2021 Summer Scholar on their digital research project examining computational and mathematical methods in detecting partisan gerrymandering to a reference letter for the Harry S. Truman Scholarship for public service leadership granted to US undergraduate students.[38] Saide Singh ('22) defended her honors thesis, "Indo-Diasporic Women Poets"—an extrapolation of her DHSS 2020 project, "Indo-Guyanese Women Poets"— to the Lafayette College Department of English in May 2022, creating a website to accompany her defense.[39] Singh used the digital platform Scalar for her DHSS project, which she spoke about in "Humanities, Digitized!": "I had so much trouble with Scalar …I had an artistic, interactive, and grungy site … it was tedious, but once I was in a groove, I stayed with it."[40] Singh goes on to say that "there are so many savvy people at Lafayette who showed us how to use tools …and other students were on this same journey, and to see where they started and what they presented at the end was amazing. We learned so much so quickly."[41] Singh provides a vision of her post-DHSS future at her thesis website, "At the University of California, Santa Barbara, I am pursuing a dissertation about 'kali pani' poetry using transnational feminist and queer methods and an environmental studies lens to continue creating digital archives for new knowledge production."[42]

Making the essential elements of the DHSS program—learning how to do independent research and gaining contact with DH and digital scholarship—more equitable is what led me and our Digital Initiatives Librarian,

Janna Avon, to create the DHSS Idea Incubator, which is open to all Lafayette College students who would like to attend. The DHSS Idea Incubator is an interdepartmental partnership and series of instructional research workshops born of the DHSS program recruitment process for the 2020 DHSS cohort;[43] we realized when reviewing the DHSS applications that we saw a recurring pattern: students had trouble conceptualizing their projects due to the need for more knowledge in understanding research data and/or traditional elements of research. Developing available educational opportunities to support students in critical thinking with a research orientation, writing project pitches and research proposals, and building problem solving skills to execute research plans—as well as answer the increased interest of Lafayette students for virtual classes and educational opportunities over the winter interim session in January—pushed the launch of this project. The DHSS Idea Incubator also covers critical and cultural approaches and perspectives in DH that there is often not enough time to cover within the DHSS program itself, including Black DH, feminist, women's, and queer DH, etc.[44] As librarians who are both members of the Lafayette Libraries' Accessibility Team, we also introduce and stress the importance of becoming cognizant of accessibility issues in creating digital projects, and provide resources in The DHSS Idea Incubator in accessibility principles, including Lafayette Technology Help tools like SiteImprove for quality control and flagging accessibility issues.[45] We presented virtually at the Digital Library Federation's fall 2021 DLF Forum, our focus being on iterative design in formulating the DHSS Idea Incubator.[46] The first iteration of the DHSS Idea Incubator was held via Zoom during the interim session in January 2021; the second was conducted during spring semester 2021 and had mandatory attendance for the newly accepted DHSS cohort; and the newest workshops will take place in the afternoons on January 12–21, 2022.

Conclusion—The Future of the DHSS Program and DH at Lafayette College

For the foreseeable future, COVID-19 and the constraints of the pandemic lifestyle, including increased virtual instruction, will continue to be a part of our lives. Even as the campus returns to physical classroom instruction being the norm, and while I do anticipate taking the DHSS program back to an in-person one, I do foresee retaining many of the virtual elements that were introduced during the pandemic in the future DHSS program. I was already interested in using remote learning in DHSS even before the

COVID-19 pandemic; having the flexibility to include workshop instructors and speakers, as well as Summer Scholars who may have to be abroad over the summer, where I may have had to exclude them before, is invaluable. The ability to use videoconferencing to continue to cultivate a "flipped classroom"[47] pedagogical style in introducing digital methods, methodology, and tools is also crucial in emphasizing how the digital affects consumption and presentation of information in research. The Lafayette College Libraries and the DHSS program will adapt and evolve its teaching and learning goals in DH and digital scholarship research in different ways to maximize student interest in and engagement with these and the Libraries itself.

Further ventures of the DHSS program into the remote learning space include enlarging the scope of topics covered in the six-week course, especially those that include more interaction with and critical understanding of transformative digital technologies, such as digital storytelling. A weekly digital storytelling assignment was introduced to the remote DHSS program in 2021 as an additional form of communication, emphasizing using audiovisual multimedia tools to construct a narrative of one's individual research journey, encompassing academic, technical, and even emotional themes. Every week, the Summer Scholars were presented with a prompt on the DHSS Slack in the form of a question. Example prompts included: What did you think you wanted to research at the beginning of the program and why? Has that changed at this point? What has the search for relevant sources been like? Are you having any trouble finding appropriate sources? What is your project timeline? Who is your project's audience? The cohort was also encouraged to document the emotional and mental facets of their process as they completed each week of their digital research project's creation. They posted audio clips, iPhone videos, etc. to their own individual thread in Slack. Their digital storytelling assignment experiences were presented at the virtual Lafayette College CITLS Assignment Design Experience in summer 2021, post-DHSS program.[48] Virtual Reality (VR) has also become a very popular topic on campus, and one that I have also introduced to the slate of topics covered during the program.

There has long been interest in the DHSS program as a credit bearing course, and indeed, a semester-long credit DH course has long been offered at the institution.[49] Both of these possibilities, and in the case of the latter option, a semester course led by the library in conjunction with pertinent faculty, are still on the table when considering the future of DH and digital scholarship at the Lafayette College Libraries, and the campus at large.

Notes

1. "About Our Program & FAQ," Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, accessed December 29, 2021, https://sites.lafayette.edu/dhss/about-our-program/.
2. Janna Avon, Anne Houston, Charlotte Nunes, and Angela Perkins, "Using Digital Scholarship Methods and Tools to Enhance and Transform Learning during COVID-19 at a Liberal Arts College," *New Review of Academic Librarianship* 27 no. 3 (2021): 302, http://doi.org/10.1080/13614533.2021.1983849.
3. Avon et al., "Using Digital Scholarship Methods," 301.
4. Brett D. Hirsch, ed., *Digital Humanities Pedagogy: Practices, Principles and Politics*, Cambridge, UK: Open Book Publishers, 2012, 115.
5. Ibid., 215.
6. "Digital Humanities Community of Practice Events and Trainings," Lafayette College Libraries' Digital Scholarship Services, accessed July 3, 2022, https://dss.lafayette.edu/dhcp-events-trainings/.
7. Hirsch, *Digital Humanities Pedagogy*, 115.
8. Modern Language Association (MLA), "Report of the Task Force on Doctoral Study in Modern Language and Literature," 2014, mla.org/Resources/Guidelines-and-Data/Reports-and-Professional-Guidelines/Report-of-the-Task-Force-on-Doctoral-Study-in-Modern-Language-and-Literature-2014.
9. Hirsch, *Digital Humanities Pedagogy*; Modern Language Association (MLA),"Report of the Task Force on Doctoral Study in Modern Language and Literature;" Roopika Risam, Justin Snow, and Susan Edwards, "Building an Ethical Digital Humanities Community: Librarian, Faculty, and Student Collaboration," *College & Undergraduate Libraries* 24, no. 2–4 (2017): 337, http://doi.org/10.1080/10691316.2017.1337530.
10. Doug Lederman, "Will Shift to Remote Teaching Be Boon or Bane for Online Learning?" *Inside Higher Ed*, March 18, 2020, https://www.insidehighered.com/digital-learning/article/2020/03/18/most-teaching-going-remote-will-help-or-hurt-online-learning.
11. Sarah Morris, "The Digital Humanities Summer Scholarship: A Model For Library-Led Undergraduate Digital Scholarship," *College & Undergraduate Libraries* 24: no 2-4 (2017): 533, http://doi.org/10.1080/10691316.2017.1338978.
12. Morris, "The Digital Humanities Summer Scholarship," 533.
13. "Research by Students," Lafayette College, accessed December 10, 2021, https://researchsupport.lafayette.edu/research-by-students/.
14. Morris, "The Digital Humanities Summer Scholarship," 533.
15. Avon et al., "Using Digital Scholarship Methods," 307.
16. Ibid., 307.
17. Ibid., 307.
18. Ian Morse, Mila Temnyalova, and William Gordon, "The Good Side of Failure: Explorative Yet Productive Failure in Digital Humanities Projects," in *Scholarship in the Sandbox: Academic Libraries as Laboratories, Forums, and Archives for Student Work*, eds. Jackson, Amy S., Cindy Pierard, and Suzanne M. Schadl (Chicago: Association of College & Research Libraries, 2019), 35.
19. Avon et al., "Using Digital Scholarship Methods," 308.
20. "Moravian Lives," Bucknell University and University of Gothenburg, accessed December 27, 2021, http://moravianlives.org/.

21. Caleb Gallemore, "Text as Data," Digital Humanities Summer Scholars Program, Lafayette College Libraries, June 10, 2020.
22. Ibid.
23. Ibid.
24. "Summer Research Colloquium: July 15, 2020," Lafayette College Center for the Integration of Teaching, Learning and Scholarship (CITLS), accessed December 29, 2021, https://media.lafayette.edu/media/Summer+Research+ColloquiumA+July+15%2C+2020/1_p8qx3s5v.
25. Katie Frost, "Original Research Through New Methodologies: A Look at the Digital Humanities Summer Scholar Program," *The Lafayette*, February 26, 2021, https://lafayettestudentnews.com/105916/arts/draft-dhss/.
26. "Projects," Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, accessed December 29, 2021, https://sites.lafayette.edu/dhss/our-projects/.
27. Avon et al., "Using Digital Scholarship Methods," 309.
28. Imane Halal, "The Social Geography of Lafayette College: Project Plan," Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 2020, accessed December 29, 2021, https://socialgeographyoflafayettecollege.wordpress.com/project-plan/.
29. "Mission and History," Lafayette College, accessed December 29, 2021, https://about.lafayette.edu/mission-and-history/.
30. Imane Halal, "Black Spaces at Lafayette College," Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 2020, accessed December 29, 2021, https://uploads.knightlab.com/storymapjs/ffa2f131379b27f1803bbfb2a02a97ad/black-spaces-at-lafayette-college/draft.html.
31. Evan Hill, Ainara Tiefenthäler, Christiaan Triebert, Drew Jordan, Haley Willis, and Robin Stein, "How George Floyd Was Killed in Police Custody," *New York Times*, May 31, 2020, last modified November 1, 2021, https://www.nytimes.com/2020/05/31/us/george-floyd-investigation.html.
32. Kurt Bresswein, "Allegations of Racism, Sexual Misconduct at Lafayette Draw Response," *Lehigh Valley Live*, July 7, 2020, https://www.lehighvalleylive.com/easton/2020/07/allegations-of-racism-sexual-misconduct-at-lafayette-draw-response.html.
33. Imane Halal, "The Social Geography of Lafayette College," Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 26, 2020, https://media.lafayette.edu/media/DHSS_2020_Final_Presentations_Day3_6.26.20/1_rg8phcw9.
34. Stephen Wilson, "Humanities, Digitized," *Lafayette News*, August 12, 2020, https://news.lafayette.edu/2020/08/12/humanities-digitized/.
35. Milena Berestko, "How to Request a Mentor (a Mini Guide)," Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, April 2020.
36. Uche Anomnachi, "Racial Perceptions of Anime in the United States," Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 2018, https://ucheanomnachidhss2018.tumblr.com.
37. Rudy Miller, "Lafayette College has Its Second Rhodes Scholar in More Than a Century," *Lehigh Valley Live*, last modified November 23, 2020, https://www.lehighvalleylive.com/news/2020/11/lafayette-college-gets-its-second-rhodes-scholar-in-more-than-a-century.html.
38. The Harry S. Truman Scholarship Foundation, "FAQ," accessed December 29, 2021, https://www.truman.gov/faq.
39. Saide Singh, "Indo-Diasporic Poets: Writings and Reorientations of the Kali Pani," Lafayette College Department of English, May 2022, https://sites.google.com/lafayette.edu/kalipanipoetics/home?pli=1&authuser=1.

40. Wilson, "Humanities, Digitized!"
41. Ibid.
42. Singh, "Indo-Diasporic Poets."
43. "DHSS Idea Incubator," Lafayette College Libraries, last modified December 17, 2021, https://scalar.lafayette.edu/dhss-idea-incubator/index.
44. "The Humanities and Humanistic Research," Lafayette College Libraries, last modified May 7, 2021, https://scalar.lafayette.edu/dhss-idea-incubator/introduction-to-digital-humanities?path=introduction-to-digital-humanities-scholarship.
45. Improving Your Website with SiteImprove," Lafayette Technology Help, accessed July 3, 2022, https://help.lafayette.edu/improving-your-website-with-siteimprove/.
46. "Workshopping the Workshop: Iterative Instructional Design in Undergraduate Focused Programming," DLF Forum, Digital Library Federation, accessed December 29, 2021, https://www.diglib.org/dlf-events/2021-dlf-forum/; https://www.youtube.com/watch?v=BFjAHRSbeRs .
47. Cynthia J. Brame, "Flipping the Classroom," Vanderbilt University Center for Teaching, 2013, accessed July 3, 2022, https://cft.vanderbilt.edu/guides-sub-pages/flipping-the-classroom/.
48. "The Virtual Assignment Design Experience," Lafayette College Center for the Integration of Teaching, Learning and Scholarship (CITLS), accessed December 29, 2021, https://citls.lafayette.edu/the-assignment-design-institute-session-descriptions/.
49. Morris, "The Digital Humanities Summer Scholarship," 534.

Bibliography

Anomnachi, Uche. "Racial Perceptions of Anime in the United States." Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 2018. https://ucheanomnachidhss2018.tumblr.com.

Avon, Janna, Anne Houston, Charlotte Nunes, and Angela Perkins. "Using Digital Scholarship Methods and Tools to Enhance and Transform Learning during COVID-19 at a Liberal Arts College." *New Review of Academic Librarianship* 27, no. 3 (2021): 301–321. http://doi.org/10.1080/13614533.2021.1983849.

Berestko, Milena. "How to Request a Mentor (A Mini Guide)." Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, April 2020.

Brame, Cynthia J. "Flipping the Classroom." Vanderbilt University Center for Teaching, 2013. Accessed July 3, 2022. https://cft.vanderbilt.edu/guides-sub-pages/flipping-the-classroom/.

Bresswein, Kurt. "Allegations of Racism, Sexual Misconduct at Lafayette Draw Response." *Lehigh Valley Live*, July 7, 2020. https://www.lehighvalleylive.com/easton/2020/07/allegations-of-racism-sexual-misconduct-at-lafayette-draw-response.html.

Frost, Katie. "Original Research Through New Methodologies: A Look at the Digital Humanities Summer Scholar Program." *The Lafayette*, February 26, 2021. https://lafayettestudentnews.com/105916/arts/draft-dhss/.

Gallemore, Caleb. "Text as Data." Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 10, 2020.

Halal, Imane. "Black Spaces at Lafayette College." Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 2020. https://uploads.knightlab.com/storymapjs/ffa2f131379b27f1803bbfb2a02a97ad/black-spaces-at-lafayette-college/draft.html.

Halal, Imane. "The Social Geography of Lafayette College." Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 26, 2020. https://media.lafayette.edu/media/DHSS_2020_Final_Presentations_Day3_6.26.20/1_rg8phcw9.

Halal, Imane. "The Social Geography of Lafayette College: Project Plan." Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 2020. https://socialgeographyoflafayettecollege.wordpress.com/project-plan/.

Hill, Evan, Ainara Tiefenthäler, Christiaan Triebert, Drew Jordan, Haley Willis and Robin Stein. "How George Floyd Was Killed in Police Custody." *New York Times*, May 31, 2020. Last modified November 1, 2021. https://www.nytimes.com/2020/05/31/us/george-floyd-investigation.html.

Hirsch, Brett D., ed. *Digital Humanities Pedagogy: Practices, Principles and Politics*. Cambridge, UK: Open Book Publishers, 2012.

Lederman, Doug. "Will Shift to Remote Teaching Be Boon or Bane for Online Learning?" *Inside Higher Ed*, March 18, 2020. https://www.insidehighered.com/digital-learning/article/2020/03/18/most-teaching-going-remote-will-help-or-hurt-online-learning.

Miller, Rudy. "Lafayette College has Its Second Rhodes Scholar in More Than a Century." *Lehigh Valley Live*. Last modified November 23, 2020. https://www.lehighvalleylive.com/news/2020/11/lafayette-college-gets-its-second-rhodes-scholar-in-more-than-a-century.html.

Modern Language Association (MLA). "Report of the Task Force on Doctoral Study in Modern Language and Literature." 2014. https://www.mla.org/Resources/Guidelines-and-Data/Reports-and-Professional-Guidelines/Report-of-the-Task-Force-on-Doctoral-Study-in-Modern-Language-and-Literature-2014.

"Moravian Lives." Bucknell University and University of Gothenburg. Accessed December 27, 2021. http://moravianlives.org/.

Morris, Sarah. "The Digital Humanities Summer Scholarship: A Model for Library-led Undergraduate Digital Scholarship." *College & Undergraduate Libraries* (2017): 532–44. http://doi.org/10.1080/10691316.2017.1338978.

Morse, Ian, Mila Temnyalova, and William Gordon. "The Good Side of Failure: Explorative Yet Productive Failure in Digital Humanities Projects." In *Scholarship in the Sandbox: Academic Libraries as Laboratories, Forums, and Archives for Student Work*, edited by Amy S. Jackson, Cindy Pierard, and Suzanne M. Schadl, 35–51. Chicago: Association of College & Research Libraries, 2019.

Risam, Roopika, Justin Snow, and Susan Edwards. "Building an Ethical Digital Humanities Community: Librarian, Faculty, and Student Collaboration." *College & Undergraduate Libraries* 24 no. 2–4 (2017): 337–349. http://doi.org/10.1080/10691316.2017.1337530.

Singh, Saide. "Indo-Diasporic Poets: Writings and Reorientations of the Kali Pani." Lafayette College Department of English. May 11, 2022. https://sites.google.com/lafayette.edu/kalipanipoetics/home?pli=1&authuser=1

Singh, Saide. "Indo-Guyanese Women Poets." Lafayette College Libraries Digital Humanities Summer Scholars (DHSS) Program, June 2020. https://scalar.lafayette.edu/indoguyanesewomenpoets/index.

Wilson, Stephen. "Humanities, Digitized!" *Lafayette News*, August 12, 2020. https://news.lafayette.edu/2020/08/12/humanities-digitized/.

"Workshopping the Workshop: Iterative Instructional Design in Undergraduate Focused Programming." DLF Forum, Digital Library Federation. Accessed December 29, 2021. https://www.diglib.org/dlf-events/2021-dlf-forum/; https://www.youtube.com/watch?v=BFjAHRSbeRs.

CHAPTER FOURTEEN

The Restorative Potential of Library-Based DH:

Reconnecting Learning Communities in the Age of COVID-19[1]

Emma Annette Wilson, Russell Hugh McConnell, Johanna Pang, and Maria Katsulos

ORIGINALLY THIS chapter intended to investigate strategies for developing thriving digital humanities (DH) initiatives and communities drawing on Emma Wilson's experience growing the DH center at the University of Alabama over a four-year period. However, since the first edition of this book, we find ourselves facing the realities of teaching DH in the library in the age of COVID-19, and doing so with years of incoming students whose experiences leading up to their entry into higher education have been shaped by the pandemic. Simultaneously, students, faculty, and staff who were already in higher education when the pandemic began in 2020 have had their expectations and experiences of academic life fundamentally altered. Consequently, and in light of recent findings about the impact of the pandemic on student learning experiences in higher education, we felt compelled to pivot the chapter to consider not how to develop thriving DH communities, but rather how to use DH instruction to support and nurture thriving learning communities across university campuses.[2] DH and digital scholarship more broadly run the risk of being siloed within specialist library units, but by working to assist students in their journeys through higher education in the time of COVID-19, we can make a campuswide difference.[3] In the 2021 *State of America's Libraries Special Report: COVID-19*, then-ALA President Julius C. Jefferson Jr. described libraries and librarians as serving as "first restorers" in the wake of the pandemic, and we see an opportunity

CC BY

for subject specialists to be just that by working with faculty and students in hands-on DH instruction sessions, activities, and assignments to help all of us to re-engage and reconnect with our studies and with one another for an enhanced learning and life experience.[4] Recent studies indicate some of the large-scale challenges students face as they return to in-person instruction after years of remote and hybrid learning. A brief literature review establishes areas of strength which librarians can draw on to address some of the issues students are experiencing as we transition back to life in person, and by drawing on the authors' experiences both as teachers (Wilson and McConnell) and as students (Katsulos and Pang), we share some effective DH teaching initiatives (e.g., class sessions and assignments) which subject specialists can implement in collaboration with faculty and students on the frontline of the restorative work done by libraries in the age of a global pandemic.

Challenges Facing Students in the Time of COVID-19

An article in the *Chronicle of Higher Education* on April 5, 2022, cites examples from professors and students from across the United States to argue that all students from all backgrounds and studying all disciplines are struggling at university as a result of the pandemic.[5] In the classroom, students are struggling to keep up with readings and preparatory work for class, to retain the information imparted in class sessions, to complete all required assignments, and to consistently attend and participate in classes. For newer students, some of these problems stem from disrupted college preparation: Ashley Shannon, chair of the Department of English at Grand Valley State University in Michigan, describes students entering college today as being "by and large tragically underprepared to meet the challenges of university life—both academically and in terms of 'adulting.'" Indeed, some students are finding that study habits they developed to handle different learning modalities in 2020 and 2021 are not effective in in-person classes. Students who were already in higher education when the pandemic hit in 2020 have had to repeatedly adjust their learning habits between in-person, virtual, and hybrid instruction sessions, and there is a sense among both professors and students surveyed that consequently everyone is at a point somewhere beyond exhaustion. Even in 2021/22 when instruction has, by and large, reverted to its normal modality for most students, both the *Chronicle* and our student authors have found faculty members still relying more on technology than they would have prior to the pandemic, requiring students to go

online to complete activities rather than having as many in-class discussion sessions or group activities. As a result of this continued use of effectively remote learning strategies, there remains a sense that everything still feels virtual or fake. Beyond the classroom an overwhelming number of students have suffered significant hardships from the pandemic, from losing family members to experiencing financial insecurity and other traumas, with underserved communities disproportionately affected. The cumulative effect of these issues is a persistent sense of existential dread, increased academic and social anxiety and stress, and emotional and physical exhaustion.

The Center for Collegiate Mental Health (CCMH), an international practice-research network of nearly 700 college and university counseling centers, released a series of blog posts analyzing the "initial impacts of COVID-19 on college students seeking treatment for mental health concerns."[6] The reports compare data from 2020–21 with that from 2017–2019 to provide preliminary insights into the ways in which students have and are experiencing psychological stress during the pandemic. The survey results make for heartbreaking reading: 94 percent of over 43,000 students who were seeking counseling services reported that at least one area of their lives had been negatively affected by the pandemic; 90 percent reported that multiple areas had been affected. CCMH reports that "the most common areas affected were mental health (72%), motivation or focus (68%), loneliness or isolation (67%), academics (66%), and missed experiences or opportunities (60%)."[7] Beyond these issues, more than 30 percent of those surveyed reported that they had been affected in terms of their career or employment, their relationships, and their finances, and 9 percent of students had experienced grief or the loss of a person and 9 percent had experienced food or housing insecurity. Students who specifically cited COVID-19 as a reason for their referral to counseling services reported higher levels of distress in every area surveyed.

The third part of CCMH's report examined the ways in which students from different demographics reported hardships or distress experienced during the pandemic.[8] The study indicates that students from almost all identity demographics consistently reported that their mental health had been negatively impacted by the COVID-19 pandemic, even if COVID-19 was not the main reason they were seeking counseling services: "The only exceptions to this trend were among African American students and students who self-identified their race/ethnicity," who identified "motivation or focus as the most common concern, followed by mental health."[9] A key recommendation from CCMH's report examining the impact of COVID-19 on different identity demographics is that "colleges and universities should be attending

to the needs of students in a developmentally differentiated manner" as we transition back to pre-pandemic learning modalities. The study notes that incoming first-year students report dealing with the loss of missed experiences, whilst seniors report distress concerning the potential impact on their career or employment prospects, suggesting that both groups need support but in different ways and for different reasons. These disparities may present in classroom environments and that is one reason why targeted, customizable DH teaching may be an effective solution to reinvigorate post-pandemic instruction.

Students are facing issues from the pandemic in nearly every aspect of their lives, and when problems are so wide-ranging, DH teaching alone cannot solve them. However, a common theme in both CCMH's data and the *Chronicle* article is that in the time of COVID-19 students everywhere are experiencing a fundamental sense of disconnection and disengagement. The ALA's 2021 special issue about libraries during the COVID-19 pandemic observed that college students in particular "needed encouragement" to use library resources intended to help them during remote and hybrid learning.[10] If we build it, they will not necessarily come—we will have to go to them and do some high-quality outreach to try to find appropriate solutions for some of the problems of academic disengagement. For the foreseeable future, the academic landscape has shifted for all students, both those currently in attendance and those yet to come. Recent articles focus on the undergraduate experience of COVID-19, but it is almost certain that graduate students are also facing issues because of the pandemic, and while nuance is required when teaching any demographic, we anticipate some of the strategies discussed in this chapter to be needed and applicable at the undergraduate and graduate level. As bleak as the picture seems, there are tangible ways in which we, as librarians, can provide help, support, hope, and some major academic and life skills by offering targeted, hands-on DH activities, workshops, and assignments to help classrooms to become reconnected. Let's go digital to get back in touch.

How Library-Based DH Can Help

DH instruction by its nature tends to be hands-on. Students typically learn and then apply technological approaches to humanities topics for the purposes of research, discovery, intellectual exploration and (hopefully!) delight, public scholarship, and gaining real-world technical, administrative, and analytical skills ready for use in the job market. Not all DH teaching initiatives will

encompass all these features; they run the gamut from one-off extracurricular workshops to in-class sessions, substantial class assignments, and participating in long-term research projects. Outputs range from the satisfaction of completing a standard exercise to test out a new skill, such as plotting points in Google's MyMaps after a workshop; producing entries in a public-facing online resource, such as curating, researching, and documenting entries for a digital archive; creating individual or small-group mini-projects in a semester, such as researching and visualizing the social networks of nineteenth-century Parisian artists; or contributing to a long-term DH research initiative, such as transcribing marginalia found in Renaissance manuscripts for inclusion in a large-scale online database used by researchers.

Common to all these approaches is that they demand hands-on and immediate student engagement. In every case, students must engage in hands-on work *applying* new skills, whether they are plotting points on a digital map, transcribing and encoding text for digital publication in an online edition of a book, generating metadata about items for an online archive or exhibit, creating spreadsheets they can use to visualize social networks, running texts through text-mining software to identify patterns and trends using big data analysis, or countless other activities. None of these tasks can be done passively by watching a lecture or even by taking good notes. Students may well need some preparatory lecture or to take notes ahead of tackling these tasks, but to do the work, they have to take what is being taught and apply it themselves. Many articles note the efficacy of DH in enhancing student engagement. Rivard uses DH strategies to help composition students understand the archives *as* data, thereby bringing students to a greater understanding of what research materials *are*,[11] while Fyfe explains how DH assignments enabled first-year honors students to acquire key research and analytical skills they can transfer to their classes later in their university careers.[12] The capability of DH assignments to equip first-year students with strong research skills they can use later in their degree is critical in thinking about how DH can mitigate some of the challenges faced by incoming students due to the pandemic. In addition to enhancing students' academic preparation, Sanders, Rodrigues, and Li share how a digitization project helped students in a literature course to engage with unfamiliar materials by bringing them into very close proximity with those materials through hands-on work, an approach which might be of special relevance at a time when it seems almost every part of academic life has become less familiar and more remote for our students.[13] McCarl found a DH project enabled students to engage in collaborative participation, and

the communal approach which is central to many if not all large-scale DH projects could play a major role in healing some of the divides and bridging some of the distance which students are feeling in the classroom today by engaging *all* of us in collaborative work and therefore conversation.[14] Alongside the academic and social benefits of these approaches, Goldenberg discusses a collaborative DH project involving graduate students and high school students which brought students into conversation with their local communities through research activities including the gathering of oral histories.[15] Denda and Hunter reported similar positive outcomes in terms of community building from their students' oral history projects at Rutgers.[16] In our current climate, after a time of necessary separation, that kind of community-building can be valuable in helping students to feel a part of their immediate surroundings. These approaches speak to ways in which DH can enhance academic and social elements of campus life, and to the special role which it can play in community healing as we slowly recover from and learn to live with the after-effects of the pandemic.

DH teaching differs from traditional essay or exam assignments in that it puts students immediately into direct contact with their research and learning materials in an active way. When a student researches and writes an essay, or studies for an exam, however immersed they may be in the ideas, all their preparatory work is ultimately secondary to their finished essay or perfect exam score. As students read and take notes for an essay or an exam, they are assembling materials that may or may not be included in the finished product, and all the notes they generate are fundamentally preparatory to the end result; it is scholarship at one remove. At times of heightened academic uncertainty, this approach may contribute to a student's sense of lacking direction and in turn potentially becoming less engaged or losing motivation, issues which are repeatedly highlighted in the above reports. Our authors witnessed first-year students in Spring 2022 work very diligently to identify an essay topic, find sources, and make notes on those sources, only to flounder when it came to knowing how to take the next step and coalesce all that hard work into a focused essay. There was a sense that the students had done a lot of work (which they had), but that for all that time invested, they didn't *have* anything that would then enable them to complete the required essay. In many cases, students did have useful resources; but there was confusion regarding quite what to do with those resources, as if the entire assignment appeared arbitrary or unreal. We see this as a prime example of the kind of academic disconnect discussed by CCMH and the *Chronicle*, and the more

hands-on approach taken by some DH teaching strategies can be a good antidote.

Another powerful benefit of DH pedagogy is that much of it—especially in the early stages of a project, when student engagement might be particularly fragile—is done "live" in the classroom. Students working on their own, typically at home or in the library, are liable to experience (or re-experience) the feelings of isolation and detachment that came with online learning during lockdown. Working on their own, they may feel aimless or uncertain, not knowing if they are doing an assignment "right," and therefore easily becoming distracted and discouraged. Struggles with motivation or focus and with loneliness or isolation were among the largest problems reported by the CCMH, and it is easy to see how the solitary labor of taking notes on an article or drafting an essay may feed these problems. In DH pedagogy, however, it is very rare (indeed, often impossible) for an instructor to simply send students away with a task to complete on their own. Overwhelmingly, DH-based activities tend to require in-person, step-by-step instruction, as students work through a series of operations, often using unfamiliar software. Students are therefore not working in isolation, but are instead grappling with new challenges literally side-by-side with their peers, with whom they may consult and to whom they may offer advice and support. Peer support of this kind benefits both parties, paying dividends in both improved knowledge and increased confidence. During this process students are guided by an instructor who is in the room with them, ready to field questions and equip them with relevant problem-solving tools and procedures as they are needed.

As students advance through a DH project, they are typically making tangible progress at every stage. Many traditional academic activities, such as brainstorming ideas, taking notes on primary and secondary texts, and preparing outlines are, despite their importance, not directly and obviously contributory to a finished product. For example, it is not uncommon for a student, particularly an inexperienced one, to spend hours finding articles in an academic database and taking notes on them, only to realize later that these particular articles are not going to be useful, that the painstaking notes will not be needed. To an extent, such pains and disappointments are inevitable, but DH projects tend to be relatively free of them. In the preparation of an online edition, for instance, every word of transcription that a student types is a small piece of the project getting completed. Every photograph uploaded to an online exhibit brings that exhibit closer to its finished state. Every point added to a digital map makes that map just a little bit more helpful for its intended users. Errors and dead ends can never be fully eliminated

from any educational process, but DH's capacity to generate instant, tangible, finished results as students go along tends to prevent the kind of discouragement that damages motivation and endangers academic success.

In summary, beyond the inherent technological and humanistic skills which it can impart, some core benefits of working with DH in the classroom which can address pandemic-era distress including disengagement and disconnection include:

- DH is inherently hands-on, not passive;
- DH work can be done "live" in the classroom promoting participation and activity in that setting;
- DH learning is inherently incremental, allowing students to see real-time gains every time they sit down to work;
- DH can generate results quickly, providing gratification and scholarly reward en route through an assignment or activity (making a point on a map, encoding a line of text, running a simple code through a text to produce a graph illustrating patterns therein, etc.);
- DH insists on direct, slow, minute engagement with the subject at hand (through transcription, metadata creation, data gathering or creation, the writing of code, etc.);
- DH is often communal and thrives as a community of practice.

How Library-Based DH Can Be Particularly Effective in the Time of COVID-19

In many ways, DH teaching activities are the makerspaces of the humanities; in Ellyssa Kroski's invaluable edited collection *Makerspaces in Practice*, Cherie Bronkar argues that the maker movement gained traction when it was seen as transforming libraries from "disseminators of knowledge to creators of knowledge," and in the classroom DH can have a similar effect, empowering students and teachers alike to become actively engaged in knowledge creation.[17] By thinking about our students as proponents of knowledge creation and generation, and teaching them accordingly, we can help them to re-engage in the academic process as active participants rather than recipients who are increasingly distant and disconnected.

Literature concerning the role of libraries during COVID-19 is still emergent, but from the articles which are currently available, one trend is very clear: that libraries are doing significant work in connecting people both with information and with one another. Spehar shares how the Ohio University

Libraries created new content request forms to assist with providing materials remotely as well as embedding librarians in courses to assist with assignments;[18] Colding and Gause explain how the University of Central Florida swiftly set up virtual reference services via Zoom and how librarians had to field many more questions concerning e-books and copyright;[19] and Kapanka discusses how a school media specialist responded during lockdown.[20] The ALA 2021 special report gives a birds-eye view of the support which public, school, and higher education libraries provided for remote learning in terms of materials and reference interactions and also in terms of trying to address the digital divide by leaving Wi-Fi services on even when physical buildings had to be shut "because libraries bring people together even when we have to stay apart."[21] It is this capacity which makes library-based DH particularly potent as our campus populations continue to move back to in-person and "normal" operations, as libraries typically have campus-wide reach and therefore the potential to renew, reinvigorate, and, indeed, create for the first time, new connections and networks of learners.[22]

Even before the pandemic, a literature review indicated the importance of community-building and community enrichment for DH in the library. Interviewing humanities scholars about their DH work done in partnership with their libraries, Senseney, Dickinson Koehl, and Nay report that the library was seen as making efforts to build campus-wide expertise to support DH and DS.[23] Cox notes that libraries are skilled in creating "deep partnerships among campus units" to enhance what DH work can be done within an institution, as well as extending that reach to a "global" level through partnerships with other institutions and also through the creation and dissemination of new knowledge.[24] Libraries are partly primed to do this work because they typically work with all academic departments across campus, thereby expanding the reach of the kinds of revitalizing teaching initiatives which we are suggesting here, and also because they possess significant in-house expertise for supporting DH.

In light of the multi-dimensional nature of DH, ranging from humanistic research and scholarly questions to imaginative IT solutions, rigorous metadata to make projects usable and discoverable, sound project management to keep multi-person teams on track, and more, Sandy, Million, and Hudson-Vitale sagely warn against seeking all necessary skills for performing DH in one "'unicorn' person."[25] Instead, and in contrast to the typical humanistic scholarly model, libraries are full of people possessing various skills, and accustomed to managing those people to work together effectively in interdisciplinary teams, and that collaborative model in tandem

with well-developed management skills and workflows enables libraries to call on their "established …values and strengths" in a new way to support DH.[26] Yin Zhang, Fangli Su, and Brenna Hubschman argue that in DH and DS libraries have the capacity to be both mediators, helping scholars and students to navigate between different materials and methods as they study, and innovators, calling on that established and variegated skillset to open up new learning pathways for campuses.[27] For our current generation of students and faculty, there has never been a moment of greater need for mediation and innovation to help everyone work their way back into their studies and into a more integrated campus life once again. We offer our example of an effective DH assignment below (see Appendix 14B) as a model for subject specialists because if it were rolled out through the library's campus-wide learning network rather than one class at a time by "hero" instructors in individual departments, the whole learning community will benefit in a timely way from this kind of active pedagogical intervention and innovation.

Our Experience: Some Effective Assignments to Rebuild a Community Of Learning

In 2016 Laura Braunstein described DH as "a community of practice that inhabits the intersection of technology and human culture."[28] In 2022, we experienced two of the most traumatic years in which that intersection has been thrust upon all of us, worldwide, and it is our proposition here that, at least in a teaching context, DH strategies offer a good way for us to reconfigure once again our human relationship with learning as a technology. Articles about classes learning the Text Encoding Initiative or TEI to transcribe and edit text for digital publication are striking in setting forth a picture of focused, almost meditative scholarship, done in a communal way among classmates, which provides students with job-market-relevant skills but in a way which prioritizes process over product. Green discusses TEI teaching at The University of Illinois at Urbana-Champaign as creating a community of practice on campus,[29] and McCarl and Giannetti both provide examples from their class sessions in which students used TEI to create editions of rare materials and in so doing developed high-level academic and management skills all whilst working productively and harmoniously together.[30] Laura Estill has also written, and recently spoken, eloquently about the benefits of teaching students to encode manuscript marginalia about or on the edges of dramatic texts copied out in early modern commonplace books, and her approach both for teaching and in her wider research project on this subject

is a bar-raising role model.[31] Our experience in TEI courses at the undergraduate and graduate level at Southern Methodist University during the pandemic chimes with these testimonies, and we see specific ways in which this form of DH instruction can act as at least a partial remedy for the teaching and learning fatigue, ennui, and disengagement which seem to be the current universal experience. In what follows we share a short account of our collective teaching and learning experience from a library-based TEI course in which we are primarily focused on the ways in which this DH technique and our model of instruction can be particularly effective in addressing and alleviating some of the learning challenges faced by students in the classroom today. We offer this account as a potential model for other teaching initiatives, along with some recommendations and resources for anyone interested in experimenting with this approach.

Editing Seventeenth-Century Texts: A Collaborative Student Initiative is a digital editing and publishing project undertaken first by undergraduate students in Spring 2020 and then continued by graduate students in English DH classes in Fall 2020 at SMU. The goal of the project is to create online, annotated editions of seventeenth-century books (or extracts thereof) from holdings in our special collections in the DeGolyer Library. By the end of the project, each student produced:

A. A digital edition of their text;
B. Scholarly annotations encoded into that edition;
C. A short critical introduction to the text arguing for its value and interest for a new reader.

By learning TEI alongside traditional archival and research skills, and applying this to their own unique projects, students had the chance to act as researchers in their own right, and to make real and new contributions to knowledge. The learning outcomes were as follows:

- Gaining hands-on experience conducting both traditional archival rare book research and modern, up-to-date digital literary research;
- Learning traditional paleographic, bibliographic, and transcription skills and technological encoding and metadata creation skills;
- Learning how to think of literary texts or creative works more broadly as a form of data;
- Gaining a wider awareness of how versatile data is in the modern world (and in the seventeenth century!);
- Experiencing how technology and literature have always been and continue to be intimately entwined;

- Marketable traditional and digital skills for a resumé.

The assignment is a realistic piece of academic work which gives students significant autonomy but with clearly-defined goals. It was completed through multiple weeks of consultative class sessions in which students worked while librarians and a faculty member checked in on them, as well as sharing their work occasionally with one another in very small groups (2–3 people) who were experiencing related issues and could "team teach" their way out of those dilemmas.

The texts selected had not received a modern edition (meaning none since the eighteenth century), and are not likely to, as they have largely slipped out of mainstream canonical memory—and in that act of cultural forgetting, our students are presented with an opportunity to do some literary reclamation, rediscovering these texts and using digital editing technology to "upgrade" them for modern readers. Transcribing and encoding these texts using TEI makes it feasible for students to make substantive contributions to knowledge while acquiring tangible digital skills which are critical for a modern workplace. The project was conceived before COVID-19 "was a thing" as one student put it, but it turned out to have benefits that directly remedied the remoteness of that initial phase of the pandemic, and which can be good rehabilitators today, because to transcribe a text which has, essentially, been unread for several hundred years, is a slow, sometimes tedious—but also intimate—hands-on endeavor. Each student "adopted" a seventeenth-century text from the DeGolyer Library from a selection which had been curated well ahead of time in consultation with rare book librarian Christina Jensen.[32] As each student was individually responsible for a text, they had the chance to take ownership of its literary future: certain standard typographic features had to be encoded (we all like line breaks when reading poetry, whether on screen or on paper, for example), as well as appropriate publication information to give the original author, themselves as the encoder, and the DeGolyer Library, suitable credit for the venture; but when it came to editorial observations and notes, their hands were free—they are the newest authority on this text, and situated to tell the reader what parts of it should be noticed. The nature of this approach means that every student project is unique; they are inherently different from one another, and after two years of seeing one another primarily as arrays of squares on small, pixelated screens, that kind of individuation in and of itself could be highly beneficial to students in terms of acknowledging their unique part to play in the classroom.

Transcribing a text takes an extremely long time, especially to people new to the task. On the one hand it can feel dreary, but on the other it is a golden opportunity to really "meet" a text, to get to know it quite literally letter by letter; and when accuracy is the name of the game there is no choice but to pay attention if you want to succeed in the assignment. At its best, transcription can be an almost meditative experience: the only task before you is to take this old book and type it up on your modern screen. Nothing else is required, no other active thought is demanded of you, just this one, manual task. Medieval monks working in scriptoria sometimes sought refuge from their days of transcription by working in their gardens; here the act of transcription gives us a kind of classroom garden in which to take some air and some space, using the peace and quiet of the rare book room to find some solace, all while getting an assignment checked off the list. Transcription is an inherently focused task, but one that is not hard to explain or to master, and as such it has the potential to bring both solace and satisfaction to students struggling to engage with more generic classwork. We are raising the stakes by making students individually responsible for particular books, and lowering them by requiring the completion of a humble manual task.

It may seem antithetical to advocate for creating digital editions as a way to reconnect post-pandemic, but there is also inherently something hands-on about transcription: true, we ended up completing our project remotely when classrooms shut down nationwide in March 2020; but we began by working with physical books in the library, and our hands were constantly physically engaged in typing these texts out, leaving less space for wandering minds and thoughts, both potential issues in the current environment. In light of the recent reports about student struggles with concentration and focus, transcription can be an unexpected antidote: it is in some ways low stakes, because you can check quite easily for yourself if you did it correctly, thereby hopefully keeping anxiety at bay; it is also vital, because if you don't do it there is no project—and more than that, your author will continue to languish in the archive, and pairing real cultural responsibility with a task which you can instantaneously check to make sure all is going well and staying on track can help students to see their achievements and progress in real time. Encoding transcribed text has similar benefits: we worked with the TEI editor oXygen which immediately flags if there is a problem with encoding, and as such any small slips can be caught early on rather than building to become major errors. These practical components of transcribing and encoding lend themselves to recovery pedagogy as students engage in individualized hands-on project work with instant and continuous but quiet

feedback from Word's squiggly underlines indicating potential transcription slips to oXygen's green light of approval or red of caution.

So far, so solitary. However, the project was highly communal and conversational. Six weeks of class time were devoted to transcription, encoding, and researching the texts, and when we shifted to remote instruction, we extended that to eight weeks to ensure sufficient time for completion. Sometimes librarians and a faculty member would circulate among students (either in person or via Zoom room check-ins) to see how work was progressing and to ask about the students' ideas for their annotations and any issues or questions they might have. In these one-on-one conversations everyone got to know one another extremely well—and we all got to know our forgotten seventeenth-century authors well too! Wilson has used this micro-project consultative approach in hundreds of teaching sessions in her prior role leading the DH center at the University of Alabama, and it is always effective in forging a real sense of partnership between faculty, librarian, and student, as we all work towards specific shared research goals. Very often when working with obscure texts, questions will arise about past cultural practices or highly specific and obscure historic events or figures which even the most seasoned faculty member or librarian may well not know the answer to—and so we all embark on a process of collective research both through traditional databases and also through the world's longest, most complex Google searches. There is a levelling effect when students see that those leading their sessions are still learning, too, and we have seen this contribute to student motivation and determination in these assignments. Students also frequently worked together in this assignment.

Following individual consultations, the faculty member and librarians would often notice common issues occurring. If these affected the whole class, of course we could pause to give whole-group instruction (everyone does in fact need line breaks; this is how to encode them). Equally, if two or three students were coming up against similar issues, we would have them talk together either in person (before lockdown), via Canvas discussion boards, or live in Zoom breakout rooms. This peer-to-peer approach helped enormously with generating community among the classroom, even at distance, and also helped every student maintain motivation and focus for the assignment, because there is an implicit language of collective enterprise. Your work will, in fact, affect more than just you; we really do all want to see it, and to read it, and to hear about it. That sense of fellowship will be critical in classrooms as we work to reunite following years spent apart.

Whilst we are advocating for students using technology extensively in this assignment, it is a responsive technology providing feedback which is crucial both for the student and for instructors. If a librarian instructor or faculty member consults with a student during a class session or office hour, any of us can immediately hone in on potential problem areas within the assignment and troubleshoot to help students to stay on track. The capacity for quick check-ins and progress reports is helpful for students as they must learn all over again how to manage their time in the latest learning environment. Furthermore, the technological element of this assignment generates real-world skills and results: students learn how to use a technology, something which they can put on their CVs, and they also see the finished product when their edition is ready for publication online. The one factor which limited that final stage in this first iteration of the project was fatigue from remote learning—but the web version of the project will be going live in the coming year as new generations of students continue to contribute to the initiative.

Some crucial factors to bear in mind with this kind of long-term assignment are:

1. Providing feedback along the way at both structured intervals and through less formal in-class consultation to prevent students from feeling lost or overwhelmed, or mistakes from snowballing;
2. Allocating sufficient time to complete the project, and being flexible if that time needs to extend during a first iteration of the project;
3. Having students keep track of their work in and out of class in a simple journal; this does not need to be elaborate (bullet point notes are fine), but it is a good, efficient way of staying on track and remembering where work started and stopped each session (numerous students specifically cited this approach as useful in end-of-term surveys);
4. It may be useful to create collective documentation such as an online discussion board in Canvas or Blackboard or a Google doc where anyone can note down "I encountered x problem and found that y and z solved it"; this is efficient and good general DH practice, and also helps build community among the group;
5. Setting realistic goals: students cannot individually transcribe, encode, and edit an entire novel in a matter of weeks, so it is important to have realistic targets (e.g., one act of a play, twenty pages of poetry, etc., as fits your timeline);

6. To access these benefits you don't necessarily need to do an assignment spanning weeks: even a couple of classes transcribing, encoding, and flash-researching a single item from an archive or one line of text from an obscure play could still be engaging, individualistic, challenging, and weird enough to keep us all awake as we return from remote instruction.

TEI assignments and classes succeed in large part because they are truly student-centered and student-driven; if you are working on a seventeenth-century play which hasn't been read or performed in hundreds of years, you know better than anyone else what edits can best showcase it to the wider world, and if we can bring some of that sense of empowerment, control, and aptitude into the modern classroom via the library's campus-wide wingspan and rich DH skillset, it will do much to uplift our returning students.

Thinking about nurturing our students, we would like to close this section with a reflection from Maria Katsulos, a co-author of this chapter and a graduate of the TEI course, who wishes to share her experiences for this article:

> English 3318: Literature as Data, which counted as my technology and mathematics requirement, was a fantastic alternative to my longtime enemy, calculus. Beyond some of my first brushes with theorists of Digital Humanities and information-gathering, in the course I was plunged into hands-on humanities research for the first time. My peers and I each chose an out-of-print 17th century text to transcribe, annotate, and encode. I chose Nathaniel Lee's play about Alexander the Great, *The Rival Queens*. My initial read—think getting thrown into the deep end of Paleography 101 without floaties—proved equally hilarious and heartbreaking. I realized that Lee was a prime example of an author intentionally shunned from the canon because of his identity. In addition to transcribing the play and encoding stage directions and historical annotations, I learned how to use the Node XL plug-in to create maps of Lee's relationships to get a better sense of the context in which he lived, worked, and eventually died in squalor and infamy. English 3318 not only opened my eyes to the issues that remain in our canon of literature—it gave me tools to help fix those problems in my own academic career.

Closing Thoughts

In her June 2021 editorial for *College and Research Libraries*, Wendi A. Kaspar extols "flexibility (or adaptability, agility, etc.) and patience" as some of the key attributes of librarians which enable them both historically and today to evolve their work effectively in "response to the [current] environment."[33] By strategically and creatively drawing students and faculty to the library to work in collaborative DH teaching activities, we see an opportunity for librarians to use those skills alongside their in-house digital and pedagogical expertise to help our learning communities to rebuild and thrive afresh in the wake of the pandemic and its challenges.

Appendix 14A. Resources

Further Reading

There are several books and journal issues providing useful guidelines and strategies for teaching with DH, as well as examples of successful DH teaching initiatives; we have found the following helpful:

Some General Works on DH in the Classroom:

- *Using Digital Humanities in the Classroom: A Practical Introduction for Teachers, Lecturers, and Students* (2nd edition) by Claire Battershill and Shawna Ross (London: Bloomsbury Academic, 2022). This updated edition of one of the founding books in this field discusses finding, creating and using digital resources, designing and managing classroom activities and assignments, working with undergraduate and graduate students, ensuring accessibility, and collaboration.
- *Quick Hits for Teaching with Digital Humanities* ed. Christopher J. Young, Michael Morrone, Thomas C. Wilson, and Emma Annette Wilson (Bloomington, IN: Indiana University FACET, 2020). This book provides 24 chapters suggesting ways of incorporating DH techniques into the classroom, from mapping to text encoding, text mining and data mining, to podcasting, creating digital archives, and considering the ways in which these activities can help students to engage in their academic work and their local and global communities.
- *Hacking the Academy: New Approaches to Scholarship and Teaching from Digital Humanities* eds. Daniel J. Cohen and Joseph Thomas Scheinfeldt (Ann Arbor, MI: University of Michigan Press, 2013). The chapters on teaching in this book examine how DH teaching compares with traditional approaches, some potential approaches in the DH classroom, ways in which DH can support digital literacy for students, and how DH can help us all to reimagine the academic archive.
- *Digital Humanities Pedagogy: Practices, Principles, and Politics* ed. by Brett D. Hirsch (Cambridge: Open Book Publishers, 2012) https://doi.org/10.2307/j.ctt5vjtt3. This relatively early book in the field includes contributions about DH and first-year writing programs, DH and the archives, DH mapping, DH text mining, and creating a DH curriculum.

Subject-Specific Works on DH In the Classroom:

- *Shakespeare and Digital Pedagogy: Case Studies and Strategies* ed. Diana E. Henderson and Kyle Sebastian Vitale (London: Bloomsbury Academic, 2021). This book shares 15 methodologies for applying digital techniques when teaching Shakespeare; much of its wisdom would be equally relevant in any literary studies or historically-oriented humanities classroom.
- *Text Analysis with R For Students of Literature* by Matthew L. Jockers (Cham: Springer, 2014). This is a detailed introduction to big data and text mining specifically geared for literature students, with practical how-to steps and clear discussions of the benefits and issues of using these methods.
- *Pastplay: Teaching and Learning History with Technology* ed. Kevin Kee (Ann Arbor, MI: University of Michigan Press, 2014). This book makes a persuasive case that engaging playfully with technology in the humanities classroom, experimenting with new techniques rooted in traditional research, can lead students to a new and more direct understanding of how history is created; as above, many of its strategies would be very effective in any humanities or social sciences classroom.
- *Teaching History in the Digital Age* by T. Mills Kelly (Ann Arbor, MI: University of Michigan Press, 2013). This book is a guide which emphasizes the transformative power of DH in the History classroom, and as above many of the pedagogical approaches advocated would apply successfully in other humanities and social sciences instruction.

Appendix 14B. Sample DH Assignment. Text Encoding Project

Overview

Assignment Text Encoding Project
% of Final Grade 40%
Due In stages as follows:

1. Weekly, entries in your project journal (from 2/24 to 3/30)
2. Cleaned-up transcription of your text and basic TEI encoding (TEI header, structure of your text, basic bibliographic features)
   show me **in class Monday 3/2**
3. Full draft of TEI encoding including draft of researched "Notes" and clear idea for your editorial introduction
   show me **in class Wednesday 3/11**
4. Full final project (encoding including annotations and editorial introduction) **12pm 3/30**

Optional Extra: present your work via your journal and encoding at the South-Central Renaissance Conference

Submission Via Canvas (there should be 3 files: the XML of your encoding, a Word document of your editorial introduction, and a Word document of your project journal)

Objective

We have thought about what it means to consider literature as data from a theoretical perspective; in this assignment, we will learn what it means to actively *treat* literature as data in a practical way through a digital editing and publishing project. Our end product will be a digital edition of rare books or extracts from rare books from the DeGolyer and Bridwell libraries which have not been published in a new edition for over 100 years.

You will learn about the process of creating and publishing books, from early printing techniques to cutting-edge digital literary editing approaches. You will "adopt" a seventeenth-century literary text from a selection put together for this class from the Bridwell and DeGolyer special collections libraries. You will learn how to create a published digital edition of this text using the Text Encoding Initiative (TEI), and you will encode the text with both bibliographic features detailing the book's printed and aesthetic components, and literary features such as character details, structural details (lines, stanzas, acts, etc.) and types of writing such as songs or verse forms. You will also conduct secondary research into the text, its author, and seventeenth-century literary culture, and encode annotations about these details in your digital edition.

You will end up with (a) a digital edition of your text; (b) scholarly annotations encoded in that edition; (c) a short critical introduction to your text in which you make an argument about the value and interest of your text, relating it to critical work about your text or your author.

This project is a combination of traditional literary analysis and digital analytical approaches, giving you an extended opportunity to learn and apply these techniques in a hands-on way with texts that have not been studied in earnest for over 100 years—you have the chance to do some real original research here, and if things go well, we may even try to put together a group scholarly publication about our discoveries.

Instruction Sessions

W 2/19	DeGolyer Library, meeting our rare books
F 2/21	DeGolyer Library, choosing and beginning work on our transcription and encoding
M 2/24	DeGolyer Library, researching your rare book to begin encoding
W 2/26	DeGolyer Library, researching your rare book and encoding
F 2/28	Fondren Library instruction room (3rd floor), introduction to critical research for your annotations and your editorial introduction
M 3/2	DeGolyer Library, rare book research (title pages, printers)

W 3/4	De Golyer Library, rare book research (typography, introductory pages)
F 3/6	Fondren Library, critical research about your author and text
M 3/9	Dallas Hall (regular classroom), encoding your annotations and researching editorial introduction
W 3/11	Dallas Hall (regular classroom), final encoding session and researching editorial introduction plus planning our South-Central Renaissance Conference session
F 3/13	No in-person class, use this time or its equivalent plus homework time over the break to work on your encoding and your editorial introduction
M 3/23	No in-person class, use this time to work on your encoding and editorial introduction
W 3/25	Final in-class consultations for your encoding and editing project

How It Works

We are lucky to have multiple rare book libraries and special collections libraries on campus, and we'll be working with a selection of books from these to create digital editions of books which haven't been published for over 100 years. You'll be encoding specific features of your book to make it into a digital edition.

I'll be introducing you to the *schema* explaining which features of your book you need to encode, and I'll teach you how to do this using TEI and oXygen.

You'll learn how to research both the bibliographic features of your book (its printer, paper, binding, type, illustrations and other graphics) and the literary and historical features of your book (I'll call this "critical research"), and you'll use that training to write and encode scholarly annotations to help someone to understand your book more clearly.

A big part of a digital edition is making a book available to a wide audience: you'll research a *specific* feature or idea within your book and use what you find to write an editorial introduction that shares the thing that you find most exciting, interesting, or just frankly annoying or befuddling about your

book with a public audience. This must be well-researched (details below) but it can be informal in tone to help someone to really become interested and engaged in your book.

Instructions

1. You will sign up to work on a specific section of a rare book from our special collections libraries. You may work in pairs if you wish, but do remember if you pair up that you will be expected to cover the same amount of material each as if you worked separately.
2. You will transcribe your book as plain text with *exactly* the same spelling and punctuation as the original text.
3. You will do basic TEI encoding for your transcribed book. This means completing the TEI header,
4. You will do some more customized TEI encoding for your transcribed book, encoding at least **2 more advanced, specific features** of your text.
5. You will research **5 notes** about your book which you will add via encoding.
6. You will write a short editorial introduction to your text (5-8 pages).
7. You will document your process in weekly journal entries to keep track of your progress and also to have ready to talk from if you participate in the South-Central Renaissance Conference.

By the end of the assignment you will have produced:

1. Basic TEI text encoding of your chosen rare book (TEI header, structural features of the text [lines, stanzas, acts, scenes, and title page as appropriate]).
2. More customized encoding specific to your text; you should encode at least 2 of the following types of feature: bibliographic features (title page, ornamental printing features like florilegia, illustrations); literary features (speakers, names, places, poetic features like a caesura, stage directions, songs, as appropriate); graphics (illustrations, maps); textual difficulties (spelling variations). **Note: this doesn't mean 2 bits of code! It means, you identify 2 of these features and encode them *throughout your book*, *whenever* they occur.** Yes, this is time consuming.
3. At least **5** researched annotations included within your TEI text encoding as "Notes". These may either be on a focused theme or

addressing a range of issues but you **MUST** show depth and variety of research within these—for example, 5 annotations providing the classical origins of Greek gods or goddesses all derived from 1 modern encyclopedia documenting these doesn't necessarily showcase the range and scope of your research abilities, not to mention sustained research, so if this kind of question interests you, call on multiple different primary and secondary sources (e.g. other early modern books, a classical encyclopedia, and a secondary critical work) to bring real value and depth to that work. Either in an annotation or in your editorial introduction you must provide some concise information about the bibliographic details of your book in the technical literary sense of that word, meaning that you must research the printer or printing history of the book.

4. A short (5-8 pages) editorial introduction to your text. In this you should explain to your reader what is interesting about your book and why they might want to read it. You can be very specific here, zooming in on a particular feature or a particular theme which interests you or which you found, honestly, difficult and puzzling about your book, and explaining these to your reader. It may help to think of this part of the assignment as a kind of analytical English essay written for a public audience—this should be approachable and accessible in style, and you can be a little informal if it helps to engage your reader and to explain a complex idea.

   This introduction **must** be supported with critical research—you should be citing at least **7** critical sources which could be critical articles about your topic, rare book catalogues, dictionaries or reference works about printers and printing, *other* rare books by the same author or publishing house, or *other* types of printed item such as the prints in the Bywaters Special Collections.

   You have a lot of freedom in this part of the assignment—the beauty of archival work is finding something that sets your mind on fire and pursuing it. So, here are some possible options:

   – If you discover that the printer who made your book made other interesting rare books, either some which we have here at SMU or which we can find online somewhere else, you might want to write a kind of survey of these to suggest what a bookshelf of his (always his, although it wouldn't be if we worked on texts from the Netherlands) works would have held on it.

- If you discover that your author was writing a very specific type of poem or play (e.g. a revenge tragedy or a Pindaric ode), you could look at other examples and suggest how yours fits in that wider literary landscape.
- If you discover that your author was really scandalous, you could research their biography in detail and relate it to your book and to other writers working in this time period.
- If you discover that your book deals with a famous character, you might want to research how they are portrayed elsewhere and compare and contrast these (e.g., looking at paintings of them from the time period, or looking at other plays, poems, etc. about them).
- If you love the illustrations or graphics in your book, maybe you want to write about how these were produced (you could even substitute this with a video recording of yourself handling some type and printing equipment in the Bywaters Special Collections if you fancy this); maybe you want to compare the prints in your rare book *with* the twentieth-century prints and techniques you saw in the Bywaters Special Collections.
- If you love the binding of your book—that is, the thing its pages are held in, you can research that and see how it compared to others from the period. You may sometimes find out that the book has been rebound over time, and that's something you might be able to research to give someone a sense of what the physical book is like even if they're reading your digital edition.

1. You will document your process in a weekly journal (provided here as an outline to complete). This is an important part of the project to help you to keep track of your research avenues (what worked, what was a dead end, what was interesting but ultimately had to be sacrificed on the altar of hard literary decisions, etc.). This is a chance to have a sense of what it's like to research like an English prof, and I'm really excited and privileged to join you on this journey. Let's see where we go.
2. You can present your work at the South-Central Renaissance Conference; this counts towards your participation grade for the class, and you are welcome to come not only to your session but to the whole conference if you like (or any parts of it that sound interesting). I really encourage you to do so—some of the people whose articles we read for part 1 of the course may well be there and would love to meet you! It's a very friendly event and will look good on your cv, too!

Notes

1. The authors would like to give thanks to Laura R. Braunstein, Arianne Hartsell-Gundy, and Liorah A. Golomb for their scholarly generosity and collegiality during the writing of this chapter. Many thanks to the Hamilton Undergraduate Research Scholars Program at Southern Methodist University for their grant support enabling the completion of this chapter, and to the Department of English at Southern Methodist University for providing funds for RA support for this work. Thanks to Research Librarian and Coordinator of the Intern Program at Southern Methodist University Libraries, Rebecca Graff, for her instrumental support of student research for the assignment detailed in this chapter; to Christina Jensen, Head of Public Services at SMU's DeGolyer Library; University Archivist Joan Gosnell for helping to find suitable texts for students to work with; and to Cindy Boeke, Assistant Director of the Norwick Center for Digital Solutions at SMU, and her team for digitizing everything for the course. We thank Charlotte D. Betts for her invaluable copyediting and bibliographic work in preparing the manuscript. We also thank Brynn Price, recent SMU alumna, for her expert and dexterous copyediting in the final stages of this chapter. Emma Annette Wilson owes deep gratitude to Dr. Tad R. Kosanovich, Heather Stuebe, and the team at Englewood Eyecare & Optical and to Dr. Kelvin Van Voorst, Lauren Foster, and Tara Jullette at Prosper Family Eyecare for supporting her post-pandemic optometrical recovery enabling her to complete this chapter.
2. Beth McMurtrie, "A Stunning Level of Student Disconnection," *The Chronicle of Higher Education*, April 5, 2022, https://www.chronicle.com/article/a-stunning-level-of-student-disconnection.
3. Hannah Matthew, Erla P. Heyns, and Rikk Mulligan, "Inclusive Infrastructure: Digital Scholarship Centers and the Academic Library Liaison," *portal: Libraries and the Academy* 20, no. 4 (2020) 693–714.
4. American Library Association, *2021 State of America's Libraries Special Report: COVID-19*, April 2021, https://www.ala.org/news/sites/ala.org.news/files/content/State-of-Americas-Libraries-Report-2021-4-21.pdf.
5. McMurtrie, "Stunning."
6. CCMH, "Fall 2021: COVID-19 Impact on College Student Mental Health," last modified March 22, 2022, https://ccmh.psu.edu/index.php?option=com_dailyplanetblog&view=entry&year=2022&month=03&day=21&id=23:fall-2021-covid-19-impact-on-college-student-mental-health. Thanks to the *Chronicle* article for drawing our attention to this crucial study.
7. CCMH, "Part 1 of 5: COVID-19 Impact on College Student Mental Health," last modified February 2, 2022, https://ccmh.psu.edu/index.php?option=com_dailyplanetblog&view=entry&year=2021&month=02&day=01&id=9:part-1-of-5-covid-19-s-impact-on-college-student-mental-health.
8. CCMH, "Part 3 of 5: Mental Health Impact of COVID-19 on Various Demographic Groups," last modified February 16, 2022, https://ccmh.psu.edu/index.php?option=com_dailyplanetblog&view=entry&year=2021&month=02&day=15&id=11:part-3-of-5-mental-health-impact-of-covid-19-on-various-demographic-groups.
9. CCMH, "Part 3."
10. American Library Association, *2021 State of America's Libraries Special Report: COVID-19*, 20. This report has great information about the wide range of services provided by libraries during the pandemic, including the provision of educational

resources such as e-books and learning packets and being a Wi-Fi hub even if physical buildings had to remain closed in an attempt to help mitigate information redlining.

11. Courtney Rivard, "Turning Archives into Data: Archival Rhetorics and Digital Literacy in the Composition Classroom," *College Composition and Communication* 70, no. 4 (June 2019): 527–559.
12. Paul Fyfe, "Reading, Making, and Metacognition," *Digital Humanities Quarterly* 12, no. 2 (2018).
13. Leslie Sanders, Liam Rodrigues, and Kay Li, "Enhanced Student Engagement and Culturally Responsive Pedagogy: Innovations in the SAGGITARIUS-ORION-Shaw Literature Digitizing Pilot Project," *Shaw*, 36, no. 2 (2016): 272–289.
14. Clayton McCarl, "Editing the Eartha M. M. White Collection: An Experiment in Engaging Students in Archival Research and Editorial Practice," *The Journal of Academic Librarianship* 44, no. 4 (2018): 527–537.
15. Barry M. Goldenberg, "Rethinking Historical Practice and Community Engagement: Research Together with 'Youth Historians,'" *Rethinking History* 23, no. 1 (2019): 52–77.
16. Kayo Denda and Jennifer Hunter, "Building 21st Century Skills and Creating Communities: A Team-based Engagement Framework for Student Employment in Academic Libraries," *Journal of Library Administration* 56 (2016): 251–265.
17. As Bronkar notes, libraries had of course long been creators of knowledge, but the maker movement underscored how they could be perceived that way in terms of technology as well as in other contexts. Cherie Bronkar, "The Current State of Library Makerspaces." In *Makerspaces In Practice: Successful Models For Implementation*, ed. Ellyssa Kroski (Chicago: ALA Editions, 2021), 2.
18. Morgan Spehar, "Libraries' Reflections after a Year of COVID-19," last modified April 14, 2021, https://www.ohio.edu/library/about/news-events/all-news/libraries-reflections-after-year-covid-19.
19. Linda K. Colding and Rich Gause, "Reimagining Services for the University Community during a Pandemic: A Tale of Two Florida Academic Libraries," last modified May 12, 2021, https://www.elsevier.com/connect/library-connect/reimagining-services-for-the-university-community-during-a-pandemic-a-tale-of-two-florida-academic-libraries.
20. Heather Kapanka, "School Library Media Specialists: An Evolving Profession in a Pandemic," *IFLA Journal*, 48, no. 1 (March 2022): 41–49.
21. American Library Association, *2021 State of America's Libraries Special Report: COVID-19,* 5.
22. Joan Lippincott, Harriette Hemmasi, and Viv Lewis, "Trends in Digital Scholarship Centers," *Educause Review*, June 16, 2014, https://er.educause.edu/articles/2014/6/trends-in-digital-scholarship-centers.
23. Megan Senseney, Eleanor Dickson Koehl, and Leanne Nay, "Collaboration, Consultation, or Transaction: Modes of Team Research in Humanities Scholarship and Strategies for Library Engagement," *College & Research Libraries* 80, no. 6 (2019): 793.
24. John Cox, "Communicating New Library Roles to Enable Digital Scholarship: A Review Article," *New Review of Academic Librarianship* 22, nos. 2–3 (2016): 137–138.
25. Heather Moulaison Sandy, A. J. Million, and Cynthia Hudson-Vitale, "Innovating Support for Research: The Coalescence of Scholarly Communication?" *College & Research Libraries* 81, no. 2 (2020): 195.
26. Shannon Lucky and Craig Harkema, "Back to Basics: Supporting Digital Humanities and Community Collaboration Using the Core Strength of the Academic Library," *Digital Library Perspectives* 34, no. 3 (2018): 193; A. Miller, "Innovative management

Strategies for Building and Sustaining a Digital Initiatives Department with Limited Resources," *Digital Library Perspectives* 34, no. 2 (2018): 118.

27. Yin Zhang, Fangi Su, and Brenna Hubschman, "A Content Analysis of Job Advertisements for Digital Humanities-Related Positions in Academic Libraries," *The Journal of Academic Librarianship* 47 (2021), 372.
28. Laura Braunstein, "6 Questions for a Digital Humanities Librarian", interview by Joshua Kim, Inside Higher Ed (blog), August 17, 2016, https://www.insidehighered.com/blogs/technology-and-learning/6-questions-digital-humanities-librarian.
29. Harriett E. Green, "Facilitating Communities of Practice in Digital Humanities: Librarian Collaborations for Research and Training in Text Encoding," *Library Quarterly: Information, Community, Policy* 84, no. 2 (2014): 219–234.
30. McCarl, "Editing the Eartha M. M. White Collection," 527–537; Francesca Gianetti, "'So Near While Apart': Correspondence Editions as Critical Library Pedagogy and Digital Humanities Methodology," *The Journal of Academic Librarianship* 45 (2019): 1–11.
31. Laura Estill, "Encoding the Edge: Manuscript Marginalia and the TEI," *Digital Literary Studies* 1, no. 1 (5 May 2016), https://journals.psu.edu/dls/article/view/59715/59912.
32. Many thanks to Christina for the weird Pindaric odes about plagues, all too appropriate, and taxation!
33. Wendi A. Kaspar, "Adapt and Thrive," *College & Research Libraries* 82, no. 4 (2021), https://crl.acrl.org/index.php/crl/article/view/24982/32848.

Bibliography

American Library Association. *2021 State of America's Libraries Special Report: COVID-19*. April 2021. https://www.ala.org/news/sites/ala.org.news/files/content/State-of-Americas-Libraries-Report-2021-4-21.pdf.

Bronkar, Cherie. "The Current State of Library Makerspaces" In *Makerspaces In Practice: Successful Models for Implementation*, edited by Ellyssa Kroski, 1–18. Chicago: ALA Editions, 2021.

Colding, Linda K. and Rich Gause. "Reimagining Services for the University Community during a Pandemic: A Tale of Two Florida Academic Libraries." Last modified May 12, 2021. https://www.elsevier.com/connect/library-connect/reimagining-services-for-the-university-community-during-a-pandemic-a-tale-of-two-florida-academic-libraries.

Cox, John. "Communicating New Library Roles to Enable Digital Scholarship: A Review-Article." *New Review of Academic Librarianship* 22, nos. 2–3 (2016): 132–147.

Denda, Kayo and Jennifer Hunter, "Building 21st Century Skills and Creating Communities: A Team-based Engagement Framework for Student Employment in Academic Libraries," *Journal of Library Administration* 56 (2016): 251–265.

Estill, Laura. "Encoding the Edge: Manuscript Marginalia and the TEI." *Digital Literary Studies* 1, no. 1 (5 May 2016), https://journals.psu.edu/dls/article/view/59715/59912.

Fyfe, Paul. "Reading, Making, and Metacognition." *Digital Humanities Quarterly* 12, no. 2 (2018).

Gianetti, Francesca. "'So Near While Apart': Correspondence Editions as Critical Library Pedagogy and Digital Humanities Methodology." *The Journal of Academic Librarianship* 45 (2019): 1–11.

Goldenberg, Barry M. "Rethinking Historical Practice and Community Engagement: Research Together with 'Youth Historians.'" *Rethinking History* 23, no. 1 (2019): 52–77.

Green, Harriett E. "Facilitating Communities of Practice in Digital Humanities: Librarian Collaborations for Research and Training in Text Encoding." *Library Quarterly: Information, Community, Policy* 84, no. 2 (2014): 219–234.

Kapanka, Heather. "School Library Media Specialists: An Evolving Profession in a Pandemic." *IFLA Journal* 48, no. 1 (March 2022): 41–49.

Kaspar, Wendi A. "Adapt and Thrive." *College & Research Libraries* 82, no. 4 (2021), https://crl.acrl.org/index.php/crl/article/view/24982/32848.

Lippincott, Joan, Harriette Hemmasi, and Viv Lewis. "Trends in Digital Scholarship Centers." *Educause Review*, June 16, 2014. https://er.educause.edu/articles/2014/6/trends-in-digital-scholarship-centers.

Lucky, Shannon and Craig Harkema, "Back to Basics: Supporting Digital Humanities and Community Collaboration Using the Core Strength of the Academic Library," *Digital Library Perspectives* 34, no. 3 (2018): 188-199.

Matthew, Hannah, Erla P. Heyns, and Rikk Mulligan. "Inclusive Infrastructure: Digital Scholarship Centers and the Academic Library Liaison." *portal: Libraries and the Academy* 20, no. 4 (2020): 693–714. https://doi.org/10.1353/pla.2020.0033.

McCarl, Clayton. "Editing the Eartha M.M. White Collection: An Experiment in Engaging Students in Archival Research and Editorial Practice." *The Journal of Academic Librarianship* 44 (2018): 527–537.

McMurtrie, Beth. "A stunning level of student disconnection." *The Chronicle of Higher Education*. April 5, 2022. https://www.chronicle.com/article/a-stunning-level-of-student-disconnection.

Miller, A. "Innovative Management Strategies for Building and Sustaining a Digital Initiatives Department with Limited Resources." *Digital Library Perspectives* 34, no. 2 (2018): 117–136.

Penn State Student Affairs Center for Collegiate Mental Health (CCMH). "Fall 2021: COVID-19 Impact on College Student Mental Health." Last modified March 22, 2022. https://ccmh.psu.edu/index.php?option=com_dailyplanetblog&view=entry&year=2022&month=03&day=21&id=23:fall-2021-covid-19-impact-on-college-student-mental-health.

———. "Part 1 of 5: COVID-19 Impact on College Student Mental Health." Last modified February 2, 2022, https://ccmh.psu.edu/index.php?option=com_dailyplanetblog&view=entry&year=2021&month=02&day=01&id=9:part-1-of-5-covid-19-s-impact-on-college-student-mental-health

———. "Part 3 of 5: Mental Health Impact of COVID-19 on Various Demographic Groups." Last modified February 16, 2022, https://ccmh.psu.edu/index.php?option=com_dailyplanetblog&view=entry&year=2021&month=02&day=15&id=11:part-3-of-5-mental-health-impact-of-covid-19-on-various-demographic-groups

Rivard, Courtney. "Turning Archives into Data: Archival Rhetorics and Digital Literacy in the Composition Classroom." *College Composition and Communication* 70, no. 4 (June 2019): 527–559.

Sanders, Leslie, Liam Rodrigues, and Kay Li. "Enhanced Student Engagement and Culturally Responsive Pedagogy: Innovations in the SAGGITARIUS-ORION-Shaw Literature Digitizing Pilot Project," *Shaw* 36, no. 2 (2016): 272–289.

Sandy, Heather Moulaison, A. J. Million, and Cynthia Hudson-Vitale. "Innovating Support for Research: The Coalescence of Scholarly Communication?" *College & Research Libraries* 81, no. 2 (2020): 193–214.

Senseney, Megan, Eleanor Dickson Koehl, and Leanne Nay. "Collaboration, Consultation, or Transaction: Modes of Team Research in Humanities Scholarship and Strategies for Library Engagement." *College & Research Libraries* 80, no. 6 (2019): 787–804.

Spehar, Morgan. "Libraries' Reflections after a Year of COVID-19." Last modified April 14, 2021. https://www.ohio.edu/library/about/news-events/all-news/libraries-reflections-after-year-covid-19.

Zhang, Yin, Fangi Su, and Brenna Hubschman. "A Content Analysis of Job Advertisements for Digital Humanities-Related Positions in Academic Libraries." *The Journal of Academic Librarianship* 47 (2021).

APPENDIX

Tools and Resources Referenced in this Book

Tools

Adobe Creative Cloud (https://www.adobe.com/creativecloud.html). Commercial cloud-based host of Adobe products such as Photoshop, Illustrator, and InDesign.

AntConc (http://www.laurenceanthony.net/software/antconc/). Freeware corpus analysis toolkit for concordancing and text analysis.

ArcGIS (https://www.arcgis.com/index.html). Commercial tool for creating interactive maps.

CanvasX (https://www.canvasgfx.com/products/canvas-x-pro). Commercial tool for creating, enhancing, and sharing technical illustration. *Updated link.*

Cascading Style Sheets (CSS) (http://www.w3.org/Style/CSS/Overview.en.html). A simple mechanism for adding styles (e.g., fonts, colors, spacing) to Web documents.

Constellate (https://constellate.org/). Text analytics service for educational institutions.

Design Thinking for Libraries (https://designthinkingforlibraries.com/). Offers toolkit to introduce a way of working for librarians to understand the needs of patrons and engage with communities.

Digital Research Tools (DiRT) (http://dirtdirectory.org/ [site discontinued]). Registry of digital research tools for scholarly use. *No longer supported; see TAPoR below.*

Dublin Core (http://dublincore.org/documents/dcmi-terms/). A set of standardized metadata terms maintained by the Dublin Core Metadata Initiative (http://dublincore.org/).

English Corpora (https://www.english-corpora.org/). Gateway to various English-language corpora.

Final Cut Pro (https://www.apple.com/final-cut-pro/). Commercial Apple software for video editing.

Gale Digital Scholar Lab (https://www.gale.com/primary-sources/digital-scholar-lab). Analyzes a library's Gale Primary Sources holdings.

Google Books (https://books.google.com/). Index of full-text public domain and open access books, with fragments of in-copyright texts.

Google Forms (https://www.google.com/forms/about/). Web-based tool for creating and analyzing forms and surveys.

Google MyMaps (https://www.google.com/maps/d/). Web-based map making tool.

Google Ngram Viewer (https://books.google.com/ngrams/). Charts frequencies of search strings using Google Books as a corpus.

HathiTrust Research Center Analytics (https://analytics.hathitrust.org/). Supports large-scale computational analysis of the works in the HathiTrust Digital Library to facilitate non-profit and educational research.

Inclusive Design Guide (https://guide.inclusivedesign.ca/). Openly-available guide that can be applied to digital design.

Inclusive Design Toolkit (http://www.inclusivedesigntoolkit.com/). Tools and information provided by the University of Cambridge.

Java (https://www.java.com/en/). Object-oriented programming language.

Jing (now TechSmith Capture). (https://www.techsmith.com/jing-tool.html). Computer service that lets you capture basic video, animation, and still images, and share them on the web. *Updated link*.

Juxta. Open-source tool for comparing and collating multiple witnesses to a single textual work. *Defunct, no longer available*.

MALLET: MAchine Learning for LanguagE Toolkit (https://mimno.github.io/Mallet/index). Java-based package for statistical natural language processing, document classification, clustering, topic

modeling, information extraction, and other machine learning applications to text.

Mukurtu (https://mukurtu.org/). Free, mobile, and open source platform built with Indigenous communities to manage and share digital cultural heritage.

NodeXL (https://www.smrfoundation.org/nodexl/). Add-ins for Microsoft Excel that support social network and content analysis.

NVivo (https://lumivero.com/products/nvivo/). Commercial qualitative analysis software.

Omeka (http://omeka.org/). Free, flexible, and open source web-publishing platform for the display of library, museum, archives, and scholarly collections and exhibitions.

oXygen (https://www.oxygenxml.com/). Suite of commercial software for web authoring and XML editing.

Poem Viewer (https://nms.kcl.ac.uk:8443/poemvis/index.html). Web-based tool for visualizing poems in support of close reading.

PowerPoint (https://www.microsoft.com/en-us/microsoft-365/powerpoint). Commercial presentation software.

Python (https://www.python.org/). Open source programming language for beginner and experienced programmers.

Scalar (https://scalar.me/anvc/scalar/). Open source publishing tool.

Semantic Web (http://www.w3.org/standards/semanticweb/). A web of data such as would be found in databases. A project of the World Wide Web Consortium (W3C) (http://www.w3.org/Consortium/).

Siteimprove (https://www.siteimprove.com/toolkit/accessibility-checker/). Website accessibility checker.

Slack (https://slack.com/). Group chat software.

Tableau (https://www.tableau.com/). Commercial visual analytics platform.

Teaching Beyond Quarantine (formerly Teaching During Quarantine) (https://teachdh.sdsu.edu/). Resources for virtual instruction.

Toolkit for Digital Scholarship (https://dsseditoolkit.hcommons.org/edi-work-in-the-acrl-digital-scholarship-section/). Provides an overview of DSS efforts to engage in anti-oppressive and anti-racist work.

Transkribus (https://readcoop.eu/transkribus/). AI-powered platform for text recognition, transcription and searching of historical documents – from any place, any time, and in any language.

Value Sensitive Design Lab toolkits (https://vsdesign.org/toolkits/). Toolkits and other creative works to enable designers, engineers, technologists, educators, researchers, and more to foreground human values in their processes.

Voyant (http://voyant-tools.org/). Web-based reading and analysis environment for digital texts.

Windows Movie Maker. *Movie editing software, now discontinued.*

WordPress (https://wordpress.com/). Free hosting platform for websites and blogs.

Zoom (https://zoom.us/). Video conferencing software.

Tutorials and Online Courses

Lynda.com. Online learning environment. Now LinkedIn Learning with Lynda (https://www.linkedin.com/learning/). Paid LinkedIn account required.

Moodle (https://moodle.org/). Open source learning platform.

Programming Historian (https://programminghistorian.org/). Tutorials that help humanists learn a wide range of digital tools, techniques, and workflows to facilitate research and teaching.

Communities and Professional Organizations

Association for Computers and the Humanities—Digital Humanities Questions and Answers (http://digitalhumanities.org/answers/) and its Twitter feed (@DHanswers). Community-run Q&A board for digital humanities questions.

Design Justice Network (https://designjustice.org/). Advances practices that center those who are normally excluded from and adversely impacted by design decisions in design processes.

dh + lib (@DHandLib). Where the digital humanities and librarianship meet. A project of the ACRL DH Interest Group.

HuMetricsHSS: Humane Metrics Initiative (https://humetricshss.org/). Creates and supports values-enacted frameworks for understanding and evaluating all aspects of the scholarly life well-lived and for promoting the nurturing of these values in scholarly practice.

Humans in User Experience at UCLA (https://hue.humspace.ucla.edu/). UCLA-based network aiming to bring humans together in UX/UI Design.

IDEO.org (https://www.ideo.org/). Nonprofit design studio offering products and services alongside organizations that are committed to creating a more just and inclusive world.

Nielsen Norman Group (https://www.nngroup.com/). UX research and consulting firm.

Roy Rosenzweig Center for History and New Media (https://rrchnm.org/). Create digital history and software that is free and fully available to all.

Scholars' Lab (https://scholarslab.lib.virginia.edu/). Lab for the practice of experimental scholarship offering mentoring, collaboration, and a safe and supportive community experience.

Stanford Literary Lab (https://litlab.stanford.edu/). Research collective that applies computational criticism to the study of literature.

Text Encoding Initiative (http://www.tei-c.org/index.xml). A consortium which collectively develops and maintains a standard for the representation of texts in digital form.

THATCamp (The Humanities and Technology Camp) (http://thatcamp.org/). Inexpensive meetings where humanists and technologists of all skill levels learn and build together in sessions proposed on the spot. *No longer active*.

Usability.gov (https://www.usability.gov/how-to-and-tools/index.html). Guidance on building better digital services in government. Archived, no longer updated. Now Digital.gov (https://digital.gov/).

User Experience Professionals Association (https://uxpa.org/about-uxpa-international/). Supports people who research, design, and evaluate the user experience (UX) of products and services.

UX at University of Arizona (https://uxua.arizona.edu/). Learning community that aims to grow UX expertise at the University of Arizona and beyond.

About the Authors

Tami Albin is an associate librarian in the Center for Faculty/Staff Initiatives and Engagement at the University of Kansas. Her areas of expertise and research include oral history, information literacy, LGBTQ studies, and digital humanities.

Laura Braunstein is the Head of Digital Scholarly Engagement at Dartmouth College, where she co-leads Digital by Dartmouth Library, the Library's digital collections program. She has a doctorate in English from Northwestern University, where she taught writing and literature classes. She is also a crossword puzzle constructor and co-founded The Inkubator, https://inkubatorcrosswords.com/, a project that publishes crosswords by women and non-binary puzzlemakers.

Mark Dahlquist is the subject liaison librarian for the departments of English and Media, Journalism & Film at Miami University. He holds PhD and MLIS degrees (both from the University of Illinois at Urbana-Champaign) and has interests in early modern literature and digital textual analysis.

Frances Devlin retired from the University of Kansas Libraries, where she was a Librarian in the Center for Faculty/Staff Initiatives and Engagement.

Heather Froehlich supports digital scholarship activities across the University of Arizona, with a particular focus on text and data mining. Prior to being Digital Scholarship Specialist at University of Arizona Libraries she was the Literary Informatics Librarian at Penn State University Libraries.

Brian Wade Garrison is the business and government documents librarian at the University of North Dakota. Prior to that he was an Assistant Librarian in the Center for Faculty/ Staff Initiatives and Engagement at the University of Kansas.

Katie Gibson is a humanities and area studies librarian and serves as liaison to language and area studies departments as well as the Individualized Studies Program at Miami University. She also serves as a member of the Digital Humanities Interest Group in the Libraries.

Liorah Golomb is associate professor and humanities librarian at the University of Oklahoma. She holds a doctorate in Drama from the University of Toronto and earned her MLIS at Pratt Institute. She has published several articles and chapters both within and outside of the field of librarianship. Her most recent publication is "'My Kind of Librarian or Your Kind of Library?': Information Seeking Behavior in *Supernatural*," in *Loremasters and Libraries in Fantasy and Science Fiction*," edited by Jason Fisher and Janet Brennan Croft.

Arianne Hartsell-Gundy is the Head, Humanities and Social Sciences Department and librarian for literature at Duke University. She was previously the humanities librarian at Miami University in Ohio. She holds a dual masters degree in Comparative Literature/Library Science at Indiana University, and a BA in English from the University of Missouri-Columbia. Her research interests include information literacy, graduate student pedagogy, collection analysis, and digital humanities. She is the co-editor of *Learning in Action: Designing Successful Graduate Student Work Experiences in Academic Libraries* and *Digital Humanities in the Library: Challenges and Opportunities for Subject Specialists*.

Kathleen A. Johnson, emerita professor of libraries, was the subject librarian for English at University Libraries, University of Nebraska-Lincoln, from the beginning of digital scholarship until her retirement in 2020. She participated in early planning and development of electronic scholarship at UNL and has continued to engage with digital humanities post-retirement.

Maria Katsulos (she/they) is an incoming History PhD student at Northwestern University studying gender, sex, and sexuality in 17th-century English and Scottish art and literature. A member of Phi Beta Kappa, Phi Alpha Theta, and Eta Sigma Phi honors societies, she graduated magna cum laude from Southern Methodist University in May 2022 with degrees in History and English. They thank Dr. Wilson for all her academic guidance throughout undergrad.

Pamella R. Lach is the digital humanities librarian at San Diego State University. She is director of the Digital Humanities Center and co-director of the campuswide digital humanities initiative. She has a PhD in US history from the University of North Carolina at Chapel Hill and a MSIS from UNC's School of Information and Library Science.

Kathleen A. Langan is professor and engagement librarian at Western Michigan University. Her work focuses on programming for students and the broader university community.

Elizabeth Lorang is associate professor of libraries at the University of Nebraska-Lincoln Libraries, where she directs and contributes to several digital humanities activities. She is the principal investigator of the NEH-funded project "Image Analysis for Archival Discovery," directs the Digital Scholarship Incubator, and co-teaches the Digital Humanities Practicum course. Liz also is a co-director of Civil War Washington.

Russell Hugh McConnell received his BA in English from the University of Calgary and his MA and PhD in English from the University of Western Ontario, with graduate funding from the Social Sciences and Humanities Research Council of Canada. He has taught at universities in Canada and the United States, working with a diverse range of students in multiple honors programs, and is currently a Lecturer at Southern Methodist University. His teaching experience includes composition, British literature, American literature, Shakespeare, comic books and graphic novels, and poetry, as well as liberal arts courses on epistemology and ethics. Recently he taught a seminar session on text mining as part of SMU's inaugural Digital Scholarship Challenge Grants supporting original research projects being conducted by students in the English Distinction program.

A. Miller is an associate professor and leads the Digital Scholarship Initiatives unit at Middle Tennessee State University. Educational and professional experience includes information science, user experience design, open access publishing, and equity, diversity, inclusion, and accessibility practices.

Ginny Moran is the special collections librarian and a liaison librarian to arts and humanities departments at Macalester College. She holds an MLIS from the University of Wisconsin-Milwaukee and a graduate certificate in Information Systems from the University of St. Thomas, and has worked at

the intersections of technology and education throughout her professional career.

Paige Morgan is the digital publishing and copyright librarian and head of digital initiatives and preservation at the University of Delaware Library, Museums & Press. Her current work focuses on the intersection of digital scholarship and scholarly communications: thinking about how digital scholarship and digital collections materials can be made as easily accessible as possible for researchers to use, adapt, and build on. You can find her writing at *DH+Lib*, *Archive Journal*, *Romanticism*, and in *College and Undergraduate Libraries*.

David Oberhelman, to whom this work is dedicated, was the W. P. Wood Professor of Library Service at the Oklahoma State University Library until his death in 2018. He earned a PhD in English with an emphasis in critical theory from the University of California, Irvine, and a MLIS degree from the University of Pittsburgh. He was the subject specialist for the humanities and performing arts, and he published and presented a range of topics including Anglo-American fantasy and science fiction, digital humanities in pedagogy, and changes in scholarly communications in the humanities.

Johanna Pang graduated from SMU in 2022 with BAs in English and human rights. She now attends SMU Dedman School of Law, where she received full funding from the Luce Scholarship. After practicing within the area of child advocacy, she hopes to become a judge in a family court. Johanna finds joy in writing meditative poetry and taking scenic walks in her free time.

Angela Perkins is a research and instruction librarian and director of the Digital Humanities Summer Scholars (DHSS) Program at the Lafayette College Libraries. She earned her masters in information studies (MSIS) in May 2017 from The University of Texas at Austin, where she specialized in archives and preservation. She also holds an MFA in screenwriting from The American Film Institute (AFI) Conservatory, and a BA in political science from Bates College.

Caro Pinto is a research and instruction librarian at Mount Holyoke College. She has written about digital humanities, critical information literacy, project management, mid-career librarianship, and mindfulness in several book chapters, *The Chronicle of Higher Education*, and *In the Library with the Lead*

Pipe among other publications. She is also an avid New England sports fan and an amateur car mechanic.

Dot Porter holds master's degrees in medieval studies and library science and started her career working on image-based digital editions of medieval manuscripts. She has worked on a variety of digital humanities projects over a decade-long career, focusing on materials as diverse as ancient texts and Russian religious folklore, providing both technical support and scholarly expertise. From 2010 until March 2013, she was the associate director for digital library content and services at the Indiana University Bloomington Libraries, where she led in planning and implementing new services to support librarians and faculty in the creation of digital projects. She has also worked for the Digital Humanities Observatory at the Royal Irish Academy, and the Collaboratory for Research in Computing for Humanities at the University of Kentucky.

Jenny Presnell is humanities/social sciences librarian with over 37 years of liaison, instruction, reference, and collection development experience in the humanities, including history, American studies and women's studies. She is a former member of the Libraries' Digital Humanities Working Group. She has published a book, *The Information-Literate Historian* (3rd ed. Oxford, 2013) to help history undergraduate students learn to do advanced research, using both traditional and digital methodologies.

Aisling Quigley is the Andrew W. Mellon Postdoctoral Fellow in Digital Liberal Arts at Macalester College, where she teaches digital humanities courses and leads digital initiatives for faculty. She holds a PhD in information culture and data stewardship from the School of Computing and Information at the University of Pittsburgh.

Brian Rosenblum is founding co-director of the Institute for Digital Research in the Humanities, and librarian for digital scholarship at the University of Kansas Libraries, where he has administrative, production, and outreach responsibilities in support of a variety of digital initiatives and publishing services. Prior to joining KU Libraries' digital initiatives program in 2005 he worked at the Scholarly Publishing Office at the University Library, University of Michigan, where he helped develop electronic journals and digital scholarly projects.

John Russell is digital humanities librarian and associate director for the Center for Virtual/Material Studies at The Pennsylvania State University. From 2015 until 2022, he taught digital humanities professional development courses for Library Juice Academy. He has written and presented on digital humanities pedagogy and digital art history and is Editor-in-Chief of *dh+lib*.

Brooke Schmolke is the digital resource center manager at Macalester College. She works with students, faculty, and staff around a wide variety of technologies.

Louann Terveer, MLIS, is the digital initiatives and scholarly communication librarian at Macalester College. Her academic interests include digital publishing and open access publishing models.

Ilse Schweitzer VanDonkelaar is an assistant professor in the Department of Writing, Rhetoric, and American Cultures at Michigan State University, where she teaches first-year writing and courses on environmental writing and on Viking history and culture. Previously, she taught courses in mythologies and environmental writing at Grand Valley State University in Michigan. She is also an editor for the journal *Medieval Ecocriticisms*, offered by Medieval Institute Publications, and is a member of the advisory board for the series Environmental Humanities in Premodern Cultures, published by Amsterdam University Press. She received her PhD in English from Western Michigan University.

Anu Vedantham is assistant university librarian for research, teaching and social sciences and liaison for Indigenous studies at the Princeton University Library. She enjoys helping individuals with research and teaching needs and being an active member of the university community. She has authored articles and book chapters about student-centered librarianship, library space design and media literacy. She holds her doctorate in higher education management from Penn's Graduate School of Education, her New Jersey principal certificate, her masters in public affairs from the School of Public and International Affairs at Princeton University and her bachelors and masters in electrical engineering and computer science from the Massachusetts Institute of Technology (MIT).

Emma Annette Wilson is assistant professor at Southern Methodist University where she started the digital humanities program in the English

department. Previously, as digital scholarship librarian at the University of Alabama, she grew the Digital Humanities Center from having five to over 140 projects and established the conference Digitorium. On the basis of her experience, she published the textbook *Digital Humanities for Librarians* (Rowman & Littlefield, 2020) and the co-edited collection *Teaching with Digital Humanities* (University of Indiana Press, 2020). In 2022 she started SMU's Digital Scholarship Challenge Grant program, engaging students in frontline research focused on DEIA questions, and she is passionate about empowering people at all different educational stages to benefit from all the collaborative, generative, creative possibilities that digital humanities offers. She gives deep thanks to the Hamilton Undergraduate Research Scholars Program at Southern Methodist University for their grant support enabling the completion of portions of this chapter.